BOUNDEDNESS OF COMPLEMENTS FOR LOG CANONICAL PAIRS

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Abstract. We prove Shokurov's conjectures on the boundedness of complements for $\mathbb{R}$-complementary pairs that are either with DCC coefficients or klt $\mathbb{R}$-complementary, without assuming the Fano type condition.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Complements were introduced by Shokurov in his construction of threefold log flips [Sho92]. See [Pro01] for an introduction. Roughly speaking, an n-complement of a pair (X, B) is a pair (X, B^+) such that (X, B^+) is lc, $n(K_X + B^+) \sim 0$, and B^+ is bounded from below in terms of B and n (Definition 2.10). Shokurov's conjecture on the boundedness of complements first appeared in [Sho00, 1.3]. It predicts that any pair of dimension d over a base Z which satisfies certain conditions has an n-complement over a neighborhood of a given point $z \in Z$ for some n in a finite set of positive integers depending only on d.

The conditions discussed in [Sho00, 1.3] are the existence of some complement for pairs with standard coefficients, that is, pairs with coefficients in $\{1 - \frac{1}{j} \mid j \in \mathbb{Z}_{>0}\} \cup \{1\}$, with the additional (weak) Fano type condition. Shokurov also suggested that "the existence of some complement" alone might suffice. In modern terminology, this stands for $\mathbb{R}$-*complementary*.

We formulate Shokurov's boundedness of complements conjecture in the form of [CHL24, Conjecture 1.3] (cf. [Sho00, 1.3] and [CH21, Conjecture 6.1]). We refer the reader to Definitions 2.1, 2.10, and 2.12 for related definitions and terminology.

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Conjecture 1.1 (Shokurov)**.** *Let d be a positive integer and $\Gamma \subset [0,1]$ a DCC set. Then there exists a positive integer n depending only on d and Γ satisfying the following. Let $(X/Z \ni z, B)$ be an $\mathbb{R}$-complementary pair of dimension d such that $B \in \Gamma$. Then $(X/Z \ni z, B)$ has an n-complement. Moreover, if $\overline{\Gamma} \subset \mathbb{Q}$, then $(X/Z \ni z, B)$ has a monotonic n-complement.*

Conjecture 1.1 considers pairs with DCC coefficients and asks for a uniform positive integer n. [Sho20, Conjecture 1] further suggests that we may remove the condition "$\Gamma \subset [0,1]$ is a DCC set". In this case, we need to assume that $(X/Z \ni z, B)$ is klt $\mathbb{R}$-complementary, and we need to choose n in a finite set instead of a single uniform n. We formulate the conjecture as follows.

Conjecture 1.2. *Let d and l be positive integers. Then there exists a finite set $\mathcal{N}$ of positive integers divisible by l, depending only on d and l, satisfying the following. Let $(X/Z \ni z, B)$ be a klt $\mathbb{R}$-complementary pair of dimension d. Then $(X/Z \ni z, B)$ has an n-complement for some $n \in \mathcal{N}$.*

Most known cases of Conjectures 1.1 and 1.2 assume that X is of Fano type over Z, that is, there exists a boundary Δ such that (X, Δ) is klt and $-(K_X + \Delta)$ is big and nef over Z [PS09, Lemma-Definition 2.6]. When X is of Fano type over Z, the following results are known.

(1) Hyperstandard coefficients. Let $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ be a finite set such that $0, 1 \in \mathfrak{R}$. It was proved in [Bir19, Theorem 1.7] that there exists a positive integer n depending only on d and $\mathfrak{R}$ such that any projective lc pair (X, B) of dimension d with X of Fano type, $B \in \Phi(\mathfrak{R})$, and $-(K_X + B)$ nef has a monotonic n-complement. The relative version over a contraction is [Bir19, Theorem 1.8].
(2) DCC coefficients. Conjecture 1.1 holds when X is of Fano type over Z, and n can moreover be taken divisible by any given positive integer l [HLS19, Theorem 1.8]. Indeed, the monotonicity in [HLS19, Theorem 1.8] holds under a rationality condition on Γ which is satisfied when the closure of Γ is contained in $\mathbb{Q}$. A version with decomposable $\mathbb{R}$-complements is [HLS26, Theorem 1.5].
(3) Generalized pairs. For generalized pairs [FM20, Definition 2.2] of Fano type, bounded complements were obtained in [Che23, Theorem 1.1] for coefficients in a DCC set, and in [Bir19, Theorem 1.10] for coefficients in a hyperstandard set $\Phi(\mathfrak{R})$.
(4) Coefficients in $[0,1]$. [Sho20, Theorem 3] proved Conjecture 1.2 when X is of Fano type over Z. The result was later extended to generalized pairs of Fano type [CHHX25, Theorems 1.1 and 8.1].

We refer the reader to [Has26] for generalizations of the aforementioned results for projective morphisms between complex analytic spaces.

Before [Bir19], the case of surfaces was treated in [Sho00] and the relative threefold case in [PS01]. It was shown in [PS09, Theorem 1.4] that, for projective klt pairs (X, B) with X of Fano type, $-(K_X + B)$ nef, and $B \in \Phi(\mathfrak{R})$, bounded complements follow from the minimal model program, the boundedness of Fano varieties, and the Prokhorov–Shokurov $\boldsymbol{b}$-semiampleness conjecture [PS09, Conjecture 7.13]. It was shown in [PS01, Theorem 3.1] that, for klt pairs $(X/Z \ni z, B)$ of dimension d with standard coefficients, $\dim Z > 0$, and $-(K_X + B)$ nef and big over Z, the existence of bounded complements reduces, assuming the minimal model program in dimension d, to the case when Z is a point in dimension $d-1$. The theorems of [Bir19] are a key input in the proof of the boundedness of Fano varieties [Bir21], and complements are used in the K-stability theory of Fano varieties, see for example [LXZ22].

We now explain where the Fano type assumption enters the proof of [Bir19, Theorem 1.7], following the outline in [Bir19, pp. 353–354]. It is used, among other places, in the following three steps. (i) Since X is of Fano type, an MMP can be run on any divisor on X, and the proof runs an MMP on the negative of a log canonical divisor. (ii) When (X, B) has a non-klt center, a complement on an lc center is lifted to X by the Kawamata–Viehweg vanishing theorem, which requires $-(K_X + B)$ to be nef and big. (iii) When X admits a fibration, the problem is reduced to a generalized pair on the base, and complements of generalized pairs of Fano type [Bir19,

Theorem 1.10] are used on the base. Both [Bir19, Theorem 1.10] and the later relative version [FM20, Theorem 1.2] require the underlying variety of the base to be of Fano type.

Without the Fano type assumption, step (i) depends on the existence of good minimal models of lc pairs. In step (ii), $-(K_X+B)$ can be the pullback of an ample divisor on a lower-dimensional variety, and then $-(K_X+B)$ is not big. In step (iii), the lower-dimensional variety in step (ii) can have non-klt points, and then it is not of Fano type.

The following example shows the situations of steps (ii) and (iii). Let C be an elliptic curve, L a line bundle on C with $\deg L>0$, $S:=\mathbb{P}_C(\mathcal{O}_C\oplus L)$, $C_0\subset S$ the section with $(C_0^2)<0$, and $C_\infty\subset S$ a section disjoint from C_0. Let $g:S\to T$ be the contraction of C_0, so that T is the projective cone over C with respect to L, and let $H:=g_*C_\infty$, which is an ample Cartier divisor on T. Since $K_C\sim 0$, we have

$$K_S\sim -C_0-C_\infty=g^*K_T-C_0,\qquad K_T\sim -H.$$

Thus $a(C_0,T,0)=0$, and $(T,0)$ is lc since (S,C_0) is lc and $K_S+C_0\sim g^*K_T$ ([Har77, V.2.9, V.2.10, V.2.11.4], [Kol13, Lemma 3.1]). In particular, T is not klt at the vertex. Let C' be an elliptic curve, $X:=T\times C'$, and $p:X\to T$ the projection. Then $-K_X\sim p^*H$ is nef and

$$(-K_X)^3=(p^*H)^3=0.$$

Thus $-K_X$ is not big, as in step (ii), and T is not of Fano type, as in step (iii).

Without the Fano type assumption, Conjecture 1.1 and related statements are known in the following cases.

(1) Curves and surfaces. When $d\le 2$, for pairs with DCC coefficients, Conjecture 1.1 was proved in [CH21, Theorem 1.1] while Conjecture 1.2 was proved in [CHX24, Theorem 1.3]. See also [Zen23, Theorem 1.4] with additional restrictions on the indices, [Liu23] for explicit bounds, and [LS23, Corollary 1.11(2)] for a better explicit bound: any $\mathbb{R}$-complementary surface is n-complementary for some $n<96I_0$, where $I_0=2\cdot 11^{128\cdot 11^5}$.

(2) Threefolds. [FMX19, Theorem 1] proved Conjecture 1.1 when Γ is a DCC set such that $\overline{\Gamma}\subset\mathbb{Q}$. This result is based on earlier work [Xu19], which relies on [Jia21]. Together with [HLS19, Theorem 8.25], we obtain Conjecture 1.1 when $d=3$ (cf. [CHL24, Theorem 2.14]). Conjecture 1.2 was proved when $d=3$ in [CHX24, Theorem 1.1]. See also [CHX24, Theorem 1.2].

(3) Higher dimensions. Assuming the existence of good minimal models, it was proved in [CHL24, Theorem 1.6] that a klt $\mathbb{R}$-complementary projective variety X of dimension d such that $-K_X$ is nef has an n-complement, where n depends only on d, the non-vanishing order b of $-h^*K_X|_F$ for a general fiber F of an Iitaka fibration $X_\infty\to Z_\infty$ of $-K_X$ with induced birational morphism $h:X_\infty\to X$, and the middle Betti number of a smooth model of the cover of F associated with the unique member of $|-bh^*K_X|_F|$. [CHL24, Theorem 1.6] assumes that X is $\mathbb{Q}$-factorial, and the general case follows by a small $\mathbb{Q}$-factorialization. Conversely, the existence of bounded complements, together with the existence of good minimal models, implies the effective log Iitaka fibration conjecture [CHL24, Theorem 1.4].

We now explain why the method of [CHX24] does not go beyond dimension three. Let $f:X\to T$ be a contraction and L an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on T such that $K_X+B\sim_{\mathbb{R}} f^*L$. The canonical bundle formula [Kaw98, FM00, Amb04] (in the form of [FG14b, 3.4] for (X,B) lc over the generic point of T, and, for $\mathbb{R}$-divisors, after writing K_X+B as a convex combination of $\mathbb{Q}$-divisors K_X+B_i such that (X,B_i) is lc over the generic point of T and $K_X+B_i\sim_{\mathbb{Q}} f^*L_i$ for some $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L_i on T) writes

$$K_X+B\sim_{\mathbb{R}} f^*(K_T+B_T+M_T),$$

where B_T is the discriminant part and M_T is the moduli part. The proof of [CHX24, Theorem 1.1] uses several results which are proved in [CHX24] only in dimension at most three: the existence of good minimal models of $-(K_X+B_\Phi)$ for a fixed hyperstandard set Φ [CHX24, Lemma 2.15],

a positive integer n depending only on Φ such that $n(K_X+B)\sim 0$ for any projective $\mathbb{Q}$-factorial dlt pair (X,B) with $K_X+B\sim_{\mathbb{R}} 0$ and $\kappa(X,B-B_\Phi)=0$ [CHX24, Lemma 2.18], a uniform bound on the Cartier index of the moduli part [CHX24, Proposition 3.3], and the base point freeness of a fixed multiple of the moduli part [CHX24, Proposition 3.4]. As explained in [CHX24, p. 3], the base point freeness of a fixed multiple of the moduli part (effective adjunction, [PS09, Conjecture 7.13, (7.13.3)]; see [HLX25, Theorem 1.4] for its equivalence with Li's version for real coefficients, which holds in relative dimension one [HLX25, Corollary 1.5]) is unknown when the relative dimension of f is at least two. To lift complements, [CHX24, Proposition 3.5] takes an n-complement of a pair $(T',B_{T'}+M_{T'})$, where T' is a crepant model of T and $M_{T'}\geq 0$ is a $\mathbb{Q}$-divisor on T' which has no common component with $B_{T'}$ and such that $pM_{T'}$ is linearly equivalent over Z to p times the moduli part on T', for a positive integer p dividing n [CHX24, Proposition 3.5(2)]. Base point freeness of a fixed multiple of the moduli part yields such $M_{T'}$. For coefficients in $[0,1]$ in any dimension, we keep the moduli part as it is and take a complement of the generalized pair on T by [CHHX25, Theorem 8.1]. For this, we need a uniform bound on the Cartier index of the moduli part on a sufficiently high birational model of T, and a lifting of complements from such a model of T to X which does not change the index, including at prime divisors of X whose image on T has codimension at least 2. See the sketch of the proof below.

Finally, we say a few words about Conjecture 1.2. First, if the coefficients of B are only assumed to be in $[0,1]$, then for $d\geq 3$ the existence of an $\mathbb{R}$-complement does not imply the existence of an n-complement with n bounded in terms of d [Sho20, Example 11]. Second, for arbitrary real coefficients, monotonic n-complements with bounded n do not exist in general, already in dimension one [HLS26, Example 5.1]. Third, for any $d\geq 1$, even under the klt hypothesis, a finite set of indices which works for all pairs of dimension d with coefficients in $[0,1]$ contains at least two elements [Sho20, Example 2].

1.1. **Main results.** The goal of our paper is to prove Conjectures 1.1 and 1.2. First, we prove Conjecture 1.1 for finite rational coefficients:

Theorem 1.3. *Let d be a positive integer and $\Gamma\subset[0,1]\cap\mathbb{Q}$ a finite set. Then there exists a positive integer N depending only on d and Γ, such that $Nb\in\mathbb{Z}$ for any $b\in\Gamma$, satisfying the following. Let $(X/Z\ni z,B)$ be an $\mathbb{R}$-complementary pair such that $\dim X\leq d$ and $B\in\Gamma$. Then $(X/Z\ni z,B)$ has a monotonic N-complement.*

Remark 1.4. A monotonic N-complement is a monotonic kN-complement for any positive integer k, so N may be replaced with $\operatorname{lcm}(N,l)$ for any positive integer l.

Combining Theorem 1.3 with [HLS19, Theorem 8.25], we obtain the following:

Theorem 1.5. *Conjecture 1.1 holds.*

Our second main result is the proof of Conjecture 1.2.

Theorem 1.6. *Conjecture 1.2 holds.*

We also remark that, in Section 12 we prove Theorem 12.2, which only requires that the pair obtained by rounding the coefficients of B down to a set of rational numbers determined by d, l, and a fixed hyperstandard set is klt $\mathbb{R}$-complementary, and Theorem 1.6 follows from Theorem 12.2 with the coefficient set $\{1-\frac{1}{j}\mid j\in\mathbb{Z}_{>0}\}\cup\{1\}$ on a small $\mathbb{Q}$-factorialization of X. Theorem 12.2 confirms [CHX24, Conjecture 2.16].

Sketch of the proof. The proof of Theorem 1.3 has three steps.

(1) By Proposition 5.3, based on Theorem 5.1, we may assume that (X,B) is a $\mathbb{Q}$-factorial dlt pair with $B\in\Gamma\cup\{1\}$ and $-(K_X+B)\sim_{\mathbb{Q}} g^*H$ for a contraction $g:X\to T$ over Z and a $\mathbb{Q}$-divisor H on T ample over Z.

(2) If all lc centers of (X, B) dominate T, then the generalized pair $(T, B_T, \mathbf{M})$ induced on T by Theorem 3.3 is generalized klt and $-(K_T + B_T + \mathbf{M}_T) \sim_{\mathbb{Q}} H$, so T is of Fano type over Z (Lemma 4.1), and a complement on T (Proposition 4.2) pulls back to X (Proposition 8.4).

(3) Otherwise, let $V \subsetneq T$ be minimal among the images of the lc centers, W a minimal lc center with image V, and $W \to V' \to V$ the Stein factorization, with $V' \to V$ Galois with group G (Lemma 7.5). Theorem 6.4 descends the canonical bundle formula of $W \to V'$ to a generalized pair $(V, B_V, \mathbf{N})$, which is generalized klt as W is a minimal lc center, with $-(K_V + B_V + \mathbf{N}_V) \sim_{\mathbb{Q}} H|_V$ (Proposition 8.5). Thus V is of Fano type over the base of the Stein factorization of $V \to Z$ (Lemma 4.1) and has a complement (Proposition 4.2). It yields a G-invariant section on W (Proposition 8.5), and its extension to X by Theorem 7.13 defines a complement by Lemma 8.7, inversion of adjunction [Kaw07], and connectedness [Fuj17, Theorem 6.3] (Proposition 8.8).

Two difficulties arise in step (3).

(a) Descent along $V' \to V$. Only G-invariant sections on W extend to X (Theorem 7.13(2)), and the proof uses no bound on $|G|$ or on the ramification in terms of d and Γ. Norms of complements on V' multiply the index by $|G|$, and by Theorem 3.3 on V', the coefficient of the descended moduli part at a prime divisor over V along which $V' \to V$ has ramification index e becomes an integer only after multiplication by e and an integer depending only on d and Γ. Moreover, elements of G lift to crepant birational self-maps of W only up to crepant self-maps over V', whose scalars on pluricanonical forms of general fibers have finite orders depending on the fiber [Gon13, Theorem B], [FG14a, Theorem 1.1]. Theorem 6.2 bounds the orders uniformly, so Hilbert's Theorem 90 yields an invariant form of bounded degree (Lemma 6.5).

Lemma 6.9 removes the factor e. At such a divisor, the moduli part on V has coefficient w/e for the weight w of an invariant form on a transverse curve (Lemma 6.10). A generator of the inertia group induces a crepant birational self-map of a projective klt lc center of a dlt model over that curve, which multiplies the residue of the form by ζ^{-aw} for a primitive e-th root of unity ζ, so Theorem 6.2 implies $e \mid pw$ for the integers a and p of Lemma 6.9, which satisfy $a \mid p$ and depend only on d and Γ. Thus the Cartier index in Theorem 6.4(4) is independent of G.

(b) Extension from a minimal lc center. Instead of Kawamata–Viehweg vanishing (step (ii) above), we use the extension theorem [Fuj17, Theorem 6.4], which needs only semi-ampleness, but extends sections only from the union $\tilde{\Sigma}_V$ of the strata with image V (Construction 7.2), whose components need not be birational to W. By Theorem 7.13(2), the sections on $\tilde{\Sigma}_V$ restrict isomorphically onto the G-invariant sections on W, since Kollár's $\mathbb{P}^1$-links [Kol16, Theorem 10, Proposition 14] and Theorem 6.2 relate the sections on any two minimal lc centers with image V (Lemma 7.11). Log canonicity passes from W to each maximal lc center W_1 with image V through an lc center in W_1 crepant birational to W and dominating the base of the Stein factorization of $W_1 \to V$ (Proposition 7.14(3), Lemma 8.7).

The complement on the base is not explicit: Proposition 4.2 constructs it from a general member of a linear system. As Γ is finite, the boundaries on all lc centers have coefficients in $\Phi(\mathfrak{R})$ for one finite set $\mathfrak{R}$ whose denominators divide N (Construction 7.3, Lemma 2.13(2)). Step (2) combines Theorem 3.3, which depends on Theorems 2.21 and 3.5, with [FM20, Theorem 1.2].

For Theorem 12.2, fix positive integers d and l and a finite set $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ with $0, 1 \in \mathfrak{R}$, and let $\Phi := \Phi(\mathfrak{R})$ and $\mathcal{N} := \mathcal{N}(d, l, \mathfrak{R})$ (Construction 11.1). Lemmas 5.2 and 2.14(4) and a small $\mathbb{Q}$-factorialization reduce Theorem 12.2 to finding an n-complement of $(X/Z \ni z, B')$ with $n \in \mathcal{N}$, where (X, B') is klt, $B' \geq B_{\Gamma(\mathcal{N},\Phi)}$, and $K_X + B' \sim_{\mathbb{Q},Z} 0$ (Set-up 11.2). Here (X, B') has no lc centers, but the complements need not be monotonic, so their indices cannot be enlarged.

For the truncations $B'_1 \leq \cdots \leq B'_d$ of B' to the sets $\Phi_1 \subset \cdots \subset \Phi_d$ of Construction 11.1, where $\mathcal{N} = \mathcal{N}_1 \cup \cdots \cup \mathcal{N}_d \cup \{n_0\}$ and $n_0 := \operatorname{lcm}\{l, I(d, \Phi_d)\}$ for the uniform lc index $I(d, \Phi_d)$ of Theorem 2.21, Lemma 11.4 leaves three cases.

(1) $B' - B'_1$ is big over Z. Then X is of Fano type over Z, and Proposition 4.3 yields an n-complement with $n \in \mathcal{N}_1$.
(2) Z is a point and $\kappa(X, B' - B'_d) = 0$. Then $n_0(K_X + B') \sim 0$ by Theorems 5.1 and 2.21 (Lemma 11.3).
(3) $B' - B'_i$ and $B' - B'_{i+1}$ have equal relative Iitaka dimensions over Z for some i. Then two MMPs yield a common base T_2 of the canonical bundle formulas of B'_i and B'_{i+1} with one moduli $\boldsymbol{b}$-divisor, whose Cartier index divides an integer p_i depending only on d and Φ_i (Proposition 9.3). Let T' be an exposing model of T_2 (Definition 10.1, Proposition 10.3), so that any prime divisor P of X not dominating T_2 is mapped onto a prime divisor Q_P of T'. Then Proposition 4.3 yields an n-complement with $n \in \mathcal{N}_{i+1}$ and $p_i \mid n$, and Proposition 10.4 lifts it to X.

Since p_i depends only on Φ_i, the sets $\mathcal{N}_{i+1}$ are defined inductively (Construction 11.1). The case that occurs determines which of $\mathcal{N}_1, \dots, \mathcal{N}_d, \{n_0\}$ contains n, but not n itself.

(c) Lifting with a fixed index. In case (3), on the exposing model, the lc threshold of Q_P bounds the coefficient of P, which controls the rounding at P (Proposition 10.4). At a prime divisor P on X whose image on T_2 has codimension at least 2 (cf. [CHX24, pp. 3–4]), the coefficient of the pullback is not determined by a coefficient on T_2, and on a higher model of T_2 the discriminant may have negative coefficients, which [CHHX25, Theorem 8.1] does not allow. By Set-up 10.2(3) and the klt property, the prime divisor over T_2 onto which such a P is mapped has generalized log discrepancy in $(0, 1]$, so [BZ16, Lemma 4.5] extracts the finitely many such prime divisors and keeps the discriminant non-negative (Proposition 10.3).

Our proofs use four recent results: the existence of good minimal models of projective lc pairs with pseudo-effective log canonical divisor and relative abundance for lc pairs [OAI26a, Theorem 11.1, Corollary 14.1], a uniform lc index for projective lc log Calabi–Yau pairs with coefficients in a DCC set of rational numbers [OAI26d, Theorem 1.2], a uniform denominator for the moduli part of the canonical bundle formula for finite coefficient sets [OAI26c, Theorem 7.2], whose main input [OAI26c, Theorem 4.2] we use for coefficients in a DCC set, and a uniform exponent for the scalars by which crepant birational self-maps of a projective klt log Calabi–Yau pair act on the pluricanonical forms of the pair [OAI26b, Theorem 6.1].

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce complements of generalized pairs, truncations with respect to hyperstandard sets, and the relative Iitaka dimension, and recall the uniform lc index of [OAI26d, Theorem 1.2].

In Section 3 we show that, for pairs whose coefficients belong to a fixed DCC set of rational numbers, the moduli part of the canonical bundle formula has a Cartier multiple bounded in terms of the dimension and that set, and we relate complements of such a pair to complements of the generalized pair induced on the base. In Section 4 we prove a criterion for a generalized pair to be of Fano type, and we prove bounds on complements of generalized pairs of Fano type whose coefficients belong to a DCC set of rational numbers with rational accumulation points, or to $[0, 1]$ when the generalized pair is klt $\mathbb{R}$-complementary. In Section 5, we run the minimal model program to replace an $\mathbb{R}$-complementary pair with finite rational coefficients with a $\mathbb{Q}$-factorial dlt pair with semi-ample anti-canonical class without changing the indices of monotonic complements.

In Section 6 we show that the canonical bundle formula descends along a finite Galois morphism of the base, and that the Cartier index of the moduli part on the quotient is bounded independently of the degree of the morphism. In Section 7, for a contraction $g : X \to T$ such that $-(K_X + B)$ is the pullback of a divisor on T and a subvariety $V \subsetneq T$ minimal among the images of the lc centers of (X, B), we identify sections of multiples of $-(K_X + B)$ over the union of the lc centers mapping onto V with invariant sections on a minimal lc center over V. In Section 8 we construct complements of $\mathbb{Q}$-factorial dlt pairs whose anti-log canonical divisor is the pullback of a divisor, ample over Z, on the base of a contraction, which is the main step in the proof of Theorem 1.3.

In Section 9 we study the boundaries obtained by rounding the coefficients of B down to two hyperstandard sets, and two MMPs whose output bases are birational. In Section 10, for a contraction $g : X \to T$, we construct a higher model T' of T such that each prime divisor of X not dominating T is mapped onto a prime divisor of T', and we lift n-complements from T' to n-complements on X. In Section 11 we consider a sequence of roundings of the coefficients of B down to hyperstandard sets and choose the sets of indices inductively.

In Section 12 we prove Theorems 1.3, 1.5, 12.2, and 1.6.

The proofs of Theorems 1.3 and 1.6 share only the following material from Sections 2–5: the preliminaries of Section 2, Definition 3.1, Lemma 3.2, Theorem 3.3, Definition 3.4, Theorem 3.5, Lemma 3.8, Lemma 4.1, Theorem 5.1, and Lemma 5.2. Sections 6, 7, and 8 are used only for Theorem 1.3, and Sections 9, 10, and 11 only for Theorems 12.2 and 1.6. Thus a reader interested only in Theorem 1.3 may skip Sections 9–11, and a reader interested only in Theorem 1.6 may skip Sections 6–8.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. **Pairs and generalized pairs.** For a prime divisor P on a normal variety X and an $\mathbb{R}$-divisor F on X, we denote by $\operatorname{mult}_P F$ the coefficient of P in F. Let E be a prime divisor over X, F an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X, and φ a non-zero rational function on X. We denote by $\operatorname{mult}_E F$ (resp. $\operatorname{mult}_E \operatorname{div}(\varphi)$) the coefficient of E in the pullback of F (resp. $\operatorname{div}(\varphi)$) to a normal variety projective and birational over X on which E is a prime divisor. If moreover $f : X \to Z$ is a morphism and F is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on Z, we write $\operatorname{mult}_E F := \operatorname{mult}_E f^*F$.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation v of X, that is, $v = c \operatorname{mult}_E \operatorname{div}(\cdot)$ on the non-zero rational functions on X for a prime divisor E over X and a real number $c > 0$, we let $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

A *contraction* is a projective morphism $f : X \to Z$ between normal quasi-projective varieties such that $f_*\mathcal{O}_X = \mathcal{O}_Z$. A *pair* $(X/Z \ni z, B)$ consists of a contraction $\pi : X \to Z$ between normal quasi-projective varieties, a point $z \in Z$, and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier over a neighborhood of z. If π is the identity morphism, then we may drop Z and say that $(X \ni z, B)$ is a pair. If $(X/Z \ni z, B)$ is a pair for any $z \in Z$, then we say that $(X/Z, B)$ is a pair and also say that (X, B) is a pair. If B is a $\mathbb{Q}$-divisor, then we say that $(X/Z \ni z, B)$ is a *$\mathbb{Q}$-pair*.

For a dlt pair (X, B) and an lc center W of (X, B), we call B_W in Lemma 2.2 below the *different* of (X, B) on W [Kol13, 4.18].

Lemma 2.2. *Let (X, B) be a dlt pair and W an lc center of (X, B). Let (W, B_W) be the pair induced by repetitive adjunction to lc places*

$$K_W + B_W := (K_X + B)|_W,$$

then:

(1) (W, B_W) is dlt. In particular, W is normal.
(2) For any lc center $V \supsetneq W$ of (X, B), let (V, B_V) be the pair induced by repetitive adjunction to lc places $K_V + B_V := (K_X + B)|_V$. Then W is an lc center of (V, B_V) and $K_W + B_W = (K_V + B_V)|_W$.
(3) Any lc center of (W, B_W) is an lc center of (X, B).

Proof. (1) and (2) are [Kol13, Theorems 4.16 and 4.19]. Since (X, B) is dlt, (W, B_W) is dlt. Thus any lc center of (W, B_W) is a stratum of $\lfloor B_W \rfloor$, hence a stratum of $\lfloor B \rfloor$, hence an lc center of (X, B). (3) follows. □

Definition 2.3. Let (X, B) be a pair. The *non-klt locus* $\mathrm{Nklt}(X, B)$ of (X, B) is the union of the centers on X of the prime divisors E over X such that $a(E, X, B) \leq 0$. If (X, B) is lc and $D \geq 0$ is an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on X, then the *lc threshold* of D with respect to (X, B) is

$$\mathrm{lct}(X, B; D) := \sup\{t \geq 0 \mid (X, B + tD) \text{ is lc}\}.$$

Definition 2.4. Let (X, B) and (X', B') be sub-pairs. A birational map $\phi : X \dashrightarrow X'$ is *crepant* for (X, B) and (X', B') if there exist a normal variety Y and projective birational morphisms $p : Y \to X$ and $q : Y \to X'$ such that $q = \phi \circ p$ and $a(E, X, B) = a(E, X', B')$ for any prime divisor E on Y. If $(X', B') = (X, B)$, then we say that ϕ is *crepant for* (X, B).

Definition 2.5. Let X be a normal variety of dimension r, $j : X_{\mathrm{reg}} \to X$ the inclusion of the regular locus, and m a non-zero integer. Let $\omega_{X_{\mathrm{reg}}}^{\otimes m}$ be the m-th tensor power of the line bundle $\omega_{X_{\mathrm{reg}}} := \Omega^r_{X_{\mathrm{reg}}/\mathbb{C}}$, where for $m < 0$ we take the $(-m)$-th tensor power of the dual of $\omega_{X_{\mathrm{reg}}}$. The *sheaf of m-canonical forms* on X is $j_*\omega_{X_{\mathrm{reg}}}^{\otimes m}$. A *rational m-canonical form* on X is a non-zero rational section ω of $j_*\omega_{X_{\mathrm{reg}}}^{\otimes m}$, that is, a non-zero section of $\omega_{X_{\mathrm{reg}}}^{\otimes m}$ over a non-empty open subset of X_{reg}, where two such sections are identified if their restrictions to some non-empty open subset are equal. Any two rational m-canonical forms ω and ω' on X satisfy $\omega' = \varphi\omega$ for a unique non-zero rational function φ on X. Let P be a prime divisor on X. Since X is normal, the generic point of P belongs to X_{reg}, so there exists a nowhere vanishing section θ of $\omega_{X_{\mathrm{reg}}}^{\otimes m}$ over an open subset of X_{reg} intersecting P, and $\omega = \varphi\theta$ for a non-zero rational function φ on X. We let

$$\mathrm{mult}_P \operatorname{div}(\omega) := \mathrm{mult}_P \operatorname{div}(\varphi),$$

which does not depend on θ. Then $\operatorname{div}(\omega)$ is the Weil divisor

$$\operatorname{div}(\omega) := \sum_P \big(\mathrm{mult}_P \operatorname{div}(\omega)\big) P$$

on X, where P runs over the prime divisors on X, and

$$\operatorname{div}(\varphi\omega) = \operatorname{div}(\varphi) + \operatorname{div}(\omega)$$

for any non-zero rational function φ on X. If $p : Y \dashrightarrow X$ is a dominant rational map from a normal variety Y of dimension r, then $p^*\omega$ is the rational m-canonical form on Y obtained by pulling back ω as a section of $\omega_{X_{\mathrm{reg}}}^{\otimes m}$ along the restriction of p to a non-empty open subset of Y_{reg} on which p is an étale morphism to X_{reg}, which exists by generic smoothness [Har77, Corollary III.10.7]. For a prime divisor E over X, we denote by $\mathrm{mult}_E \operatorname{div}(\omega)$ the coefficient of E in $\operatorname{div}(p^*\omega)$, where $p : X' \to X$ is a projective birational morphism from a normal variety X' on which E is a prime divisor.

Definition 2.6. Let k be a field of characteristic zero, $\overline{k}$ an algebraic closure of k, V a normal geometrically integral variety over k, and $B_V \geq 0$ an $\mathbb{R}$-divisor on V such that $K_V + B_V$ is $\mathbb{R}$-Cartier. Let $V_{\overline{k}} := V \times_k \overline{k}$ and $B_{V,\overline{k}}$ the pullback of B_V to $V_{\overline{k}}$. We say that (V, B_V) is *geometrically lc* if $(V_{\overline{k}}, B_{V,\overline{k}})$ is lc. We say that (V, B_V) is *geometrically klt* if $(V_{\overline{k}}, B_{V,\overline{k}})$ is klt.

Definition 2.7. Let X be a normal quasi-projective variety. A $\mathbb{Q}$-***b***-*divisor* $\mathbf{M}$ over X is a collection of $\mathbb{Q}$-divisors $\mathbf{M}_{X'}$ on all normal varieties X' projective and birational over X, such that $g_*\mathbf{M}_{X''} = \mathbf{M}_{X'}$ for any projective birational morphism $g : X'' \to X'$ over X. The $\mathbb{Q}$-divisor $\mathbf{M}_{X'}$ is the *trace* of $\mathbf{M}$ on X'. We say that $\mathbf{M}$ *descends* to X' if $\mathbf{M}_{X'}$ is $\mathbb{Q}$-Cartier and $\mathbf{M}_{X''} = g^*\mathbf{M}_{X'}$ for any such g, and in this case X' is called a *determining model* of $\mathbf{M}$. We say that $\mathbf{M}$ is ***b***-$\mathbb{Q}$-*Cartier* if $\mathbf{M}$ descends to some X', and ***b***-*Cartier* if moreover $\mathbf{M}_{X'}$ is Cartier. If $X \to U$ is a projective morphism, then $\mathbf{M}$ is *nef over* U if $\mathbf{M}$ descends to some X' such that $\mathbf{M}_{X'}$ is nef over U. For a rational function φ on X, $\mathrm{div}(\varphi)$ also denotes the $\mathbb{Q}$-***b***-divisor with traces $\mathrm{div}_{X'}(\varphi)$. Let $\rho : X_0 \to X$ be a finite surjective morphism from a normal quasi-projective variety X_0, and assume that $\mathbf{M}$ descends to X'. Let X_0' be the normalization of the irreducible component of $X' \times_X X_0$ dominating X_0, which is unique since $X' \to X$ is birational. We call X_0' the *normalization of* X' *in* X_0. Then $\rho^*\mathbf{M}$ denotes the $\mathbb{Q}$-***b***-divisor over X_0 which descends to X_0' and whose trace on X_0' is the pullback of $\mathbf{M}_{X'}$ by the finite morphism $X_0' \to X'$. If a finite group G acts on X_0, then a ***b***-$\mathbb{Q}$-Cartier $\mathbb{Q}$-***b***-divisor $\mathbf{M}_0$ over X_0 is G-*invariant* if $\sigma^*\mathbf{M}_0 = \mathbf{M}_0$ for any $\sigma \in G$.

Definition 2.8. A *generalized pair* $(X, B, \mathbf{M})/U$ consists of a normal quasi-projective variety X, a projective morphism $\pi : X \to U$ to a normal quasi-projective variety U, an $\mathbb{R}$-divisor $B \geq 0$ on X, and a ***b***-$\mathbb{Q}$-Cartier $\mathbb{Q}$-***b***-divisor $\mathbf{M}$ over X which is nef over U, such that $K_X + B + \mathbf{M}_X$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then $(X, B, \mathbf{M})/U$ is a *generalized sub-pair*. Let $h : X' \to X$ be a log resolution of $(X, \operatorname{Supp} B)$ such that $\mathbf{M}$ descends to X', E a prime divisor on X', and write $K_{X'} + B' + \mathbf{M}_{X'} := h^*(K_X + B + \mathbf{M}_X)$. Then $a(E, X, B, \mathbf{M}) := 1 - \mathrm{mult}_E B'$ is the *generalized log discrepancy* of E with respect to $(X, B, \mathbf{M})$. We say that $(X, B, \mathbf{M})/U$ is *generalized (sub-)lc* if $a(E, X, B, \mathbf{M}) \geq 0$ for any prime divisor E over X, and *generalized (sub-)klt* if $a(E, X, B, \mathbf{M}) > 0$ for any prime divisor E over X. For a point $u \in U$, we write $(X/U \ni u, B, \mathbf{M})$ for $(X, B, \mathbf{M})/U$ considered over neighborhoods of u; for $(X/U \ni u, B, \mathbf{M})$, we only require that $K_X + B + \mathbf{M}_X$ is $\mathbb{R}$-Cartier over a neighborhood of u.

Definition 2.9 (cf. [CHHX25, Definition 2.8])**.** Let $\pi : X \to U$ be a projective morphism from a normal quasi-projective variety X to a normal quasi-projective variety U. For $\mathbb{R}$-Cartier $\mathbb{R}$-divisors D_1, D_2 on X, we write $D_1 \sim_{\mathbb{R},U} D_2$ if $D_1 - D_2 \sim_{\mathbb{R}} \pi^*L$ for some $\mathbb{R}$-Cartier $\mathbb{R}$-divisor L on U, and $D_1 \sim_{\mathbb{Q},U} D_2$ if $D_1 - D_2 \sim_{\mathbb{Q}} \pi^*L$ for some $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on U. An $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X is *big over* U if $D \sim_{\mathbb{R},U} H + E$ for an $\mathbb{R}$-divisor H which is ample over U and an $\mathbb{R}$-divisor $E \geq 0$ (cf. [KMM87, Corollary 0-3-4]). We say that X is *of Fano type over* U if there exists an $\mathbb{R}$-divisor $\Delta \geq 0$ on X such that (X, Δ) is klt and $-(K_X + \Delta)$ is nef and big over U. For a point $u \in U$, we say that X is of Fano type over a neighborhood of u if $X \times_U U^0$ is of Fano type over U^0 for some open neighborhood U^0 of u.

2.2. **Complements.**

Definition 2.10 (cf. [CHX24, Definition 2.12])**.** For a real number b, let $\{b\} := b - \lfloor b \rfloor$. For an $\mathbb{R}$-divisor D, $\lfloor D \rfloor$ and $\{D\} := D - \lfloor D \rfloor$ are the round-down and the fractional part of D, taken coefficientwise. Let $(X/Z \ni z, B)$ be a pair, where the point $z \in Z$ is not necessarily closed (cf. [CH21, Conjecture 6.1] and the paragraph after [CHX24, Definition 2.12]), and n a positive integer.

(1) An $\mathbb{R}$-*complement* of $(X/Z \ni z, B)$ is a pair $(X/Z \ni z, B^+)$ such that $B^+ \geq B$ and, over some neighborhood of z, (X, B^+) is lc and $K_X + B^+ \sim_{\mathbb{R}} 0$. If moreover (X, B^+) is klt over some neighborhood of z, then $(X/Z \ni z, B^+)$ is a *klt* $\mathbb{R}$-*complement* of $(X/Z \ni z, B)$. An $\mathbb{R}$-complement $(X/Z \ni z, B^+)$ of $(X/Z \ni z, B)$ such that B^+ is a $\mathbb{Q}$-divisor is a

$\mathbb{Q}$-*complement* of $(X/Z \ni z, B)$. We say that $(X/Z \ni z, B)$ is $\mathbb{R}$-*complementary* if it has an $\mathbb{R}$-complement, and *klt* $\mathbb{R}$-*complementary* if it has a klt $\mathbb{R}$-complement.

(2) An n-*complement* of $(X/Z \ni z, B)$ is a pair $(X/Z \ni z, B^+)$ such that B^+ is a $\mathbb{Q}$-divisor and, over some neighborhood of z, (X, B^+) is lc, $n(K_X + B^+) \sim 0$, and $nB^+ \geq n\lfloor B \rfloor + \lfloor (n+1)\{B\} \rfloor$. An n-complement $(X/Z \ni z, B^+)$ is *monotonic* if moreover $B^+ \geq B$.

(3) If $(X/Z \ni z, B^+)$ is an $\mathbb{R}$-complement (resp. n-complement) of $(X/Z \ni z, B)$ for any $z \in Z$, then we say that $(X/Z, B^+)$ is an $\mathbb{R}$-complement (resp. n-complement) of $(X/Z, B)$.

Definition 2.11 (cf. [CHHX25, Definition 2.17])**.** Let $(X/U \ni u, B, \mathbf{M})$ be a generalized pair and n a positive integer.

(1) An $\mathbb{R}$-*complement* of $(X/U \ni u, B, \mathbf{M})$ is a generalized pair $(X/U \ni u, B^+, \mathbf{M})$ such that $B^+ \geq B$ and, over some neighborhood of u, $(X, B^+, \mathbf{M})$ is generalized lc and $K_X + B^+ + \mathbf{M}_X \sim_{\mathbb{R}} 0$. If moreover $(X, B^+, \mathbf{M})$ is generalized klt over some neighborhood of u, then $(X/U \ni u, B^+, \mathbf{M})$ is a *klt* $\mathbb{R}$-*complement* of $(X/U \ni u, B, \mathbf{M})$. We say that $(X/U \ni u, B, \mathbf{M})$ is $\mathbb{R}$-*complementary* if it has an $\mathbb{R}$-complement, and *klt* $\mathbb{R}$-*complementary* if it has a klt $\mathbb{R}$-complement.

(2) An n-*complement* of $(X/U \ni u, B, \mathbf{M})$ is a generalized pair $(X/U \ni u, B^+, \mathbf{M})$ such that B^+ is a $\mathbb{Q}$-divisor, $n\mathbf{M}$ is $\boldsymbol{b}$-Cartier, and, over some neighborhood of u, $(X, B^+, \mathbf{M})$ is generalized lc, $n(K_X + B^+ + \mathbf{M}_X) \sim 0$, and $nB^+ \geq n\lfloor B \rfloor + \lfloor (n+1)\{B\} \rfloor$. An n-complement $(X/U \ni u, B^+, \mathbf{M})$ is *monotonic* if moreover $B^+ \geq B$.

Definition 2.12 (cf. [CHX24, Definition 2.1], [PS09, 3.2])**.** A set $\Gamma \subset \mathbb{R}$ satisfies the *descending chain condition* (DCC) if any non-increasing sequence in Γ stabilizes. For a set $\Gamma \subset \mathbb{R}$ and an $\mathbb{R}$-divisor B, we write $B \in \Gamma$ if the coefficients of B belong to Γ. Let $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ be a finite set such that $0, 1 \in \mathfrak{R}$. The *hyperstandard set* associated with $\mathfrak{R}$ is

$$\Phi(\mathfrak{R}) := \left\{ 1 - \frac{r}{j} \;\middle|\; r \in \mathfrak{R},\ j \in \mathbb{Z}_{>0} \right\}.$$

For $\Phi := \Phi(\mathfrak{R})$ and a finite set $\mathcal{N}$ of positive integers, let

$$\Gamma(\mathcal{N}, \Phi) := \left\{ 1 - \frac{r}{j} + \frac{1}{j} \sum_{n \in \mathcal{N}} \frac{u_n}{n+1} \;\middle|\; r \in \mathfrak{R},\ j \in \mathbb{Z}_{>0},\ u_n \in \mathbb{Z}_{\geq 0} \right\} \cap [0,1].$$

For a set $\Gamma \subset [0,1]$ such that $0 \in \Gamma$ and $\{\gamma \in \Gamma \mid \gamma \leq b\}$ has a maximum for any $b \in [0,1]$, and for an $\mathbb{R}$-divisor $B = \sum_i b_i B_i$ with $b_i \in [0,1]$ and B_i distinct prime divisors, let

$$B_\Gamma := \sum_i \max\{\gamma \in \Gamma \mid \gamma \leq b_i\} B_i.$$

Lemma 2.13. *Let n be a positive integer.*

(1) For any real number $0 \leq b < 1$,

$$n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor = \lfloor (n+1)b \rfloor \leq \lceil nb \rceil.$$

(2) Let $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ be a finite set such that $0, 1 \in \mathfrak{R}$ and $nr \in \mathbb{Z}$ for any $r \in \mathfrak{R}$. Then $n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor = \lceil nb \rceil$ for any $b \in \Phi(\mathfrak{R})$.

Proof. (1) Since $0 \leq b < 1$, we have $\lfloor b \rfloor = 0$ and $\{b\} = b$, so

$$n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor = \lfloor (n+1)b \rfloor.$$

If $nb \in \mathbb{Z}$, then $\lfloor (n+1)b \rfloor = nb + \lfloor b \rfloor = nb = \lceil nb \rceil$. Otherwise, write $nb = m + s$ with $m \in \mathbb{Z}$ and $0 < s < 1$. Since $0 < s + b < 2$,

$$\lfloor (n+1)b \rfloor = m + \lfloor s + b \rfloor \leq m + 1 = \lceil nb \rceil.$$

(2) Let $b \in \Phi(\mathfrak{R})$. If $b = 1$, then

$$n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor = n = \lceil nb \rceil.$$

Otherwise, $b = 1 - r/j$ for some $r \in \mathfrak{R}$ with $0 < r \leq 1$ and some $j \in \mathbb{Z}_{>0}$, and $0 \leq b < 1$. Let $t := nr \in \mathbb{Z}_{\geq 0}$, and write $t = mj + s$ with $m, s \in \mathbb{Z}$ and $0 \leq s \leq j - 1$. Since $0 < r \leq 1$,

$$\frac{(n+1)r}{j} = m + \frac{s+r}{j} \quad \text{and} \quad 0 < \frac{s+r}{j} \leq 1,$$

so $\lceil (n+1)r/j \rceil = m + 1$. Thus

$$\lfloor (n+1)b \rfloor = n + 1 - \left\lceil \frac{(n+1)r}{j} \right\rceil = n - m = n - \left\lfloor \frac{nr}{j} \right\rfloor = \lceil nb \rceil,$$

and (1) implies (2) (cf. [CHX24, Lemma 2.3], [PS09, Lemma 3.5]). □

Lemma 2.14 (cf. [CHX24, Lemmas 2.4 and 2.14])**.** *Let $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ be a finite set such that $0, 1 \in \mathfrak{R}$. Let $\Phi := \Phi(\mathfrak{R})$, and let $\mathcal{N}$ be a finite set of positive integers.*

(1) There exists a finite set $\mathfrak{R}' \subset [0,1] \cap \mathbb{Q}$ such that $0, 1 \in \mathfrak{R}'$ and $\Gamma(\mathcal{N}, \Phi) = \Phi(\mathfrak{R}')$. In particular, $\Gamma(\mathcal{N}, \Phi)$ satisfies the DCC, and $B_{\Gamma(\mathcal{N},\Phi)}$ is defined for any $\mathbb{R}$-divisor $B \in [0,1]$.

(2) For any positive integer n and any real numbers $0 \leq b \leq b' \leq 1$, we have

$$n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor \leq n\lfloor b' \rfloor + \lfloor (n+1)\{b'\} \rfloor.$$

(3) Let $n \in \mathcal{N}$, $b \in [0,1]$, and b' the largest element of $\Gamma(\mathcal{N}, \Phi)$ such that $b' \leq b$. Then

$$n\lfloor b' \rfloor + \lfloor (n+1)\{b'\} \rfloor = n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor.$$

(4) Let $(X/Z \ni z, B)$ be a pair such that $B \in [0,1]$. Let $n \in \mathcal{N}$, and let $B' \in [0,1]$ be an $\mathbb{R}$-divisor on X such that $B' \geq B_{\Gamma(\mathcal{N},\Phi)}$. Then for any $\mathbb{Q}$-divisor $B^+ \geq 0$ on X such that, over some neighborhood of z, (X, B^+) is lc, $n(K_X + B^+) \sim 0$, and $nB^+ \geq n\lfloor B' \rfloor + \lfloor (n+1)\{B'\} \rfloor$, the pair $(X/Z \ni z, B^+)$ is an n-complement of $(X/Z \ni z, B)$.

Proof. (1) Let

$$\mathfrak{R}' := \left\{ r - \sum_{n \in \mathcal{N}} \frac{u_n}{n+1} \;\middle|\; r \in \mathfrak{R},\ u_n \in \mathbb{Z}_{\geq 0} \right\} \cap [0,1].$$

For an element $r - \sum_{n \in \mathcal{N}} u_n/(n+1)$ of $\mathfrak{R}'$, we have $u_n \leq (n+1)r \leq n+1$ for any $n \in \mathcal{N}$. Thus $\mathfrak{R}' \subset [0,1] \cap \mathbb{Q}$ is finite, and taking all $u_n = 0$ and $r = 0, 1$ shows that $0, 1 \in \mathfrak{R}'$. For $r \in \mathfrak{R}$, $j \in \mathbb{Z}_{>0}$, and $u_n \in \mathbb{Z}_{\geq 0}$, let $r' := r - \sum_{n \in \mathcal{N}} u_n/(n+1)$. Then

$$1 - \frac{r}{j} + \frac{1}{j} \sum_{n \in \mathcal{N}} \frac{u_n}{n+1} = 1 - \frac{r'}{j}.$$

Since $r' \leq r \leq 1 \leq j$, we have $1 - r'/j \geq 0$, and $1 - r'/j \leq 1$ if and only if $r' \geq 0$. Thus $\Gamma(\mathcal{N}, \Phi) = \Phi(\mathfrak{R}')$.

Let $0 \leq b < 1$. If $1 - r'/j \leq b$ for some $r' \in \mathfrak{R}'$ and $j \in \mathbb{Z}_{>0}$, then $r' \geq (1-b)j > 0$ and $j \leq r'/(1-b) \leq 1/(1-b)$. Hence $\{\gamma \in \Gamma(\mathcal{N}, \Phi) \mid \gamma \leq b\}$ is a finite set, and it contains 0. Since $1 \in \Gamma(\mathcal{N}, \Phi)$, the set $\{\gamma \in \Gamma(\mathcal{N}, \Phi) \mid \gamma \leq b\}$ has a maximum for any $b \in [0,1]$, so $B_{\Gamma(\mathcal{N},\Phi)}$ is defined for any $\mathbb{R}$-divisor $B \in [0,1]$. A non-increasing sequence in $\Gamma(\mathcal{N}, \Phi)$ either has all terms equal to 1, or has all terms from some index on in the finite set $\{\gamma \in \Gamma(\mathcal{N}, \Phi) \mid \gamma \leq b\}$ for some $b < 1$. In both cases the sequence stabilizes, so $\Gamma(\mathcal{N}, \Phi)$ satisfies the DCC.

(2) By Lemma 2.13(1), for $0 \leq c < 1$ we have

$$n\lfloor c \rfloor + \lfloor (n+1)\{c\} \rfloor = \lfloor (n+1)c \rfloor \leq n,$$

and for $c = 1$ we have $n\lfloor c \rfloor + \lfloor (n+1)\{c\} \rfloor = n$. If $b' < 1$, then (2) reads $\lfloor (n+1)b \rfloor \leq \lfloor (n+1)b' \rfloor$, which holds since $b \leq b'$. If $b' = 1$, then the right-hand side of (2) equals n, and the left-hand side is at most n.

(3) We first show that $\frac{k}{n+1} \in \Gamma(\mathcal{N}, \Phi)$ for any integer $0 \leq k \leq n+1$: take $r = 1$, $j = 1$, $u_n = k$, and $u_m = 0$ for any $m \in \mathcal{N} \backslash \{n\}$. If $b = 1$, then $b' = 1$. Assume that $b < 1$, and let $k := \lfloor (n+1)b \rfloor$. Then $\frac{k}{n+1} \leq b$, so $\frac{k}{n+1} \leq b' \leq b$, and

$$k \leq (n+1)b' \leq (n+1)b < k + 1.$$

Thus $\lfloor (n+1)b' \rfloor = k$ and $b' < 1$, and Lemma 2.13(1) implies that both sides of (3) equal k.

(4) Let P be a prime divisor on X, and let b and b' be the coefficients of P in B and B' respectively. The coefficient of P in $B_{\Gamma(\mathcal{N},\Phi)}$ is the largest element b_0 of $\Gamma(\mathcal{N},\Phi)$ such that $b_0 \le b$, and $b' \ge b_0$. By (2) and (3),

$$n\lfloor b' \rfloor + \lfloor (n+1)\{b'\} \rfloor \ge n\lfloor b_0 \rfloor + \lfloor (n+1)\{b_0\} \rfloor = n\lfloor b \rfloor + \lfloor (n+1)\{b\} \rfloor.$$

Thus, over some neighborhood of z,

$$nB^+ \ge n\lfloor B' \rfloor + \lfloor (n+1)\{B'\} \rfloor \ge n\lfloor B \rfloor + \lfloor (n+1)\{B\} \rfloor,$$

and $(X/Z \ni z, B^+)$ is an n-complement of $(X/Z \ni z, B)$. □

Section 11 uses Lemma 2.14(4) to replace a boundary with its truncations when constructing n-complements.

Lemma 2.15 (cf. [Bir19, 6.1(2)(3)], [CHX24, Lemma 2.13]). *Let $f : X \to Z$ and $f' : X' \to Z$ be contractions and $g : X \dashrightarrow X'$ a birational map over Z. Let $B \ge 0$ and $B' \ge 0$ be $\mathbb{Q}$-divisors on X and X' respectively such that $K_X + B$ and $K_{X'} + B'$ are $\mathbb{Q}$-Cartier. Let $p : W \to X$ and $q : W \to X'$ be a common resolution such that $q = g \circ p$, and assume that $p_*\big(q^*(K_{X'} + B') - p^*(K_X + B)\big) \ge 0$. Let n be a positive integer, L a Cartier divisor on Z, φ a rational function on X', and $B'^+ \ge B'$ a $\mathbb{Q}$-divisor on X' such that (X', B'^+) is lc and $n(K_{X'} + B'^+) = f'^*L + \operatorname{div}_{X'}(\varphi)$. Let B^+ be the $\mathbb{Q}$-divisor on X such that $K_X + B^+ = p_*q^*(K_{X'} + B'^+)$. Then $B^+ \ge B$, (X, B^+) is lc, and $n(K_X + B^+) = f^*L + \operatorname{div}_X(g^*\varphi)$. In particular, $(X/Z \ni z, B^+)$ is a monotonic n-complement of $(X/Z \ni z, B)$ for any $z \in Z$.*

Proof. Since g is a birational map over Z, $f' \circ q = f \circ p$. Pulling back

$$n(K_{X'} + B'^+) = f'^*L + \operatorname{div}_{X'}(\varphi)$$

to W, we obtain

$$nq^*(K_{X'} + B'^+) = (f \circ p)^*L + \operatorname{div}_W(q^*\varphi).$$

Since $q^*\varphi = p^*g^*\varphi$, pushing forward to X implies

$$n(K_X + B^+) = f^*L + \operatorname{div}_X(g^*\varphi).$$

In particular, $n(K_X + B^+)$ is Cartier, and pulling back the last equality to W implies

$$p^*(K_X + B^+) = q^*(K_{X'} + B'^+).$$

Hence $a(E, X, B^+) = a(E, X', B'^+)$ for any prime divisor E over X [KM98, Lemma 2.30].

Since $K_{X'} + B'^+$ and $K_{X'} + B'$ are $\mathbb{Q}$-Cartier, $B'^+ - B' \ge 0$ is $\mathbb{Q}$-Cartier, so $q^*(B'^+ - B') \ge 0$. Thus

$$B^+ - B = p_*\big(q^*(K_{X'} + B') - p^*(K_X + B)\big) + p_*q^*(B'^+ - B') \ge 0.$$

In particular, $B^+ \ge B \ge 0$, and (X, B^+) is lc because (X', B'^+) is lc.

Let $z \in Z$. Since L is Cartier, there exists an open neighborhood Z^0 of z such that $L|_{Z^0} \sim 0$, so $n(K_X + B^+) \sim 0$ over Z^0. Thus $(X/Z \ni z, B^+)$ is a monotonic n-complement of $(X/Z \ni z, B)$. □

2.3. **Prime divisors over the base and contractions.**

Lemma 2.16. *Let $g : Y \to T$ be a dominant morphism between normal varieties and E a prime divisor over Y whose center on Y does not dominate T, or equivalently, such that $\operatorname{mult}_E \operatorname{div}(g^*\varphi) \ne 0$ for some non-zero rational function φ on T. Then there exist a positive integer e, a smooth variety T' with a projective birational morphism $\tau : T' \to T$, and a prime divisor Q on T' such that*

$$\operatorname{mult}_E \operatorname{div}(g^*\varphi) = e \operatorname{mult}_Q \operatorname{div}(\tau^*\varphi)$$

for any non-zero rational function φ on T.

Moreover, let $E_1, \dots, E_r$ be prime divisors over Y whose centers on Y do not dominate T, and let T_0 be a variety projective and birational over T. Then there exist a smooth variety T' with a

projective birational morphism $\tau : T' \to T$ such that T' dominates T_0, prime divisors $Q_1, \dots, Q_r$ on T', and positive integers $e_1, \dots, e_r$, such that

$$\operatorname{mult}_{E_i} \operatorname{div}(g^*\varphi) = e_i \operatorname{mult}_{Q_i} \operatorname{div}(\tau^*\varphi)$$

for any i and any non-zero rational function φ on T.

Proof. Let Y_E be a normal variety with a projective birational morphism $Y_E \to Y$ such that E is a prime divisor on Y_E. Let $g_E : Y_E \to T$ be the induced morphism, and V_E the closure of $g_E(E)$, which is the closure of the image under g of the center of E on Y.

We first show that the condition $V_E \neq T$ is equivalent to the existence of a non-zero rational function φ on T such that $\operatorname{mult}_E \operatorname{div}(g^*\varphi) \neq 0$. If $V_E = T$, then any non-zero rational function φ on T is regular and nowhere vanishing on a non-empty open subset $T^0 \subset T$, $E \cap g_E^{-1}(T^0) \neq \emptyset$, and $g_E^*\varphi$ is regular and nowhere vanishing on $g_E^{-1}(T^0)$, so $\operatorname{mult}_E \operatorname{div}(g^*\varphi) = 0$. If $V_E \neq T$, then for an affine open subset $T^0 \subset T$ such that $V_E \cap T^0 \neq \emptyset$ and a non-zero regular function φ on T^0 which vanishes on $V_E \cap T^0$, we have $E \cap g_E^{-1}(T^0) \neq \emptyset$, and $g_E^*\varphi$ is regular on $g_E^{-1}(T^0)$ and vanishes on $E \cap g_E^{-1}(T^0)$, so $\operatorname{mult}_E \operatorname{div}(g^*\varphi) > 0$.

Assume that $V_E \neq T$. By generic flatness [Gro65, Théorème 6.9.1], there exists a non-empty open subset $U \subset T$ over which g_E is flat. By flattening by blow-ups [RG71, Théorème 5.2.2], there exists a projective birational morphism $\tau_1 : T_1 \to T$ from a variety T_1 which is an isomorphism over U, such that the closure Y_1 of $g_E^{-1}(U)$ in $Y_E \times_T T_1$ is flat over T_1. Let $\tau_2 : T' \to T_1$ be a resolution of singularities and $\tau := \tau_1 \circ \tau_2$. Since Y_E is normal, the local ring of Y_E at the generic point of E is a discrete valuation ring, and since τ_1 and τ are proper, the valuative criterion of properness [Har77, Theorem II.4.7] implies that the rational maps $Y_E \dashrightarrow T_1$ and $Y_E \dashrightarrow T'$ over T are morphisms $g_1 : Y_E^0 \to T_1$ and $g' : Y_E^0 \to T'$ on an open subset $Y_E^0 \subset Y_E$ such that $E \cap Y_E^0 \neq \emptyset$, and $g_1 = \tau_2 \circ g'$.

Let $\mu : Y_1 \to Y_E$ be the projection, which is projective and birational. The morphism $\sigma : Y_E^0 \to Y_E \times_T T_1$ with components the inclusion and g_1 maps Y_E^0 into Y_1, since Y_E^0 is irreducible and σ maps $Y_E^0 \cap g_E^{-1}(U)$ into Y_1. Since $\mu \circ \sigma$ is the inclusion of Y_E^0, and $\sigma \circ \mu$ agrees with the identity of the variety $\mu^{-1}(Y_E^0)$ over U, μ is an isomorphism over Y_E^0. Let E_{Y_1} be the strict transform of E on Y_1, η_1 the generic point of E_{Y_1}, and $\xi_1 := g_1(\mu(\eta_1))$, the image of η_1 under $Y_1 \to T_1$. Since μ is an isomorphism near η_1, $\dim \mathcal{O}_{Y_1,\eta_1} = 1$. Since Y_1 is flat over T_1, the local homomorphism $\mathcal{O}_{T_1,\xi_1} \to \mathcal{O}_{Y_1,\eta_1}$ is flat, and the dimension formula for flat local homomorphisms [Mat89, Theorem 15.1] implies $\dim \mathcal{O}_{T_1,\xi_1} \leq 1$. Since $\tau_1(\xi_1)$ is the generic point of $V_E \neq T$, ξ_1 is not the generic point of T_1. Thus the closure Q_{T_1} of ξ_1 is a prime divisor on T_1, and $g_1(E \cap Y_E^0)$ is dense in Q_{T_1}. Let Q be the closure of $g'(E \cap Y_E^0)$ in T'. Since τ_2 is proper and $g_1 = \tau_2 \circ g'$, $\tau_2(Q) = Q_{T_1}$, so $\dim Q \geq \dim T - 1$. Since $\tau(Q) \subset V_E \neq T$, $Q \neq T'$. Thus Q is a prime divisor on T', and $g'(E \cap Y_E^0)$ is dense in Q.

Since T' is smooth, there exist an open subset $T'^0 \subset T'$ such that $Q \cap T'^0 \neq \emptyset$ and a regular function x on T'^0 such that $\operatorname{div}(x) = Q|_{T'^0}$. Since $g'(E \cap Y_E^0)$ is dense in Q, $E \cap g'^{-1}(T'^0) \neq \emptyset$, and g'^*x is regular on $g'^{-1}(T'^0)$ and vanishes on $E \cap g'^{-1}(T'^0)$. Hence $e := \operatorname{mult}_E \operatorname{div}(g'^*x)$ is a positive integer. Let φ be a non-zero rational function on T and $k := \operatorname{mult}_Q \operatorname{div}(\tau^*\varphi)$. Then $x^{-k}\tau^*\varphi$ is regular and nowhere vanishing on an open subset of T'^0 intersecting Q, so $g'^*(x^{-k}\tau^*\varphi)$ is regular and nowhere vanishing on an open subset of Y_E^0 intersecting E. Since $g_E^*\varphi = g'^*\tau^*\varphi$ on Y_E^0, we obtain

$$\operatorname{mult}_E \operatorname{div}(g^*\varphi) = k \operatorname{mult}_E \operatorname{div}(g'^*x) = e \operatorname{mult}_Q \operatorname{div}(\tau^*\varphi).$$

We prove the second statement. By the first statement, for any $1 \leq i \leq r$ there exist a smooth variety T_i' with a projective birational morphism $\tau_i' : T_i' \to T$, a prime divisor Q_i' on T_i', and a positive integer e_i, such that

$$\operatorname{mult}_{E_i} \operatorname{div}(g^*\varphi) = e_i \operatorname{mult}_{Q_i'} \operatorname{div}(\tau_i'^*\varphi)$$

for any non-zero rational function φ on T. Let T' be a resolution of singularities of the irreducible component of $T_0 \times_T T_1' \times_T \cdots \times_T T_r'$ that dominates T, and $\tau : T' \to T$ the induced morphism. Then T' is a smooth variety projective and birational over T that dominates $T_0, T_1', \ldots, T_r'$. For any i, let Q_i be the strict transform of Q_i' on T'. Since T_i' is normal, the morphism $T' \to T_i'$ is an isomorphism over the generic point of Q_i', hence

$$\operatorname{mult}_{Q_i} \operatorname{div}(\tau^*\varphi) = \operatorname{mult}_{Q_i'} \operatorname{div}(\tau_i'^*\varphi)$$

for any non-zero rational function φ on T. The second statement follows. □

Lemma 2.17. *Let $g : Y \to T$ be a contraction.*

(1) A $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on T satisfies $D \geq 0$ if and only if $g^*D \geq 0$.
(2) Let D be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T and φ a rational function on Y such that $g^*D = \operatorname{div}(\varphi)$. Then $\varphi = g^*\varphi_0$ for a rational function φ_0 on T with $D = \operatorname{div}(\varphi_0)$.
(3) Let $G \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y such that $G \sim_{\mathbb{Q}} g^*H$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H on T. Then there exists a unique $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $P \geq 0$ on T such that $G = g^*P$, and $P \sim_{\mathbb{Q}} H$.
(4) If $g = \beta \circ \alpha$ for a dominant rational map $\alpha : Y \dashrightarrow T_1$ to a variety T_1 and a generically finite morphism $\beta : T_1 \to T$, then β is birational. Moreover, any non-zero rational function ψ on Y such that no component of $\operatorname{div}(\psi)$ dominates T is of the form $g^*\psi_0$ for a rational function ψ_0 on T. Finally, there exists a non-empty open subset $T^0 \subset T$ such that g is flat over T^0, the fiber of g over any point of T^0 is geometrically normal and geometrically integral, and the fiber of g over any closed point of T^0 is a normal projective variety.

Proof. We first show the following two properties.

(a) Let $T^0 \subset T$ be a non-empty open subset and ψ a non-zero rational function on Y such that $\operatorname{div}(\psi) \geq 0$ on $g^{-1}(T^0)$. Then $\psi = g^*\psi_0$ for a rational function ψ_0 on T which is regular on T^0.
(b) For any prime divisor S on T, there exist a prime divisor S_Y on Y and a positive integer a such that $\operatorname{mult}_{S_Y} D = a \operatorname{mult}_S D$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on T.

For (a), since Y is normal and $\operatorname{div}(\psi) \geq 0$ on $g^{-1}(T^0)$, ψ is regular on $g^{-1}(T^0)$ [Har77, Proposition II.6.3A]. Since $g_*\mathcal{O}_Y = \mathcal{O}_T$, the regular functions on $g^{-1}(T^0)$ are the pullbacks of the regular functions on T^0, so $\psi = g^*\psi_0$ for a regular function ψ_0 on T^0, that is, a rational function ψ_0 on T which is regular on T^0.

For (b), since T is normal, the local ring $\mathcal{O}_{T,S}$ of T at the generic point of S is a discrete valuation ring. Let t be a uniformizer of $\mathcal{O}_{T,S}$ and $Y_S := Y \times_T \operatorname{Spec} \mathcal{O}_{T,S}$. Let s be the closed point of $\operatorname{Spec} \mathcal{O}_{T,S}$ and Y_s the fiber of $Y_S \to \operatorname{Spec} \mathcal{O}_{T,S}$ over s. Since g is surjective, Y_s is non-empty. The fiber Y_s is the zero locus of g^*t, so by Krull's principal ideal theorem Y_s has an irreducible component S_0 of codimension one in Y_S. Let S_Y be the closure of S_0 in Y and $a := \operatorname{mult}_{S_Y} \operatorname{div}(g^*t)$. Then S_Y is a prime divisor on Y dominating S, and a is a positive integer. For any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on T, we have $D = (\operatorname{mult}_S D) \operatorname{div}(t)$ near the generic point of S, hence $\operatorname{mult}_{S_Y} D = a \operatorname{mult}_S D$.

In particular, (b) implies that g^* is injective on $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on T: if $g^*D = g^*D'$, then $\operatorname{mult}_S(D - D') = a^{-1} \operatorname{mult}_{S_Y}(D - D') = 0$ for any prime divisor S on T.

(1) If $D \geq 0$, then $g^*D \geq 0$. If $g^*D \geq 0$, then (b) implies $\operatorname{mult}_S D = a^{-1} \operatorname{mult}_{S_Y} D \geq 0$ for any prime divisor S on T, hence $D \geq 0$.

(2) Let $T^0 := T \setminus \operatorname{Supp} D$. Then $\operatorname{div}(\varphi) = g^*D = 0$ on $g^{-1}(T^0)$, and (a) implies $\varphi = g^*\varphi_0$ for a rational function φ_0 on T. Thus

$$g^*D = \operatorname{div}(g^*\varphi_0) = g^* \operatorname{div}(\varphi_0),$$

and the injectivity of g^* implies $D = \operatorname{div}(\varphi_0)$.

(3) There exist a positive integer r and a rational function u on Y such that

$$G = g^*H + \frac{1}{r}\operatorname{div}(u).$$

Let $T^0 := T \setminus \operatorname{Supp} H$. Since $g^*H = 0$ on $g^{-1}(T^0)$, we have $\frac{1}{r}\operatorname{div}(u) = G \geq 0$ on $g^{-1}(T^0)$, and (a) implies $u = g^*u_0$ for a rational function u_0 on T. Let

$$P := H + \frac{1}{r}\operatorname{div}(u_0).$$

Then P is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T such that $G = g^*P$ and $P \sim_{\mathbb{Q}} H$. By (1), $P \geq 0$, and the injectivity of g^* implies the uniqueness of P.

(4) Let α and β be as in (4). Since g is surjective and α is dominant, β is dominant, so there exists a non-empty affine open subset $T^0 \subset T$ over which β is finite [Har77, Exercise II.3.7]. Then $T_1^0 := \beta^{-1}(T^0)$ is affine. Let s be a regular function on T_1^0. Since β is finite over T^0, s satisfies a monic polynomial equation whose coefficients are regular functions on T^0. Hence the rational function α^*s on Y satisfies a monic polynomial equation whose coefficients are regular functions on $g^{-1}(T^0)$. Since Y is normal, α^*s is regular on $g^{-1}(T^0)$, so (a) implies $\alpha^*s = g^*s_0 = \alpha^*\beta^*s_0$ for a regular function s_0 on T^0. Since α is dominant, $s = \beta^*s_0$. Thus the pullback by β from the regular functions on T^0 to the regular functions on T_1^0 is surjective, and it is injective since β is dominant. Since T^0 and T_1^0 are affine, β is an isomorphism over T^0, hence β is birational.

If no component of $\operatorname{div}(\psi)$ dominates T, then $g(\operatorname{Supp}\operatorname{div}(\psi))$ is a proper closed subset of T, since g is projective. Let $T^0 := T \setminus g(\operatorname{Supp}\operatorname{div}(\psi))$. Then $\operatorname{div}(\psi) = 0$ on $g^{-1}(T^0)$, so (a) implies $\psi = g^*\psi_0$ for a rational function ψ_0 on T.

For the last statement, by generic flatness [Gro65, Théorème 6.9.1], g is flat over a non-empty open subset $T^1 \subset T$. Since the local rings of the fiber of g over the generic point of T are local rings of Y, the fiber of g over the generic point of T is normal, hence geometrically normal since the characteristic is zero [Gro65, Proposition 6.7.7]. By [Gro66, Théorème 12.2.4(iv)], applied to the proper flat morphism $g^{-1}(T^1) \to T^1$, the points of T^1 over which the fiber of g is geometrically normal form an open subset $T^0 \subset T^1$, which is non-empty since it contains the generic point of T. Any fiber of g is projective since g is projective, and connected since $g_*\mathcal{O}_Y = \mathcal{O}_T$ [Har77, Corollary III.11.3]. The local rings of a normal scheme are integral domains, so two distinct irreducible components of a normal noetherian scheme do not meet, and a connected normal noetherian scheme is irreducible. Thus the fiber of g over any closed point of T^0 is a normal projective variety, hence geometrically integral since $\mathbb{C}$ is algebraically closed. By [Gro66, Théorème 12.2.4(viii)], applied to the proper flat morphism $g^{-1}(T^0) \to T^0$, the points of T^0 over which the fiber of g is geometrically integral form an open subset of T^0, which contains all closed points of T^0 and hence is T^0. Thus the fiber of g over any point of T^0 is geometrically normal and geometrically integral. □

2.4. **Relative Iitaka dimension and a uniform index.**

Definition 2.18. Let $\pi : X \to Z$ be a contraction. We say that a property holds for a *general fiber* of π if there exists a non-empty open subset $Z^0 \subset Z$ such that the fiber of π over any closed point of Z^0 has the property. We say that a property holds for a *very general fiber* of π if there exist countably many proper closed subsets of Z such that the fiber of π over any closed point of Z outside their union has the property. Let $D \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X and m_0 a positive integer such that m_0D is Cartier. The *relative Iitaka dimension* of D over Z is

$$\kappa(X/Z, D) := \kappa(F, D|_F) := \kappa(F, m_0D|_F)$$

for a very general fiber F of π, where $\kappa(F, m_0D|_F)$ is the Iitaka dimension of the Cartier divisor $m_0D|_F$ on the normal projective variety F [Laz04, Definition 2.1.3]. By Lemma 2.19, F is a normal projective variety such that $F \not\subset \operatorname{Supp} D$, and $\kappa(F, D|_F)$ depends neither on F nor on m_0. If Z is a point, then $F = X$ and $\kappa(X/Z, D) = \kappa(X, D)$. If X is $\mathbb{Q}$-factorial, then a *D-MMP*

over Z is a finite sequence of D-negative divisorial contractions and D-flips over Z between $\mathbb{Q}$-factorial varieties, where after each step D is replaced by the strict transform of D.

Lemma 2.19. *Let $\pi : X \to Z$ be a contraction, $D \geq 0$ a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, and m_0 a positive integer such that m_0D is Cartier. For any positive integer l, let r_l be the rank of the coherent sheaf $\pi_*\mathcal{O}_X(lm_0D)$ on Z.*

(1) There exist non-empty open subsets $Z_1, Z_2, \dots$ of Z such that, for any closed point $t \in \bigcap_{l\geq 1} Z_l$, the fiber $F := \pi^{-1}(t)$ is a normal projective variety of dimension $\dim X - \dim Z$, $F \not\subset \operatorname{Supp} D$, and

$$h^0(F, \mathcal{O}_F(lm_0D|_F)) = r_l$$

for any positive integer l.

(2) For any F as in (1), $\kappa(F, m_0D|_F)$ is the unique integer $\kappa \geq 0$ such that $al^\kappa \leq r_l \leq Al^\kappa$ for some real numbers $a, A > 0$ and any integer $l \gg 0$. In particular, $\kappa(X/Z, D)$ depends neither on the choice of the very general fiber nor on m_0, and it depends only on the ranks r_l.

(3) $\kappa(X/Z, D) = \dim X - \dim Z$ if and only if m_0D is π-big in the sense of [KMM87, Definition 0-3-2].

Proof. (1) By Lemma 2.17(4), there exists a non-empty open subset $Z^0 \subset Z$ such that π is flat over Z^0 and $\pi^{-1}(t)$ is a normal projective variety for any closed point $t \in Z^0$. Then $\dim \pi^{-1}(t) = \dim X - \dim Z$ for any such t [Har77, Corollary III.9.6]. Any component P of $\operatorname{Supp} D$ satisfies $\dim P < \dim X$, so over the closed points of a non-empty open subset of Z, the fibers of $P \to Z$ are empty or of dimension $\dim P - \dim Z < \dim X - \dim Z$ [Har77, Exercise II.3.22]. Shrinking Z^0, we may assume that $\pi^{-1}(t) \not\subset \operatorname{Supp} D$ for any closed point $t \in Z^0$. Then for any such t and any positive integer l, the restriction $D|_{\pi^{-1}(t)} \geq 0$ is defined, $lm_0D|_{\pi^{-1}(t)}$ is Cartier, and

$$\mathcal{O}_X(lm_0D)|_{\pi^{-1}(t)} = \mathcal{O}_{\pi^{-1}(t)}(lm_0D|_{\pi^{-1}(t)}).$$

Fix a positive integer l. Since $\mathcal{O}_X(lm_0D)$ is invertible and π is flat over Z^0, the function

$$t \to h^0\big(\pi^{-1}(t), \mathcal{O}_X(lm_0D)|_{\pi^{-1}(t)}\big)$$

on Z^0 is upper semicontinuous [Har77, Theorem III.12.8]. Let h_l be the minimum value of the function. Then there exists a non-empty open subset $Z_l \subset Z^0$ such that the function is equal to h_l at any point of Z_l. By Grauert's theorem [Har77, Corollary III.12.9], $\pi_*\mathcal{O}_X(lm_0D)$ is locally free of rank h_l over Z_l, so $h_l = r_l$. At a closed point of $Z^0 \backslash Z_l$ the value may be larger than r_l, and Z_l depends on l. Thus, for a fixed l, the equality in (1) holds on the fiber of π over any closed point of Z_l, that is, on a general fiber of π. The equality holds for all l at once on the fiber of π over any closed point of $\bigcap_{l\geq 1} Z_l$, that is, on a very general fiber of π, since $Z \backslash \bigcap_{l\geq 1} Z_l$ is the union of the countably many proper closed subsets $Z \backslash Z_l$.

(2) Let F be as in (1). Since $D|_F \geq 0$, $h^0(F, \mathcal{O}_F(lm_0D|_F)) \geq 1$ for any positive integer l. By (1) and [Laz04, Corollary 2.1.38] applied to the line bundle $\mathcal{O}_F(m_0D|_F)$ on the normal projective variety F, there exist real numbers $a, A > 0$ such that $al^\kappa \leq r_l \leq Al^\kappa$ for any integer $l \gg 0$, where $\kappa := \kappa(F, m_0D|_F)$. If $\kappa' \neq \kappa$ is a non-negative integer, then $l^{\kappa'-\kappa}$ tends to 0 or to $+\infty$, so no real numbers $a', A' > 0$ satisfy $a'l^{\kappa'} \leq r_l \leq A'l^{\kappa'}$ for any integer $l \gg 0$. This proves the uniqueness of κ and shows that $\kappa(F, m_0D|_F)$ depends only on the ranks r_l. If m_0 is replaced by em_0 for a positive integer e, then r_l is replaced by r_{el}, and $ae^\kappa l^\kappa \leq r_{el} \leq Ae^\kappa l^\kappa$ for any integer $l \gg 0$. Thus $\kappa(F, m_0D|_F)$ does not depend on m_0.

(3) The generic fiber of π, that is, the fiber of π over the generic point of Z, is a normal projective variety of dimension $\dim X - \dim Z$ over the function field of Z. By flat base change [Har77, Proposition III.9.3], the global sections of the restriction of lm_0D to the generic fiber of π form a vector space of dimension r_l over the function field of Z. By [KMM87, Definition 0-3-2], m_0D is π-big if and only if the restriction of m_0D to the generic fiber of π has Iitaka

dimension $\dim X - \dim Z$, that is, $r_l \geq al^{\dim X - \dim Z}$ for some real number $a > 0$ and any integer $l \gg 0$ (cf. [Laz04, Lemma 2.2.3]). By (2), $r_l \leq Al^{\kappa(X/Z,D)}$ for any integer $l \gg 0$, and $\kappa(X/Z, D) \leq \dim F = \dim X - \dim Z$ for any F as in (1). Together with the lower bound in (2), this implies (3). □

Lemma 2.20. *Let $\pi : X \to Z$ be a contraction and $D_1 \geq D_2 \geq 0$ $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on X.*

(1) $\kappa(X/Z, D_1) \geq \kappa(X/Z, D_2)$, and equality holds if $\operatorname{Supp} D_1 = \operatorname{Supp} D_2$.

(2) Assume that X is $\mathbb{Q}$-factorial, and let $X \dashrightarrow X'$ be a D_2-MMP over Z. Let $D_{1,X'}$ and $D_{2,X'}$ be the strict transforms of D_1 and D_2 on X'. Then $\kappa(X'/Z, D_{1,X'}) = \kappa(X/Z, D_1)$ and $\kappa(X'/Z, D_{2,X'}) = \kappa(X/Z, D_2)$.

Proof. (1) Let m be a positive integer such that mD_1 and mD_2 are Cartier. By Lemma 2.19(1) applied to D_1 and to D_2 with $m_0 = m$, there exist countably many proper closed subsets of Z such that, for any closed point t of Z outside their union, the fiber $F := \pi^{-1}(t)$ is a normal projective variety, $F \not\subset \operatorname{Supp} D_1$, and $\kappa(X/Z, D_i) = \kappa(F, mD_i|_F)$ for $i = 1, 2$. We need a very general fiber here, since the definition of $\kappa(X/Z, D_i)$ involves the sections of lmD_i for all l. Fix such a fiber F. Since $\operatorname{Supp} D_2 \subset \operatorname{Supp} D_1$ and $mD_1 - mD_2 \geq 0$ is Cartier, $lm(D_1 - D_2)|_F \geq 0$ is a Cartier divisor on F for any positive integer l, and multiplication by the section of $\mathcal{O}_F(lm(D_1 - D_2)|_F)$ defined by $lm(D_1 - D_2)|_F$ induces an inclusion

$$H^0(F, \mathcal{O}_F(lmD_2|_F)) \subset H^0(F, \mathcal{O}_F(lmD_1|_F)).$$

By Lemma 2.19(2), $\kappa(X/Z, D_1) \geq \kappa(X/Z, D_2)$. If $\operatorname{Supp} D_1 = \operatorname{Supp} D_2$, then there exists a positive integer N such that $ND_2 \geq D_1$, hence $\kappa(X/Z, D_2) = \kappa(X/Z, ND_2) \geq \kappa(X/Z, D_1)$.

(2) The equality for D_1 is [CHX24, Lemma 2.11]. We explain both equalities.

We first show that a step preserves the relative Iitaka dimension of the divisor with respect to which it is negative. Let V be a $\mathbb{Q}$-factorial variety with a contraction $\pi_V : V \to Z$, $D \geq 0$ a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on V, and $V \dashrightarrow V'$ a D-negative divisorial contraction or D-flip over Z, with contraction $V \to Y_V$ over Z. Let $D_{V'}$ be the strict transform of D on V' and $p : W \to V$, $q : W \to V'$ a common resolution. Let

$$F := p^*D - q^*D_{V'}.$$

Since $V \dashrightarrow V'$ does not extract divisors, $q_*F = D_{V'} - D_{V'} = 0$, and the pushforward of F to Y_V is zero. The $\mathbb{Q}$-divisor $-p^*D$ is nef over Y_V since $-D$ is ample over Y_V, and $q^*D_{V'}$ is nef over Y_V since $D_{V'}$ is either the pullback of a $\mathbb{Q}$-divisor on $Y_V = V'$ or ample over Y_V. Thus $-F$ is nef over Y_V, and the negativity lemma [KM98, Lemma 3.39(1)] implies $F \geq 0$. Let m be a positive integer such that mD, $mD_{V'}$, and mF are Cartier, and let l be a positive integer. Since $F \geq 0$ is q-exceptional and V' is normal, $q_*\mathcal{O}_W(lmF) = \mathcal{O}_{V'}$ (cf. [Har77, Proposition II.6.3A]), and the projection formula implies $q_*\mathcal{O}_W(lmq^*D_{V'} + lmF) = \mathcal{O}_{V'}(lmD_{V'})$. Since V is normal, $p_*\mathcal{O}_W = \mathcal{O}_V$, and the projection formula implies $p_*\mathcal{O}_W(lmp^*D) = \mathcal{O}_V(lmD)$. Let $\pi_{V'} : V' \to Z$ be the induced morphism. Since $V \dashrightarrow V'$ is over Z, $\pi_V \circ p = \pi_{V'} \circ q$, and $\pi_{V'*}\mathcal{O}_{V'} = \pi_{V'*}q_*\mathcal{O}_W = \pi_{V*}p_*\mathcal{O}_W = \mathcal{O}_Z$, so $\pi_{V'}$ is a contraction. Pushing forward to Z and using $lmp^*D = lmq^*D_{V'} + lmF$, we obtain

$$\pi_{V*}\mathcal{O}_V(lmD) = (\pi_V \circ p)_*\mathcal{O}_W(lmp^*D) = (\pi_{V'} \circ q)_*\mathcal{O}_W(lmp^*D) = \pi_{V'*}\mathcal{O}_{V'}(lmD_{V'}).$$

Since l is arbitrary, Lemma 2.19(2) implies $\kappa(V'/Z, D_{V'}) = \kappa(V/Z, D)$.

Let $X = X_0 \dashrightarrow X_1 \dashrightarrow \cdots \dashrightarrow X_r = X'$ be the steps of the D_2-MMP, and for any $0 \leq j \leq r$ let $D_{1,j}$ and $D_{2,j}$ be the strict transforms of D_1 and D_2 on X_j. By the previous paragraph and induction on j, the induced morphisms $X_j \to Z$ are contractions. Applying the equality $\kappa(V'/Z, D_{V'}) = \kappa(V/Z, D)$ to the steps $X_j \dashrightarrow X_{j+1}$ and $D = D_{2,j}$ for $j = 0, \ldots, r-1$, we obtain $\kappa(X'/Z, D_{2,X'}) = \kappa(X/Z, D_2)$.

For any $0 \leq j < r$, let $X_j \to Y_j$ be the contraction over Z of the j-th step. Since $-D_{2,j}$ is ample over Y_j, $D_{2,j+1}$ is ample over Y_j when the j-th step is a flip, ampleness over Y_j is an open condition on the finite-dimensional space $N^1(\cdot/Y_j)$ [KM98, Theorem 1.44], and the

sequence has finitely many steps, for any rational number $0 < \epsilon \ll 1$ the sequence $X \dashrightarrow X'$ is also a $(D_2 + \epsilon D_1)$-MMP over Z. Applying the equality $\kappa(V'/Z, D_{V'}) = \kappa(V/Z, D)$ to the steps $X_j \dashrightarrow X_{j+1}$ and $D = D_{2,j} + \epsilon D_{1,j}$ for $j = 0, \dots, r-1$, we obtain

$$\kappa(X'/Z, D_{2,X'} + \epsilon D_{1,X'}) = \kappa(X/Z, D_2 + \epsilon D_1).$$

Since $D_1 \geq D_2 \geq 0$ and $D_{1,X'} \geq D_{2,X'} \geq 0$, we have $(1+\epsilon)D_1 \geq D_2 + \epsilon D_1 \geq 0$ and $(1+\epsilon)D_{1,X'} \geq D_{2,X'} + \epsilon D_{1,X'} \geq 0$, and

$$\mathrm{Supp}((1+\epsilon)D_1) = \mathrm{Supp}(D_2 + \epsilon D_1) \quad \text{and} \quad \mathrm{Supp}((1+\epsilon)D_{1,X'}) = \mathrm{Supp}(D_{2,X'} + \epsilon D_{1,X'}).$$

Thus (1) implies

$$\kappa(X/Z, D_2 + \epsilon D_1) = \kappa(X/Z, D_1) \quad \text{and} \quad \kappa(X'/Z, D_{2,X'} + \epsilon D_{1,X'}) = \kappa(X'/Z, D_{1,X'}).$$

Thus $\kappa(X'/Z, D_{1,X'}) = \kappa(X/Z, D_1)$. □

Theorem 2.21 ([OAI26d, Theorem 1.2])**.** *Let d be a positive integer and $\Gamma \subset [0,1] \cap \mathbb{Q}$ a DCC set. Then there exists a positive integer $I(d, \Gamma)$ satisfying the following. Let (X, B) be a projective lc pair of dimension d such that $B \in \Gamma$ and $K_X + B \sim_{\mathbb{Q}} 0$. Then $I(d, \Gamma)(K_X + B) \sim 0$.*

Theorem 2.21, Theorem 3.5, and Theorem 6.2 are the three uniform results that we use as inputs. We use Theorem 2.21 for the uniform degree in the canonical bundle formula (Theorem 3.3) and in the cases where the base is a point.

3. Canonical bundle formulas with bounded denominators

In this section, we prove Theorem 3.3, a canonical bundle formula for lc pairs with rational DCC coefficients in which both the degree of a trivialization and the Cartier index of the moduli part are bounded in terms of the dimension and the coefficient set (cf. [OAI26c, Proposition 7.1, Theorem 7.2] for finite coefficient sets and projective varieties). For relative dimension one and two, a uniform bound on the Cartier index of the moduli part was obtained in [PS09, Theorem 8.1] and [TX09, Theorem 1.3] (cf. [Flo14]). The index of a complement of the generalized pair induced on the base of a fibration has to be divisible by the Cartier index of the moduli part of the canonical bundle formula. Thus we need a degree of a trivialization and a Cartier index that depend only on the dimension and the coefficient set.

3.1. Trivializations.

In this subsection, we define the discriminant and the moduli part of the canonical bundle formula with respect to a trivialization.

Definition 3.1. Let $f : X \to Z$ be a contraction and (X, B) a sub-pair such that B is a $\mathbb{Q}$-divisor, (X, B) is sub-lc over the generic point of Z, and $K_X + B \sim_{\mathbb{Q}} f^*L$ for some $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on Z.

(1) A *trivialization of degree m* of $K_X + B$ over Z is an equality of $\mathbb{Q}$-divisors

$$K_X + B + \frac{1}{m}\mathrm{div}(\psi) = f^*D_Z,$$

where m is a positive integer, ψ is a rational function on X, and D_Z is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Z.

(2) Let $h : Z' \to Z$ be a projective birational morphism from a normal variety and P a prime divisor on Z'. Let X' be a normal variety with a projective birational morphism $h_X : X' \to X$ such that $f \circ h_X$ factors through a morphism $f' : X' \to Z'$, and let B' be the $\mathbb{Q}$-divisor such that $K_{X'} + B' = h_X^*(K_X + B)$. The *lc threshold* t_P of P is the supremum of the real numbers t such that $(X', B' + tf'^*P)$ is sub-lc over a neighborhood of the generic point of P on which P is Cartier. The number t_P is rational and depends only on P as a prime divisor over Z, and $t_P = 1$ for all but finitely many prime divisors P on Z' (cf. [FG14b, 3.4]). Let $B_{Z'} := \sum_P (1 - t_P)P$. The $\mathbb{Q}$-divisors $B_{Z'}$ form a $\mathbb{Q}$-$\boldsymbol{b}$-divisor, the *discriminant $\boldsymbol{b}$-divisor* of (X, B) over Z.

(3) For a trivialization as in (1), the *moduli* $\boldsymbol{b}$*-divisor* is the $\mathbb{Q}$-$\boldsymbol{b}$-divisor $\mathbf{M}$ with traces $\mathbf{M}_{Z'} := h^*D_Z - K_{Z'} - B_{Z'}$, so that $K_{Z'} + B_{Z'} + \mathbf{M}_{Z'} = h^*D_Z$. We call $(Z, B_Z, \mathbf{M})$ the *generalized sub-pair induced by the trivialization.* If $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U for a projective morphism $Z \to U$ to a normal quasi-projective variety U, then $(Z, B_Z, \mathbf{M})/U$ is a generalized sub-pair in the sense of Definition 2.8, since $K_Z + B_Z + \mathbf{M}_Z = D_Z$ is $\mathbb{Q}$-Cartier.

In [OAI26c, Proposition 7.1], a trivialization is called an exact presentation. The moduli $\boldsymbol{b}$-divisor depends on the trivialization. Fixing a trivialization of degree m determines the moduli part as a $\mathbb{Q}$-$\boldsymbol{b}$-divisor, not only up to $\mathbb{Q}$-linear equivalence. Thus, for a positive integer p with $m \mid p$ such that $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier, a monotonic generalized complement on the base whose index is divisible by p induces a monotonic complement on X of that index (Proposition 3.9(2)), and two boundaries that differ by the pullback of a divisor on the base induce equal moduli parts (Lemma 3.8(1)), which is used in Lemma 9.2(5).

Lemma 3.2. *Let $f : X \to Z$ and (X, B) be as in Definition 3.1, and let $h : Z' \to Z$, P, $h_X : X' \to X$, $f' : X' \to Z'$, and B' be as in Definition 3.1(2). Let $Z'_P \subset Z'$ be an open neighborhood of the generic point of P on which P is Cartier, and for any prime divisor E on X' such that $f'(E)$ is dense in P, let e_E be the coefficient of E in the pullback of $P|_{Z'_P}$ to $f'^{-1}(Z'_P)$.*

(1) For any prime divisor E on X' such that $f'(E)$ is dense in P, e_E is a positive integer and $\operatorname{mult}_E B' + e_E t_P \le 1$, that is, $a(E, X, B) \ge e_E t_P$.

(2) Assume that X' is smooth and that, over Z'_P, $\operatorname{Supp} B' \cup \operatorname{Supp} f'^*P$ is a simple normal crossing divisor, any component of f'^*P dominates P, and any component of B' either is a component of f'^*P or dominates Z'. Let $E_1, \dots, E_k$ be the components of f'^*P over Z'_P. Then $k \ge 1$ and

$$t_P = \min_{1 \le i \le k} \frac{a(E_i, X, B)}{e_{E_i}}.$$

(3) If (X, B) is lc, then $t_P \ge 0$. If moreover $Z' = Z$, then $t_P \le 1$. In particular, if (X, B) is lc, then $B_Z \ge 0$ and $B_Z \in [0, 1]$.

Proof. (1) Since $f'(E)$ is dense in P, E intersects $f'^{-1}(Z'_P)$ and is contained in the support of the pullback of the Cartier divisor $P|_{Z'_P}$. Thus e_E is a positive integer. Let s be a real number such that $(X', B' + sf'^*P)$ is sub-lc over an open neighborhood $Z'' \subset Z'$ of the generic point of P on which P is Cartier. Since $f'(E)$ is dense in P, E intersects the preimage of $Z'' \cap Z'_P$, and the coefficient of E in $B' + sf'^*P$ is $\operatorname{mult}_E B' + se_E$. Thus $\operatorname{mult}_E B' + se_E \le 1$. Taking the supremum over s, we obtain $\operatorname{mult}_E B' + e_E t_P \le 1$. Since $a(E, X, B) = 1 - \operatorname{mult}_E B'$, this is the inequality $a(E, X, B) \ge e_E t_P$.

(2) Since $h \circ f' = f \circ h_X$ is proper and h is separated, f' is proper [Har77, Corollary II.4.8(e)]. Since $f \circ h_X$ is surjective and h is birational, the image of f' is dense, so f' is surjective. Thus the images of the finitely many irreducible components of $f'^{-1}(P)$ are closed and cover P, and some component E of $f'^{-1}(P)$ satisfies $f'(E) = P$. By [Har77, Exercise II.3.22(a)], $\dim E \ge \dim P + \dim X' - \dim Z' = \dim X' - 1$. Thus E is a component of f'^*P over Z'_P, and $k \ge 1$. Since (X, B) is sub-lc over the generic point of Z (Definition 3.1), the components of B' dominating Z' have coefficients at most 1. Let s be a real number. Over Z'_P, the components of $B' + sf'^*P$ are $E_1, \dots, E_k$, with coefficients $\operatorname{mult}_{E_i} B' + se_{E_i}$, and components of B' dominating Z'. By [Kol13, 2.20 and Corollary 2.11], $(X', B' + sf'^*P)$ is sub-lc over Z'_P if and only if

$$\operatorname{mult}_{E_i} B' + se_{E_i} \le 1 \quad \text{for all } i.$$

Since any E_i dominates P, for any open neighborhood $Z'' \subset Z'_P$ of the generic point of P, $(X', B' + sf'^*P)$ is sub-lc over Z'' if and only if $\operatorname{mult}_{E_i} B' + se_{E_i} \le 1$ for all i. Since a sub-pair that is sub-lc over an open neighborhood Z'' of the generic point of P is sub-lc over $Z'' \cap Z'_P$, we obtain

$$t_P = \min_{1 \le i \le k} \frac{1 - \operatorname{mult}_{E_i} B'}{e_{E_i}} = \min_{1 \le i \le k} \frac{a(E_i, X, B)}{e_{E_i}}.$$

(3) If (X, B) is lc, then (X', B') is sub-lc, so $s = 0$ satisfies the condition of Definition 3.1(2), and $t_P \geq 0$. Assume moreover that $Z' = Z$. Since t_P depends only on P as a prime divisor over Z, we may take $X' = X$. Since f is surjective, there exists a prime divisor E on X such that $f(E)$ is dense in P [Har77, Exercise II.3.22(a)]. By (1) and $\operatorname{mult}_E B \geq 0$,

$$t_P \leq \frac{1 - \operatorname{mult}_E B}{e_E} \leq \frac{1}{e_E} \leq 1.$$

Thus $1 - t_P \in [0, 1]$ for any prime divisor P on Z, so $B_Z \geq 0$ and $B_Z \in [0, 1]$. □

3.2. **Uniform denominators.** In this subsection, we prove Theorem 3.3.

Theorem 3.3. *Let d be a positive integer and $\Gamma \subset [0, 1] \cap \mathbb{Q}$ a DCC set. Then there exist positive integers m and p with $m \mid p$, and a DCC set $\Lambda \subset [0, 1] \cap \mathbb{Q}$, depending only on d and Γ, satisfying the following. Let $f : X \to Z$ be a contraction such that $0 < \dim Z \leq \dim X \leq d$. Let (X, B) be an lc pair such that $B \in \Gamma$ and $K_X + B \sim_{\mathbb{Q}} f^*L$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L on Z. Then:*

(1) $K_X + B$ has a trivialization $K_X + B + \frac{1}{m}\operatorname{div}(\psi) = f^*D_Z$ of degree m over Z with $D_Z \sim_{\mathbb{Q}} L$;

(2) for any trivialization of degree m of $K_X + B$ over Z, with induced generalized sub-pair $(Z, B_Z, \mathbf{M})$, we have $B_Z \geq 0$, $B_Z \in \Lambda$, $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, and $p\mathbf{M}_{Z'}$ is a Cartier divisor for any smooth variety Z' projective and birational over Z. If moreover $Z \to U$ is a projective morphism to a normal quasi-projective variety, then $\mathbf{M}$ is nef over U.

We recall the weight of a rational relative pluricanonical form from [OAI26c, Definition 4.1].

Definition 3.4. Let C be a smooth projective curve, $c \in C$ a closed point, z a rational function on C with $\operatorname{mult}_c \operatorname{div}(z) = 1$, V a normal geometrically integral variety of dimension s over the function field $\mathbb{C}(C)$ of C, and m a positive integer. A *rational relative m-canonical form* on V is a rational m-canonical form on V in the sense of Definition 2.5, with $\Omega_{V_{\mathrm{reg}}/\mathbb{C}(C)}$ in place of $\Omega_{X_{\mathrm{reg}}/\mathbb{C}}$, and for such a form φ, $\operatorname{div}(\varphi)$ is the Weil divisor on V defined as in Definition 2.5. Let φ be a rational relative m-canonical form on V. Let $C^o \subset C$ be an open subset containing c, and $\pi : W \to C^o$ a projective morphism from a normal variety W whose generic fiber is birational to V over $\mathbb{C}(C)$ (for instance, the normalization of the closure in $\mathbb{P}^N_C$ of a projective variety in $\mathbb{P}^N_{\mathbb{C}(C)}$ birational to V, with $C^o = C$). Let $W^o \subset W$ be the open subset on which π is smooth. Since $\Omega^s_{W^o/C^o}$ restricts to the sheaf of relative differentials of the generic fiber over $\mathbb{C}(C)$ [Har77, Proposition II.8.10], the form on the generic fiber which corresponds to φ under the birational map to V is the restriction of a unique rational section of $(\Omega^s_{W^o/C^o})^{\otimes m}$, which we also denote by φ. Let $\Omega_\varphi := \varphi \wedge (dz/z)^{\otimes m}$ be the rational m-canonical form on W given by $\varphi \otimes \pi^*(dz/z)^{\otimes m}$ and the isomorphism

$$\Omega^{s+1}_{W^o} \cong \Omega^s_{W^o/C^o} \otimes \pi^*\Omega^1_{C^o}.$$

We define the *weight* of φ at c as

$$w_c(\varphi) := \inf_E \frac{\operatorname{mult}_E \operatorname{div}(\Omega_\varphi) + m}{m \operatorname{mult}_E \operatorname{div}(\pi^* z)},$$

where E runs over the prime divisors over W whose center on W is contained in $\pi^{-1}(c)$.

The weight does not depend on the choice of $\pi : W \to C^o$. For two choices $\pi_1 : W_1 \to C^o_1$ and $\pi_2 : W_2 \to C^o_2$, the birational map $W_1 \dashrightarrow W_2$ over C pulls back the form Ω_φ on W_2 to the form Ω_φ on W_1 and $\pi_2^* z$ to $\pi_1^* z$. Since π_1 and π_2 are proper, the valuative criterion of properness [Har77, Theorem II.4.7] implies that the prime divisors over W_1 whose centers on W_1 are contained in $\pi_1^{-1}(c)$ are the prime divisors over W_2 whose centers on W_2 are contained in $\pi_2^{-1}(c)$. The valuative criterion of properness also shows that, for any choice of π, the prime divisors over W whose centers on W are contained in $\pi^{-1}(c)$ are exactly the prime divisors E on normal varieties birational to W such that $\pi^* z$ vanishes along E and $\pi^* u$ is regular and non-zero

at the generic point of E for any rational function u on C which is regular and non-zero at c. Thus $w_c(\varphi)$ is the weight of [OAI26c, Definition 4.1].

The ratio defining $w_c(\varphi)$ is computed with the multiplicities along E itself, and it changes when both multiplicities are multiplied by a positive constant other than 1. In the proof of Theorem 3.3, the coefficient of the moduli part along a prime divisor P on a smooth model Z' of the base is read off, up to the integer $\operatorname{mult}_P K_{Z'}$, as the weight, at the point where a general transverse curve intersects P, of the rational relative m-canonical form determined by the trivialization. Hence the uniform bound of Theorem 3.5 bounds the denominators of the moduli part. Section 6 also reads the coefficients along ramification divisors as such weights.

Theorem 3.5 ([OAI26c, Theorem 4.2])**.** *For integers $s \geq 0$ and $m \geq 1$ there exists a positive integer $q(s,m)$ satisfying the following. Let C, c, and z be as in Definition 3.4, V a normal geometrically integral projective variety of dimension s over $\mathbb{C}(C)$, and $B_V \geq 0$ a $\mathbb{Q}$-divisor on V such that (V, B_V) is geometrically lc. Let φ be a rational relative m-canonical form on V such that $\operatorname{div}(\varphi) + mB_V = 0$. Then $q(s,m)w_c(\varphi) \in \mathbb{Z}$.*

Proof of Theorem 3.3. The proof is long, so we divide it into several steps.

Let m be a positive common multiple of the integers $I(s,\Gamma)$ of Theorem 2.21 for $1 \leq s \leq d-1$, where $m := 1$ if $d = 1$. Let p be a positive common multiple of m and of the integers $q(s,m)$ of Theorem 3.5 for $0 \leq s \leq d-1$. Let $\mathcal{T}$ be the set of the rational numbers $\operatorname{lct}(X_0, B_0; G_0) \in [0,1]$, where (X_0, B_0) is an lc pair such that $\dim X_0 \leq d$, B_0 is a $\mathbb{Q}$-divisor, and $B_0 \in \Gamma$, and $G_0 \geq 0$ is a non-zero Cartier divisor on X_0. Let $\Lambda := \{1 - t \mid t \in \mathcal{T}\}$. The set $\mathcal{T}$ is contained in the union of the sets $\operatorname{LCT}_n(\Gamma \cup \{0\}, \mathbb{Z}_{>0})$ of [HMX14] for $1 \leq n \leq d$. Since $\Gamma \cup \{0\}$ and $\mathbb{Z}_{>0}$ satisfy the DCC, $\mathcal{T}$ satisfies the ACC by [HMX14, Theorem 1.1]. Thus $\Lambda \subset [0,1] \cap \mathbb{Q}$ satisfies the DCC. The integers $m \mid p$ and the set Λ depend only on d and Γ.

Let $s := \dim X - \dim Z$. Then $0 \leq s \leq d-1$.

Step 1. In this step we construct a trivialization of degree m as in (1).

We claim that there exists a non-zero rational function ψ on X such that

$$m(K_X + B) + \operatorname{div}(\psi) = G_\psi \tag{3.1}$$

for a $\mathbb{Q}$-divisor G_ψ on X no component of which dominates Z. If $s = 0$, then f is birational, no prime divisor on X dominates Z, and $\psi := 1$ satisfies (3.1). Assume that $s \geq 1$. We prove the claim using the general fibers of f, that is, the fibers over the closed points of a non-empty open subset of Z. General fibers suffice: each property of the fibers used below holds for the fibers over all closed points of one non-empty open subset of Z, and Theorem 2.21 implies $m(K_F + B_F) \sim 0$ on each such fiber F, where m does not depend on F.

Let τ be a rational 1-canonical form on X with $K_X = \operatorname{div}_X(\tau)$ (Definition 2.5), $\mu : X'' \to X$ a log resolution of (X, B), $K_{X''} := \operatorname{div}_{X''}(\mu^*\tau)$, and B'' the $\mathbb{Q}$-divisor such that $K_{X''} + B'' = \mu^*(K_X + B)$, so that $\mu_* K_{X''} = K_X$ and $\mu_* B'' = B$. Let R'' be the snc divisor formed by the μ-exceptional divisors and the strict transform of $\operatorname{Supp} B$. Let B''_h be the sum of the components of B'' dominating Z, with their coefficients in B'', and let $f'' := f \circ \mu$. Since X is normal, $\mu_* \mathcal{O}_{X''} = \mathcal{O}_X$, so $f''_* \mathcal{O}_{X''} = f_* \mathcal{O}_X = \mathcal{O}_Z$.

There exists a non-empty smooth open subset $Z^0 \subset Z$ satisfying the following.

(a) f and f'' are flat over Z^0 [Gro65, Théorème 6.9.1], and the fibers of f over the points of Z^0 are geometrically normal and geometrically integral.

(b) f'' and the restriction of f'' to any stratum of R'' dominating Z are smooth over Z^0 [Har77, Corollary III.10.7].

(c) No stratum of R'' which does not dominate Z, and no component of B'' which does not dominate Z, intersects $f''^{-1}(Z^0)$. For any μ-exceptional prime divisor P dominating Z, $\mu(P) \to Z$ is flat over Z^0, so the fibers of $\mu(P) \to Z$ over Z^0 have pure dimension $\dim \mu(P) - \dim Z \leq s - 2$ [Gro65, Théorème 6.9.1], [Har77, Corollary III.9.6], where the inequality holds since $\mu(P)$ has codimension at least 2 in the normal variety X.

By Lemma 2.17(4), the second statement of (a) holds after shrinking Z^0.

Let $z \in Z^0$ be a closed point, $F := f^{-1}(z)$, $F'' := f''^{-1}(z)$, and $\mu_F : F'' \to F$ the restriction of μ. By (a), (b), and [Har77, Corollary III.9.6], F is a normal projective variety of dimension s, and F'' is a smooth projective scheme, which is connected since $f''_*\mathcal{O}_{X''} = \mathcal{O}_Z$ [Har77, Corollary III.11.3]. Since the local rings of F'' are regular, hence integral domains, and F'' is connected, F'' is integral, that is, a smooth projective variety. Since μ is an isomorphism over the complement of the images of the μ-exceptional divisors, (c) implies that μ_F is birational and that the restriction to F'' of any μ-exceptional prime divisor is μ_F-exceptional. By (b) and (c), $R''|_{F''}$ is an snc divisor, the restriction to F'' of any component of R'' is a reduced divisor, the restrictions of distinct components of R'' have no common component, and the components of B'' intersecting F'' dominate Z. Thus $B''|_{F''} = B''_h|_{F''}$, and the coefficients of $B''|_{F''}$ are coefficients of B'', which are at most 1 since (X, B) is lc. Let $B_F := \mu_{F*}(B''|_{F''})$. Any component Q of $B''|_{F''}$ which is not μ_F-exceptional is a component of the restriction to F'' of the strict transform of a component B_1 of B, and $\mathrm{mult}_Q(B''|_{F''}) = \mathrm{mult}_{B_1} B$. Hence $B_F \geq 0$ and $B_F \in \Gamma$.

Let l be a positive integer such that lL is Cartier and $l(K_X + B) \sim lf^*L$. Then $l(K_{X''} + B'') = \mu^*(l(K_X + B))$ is a Cartier divisor, and $\mathcal{O}_{X''}(l(K_{X''} + B'')) \cong f''^*\mathcal{O}_Z(lL)$ is trivial on F''. Since f'' is smooth over the smooth variety Z^0, the cotangent sequence of the smooth morphism $f''^{-1}(Z^0) \to Z^0$ and [Har77, Proposition II.8.10] imply $\omega_{X''}|_{F''} \cong \omega_{F''}$. Hence $l(K_{F''} + B''|_{F''}) = \mathrm{div}(u)$ for a canonical divisor $K_{F''}$ of F'' and a rational function u on F''. Let $K_F := \mu_{F*}K_{F''}$, a canonical divisor of F. Pushing forward by μ_F, we obtain $l(K_F + B_F) = \mathrm{div}(u)$, so $K_F + B_F \sim_{\mathbb{Q}} 0$ and $K_{F''} + B''|_{F''} = \mu_F^*(K_F + B_F)$. Since $\mathrm{Supp}\, B''|_{F''}$ is contained in the snc divisor $R''|_{F''}$ and the coefficients of $B''|_{F''}$ are at most 1, (F, B_F) is lc [KM98, Lemma 2.30, Corollary 2.31(3)]. Thus (F, B_F) is a projective lc pair of dimension s with $B_F \in \Gamma$ and $K_F + B_F \sim_{\mathbb{Q}} 0$. By Theorem 2.21 and $I(s, \Gamma) \mid m$, we have $m(K_F + B_F) = \mathrm{div}(g)$ for a rational function g on F, hence

$$m(K_{F''} + B''|_{F''}) = \mathrm{div}(\mu_F^* g).$$

In particular, $mB''|_{F''}$ is a divisor. Any component D of B''_h dominates Z and is projective over Z, so D intersects F'', and $D|_{F''}$ is a reduced divisor. Hence mB''_h is a divisor, and

$$A'' := m(K_{X''} + B''_h)$$

is a Cartier divisor on the smooth variety X''. Since $z \in Z^0$ is an arbitrary closed point, for any closed point $z \in Z^0$ we have

$$\mathcal{O}_{X''}(A'')|_{f''^{-1}(z)} \cong \omega_{f''^{-1}(z)}^{\otimes m}\big(mB''|_{f''^{-1}(z)}\big) \cong \mathcal{O}_{f''^{-1}(z)},$$

and $h^0(f''^{-1}(z), \mathcal{O}_{X''}(A'')|_{f''^{-1}(z)}) = 1$, since $f''^{-1}(z)$ is an integral projective variety [Har77, Theorem I.3.4(a)].

Since f'' is flat over Z^0, the invertible sheaf $\mathcal{O}_{X''}(A'')$ is flat over Z^0. By the semicontinuity theorem [Har77, Theorem III.12.8], the function

$$y \to h^0\big(f''^{-1}(y), \mathcal{O}_{X''}(A'')|_{f''^{-1}(y)}\big)$$

on Z^0 is upper semicontinuous. Thus the points of Z^0 at which the function is at least 1 form a closed subset of Z^0 containing all closed points of Z^0, and the points of Z^0 at which the function is at least 2 form a closed subset of Z^0 containing no closed point of Z^0. Since a closed subset of Z^0 containing all closed points of Z^0 is Z^0, and a closed subset of Z^0 containing no closed point of Z^0 is empty, the function is equal to 1 at any point of Z^0.

By Grauert's theorem [Har77, Corollary III.12.9], $f''_*\mathcal{O}_{X''}(A'')$ is an invertible sheaf on Z^0, and for any closed point $z \in Z^0$ with residue field $\kappa(z)$ the natural map

$$f''_*\mathcal{O}_{X''}(A'') \otimes \kappa(z) \to H^0(f''^{-1}(z), \mathcal{O}_{X''}(A'')|_{f''^{-1}(z)})$$

is an isomorphism. Let $Z^1 \subset Z^0$ be a non-empty open subset on which $f''_*\mathcal{O}_{X''}(A'')$ is trivial. Let ς be a generator of the free $\mathcal{O}_{Z^1}$-module $f''_*\mathcal{O}_{X''}(A'')|_{Z^1}$ of rank one. Then ς is a section of $\mathcal{O}_{X''}(A'')$ over $f''^{-1}(Z^1)$. For any closed point $z \in Z^1$, the image of ς in $f''_*\mathcal{O}_{X''}(A'') \otimes \kappa(z)$

is non-zero, so, by the isomorphism of Grauert's theorem, the restriction of ς to $f''^{-1}(z)$ is a non-zero section of a trivial invertible sheaf on an integral projective variety, hence nowhere zero. Let ψ be the non-zero rational function on X'' such that the restriction of $A'' + \operatorname{div}(\psi)$ to $f''^{-1}(Z^1)$ is the divisor of zeros of ς. Then, since f'' maps any closed point of $f''^{-1}(Z^1)$ to a closed point of Z^1, the closed subset $\operatorname{Supp}(A'' + \operatorname{div}(\psi)) \cap f''^{-1}(Z^1)$ of $f''^{-1}(Z^1)$ contains no closed point, so it is empty. Thus no component of

$$m(K_{X''} + B'') + \operatorname{div}(\psi) = A'' + \operatorname{div}(\psi) + m(B'' - B''_h)$$

dominates Z. We regard ψ as a rational function on X. Since $\mu_* K_{X''} = K_X$ and $\mu_* B'' = B$, pushing forward by μ shows that no component of $m(K_X + B) + \operatorname{div}(\psi)$ dominates Z. This proves the claim.

Let k be a positive integer and σ a non-zero rational function on X such that $k(K_X + B - f^*L) = \operatorname{div}(\sigma)$. Then

$$\operatorname{div}(\sigma^m \psi^k) = km\left(K_X + B + \frac{1}{m}\operatorname{div}(\psi) - f^*L\right),$$

and the right-hand side equals $kG_\psi - kmf^*L$ by (3.1). Thus no component of $\operatorname{div}(\sigma^m\psi^k)$ dominates Z, and Lemma 2.17(4) implies that $\sigma^m\psi^k = f^*\sigma_0$ for a non-zero rational function σ_0 on Z. Let

$$D_Z := L + \frac{1}{km}\operatorname{div}(\sigma_0).$$

Then D_Z is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, $D_Z \sim_{\mathbb{Q}} L$, and

$$K_X + B + \frac{1}{m}\operatorname{div}(\psi) = f^*L + \frac{1}{km}\operatorname{div}(\sigma^m\psi^k) = f^*D_Z.$$

This proves (1).

Step 2. In this step we prove that $B_Z \geq 0$, $B_Z \in \Lambda$, and $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, and that $\mathbf{M}$ is nef over U if $Z \to U$ is as in (2).

Fix a trivialization

$$K_X + B + \frac{1}{m}\operatorname{div}(\psi) = f^*D_Z \tag{3.2}$$

of degree m, not necessarily the one constructed in **Step 1**, with induced generalized sub-pair $(Z, B_Z, \mathbf{M})$.

Let P be a prime divisor on Z, $Z_1 \subset Z$ an open neighborhood of the generic point of P on which P is Cartier, and f^*P the pullback of $P|_{Z_1}$ to $f^{-1}(Z_1)$. By Lemma 3.2(3), $1 - t_P \in [0, 1]$. In particular, $B_Z \geq 0$. On a log resolution of $(f^{-1}(Z_1), B|_{f^{-1}(Z_1)} + f^*P)$, the lc threshold of f^*P over any open neighborhood $Z_0 \subset Z_1$ of the generic point of P is the minimum of the ratios $a(E', X, B)/\operatorname{mult}_{E'} P$ over the finitely many components E' of the pullback of f^*P whose image intersects Z_0 [Kol13, 2.20 and Corollary 2.11]. Thus the lc thresholds of f^*P over such neighborhoods Z_0 take finitely many values, and t_P is the supremum of them. Hence there exists such a neighborhood Z_P with

$$t_P = \operatorname{lct}\left(f^{-1}(Z_P), B|_{f^{-1}(Z_P)}; f^*P|_{f^{-1}(Z_P)}\right).$$

Since $(f^{-1}(Z_P), B|_{f^{-1}(Z_P)})$ is an lc pair such that $B|_{f^{-1}(Z_P)} \in \Gamma$ and $f^*P|_{f^{-1}(Z_P)} \geq 0$ is a non-zero Cartier divisor, $t_P \in \mathcal{T}$. Thus $1 - t_P \in \Lambda$ for any prime divisor P on Z, that is, $B_Z \in \Lambda$.

Next, $f : (X, B) \to Z$ is an lc-trivial fibration in the sense of [FG14b, Definition 3.2]. Condition (1) of [FG14b, Definition 3.2] holds since (X, B) is lc. Condition (2) holds since $\mathcal{O}_X(\lceil \mathbf{A}^*(X, B)\rceil) \cong \mathcal{O}_X$ for the $\boldsymbol{b}$-divisor $\mathbf{A}^*(X, B)$ of [FG14b, 2.3], so that

$$f_*\mathcal{O}_X(\lceil \mathbf{A}^*(X, B)\rceil) = f_*\mathcal{O}_X = \mathcal{O}_Z$$

has rank one. Condition (3) holds with $D := D_Z$, since $K_X + B \sim_{\mathbb{Q}} f^*D_Z$. With the trivialization (3.2), the discriminant and moduli $\mathbb{Q}$-$\boldsymbol{b}$-divisors of [FG14b, 3.4] are the discriminant $\boldsymbol{b}$-divisor and the moduli $\boldsymbol{b}$-divisor of Definition 3.1: the thresholds are rational, so the maximum over rational numbers in [FG14b, 3.4] equals the supremum in Definition 3.1(2).

If no projective morphism $Z \to U$ is given, then let $U := Z$ and $Z \to U$ the identity. By [FG14b, Theorem 3.6] (cf. [Amb04, Theorem 0.2]), applied to the proper morphism $Z \to U$, there exists a proper birational morphism $Z_1' \to Z$ from a normal variety such that $\mathbf{M}$ descends to Z_1' and $\mathbf{M}_{Z_1'}$ is nef over U. By Chow's lemma [Har77, Exercise II.4.10] and a resolution, there exists a smooth variety Z_2' with a morphism $Z_2' \to Z_1'$ such that $Z_2' \to Z$ is projective and birational. Then $\mathbf{M}$ descends to Z_2', and $\mathbf{M}_{Z_2'}$, the pullback of $\mathbf{M}_{Z_1'}$, is nef over U by the projection formula (cf. [Laz04, Example 1.4.4(i)]). Thus $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, and $\mathbf{M}$ is nef over U.

Step 3. In this step we prove that $p\mathbf{M}_{Z'}$ is Cartier for any smooth variety Z' projective and birational over Z.

Recall the trivialization (3.2) of **Step 2**. Let $h : Z' \to Z$ be a projective birational morphism from a smooth variety and P a prime divisor on Z'. Let $D_{Z'} := h^*D_Z$ and $\alpha := \operatorname{mult}_P D_{Z'}$. Since $q(s,m) \mid p$ and Z' is smooth, it suffices to show that $q(s,m) \operatorname{mult}_P \mathbf{M}_{Z'} \in \mathbb{Z}$: then $p\mathbf{M}_{Z'}$ is a divisor, hence Cartier.

Let Y be a smooth variety with a projective birational morphism $\rho : Y \to X$ such that $f \circ \rho$ factors through a morphism $g : Y \to Z'$. Let θ be a rational 1-canonical form on X (Definition 2.5) with $K_X = \operatorname{div}_X(\theta)$, denote the pullback $\rho^*\theta$ to Y also by θ, let $K_Y := \operatorname{div}_Y(\theta)$, and let B_Y be the $\mathbb{Q}$-divisor such that $K_Y + B_Y = \rho^*(K_X + B)$. Pulling back (3.2) by ρ, we obtain

$$\operatorname{div}_Y(\theta) + B_Y + \frac{1}{m}\operatorname{div}_Y(\psi) = g^*D_{Z'}. \tag{3.3}$$

By [KM98, Theorem 0.2], we may choose Y such that the union R of $\operatorname{Exc}(\rho)$ and the supports of B_Y, g^*P, $\operatorname{div}_Y(\theta)$, $\operatorname{div}_Y(\psi)$, and $g^*D_{Z'}$ is an snc divisor. The strata of R are Y and the irreducible components of the intersections of components of R. Since (X, B) is lc, the coefficients of B_Y are at most 1. Since $h \circ g = f \circ \rho$ is projective and h is separated, g is projective [Har77, Exercises II.4.8(e) and II.4.9]. Let $Y \to Y'' \to Z'$ be the Stein factorization of g [Har77, Corollary III.11.5]. By Lemma 2.17(4), applied to the contraction f, the dominant rational map $X \dashrightarrow Y''$ induced by ρ^{-1} and $Y \to Y''$, and the generically finite morphism $Y'' \to Z' \to Z$, the morphism $Y'' \to Z$ is birational. Since h is birational, the finite morphism $Y'' \to Z'$ is birational, hence an isomorphism since Z' is normal. Thus g has connected fibers.

Let $E_1, \dots, E_l$ be the components of g^*P dominating P, $e_i := \operatorname{mult}_{E_i} P$, and $b_i := \operatorname{mult}_{E_i} B_Y$. Let Z_P' be a neighborhood of the generic point of P over which any component of g^*P is one of $E_1, \dots, E_l$ and any component of B_Y is one of $E_1, \dots, E_l$ or dominates Z' (such Z_P' exists since g is projective and the image of any component of g^*P or of B_Y which is not one of $E_1, \dots, E_l$ and does not dominate Z' is a closed subset of Z' not containing the generic point of P). Since R is a simple normal crossing divisor, Lemma 3.2(2), applied with $X' = Y$ and this Z_P', implies that

$$t_P = \min_i \frac{1 - b_i}{e_i}.$$

We construct a curve through a general point of P. Let $Z'^o \subset Z'$ be a dense open subset such that $h|_{Z'^o}$ is an open immersion, f is flat over $h(Z'^o)$ with geometrically normal and geometrically integral fibers over all points of $h(Z'^o)$, closed or not (Lemma 2.17(4)), the restriction of g to any stratum of R dominating Z' is smooth over Z'^o [Har77, Corollary III.10.7], the centers on X of the components of $\operatorname{Exc}(\rho)$ not dominating Z do not intersect $f^{-1}(h(Z'^o))$, and the centers of the other components of $\operatorname{Exc}(\rho)$ intersect any fiber of f over $h(Z'^o)$ in codimension at least 2 [Gro65, Théorème 6.9.1], [Har77, Corollary III.9.6].

If $\dim Z = 1$, then let $C^o := Z'$ and $c := P$. If $\dim Z \geq 2$, then embed Z' into a projective space, let $H_1, \dots, H_{\dim Z - 1}$ be general hyperplane sections of Z', and let $C^o := H_1 \cap \cdots \cap H_{\dim Z - 1}$, a smooth irreducible curve [Jou83, Corollaire 6.11].

The set of points $z' \in Z'$ with $\dim g^{-1}(z') > s$ is closed, being the image under the projective morphism g of the closed set E_{s+1} of [Har77, Exercise II.3.22(d)] (cf. [Gro66, Corollaire 13.1.5]), and it has codimension at least 2: if it contained a prime divisor Q, then $\dim g^{-1}(Q) \geq$

$\dim Q + s + 1 = \dim Y$. Since a general complete intersection curve on Z' is disjoint from a given closed subset of codimension at least 2, intersects a given prime divisor transversally at general points, and is not contained in a given proper closed subset (Kleiman's transversality theorem [Har77, Theorem III.10.8]), we may assume the following: C^o is disjoint from the set of points $z' \in Z'$ with $\dim g^{-1}(z') > s$ and from $g(W)$ for any stratum W of R with $\operatorname{codim}_{Z'} g(W) \geq 2$, C^o intersects $g(W)$ transversally at points over which $W \to g(W)$ is smooth for any stratum W of R such that $g(W)$ is a prime divisor, C^o intersects P at a point c not contained in any prime divisor $g(W) \neq P$ nor in any component of $\operatorname{Supp} D_{Z'}$ other than P, and the generic point ξ of C^o belongs to Z'^o.

By Kleiman's transversality theorem [Har77, Theorem III.10.8], applied to the morphisms from Y and from any stratum of R to the projective space containing Z', $Y_C := g^{-1}(C^o)$ is smooth and $R|_{Y_C}$ is an snc divisor whose strata are the restrictions of the strata of R. Since g has connected fibers and C^o is connected, Y_C is connected, hence integral. Let $g_C : Y_C \to C^o$ be the restriction of g, a projective morphism with connected fibers. Let C be the smooth projective curve containing C^o as a dense open subset, and z a rational function on C with $\operatorname{mult}_c \operatorname{div}(z) = 1$.

Let $B_{Y_C} := B_Y|_{Y_C}$, which may have negative coefficients. By the choice of C^o, near $g_C^{-1}(c)$ we have $g_C^* c = \sum_i e_i E_i|_{Y_C}$, where $E_i|_{Y_C}$ is smooth and has coefficient b_i in B_{Y_C}, and the other components of $R|_{Y_C}$ have coefficients at most 1 in B_{Y_C}. For a prime divisor E over Y_C whose center on Y_C is contained in $g_C^{-1}(c)$, we have $\operatorname{mult}_E c = \operatorname{mult}_E \operatorname{div}(g_C^* z)$, since $\operatorname{div}(z) = c$ near c, and hence

$$a(E, Y_C, B_{Y_C} + t g_C^* c) = a(E, Y_C, B_{Y_C}) - t \operatorname{mult}_E \operatorname{div}(g_C^* z).$$

Thus [Kol13, 2.20 and Corollary 2.11] implies that

$$t_P = \min_i \frac{1 - b_i}{e_i} = \inf_E \frac{a(E, Y_C, B_{Y_C})}{\operatorname{mult}_E \operatorname{div}(g_C^* z)}, \tag{3.4}$$

where E runs over the prime divisors over Y_C whose center on Y_C is contained in $g_C^{-1}(c)$.

We restrict (3.3) to Y_C. If $\dim Z = 1$, then let $\theta_C := \theta$. If $\dim Z \geq 2$, then for any j let H'_j be a hyperplane section of Z' linearly equivalent to H_j such that $c \notin H'_j$ and $\xi \notin H'_j$, and let a_j be a non-zero rational function on Z' with $\operatorname{div}(a_j) = H'_j - H_j$. The form $\theta \prod_j g^* a_j$ has simple poles along the general members $g^* H_j$. Taking Poincaré residues along $g^* H_1, \dots, g^* H_{\dim Z - 1}$ one at a time, the adjunction formula [Har77, Proposition II.8.20] implies that there exists a rational 1-canonical form θ_C on Y_C with

$$\operatorname{div}_{Y_C}(\theta_C) = \Big(\operatorname{div}_Y(\theta) + \sum_j g^* H'_j \Big)\Big|_{Y_C}.$$

Let $\psi_C := \psi|_{Y_C}$ and $D_C := (D_{Z'} + \sum_j H'_j)|_{C^o}$. Then (3.3) implies that

$$\operatorname{div}_{Y_C}(\theta_C) + B_{Y_C} + \frac{1}{m} \operatorname{div}_{Y_C}(\psi_C) = g_C^* D_C. \tag{3.5}$$

Since $c \notin H'_j$ for any j, C^o intersects P transversally at c, and c is not contained in any other component of $\operatorname{Supp} D_{Z'}$, we have $\operatorname{mult}_c D_C = \alpha$.

Let V be the fiber of f over $h(\xi)$ and B_V the restriction of B to V. By the choice of Z'^o, V is a normal geometrically integral projective variety over the residue field $\mathbb{C}(\xi)$ of ξ, which is the function field $\mathbb{C}(C)$ of C, $\dim V = s$, and $B_V \geq 0$ is a $\mathbb{Q}$-divisor. The fiber Y_ξ of g over ξ is the generic fiber of g_C, and ρ restricts to a projective birational morphism $\rho_\xi : Y_\xi \to V$. By the choice of Z'^o, the strata of $R|_{Y_\xi}$ stay smooth after the base change to an algebraic closure of $\mathbb{C}(C)$, and the ρ_ξ-exceptional divisors are the restrictions of the components of $\operatorname{Exc}(\rho)$ dominating Z. Thus the relation $K_Y + B_Y = \rho^*(K_X + B)$ restricts to the crepant relation between $(Y_\xi, B_{Y_C}|_{Y_\xi})$ and (V, B_V), and (V, B_V) is geometrically lc by [KM98, Corollary 2.31(3)], since the coefficients of B_Y are at most 1.

Since $c \in C^o$, $g_C : Y_C \to C^o$ is projective, and the generic fiber Y_ξ of g_C is birational to V over $\mathbb{C}(C)$ via ρ_ξ, we may take $W = Y_C$ and $\pi = g_C$ in Definition 3.4. Let $Y_C^o \subset Y_C$ be the open subset

on which g_C is smooth, θ_C/dz the rational section of $\Omega^s_{Y_C^o/C^o}$ such that $(\theta_C/dz)\wedge g_C^*dz=\theta_C$ under the isomorphism of Definition 3.4, and

$$\varphi:=(\theta_C/dz)^{\otimes m}\psi_C.$$

The restriction of φ to Y_ξ, transported to V by ρ_ξ, is a rational relative m-canonical form on V, which we also denote by φ. The $\mathbb{Q}$-divisor $g_C^*D_C$ and the divisor $\operatorname{div}(dz)$ restrict to zero on Y_ξ, and $\rho_{\xi*}(B_{Y_C}|_{Y_\xi})=B_V$. Thus restricting (3.5) to Y_ξ and pushing forward by ρ_ξ, we obtain $\operatorname{div}_V(\varphi)+mB_V=0$. In particular, V, B_V, and φ satisfy the assumptions of Theorem 3.5, so

$$q(s,m)w_c(\varphi)\in\mathbb{Z}.$$

We compute $w_c(\varphi)$. By definition,

$$\Omega_\varphi=\varphi\wedge(dz/z)^{\otimes m}=(g_C^*z)^{-m}\theta_C^{\otimes m}\psi_C.$$

Let E be a prime divisor over Y_C whose center on Y_C is contained in $g_C^{-1}(c)$, that is, one of the prime divisors in Definition 3.4 with $W=Y_C$ and $\pi=g_C$. Let $e_E:=\operatorname{mult}_E\operatorname{div}(g_C^*z)>0$. Pulling back (3.5) to a higher model of Y_C on which E is a divisor, and using $\operatorname{mult}_c D_C=\alpha$, we obtain

$$\operatorname{mult}_E\operatorname{div}(\theta_C)+1-a(E,Y_C,B_{Y_C})+\frac{1}{m}\operatorname{mult}_E\operatorname{div}(\psi_C)=\alpha e_E.$$

Since

$$\frac{1}{m}\operatorname{mult}_E\operatorname{div}(\Omega_\varphi)=-e_E+\operatorname{mult}_E\operatorname{div}(\theta_C)+\frac{1}{m}\operatorname{mult}_E\operatorname{div}(\psi_C),$$

we obtain

$$\frac{1}{m}\operatorname{mult}_E\operatorname{div}(\Omega_\varphi)+1=(\alpha-1)e_E+a(E,Y_C,B_{Y_C}),$$

that is,

$$\frac{\operatorname{mult}_E\operatorname{div}(\Omega_\varphi)+m}{m\operatorname{mult}_E\operatorname{div}(g_C^*z)}=\alpha-1+\frac{a(E,Y_C,B_{Y_C})}{e_E}.$$

Taking the infimum over E, (3.4) implies that $w_c(\varphi)=\alpha-1+t_P$. Since $\mathbf{M}_{Z'}=D_{Z'}-K_{Z'}-B_{Z'}$ and $\operatorname{mult}_P B_{Z'}=1-t_P$, we obtain

$$\operatorname{mult}_P\mathbf{M}_{Z'}+\operatorname{mult}_P K_{Z'}=w_c(\varphi). \tag{3.6}$$

Since $K_{Z'}$ is a divisor and $q(s,m)w_c(\varphi)\in\mathbb{Z}$, (3.6) implies that $q(s,m)\operatorname{mult}_P\mathbf{M}_{Z'}\in\mathbb{Z}$. Thus $p\mathbf{M}_{Z'}$ is Cartier.

Steps 2 and 3 prove (2). □

3.3. **Hyperstandard discriminants.** In this subsection, we show that, for a trivialization whose degree divides p and such that $p\mathbf{M}_Z$ has integer coefficients, the discriminant of a pair with hyperstandard coefficients has hyperstandard coefficients with respect to an explicit finite set.

Definition 3.6. Let p be a positive integer, and $\mathfrak{R}\subset[0,1]\cap\mathbb{Q}$ a finite set such that $0,1\in\mathfrak{R}$. We define

$$\mathfrak{R}[p]:=\mathfrak{R}\cup\left\{\frac{r}{j}-\frac{i}{p}\;\middle|\;r\in\mathfrak{R},\ i,j\in\mathbb{Z},\ 1\le i\le p,\ 1\le j\le\frac{pr}{i}\right\}.$$

Theorem 3.3 only shows that the coefficients of the discriminant belong to a DCC set Λ, while Proposition 4.2 needs a DCC set of rational numbers with rational accumulation points. The following lemma shows that the coefficients of the discriminant belong to $\Phi(\mathfrak{R}[p])$, whose only accumulation point is 1. In the proof, we cannot assume that pB has integer coefficients, since the denominators of the elements $1-\frac{r}{j}$ of $\Phi(\mathfrak{R})$ are unbounded.

Lemma 3.7. *Let $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ be a finite set such that $0, 1 \in \mathfrak{R}$, p a positive integer, $f : X \to Z$ a contraction, (X, B) an lc pair, and*

$$K_X + B + \frac{1}{m} \operatorname{div}(\psi) = f^*(K_Z + B_Z + \mathbf{M}_Z)$$

a trivialization of degree m with induced generalized sub-pair $(Z, B_Z, \mathbf{M})$. Assume that $\dim Z > 0$, $B \in \Phi(\mathfrak{R})$, $m \mid p$, and $p\mathbf{M}_Z$ has integer coefficients. Then $\mathfrak{R}[p]$ is a finite subset of $[0,1] \cap \mathbb{Q}$ containing 0 and 1, and $B_Z \in \Phi(\mathfrak{R}[p])$.

Proof. Let $r \in \mathfrak{R}$, and i, j integers such that $1 \le i \le p$ and $1 \le j \le \frac{pr}{i}$. Then $j \le p$, and

$$0 \le \frac{r}{j} - \frac{i}{p} \le \frac{r}{j} \le 1.$$

Since $\mathfrak{R}$ is finite, $\mathfrak{R}[p]$ is a finite subset of $[0,1] \cap \mathbb{Q}$, and it contains 0 and 1 since $\mathfrak{R}$ does.

Let Q be a prime divisor on Z and t the lc threshold t_Q of Q, so that $\operatorname{mult}_Q B_Z = 1 - t$. Let

$$\alpha := \operatorname{mult}_Q(K_Z + B_Z + \mathbf{M}_Z).$$

By [Har77, Exercise II.3.22(a)], there exists a prime divisor E on X such that $f(E)$ is dense in Q. Let $e := e_E$ be the coefficient of E in the pullback of Q over an open neighborhood of the generic point of Q on which Q is Cartier, which is a positive integer (Lemma 3.2 with Q in place of P, $Z' = Z$, and $X' = X$), and $b := \operatorname{mult}_E B$. Since $f(E)$ is dense in Q, we have $\operatorname{mult}_E D = e \operatorname{mult}_Q D$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on Z. Since $1 \in \mathfrak{R}$, we have $0 = 1 - \frac{1}{1} \in \Phi(\mathfrak{R})$, so $b \in \Phi(\mathfrak{R})$ also when $E \not\subset \operatorname{Supp} B$.

By Lemma 3.2(1) and (3), applied with $Z' = Z$ and $X' = X$, we have $et + b \le 1$ and $t \ge 0$. Since $b \ge 0$,

$$0 \le et + b \le 1.$$

In particular, $\operatorname{mult}_Q B_Z = 1 - t \in [0,1]$.

Comparing the coefficients of E in p times the trivialization, we obtain

$$p \operatorname{mult}_E K_X + pb + \frac{p}{m} \operatorname{mult}_E \operatorname{div}(\psi) = pe\alpha.$$

Since $\operatorname{mult}_E K_X \in \mathbb{Z}$ and $m \mid p$, we have $pe\alpha - pb \in \mathbb{Z}$. On the other hand, since $\operatorname{mult}_Q B_Z = 1 - t$,

$$\alpha = \operatorname{mult}_Q K_Z + (1 - t) + \operatorname{mult}_Q \mathbf{M}_Z, \quad \text{so} \quad p(\alpha + t) = p \operatorname{mult}_Q K_Z + p + p \operatorname{mult}_Q \mathbf{M}_Z \in \mathbb{Z},$$

since $p\mathbf{M}_Z$ has integer coefficients. Thus

$$k := p(et + b) = e \cdot p(\alpha + t) - (pe\alpha - pb)$$

is an integer, and $0 \le k \le p$.

Since $b \in \Phi(\mathfrak{R})$, we may write $b = 1 - \frac{r}{j}$ for some $r \in \mathfrak{R}$ and some positive integer j. Let $i := p - k$. Then i is an integer, $0 \le i \le p$, and

$$t = \frac{1}{e}\left(\frac{k}{p} - b\right) = \frac{1}{e}\left(\frac{r}{j} - \frac{i}{p}\right).$$

If $i = 0$, then

$$\operatorname{mult}_Q B_Z = 1 - t = 1 - \frac{r}{je} \in \Phi(\mathfrak{R}) \subset \Phi(\mathfrak{R}[p]).$$

If $i \ge 1$, then $t \ge 0$ implies $\frac{i}{p} \le \frac{r}{j}$, that is, $j \le \frac{pr}{i}$. Thus $r' := \frac{r}{j} - \frac{i}{p} \in \mathfrak{R}[p]$, and

$$\operatorname{mult}_Q B_Z = 1 - t = 1 - \frac{r'}{e} \in \Phi(\mathfrak{R}[p]).$$

Since Q is an arbitrary prime divisor on Z, $B_Z \in \Phi(\mathfrak{R}[p])$. □

3.4. **Complements through the canonical bundle formula.** In this subsection, we show that monotonic n-complements of (X, B) correspond to monotonic n-complements of the generalized pair $(Z, B_Z, \mathbf{M})$ induced on the base by a trivialization of degree m, for any positive integer n divisible by a positive integer p such that $m \mid p$ and $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier (Proposition 3.9(2)). The following lemma describes the discriminant after adding the pullback of a divisor on the base (cf. [CHX24, Lemma 3.1(2) and Lemma 3.2]). We state it for sub-pairs and for any trivialization.

Lemma 3.8. *Let $f : X \to Z$ be a contraction, (X, B) a sub-pair, and $K_X + B + \frac{1}{m}\operatorname{div}(\psi) = f^*D_Z$ a trivialization of degree m with induced generalized sub-pair $(Z, B_Z, \mathbf{M})$. Let P be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Z. Then:*

1. *$K_X + B + f^*P + \frac{1}{m}\operatorname{div}(\psi) = f^*(D_Z + P)$ is a trivialization of degree m of $K_X + B + f^*P$ over Z, whose discriminant $\boldsymbol{b}$-divisor has trace $B_{Z'} + h^*P$ on any normal variety Z' with a projective birational morphism $h : Z' \to Z$, and whose moduli $\boldsymbol{b}$-divisor is $\mathbf{M}$;*
2. *$(X, B + f^*P)$ is sub-lc if and only if the coefficients of $B_{Z'} + h^*P$ are at most 1 for any such $h : Z' \to Z$; if moreover $Z \to U$ is a projective morphism to a normal quasi-projective variety U and $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U, then this holds if and only if $(Z, B_Z + P, \mathbf{M})/U$ is generalized sub-lc;*
3. *if $a(E, X, B) > 0$ for any prime divisor E over X whose center on X does not dominate Z, for instance if (X, B) is sub-klt, or if (X, B) is lc and any lc center of (X, B) dominates Z, then the coefficients of $B_{Z'}$ are less than 1 for any such $h : Z' \to Z$.*

Proof. (1) We have $K_X + B + f^*P \sim_{\mathbb{Q}} f^*(D_Z + P)$, and $D_Z + P$ is $\mathbb{Q}$-Cartier. Since $f^*P = 0$ over $Z \setminus \operatorname{Supp} P$, the sub-pair $(X, B + f^*P)$ is sub-lc over the generic point of Z. Thus $K_X + B + f^*P + \frac{1}{m}\operatorname{div}(\psi) = f^*(D_Z + P)$ is a trivialization of degree m of $K_X + B + f^*P$ over Z.

Let $h : Z' \to Z$, Q, $h_X : X' \to X$, $f' : X' \to Z'$, and B' be as in Definition 3.1(2), with Q in place of P, and let t_Q be the lc threshold of Q. Then

$$K_{X'} + B' + f'^*h^*P = h_X^*(K_X + B + f^*P).$$

Let $c := \operatorname{mult}_Q P$. There exists a neighborhood V of the generic point of Q on which Q is Cartier and $h^*P|_V = cQ|_V$. Over V, we have

$$B' + f'^*h^*P + sf'^*Q = B' + (s + c)f'^*Q$$

for any real number s. Thus the lc threshold of Q with respect to $(X, B + f^*P)$ is $t_Q - c$, and the coefficient of Q in the discriminant $\boldsymbol{b}$-divisor of $(X, B + f^*P)$ over Z is

$$1 - (t_Q - c) = \operatorname{mult}_Q(B_{Z'} + h^*P).$$

Hence the discriminant $\boldsymbol{b}$-divisor of $(X, B + f^*P)$ over Z has trace $B_{Z'} + h^*P$ on Z'. By Definition 3.1(3), the corresponding moduli $\boldsymbol{b}$-divisor has trace

$$h^*(D_Z + P) - K_{Z'} - (B_{Z'} + h^*P) = h^*D_Z - K_{Z'} - B_{Z'} = \mathbf{M}_{Z'}$$

on Z'. This implies (1).

(2) For a projective birational morphism $h : Z' \to Z$ from a normal variety and a prime divisor Q on Z', by (1),

$$\operatorname{mult}_Q(B_{Z'} + h^*P) = 1 - t'_Q,$$

where t'_Q is the lc threshold of Q with respect to $(X, B + f^*P)$. If $(X, B + f^*P)$ is sub-lc, then so is $(X', B' + f'^*h^*P)$, hence $t'_Q \geq 0$, that is, $\operatorname{mult}_Q(B_{Z'} + h^*P) \leq 1$.

Conversely, assume that the coefficients of $B_{Z'} + h^*P$ are at most 1 for any $h : Z' \to Z$, and let E be a prime divisor over X. Suppose first that the center of E on X dominates Z. Since $f^*P = 0$ over $Z \setminus \operatorname{Supp} P$, we have $\operatorname{mult}_E P = 0$. Since (X, B) is sub-lc over the generic point of Z (Definition 3.1), we have

$$a(E, X, B + f^*P) = a(E, X, B) \geq 0.$$

Suppose now that the center of E on X does not dominate Z. By Lemma 2.16, there exist a positive integer e, a smooth variety Z' with a projective birational morphism $h : Z' \to Z$, and a prime divisor Q on Z', such that

$$\operatorname{mult}_E \operatorname{div}(f^*\varphi) = e \operatorname{mult}_Q \operatorname{div}(h^*\varphi)$$

for any non-zero rational function φ on Z. By [KM98, Lemma 2.45] (after normalizing the variety given there, which is an isomorphism near the generic point of the center of E on that variety, since the local ring of that variety at that point is a discrete valuation ring), E is a prime divisor on some normal variety projective and birational over X, and we take for X' the normalization of the closure of the graph of the rational map from that variety to Z', with $h_X : X' \to X$ and $f' : X' \to Z'$ as in Definition 3.1(2), so that E is a prime divisor on X'. Let ε be the generic point of E, $t := f'(\varepsilon)$, and q the generic point of Q. Let φ be a non-zero rational function on Z such that $h^*\varphi$ is regular at t. Then $f'^*h^*\varphi$, the pullback of $f^*\varphi$ to X', is regular at ε, so $\operatorname{mult}_Q \operatorname{div}(h^*\varphi) \geq 0$ by the choice of e and Q, and $\operatorname{mult}_Q \operatorname{div}(h^*\varphi) > 0$ if $h^*\varphi$ vanishes at t. Since h is birational, any rational function on Z' is of the form $h^*\varphi$, and since Z' is smooth, the local ring $\mathcal{O}_{Z',q}$ consists of the rational functions u on Z' with $\operatorname{mult}_Q \operatorname{div}(u) \geq 0$, and the maximal ideal of $\mathcal{O}_{Z',q}$ consists of those with $\operatorname{mult}_Q \operatorname{div}(u) > 0$. Thus the local ring $\mathcal{O}_{Z',t}$ is contained in $\mathcal{O}_{Z',q}$, and the maximal ideal of $\mathcal{O}_{Z',t}$ is contained in the maximal ideal of $\mathcal{O}_{Z',q}$. The induced morphism $\operatorname{Spec} \mathcal{O}_{Z',q} \to \operatorname{Spec} \mathcal{O}_{Z',t} \to Z'$ and the canonical morphism $\operatorname{Spec} \mathcal{O}_{Z',q} \to Z'$ are equal at the generic point of $\operatorname{Spec} \mathcal{O}_{Z',q}$, hence equal, since Z' is separated [Har77, Exercise II.4.2]. Comparing the images of the closed point, $t = q$, that is, $f'(E)$ is dense in Q. Taking for φ a rational function on Z such that $h^*\varphi$ is a local equation of Q near q, we obtain $\operatorname{mult}_E Q = \operatorname{mult}_E \operatorname{div}(f^*\varphi) = e$. By Lemma 3.2(1), with Q in place of P,

$$a(E, X, B) \geq e t_Q.$$

Moreover, $\operatorname{mult}_E P = e \operatorname{mult}_Q P$. Thus

$$\begin{aligned} a(E, X, B + f^*P) &= a(E, X, B) - \operatorname{mult}_E P \geq e\,(t_Q - \operatorname{mult}_Q P) \\ &= e\,(1 - \operatorname{mult}_Q(B_{Z'} + h^*P)) \geq 0. \end{aligned}$$

Hence $(X, B + f^*P)$ is sub-lc.

Assume moreover that $Z \to U$ is a projective morphism to a normal quasi-projective variety U and $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U. Since $K_Z + B_Z + P + \mathbf{M}_Z = D_Z + P$ is $\mathbb{Q}$-Cartier, $(Z, B_Z + P, \mathbf{M})/U$ is a generalized sub-pair. Let E be a prime divisor over Z. By [KM98, Lemma 2.45, Theorem 0.2], applied to a resolution of a higher model of Z on which E is a divisor and to which $\mathbf{M}$ descends, there exists a log resolution $h : Z' \to Z$ of $(Z, \operatorname{Supp}(B_Z + P))$ such that $\mathbf{M}$ descends to Z' and E is a prime divisor on Z'. By Definition 3.1(3),

$$K_{Z'} + B_{Z'} + h^*P + \mathbf{M}_{Z'} = h^*(K_Z + B_Z + P + \mathbf{M}_Z),$$

so

$$a(E, Z, B_Z + P, \mathbf{M}) = 1 - \operatorname{mult}_E(B_{Z'} + h^*P).$$

Since t_Q depends only on Q as a prime divisor over Z (Definition 3.1(2)), the coefficient of a prime divisor Q on Z' in $B_{Z'} + h^*P$ depends only on Q as a prime divisor over Z. Thus the coefficients of $B_{Z'} + h^*P$ are at most 1 for any $h : Z' \to Z$ if and only if $a(E, Z, B_Z + P, \mathbf{M}) \geq 0$ for any prime divisor E over Z, that is, if and only if $(Z, B_Z + P, \mathbf{M})/U$ is generalized sub-lc.

(3) Let $h : Z' \to Z$ be a projective birational morphism from a normal variety and Q a prime divisor on Z'. Since Z' is normal, Q is Cartier near the generic point of Q. By [KM98, Theorem 0.2], applied to the normalization of the main component of $X \times_Z Z'$, there exist a smooth variety X' with a projective birational morphism $h_X : X' \to X$, such that $f \circ h_X$ factors through a morphism $f' : X' \to Z'$, and an open neighborhood V of the generic point of Q on which Q is Cartier, such that $\operatorname{Supp} B' \cup \operatorname{Supp} f'^*Q$ is snc over V, where $K_{X'} + B' = h_X^*(K_X + B)$. The closure of the image in Z' of any component of f'^*Q or of B' is either Z', or Q, or a closed subset not containing the generic point of Q. Thus, possibly shrinking V, we may assume that

any component of f'^*Q over V dominates Q, and that any component of B' over V either is a component of f'^*Q or dominates Z'.

Let $E_1, \dots, E_k$ be the components of f'^*Q over V, and $e_i := \operatorname{mult}_{E_i} Q$. By Lemma 3.2(2), with Q in place of P and V in place of Z'_P, we have $k \geq 1$ and

$$t_Q = \min_{1 \leq i \leq k} \frac{a(E_i, X, B)}{e_i}.$$

For any i, $f(h_X(E_i)) = h(f'(E_i))$ is contained in the proper closed subset $h(Q)$ of Z, so the center of E_i on X does not dominate Z. By assumption, $a(E_i, X, B) > 0$ for any i, so $t_Q > 0$, that is, $\operatorname{mult}_Q B_{Z'} = 1 - t_Q < 1$.

If (X, B) is sub-klt, then $a(E, X, B) > 0$ for any prime divisor E over X. If (X, B) is lc and any lc center of (X, B) dominates Z, then $a(E, X, B) \geq 0$ for any prime divisor E over X, and if $a(E, X, B) = 0$, then the center of E on X is an lc center of (X, B), which dominates Z. In both cases, the assumption of (3) holds. □

Proposition 3.9. *Let $f : X \to Z$ and $Z \to U$ be contractions such that $\dim Z > 0$. Let (X, B) be an lc pair such that B is a $\mathbb{Q}$-divisor and $-(K_X + B) \sim_{\mathbb{Q}} f^*H$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H on Z. Let $K_X + B + \frac{1}{m}\operatorname{div}(\psi) = f^*(K_Z + B_Z + \mathbf{M}_Z)$ be a trivialization of degree m with induced generalized sub-pair $(Z, B_Z, \mathbf{M})$ such that $\mathbf{M}$ is nef over U and $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier for a positive integer p with $m \mid p$. Then:*

(1) for any $\mathbb{Q}$-divisor $B^+ \geq B$ on X such that (X, B^+) is lc and $K_X + B^+ \sim_{\mathbb{Q},U} 0$, there exists a unique $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $P \geq 0$ on Z such that $B^+ = B + f^*P$, and moreover $P \sim_{\mathbb{Q},U} H$, $(Z, B_Z + P, \mathbf{M})/U$ is generalized lc, and $K_Z + B_Z + P + \mathbf{M}_Z \sim_{\mathbb{Q},U} 0$. Conversely, for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $P \geq 0$ on Z such that $(Z, B_Z + P, \mathbf{M})/U$ is generalized lc and $K_Z + B_Z + P + \mathbf{M}_Z \sim_{\mathbb{Q},U} 0$, the pair $(X, B + f^*P)$ is lc and $K_X + B + f^*P \sim_{\mathbb{Q},U} 0$;

(2) for any $u \in U$, any positive integer n with $p \mid n$, and any $\mathbb{Q}$-divisor B^+ on X, $(X/U \ni u, B^+)$ is a monotonic n-complement of $(X/U \ni u, B)$ if and only if $B^+ \geq B$ and there exists a $\mathbb{Q}$-divisor $P \geq 0$ on Z such that, over some neighborhood of u, P is $\mathbb{Q}$-Cartier and $B^+ = B + f^*P$, and $(Z/U \ni u, B_Z + P, \mathbf{M})$ is a monotonic n-complement of $(Z/U \ni u, B_Z, \mathbf{M})$.

Proof. Let $\pi : Z \to U$ be the given contraction and $D_Z := K_Z + B_Z + \mathbf{M}_Z$. The assumption $-(K_X + B) \sim_{\mathbb{Q}} f^*H$ is the condition of Definition 3.1 under which trivializations are defined. By Lemma 2.17(2), applied to a multiple of $f^*(H + D_Z) \sim_{\mathbb{Q}} 0$, we have $H \sim_{\mathbb{Q}} -D_Z$, so H is determined up to $\mathbb{Q}$-linear equivalence by the trivialization. By Lemma 3.2(3), $B_Z \geq 0$. Since $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier, $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier. Since moreover $K_Z + B_Z + \mathbf{M}_Z = D_Z$ is $\mathbb{Q}$-Cartier, $(Z, B_Z, \mathbf{M})/U$ is a generalized pair.

(1) Let $B^+ \geq B$ be a $\mathbb{Q}$-divisor on X such that (X, B^+) is lc and $K_X + B^+ \sim_{\mathbb{Q},U} 0$. There exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L_U on U such that $K_X + B^+ \sim_{\mathbb{Q}} (\pi \circ f)^*L_U$. Then $B^+ - B \geq 0$ is $\mathbb{Q}$-Cartier, and

$$B^+ - B = (K_X + B^+) - (K_X + B) \sim_{\mathbb{Q}} f^*(H + \pi^*L_U).$$

By Lemma 2.17(3), there exists a unique $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $P \geq 0$ on Z such that $B^+ = B + f^*P$, and $P \sim_{\mathbb{Q}} H + \pi^*L_U$. In particular, $P \sim_{\mathbb{Q},U} H$. Since $(X, B + f^*P) = (X, B^+)$ is lc, $(Z, B_Z + P, \mathbf{M})/U$ is generalized sub-lc by Lemma 3.8(2). Since $B_Z + P \geq 0$, $(Z, B_Z + P, \mathbf{M})/U$ is generalized lc.

There exist a positive integer N with $m \mid N$ and a rational function φ on X such that $N(K_X + B^+ - (\pi \circ f)^*L_U) = \operatorname{div}(\varphi)$. By the trivialization,

$$Nf^*(D_Z + P - \pi^*L_U) = N(K_X + B^+ - (\pi \circ f)^*L_U) + \frac{N}{m}\operatorname{div}(\psi) = \operatorname{div}\left(\varphi\psi^{N/m}\right).$$

By Lemma 2.17(2), $N(D_Z + P - \pi^*L_U)$ is a principal divisor on Z. Thus

$$K_Z + B_Z + P + \mathbf{M}_Z = D_Z + P \sim_{\mathbb{Q}} \pi^*L_U \sim_{\mathbb{Q},U} 0.$$

Conversely, let $P \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Z such that $(Z, B_Z + P, \mathbf{M})/U$ is generalized lc and $K_Z + B_Z + P + \mathbf{M}_Z \sim_{\mathbb{Q},U} 0$. By Lemma 3.8(2), $(X, B + f^*P)$ is sub-lc, hence lc, since $B + f^*P \geq 0$. Moreover,

$$K_X + B + f^*P = f^*(D_Z + P) - \frac{1}{m}\operatorname{div}(\psi) \sim_{\mathbb{Q}} f^*(D_Z + P) \sim_{\mathbb{Q},U} 0.$$

(2) Since $p \mid n$ and $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier, $n\mathbf{M}$ is $\boldsymbol{b}$-Cartier. For an open neighborhood U_0 of u, let $Z_0 := \pi^{-1}(U_0)$ and $X_0 := f^{-1}(Z_0)$. Then $f|_{X_0} : X_0 \to Z_0$ and $\pi|_{Z_0} : Z_0 \to U_0$ are contractions, and the assumptions of the proposition hold for $(X_0, B|_{X_0})$, $H|_{Z_0}$, and the restriction of the trivialization to X_0 (discriminant and moduli parts are determined by the lc thresholds of prime divisors over the base, Definition 3.1, hence restrict).

First assume that $(X/U \ni u, B^+)$ is a monotonic n-complement of $(X/U \ni u, B)$. Then $B^+ \geq B$. Let U_0 be an affine open neighborhood of u such that $(X_0, B^+|_{X_0})$ is lc and $n(K_{X_0} + B^+|_{X_0}) = \operatorname{div}(\varphi)$ for a rational function φ on X. By (1), applied to $X_0 \to Z_0 \to U_0$, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $P_0 \geq 0$ on Z_0 such that $B^+|_{X_0} = B|_{X_0} + f^*P_0$, and $(Z_0, B_Z|_{Z_0} + P_0, \mathbf{M})/U_0$ is generalized lc. Let $P \geq 0$ be the closure of P_0 in Z. Then, over U_0, P is $\mathbb{Q}$-Cartier and $B^+ = B + f^*P$. Since $m \mid p \mid n$, over U_0 we have

$$nf^*(D_Z + P) = n(K_X + B^+) + \frac{n}{m}\operatorname{div}(\psi) = \operatorname{div}\left(\varphi\psi^{n/m}\right).$$

By Lemma 2.17(2), $n(K_Z + B_Z + P + \mathbf{M}_Z) = n(D_Z + P)$ is a principal divisor over U_0. Since $B_Z + P \geq B_Z$, $(Z/U \ni u, B_Z + P, \mathbf{M})$ is a monotonic n-complement of $(Z/U \ni u, B_Z, \mathbf{M})$.

Now assume that $B^+ \geq B$, and that $P \geq 0$ is a $\mathbb{Q}$-divisor on Z such that, over some neighborhood of u, P is $\mathbb{Q}$-Cartier and $B^+ = B + f^*P$, and $(Z/U \ni u, B_Z + P, \mathbf{M})$ is a monotonic n-complement of $(Z/U \ni u, B_Z, \mathbf{M})$. Let U_0 be an affine open neighborhood of u over which P is $\mathbb{Q}$-Cartier, $B^+ = B + f^*P$, $(Z, B_Z + P, \mathbf{M})$ is generalized lc, and $n(D_Z + P) = \operatorname{div}(\varphi_0)$ for a rational function φ_0 on Z. By Lemma 3.8(2), applied to $X_0 \to Z_0 \to U_0$, $(X_0, B^+|_{X_0})$ is sub-lc, hence lc, since $B^+ \geq B \geq 0$. Over U_0, we have

$$n(K_X + B^+) = nf^*(D_Z + P) - \frac{n}{m}\operatorname{div}(\psi) = \operatorname{div}\left(f^*\varphi_0 \cdot \psi^{-n/m}\right).$$

Since $B^+ \geq B$, $(X/U \ni u, B^+)$ is a monotonic n-complement of $(X/U \ni u, B)$. □

4. Complements of generalized pairs of Fano type

In this section, we prove two boundedness results for complements of generalized pairs of Fano type, Propositions 4.2 and 4.3. The proof of Theorem 1.3 uses Proposition 4.2, which is about DCC coefficients and monotonic complements and follows from [FM20, Theorem 1.2]. The proof of Theorem 1.6 uses Proposition 4.3, which is about coefficients in $[0, 1]$ and indices divisible by given integers and follows from [CHHX25, Theorem 8.1]. In the applications of Propositions 4.2 and 4.3 in Sections 8 and 11, the varieties are not assumed to be of Fano type in advance. Instead, the Fano type property is deduced from Lemma 4.1, using that the generalized pair $(X, B, \mathbf{M})$ is generalized klt and that $-(K_X + B + \mathbf{M}_X)$ is nef and big.

4.1. **A criterion for Fano type.**

In this subsection, we prove Lemma 4.1, a relative version of [Bir19, 2.13(6)], which treats projective generalized pairs. [HL22, Lemma 3.4] covers the case when $-(K_X + B + \mathbf{M}_X)$ is ample over U, and the nef and big case needs the perturbation in the proof below.

Lemma 4.1. *Let $(X, B, \mathbf{M})/U$ be a generalized klt pair such that U is affine and $-(K_X + B + \mathbf{M}_X)$ is nef and big over U. Then X is of Fano type over U.*

Proof. Let $A := -(K_X + B + \mathbf{M}_X)$. By Definition 2.9, $A \sim_{\mathbb{R},U} H + E$, where H is ample over U and $E \geq 0$. Since A and H are $\mathbb{R}$-Cartier, so is E. Let $\mu : X' \to X$ be a log resolution of $(X, \operatorname{Supp} B \cup \operatorname{Supp} E)$ to which $\mathbf{M}$ descends, and write $K_{X'} + B' + \mathbf{M}_{X'} := \mu^*(K_X + B + \mathbf{M}_X)$. Since $(X, B, \mathbf{M})$ is generalized klt, the coefficients of B' are < 1, so the coefficients of $B' + \epsilon\mu^*E$

are < 1 for some rational number $0 < \epsilon < 1$. Since $\operatorname{Supp} B' \cup \operatorname{Supp} \mu^* E$ is contained in a simple normal crossing divisor and

$$K_{X'} + B' + \epsilon \mu^* E + \mathbf{M}_{X'} = \mu^*(K_X + B + \epsilon E + \mathbf{M}_X),$$

$(X, B + \epsilon E, \mathbf{M})$ is generalized klt [Kol13, 2.20 and Corollary 2.11]. Then

$$-(K_X + B + \epsilon E + \mathbf{M}_X) \sim_{\mathbb{R}, U} (1 - \epsilon)A + \epsilon H$$

is ample over U. Let H_0 be a Cartier divisor which is ample over U, and let $\delta > 0$ be a rational number such that $(1 - \epsilon)A + \epsilon H - \delta H_0$ is ample over U. By [HL22, Lemma 3.4], applied to $(X, B + \epsilon E, \mathbf{M})$ and δH_0, there exists $\Delta \geq 0$ such that $\Delta \sim_{\mathbb{R}, U} B + \epsilon E + \mathbf{M}_X + \delta H_0$ and (X, Δ) is klt. Thus

$$-(K_X + \Delta) \sim_{\mathbb{R}, U} (1 - \epsilon)A + \epsilon H - \delta H_0$$

is ample over U, and X is of Fano type over U. □

In the applications, $\mathbf{M}$ is kept, and Δ is used only to show that X is of Fano type.

4.2. **Coefficients in a DCC set.** In this subsection, we prove the boundedness of monotonic complements of generalized pairs of Fano type with DCC coefficients, over a projective morphism which is not necessarily a contraction.

In Section 8, Proposition 4.2 is applied with X and S replaced by a minimal lc image V (Set-up 7.1(5)), that is, a subvariety of the base of a contraction minimal among the images of the lc centers, and by U_V, where $V \to U_V \to U$ is the Stein factorization. The proof does not use any upper bound on the number of points of U_V over u in terms of the dimension and the coefficients. A complement over a neighborhood of u is defined over an open subset of U_V containing all the points of U_V over u, and gluing the complements given by [FM20, Theorem 1.2] near the points of U_V over u by norms multiplies the index by the degree of $U_V \to U$. Instead, we take a general global section of a divisorial sheaf, so that N does not depend on the degree of $U_V \to U$.

Proposition 4.2. *Let d, p, and l be positive integers and $\Lambda \subset [0, 1]$ a DCC set such that $\overline{\Lambda} \subset \mathbb{Q}$. Then there exists a positive integer N divisible by p and l, depending only on d, p, l, and Λ, satisfying the following. Let $\pi : X \to U$ be a projective morphism between normal quasi-projective varieties such that $1 \leq \dim X \leq d$, let $X \to S \to U$ be the Stein factorization of π, and let $u \in \pi(X)$. Let $(X, B, \mathbf{M})/S$ be a generalized lc pair such that $B \in \Lambda$, $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier, $-(K_X + B + \mathbf{M}_X)$ is nef over S, and X is of Fano type over S. Then there exist an open neighborhood U^0 of u and a $\mathbb{Q}$-divisor $B^+ \geq B$ on $\pi^{-1}(U^0)$ such that $(\pi^{-1}(U^0), B^+, \mathbf{M})$ is generalized lc, $N(K_X + B^+ + \mathbf{M}_X) \sim 0$ on $\pi^{-1}(U^0)$, and $NB^+ \geq N\lfloor B \rfloor + \lfloor (N + 1)\{B\} \rfloor$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we define N.

Since $\Lambda \cup \{0, 1\}$ satisfies the DCC and any accumulation point of $\Lambda \cup \{0, 1\}$ is an accumulation point of Λ, possibly replacing Λ with $\Lambda \cup \{0, 1\}$, we may assume that $0, 1 \in \Lambda$. We use [FM20, Theorem 1.2], that is, [FM20, Conjecture 1.1] with $\delta = \epsilon = 0$, see [FM20, Definitions 2.2, 2.10, and 2.11] for generalized pairs, Fano type, and complements there. For any positive integer e, [FM20, Theorem 1.2] implies that there exists a positive integer n_e, depending only on e, p, and Λ, satisfying the following. Let $Z \to T$ be a contraction between normal quasi-projective varieties such that $\dim Z = e$, let $t \in T$ be a point, let $B_Z \geq 0$ be a $\mathbb{Q}$-divisor on Z such that $B_Z \in \Lambda$, and let $\mathbf{M}$ be a $\mathbb{Q}$-$\boldsymbol{b}$-divisor over Z which descends to a higher model Z' of Z such that $\mathbf{M}_{Z'}$ is nef and $p\mathbf{M}_{Z'}$ is Cartier. Assume that $(Z, B_Z, \mathbf{M})/T$ is generalized lc, $-(K_Z + B_Z + \mathbf{M}_Z)$ is nef over T, and Z is of Fano type over T. Then there exist an open neighborhood T^0 of t and a $\mathbb{Q}$-divisor $B_Z^+ \geq B_Z$ on the preimage Z^0 of T^0 such that $(Z^0, B_Z^+, \mathbf{M})$ is generalized lc, $n_e(K_Z + B_Z^+ + \mathbf{M}_Z) \sim 0$ on Z^0, and

$$n_e B_Z^+ \geq n_e \lfloor B_Z \rfloor + \lfloor (n_e + 1)\{B_Z\} \rfloor.$$

In [FM20], the linear equivalence $n_e(K_Z + B_Z^+ + \mathbf{M}_Z) \sim 0$ is relative over T, that is, up to the pullback of a Cartier divisor on T, and we obtain linear equivalence on Z^0 by shrinking T^0 so that the Cartier divisor on T is principal on T^0. Let

$$N := \operatorname{lcm}(p, l, n_1, \dots, n_d).$$

Then N depends only on d, p, l, and Λ, and N is divisible by p and l.

Step 2. In this step, we reduce to the case when u is a closed point and U is affine, and we take complements near the points of S over u.

Let $g : X \to S$ and $h : S \to U$ be the morphisms of the Stein factorization of π. Then $\pi = h \circ g$, $g_*\mathcal{O}_X = \mathcal{O}_S$, and h is finite ([Har77, Corollary III.11.5]). Since X is normal and $\mathcal{O}_S = g_*\mathcal{O}_X$, S is normal, so g is a contraction. Since π is projective, $\pi(X)$ is closed in U. Let $u_0 \in \pi(X)$ be a closed point in the closure of u. Then any open neighborhood of u_0 contains u, so the proposition for u_0 implies the proposition for u. Thus, possibly replacing u with u_0 and U with an affine open neighborhood of u_0, we may assume that u is a closed point and U is affine. Then S is affine since h is finite. Let $s_1, \dots, s_r$ be the points of $h^{-1}(u)$. Then $r \geq 1$ since $u \in \pi(X)$.

We check the conditions of **Step 1** for $g : X \to S$ and $e := \dim X$. Since $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier, $\mathbf{M}$ descends to a higher model X' of X such that $p\mathbf{M}_{X'}$ is Cartier (Definition 2.7, on a common higher model), and $\mathbf{M}_{X'}$ is nef over S. Since S is affine, any complete curve C on X' is contracted to a point of S, so $(\mathbf{M}_{X'} \cdot C) \geq 0$. Thus $\mathbf{M}_{X'}$ is nef. The other conditions are hypotheses of the proposition. Since $1 \leq e \leq d$, we have $n_e \mid N$. Thus by **Step 1**, possibly shrinking the neighborhoods, for any $1 \leq i \leq r$ there exist an affine open neighborhood S_i of s_i, a $\mathbb{Q}$-divisor $B_i^+ \geq B$ on $g^{-1}(S_i)$, and a non-zero rational function a_i on X such that $(g^{-1}(S_i), B_i^+, \mathbf{M})$ is generalized lc and

$$N(K_X + B_i^+ + \mathbf{M}_X) = \operatorname{div}_X(a_i) \text{ on } g^{-1}(S_i).$$

Step 3. In this step, we construct a divisorial sheaf $\mathcal{G}$ on X and a $\mathbb{Q}$-divisor Ξ on a higher model of X.

Since $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier and $p \mid N$, $N\mathbf{M}$ is $\boldsymbol{b}$-Cartier, so $N\mathbf{M}_X$ is a divisor. Let

$$\mathcal{G} := \mathcal{O}_X(-NK_X - \lceil NB \rceil - N\mathbf{M}_X),$$

that is, the coherent sheaf whose sections over an open subset $W \subset X$ are 0 and the non-zero rational functions a on X such that

$$\operatorname{div}_X(a) - NK_X - \lceil NB \rceil - N\mathbf{M}_X \geq 0 \text{ on } W.$$

For any i, on $g^{-1}(S_i)$,

$$\operatorname{div}_X(a_i) - NK_X - N\mathbf{M}_X = NB_i^+,$$

which is a divisor and $NB_i^+ \geq NB$, so $NB_i^+ \geq \lceil NB \rceil$. Thus a_i is a section of $\mathcal{G}$ over $g^{-1}(S_i)$.

Since g is projective, $g_*\mathcal{G}$ is coherent. Since S is affine, there exist finitely many non-zero global sections $a'_1, \dots, a'_t$ of $\mathcal{G}$ on X which generate $g_*\mathcal{G}$ as an $\mathcal{O}_S$-module [Har77, Example II.5.16.2]. Here $t \geq 1$ since $g_*\mathcal{G}$ has the non-zero section a_1 over S_1. Let $\mathfrak{a} \subset \mathcal{G}$ be the $\mathcal{O}_X$-submodule generated by $a'_1, \dots, a'_t$. Since X is projective over the affine variety S, there exists a Cartier divisor $D \geq 0$ on X such that $\operatorname{div}_X(a'_j) + D \geq 0$ for any j (Serre's theorem, [Har77, Theorem II.5.17]). Then $\mathfrak{a} \cdot \mathcal{O}_X(-D) \subset \mathcal{O}_X$ is an ideal sheaf. By principalization [Kol07, Theorem 3.26], applied on a log resolution of a higher model to which $\mathbf{M}$ descends, there exists a log resolution $\mu : Y \to X$ of X such that $\mathbf{M}$ descends to Y and $\mathfrak{a} \cdot \mathcal{O}_X(-D) \cdot \mathcal{O}_Y$ is invertible. Hence the $\mathcal{O}_Y$-module generated by $a'_1, \dots, a'_t$, viewed as rational functions on Y, is $\mathcal{O}_Y(-F)$ for some divisor F on Y. In particular, for any prime divisor Q on Y,

$$\operatorname{mult}_Q F = \min_j \operatorname{mult}_Q \operatorname{div}_Y(a'_j).$$

Let $V_{\mathfrak{a}}$ be the $\mathbb{C}$-vector space spanned by $a'_1, \dots, a'_t$. For any non-zero $a \in V_{\mathfrak{a}}$, let

$$L_a := \operatorname{div}_Y(a) - F \geq 0.$$

Since $a'_1, \dots, a'_t$ generate $\mathcal{O}_Y(-F)$, for any point $y \in Y$ there exists j such that $y \notin \operatorname{Supp} L_{a'_j}$. Thus the linear system $\{L_a \mid 0 \neq a \in V_{\mathfrak{a}}\}$ is base point free.

Let

$$\Xi := \frac{1}{N}F - K_Y - \mathbf{M}_Y,$$

where K_Y is the canonical divisor with $\mu_* K_Y = K_X$. Possibly replacing Y with a log resolution $\psi : Y_1 \to Y$ of $(Y, \operatorname{Supp}\Xi)$, we may assume that $\operatorname{Supp}\Xi$ is a simple normal crossing divisor: $\mathbf{M}$ descends to Y_1, the $\mathcal{O}_{Y_1}$-module generated by $a'_1, \dots, a'_t$, viewed as rational functions on Y_1, is $\mathcal{O}_{Y_1}(-\psi^* F)$, and $\Xi_1 := \frac{1}{N}\psi^* F - K_{Y_1} - \mathbf{M}_{Y_1}$ satisfies $K_{Y_1} + \Xi_1 = \psi^*(K_Y + \Xi)$, so $\operatorname{Supp}\Xi_1$ is contained in the union of the strict transform of $\operatorname{Supp}\Xi$ and the exceptional locus of ψ.

Step 4. In this step, we show that there exists an open neighborhood S^0 of $\{s_1, \dots, s_r\}$ in S such that the coefficients of Ξ over S^0 are ≤ 1.

Let Q be a prime divisor on Y such that $s_i \in (g \circ \mu)(Q)$ for some i. Recall from **Step 2** that S_i is an affine open neighborhood of s_i and $N(K_X + B_i^+ + \mathbf{M}_X) = \operatorname{div}_X(a_i)$ on $g^{-1}(S_i)$. Since $a'_1, \dots, a'_t$ generate $g_*\mathcal{G}$ and S_i is affine,

$$a_i = \sum_j c_{ij} a'_j$$

for some regular functions c_{ij} on S_i. Since $(g \circ \mu)^{-1}(S_i)$ is open and intersects Q, the generic point of Q is mapped into S_i, so $c_{ij} \circ g \circ \mu$ is regular at the generic point of Q for any j. Thus

$$\operatorname{mult}_Q F = \min_j \operatorname{mult}_Q \operatorname{div}_Y(a'_j) \leq \operatorname{mult}_Q \operatorname{div}_Y(a_i).$$

On $(g \circ \mu)^{-1}(S_i)$, let $B^+_{i,Y} := \frac{1}{N}\operatorname{div}_Y(a_i) - K_Y - \mathbf{M}_Y$. Then

$$K_Y + B^+_{i,Y} + \mathbf{M}_Y = \frac{1}{N}\operatorname{div}_Y(a_i) = \mu^*(K_X + B_i^+ + \mathbf{M}_X) \text{ on } (g \circ \mu)^{-1}(S_i).$$

Since $\mathbf{M}$ descends to Y and $(g^{-1}(S_i), B_i^+, \mathbf{M})$ is generalized lc, $\operatorname{mult}_Q B^+_{i,Y} = 1 - a(Q, X, B_i^+, \mathbf{M})$, which is at most 1. Thus

$$\begin{aligned}\operatorname{mult}_Q \Xi &= \frac{1}{N}\operatorname{mult}_Q F - \operatorname{mult}_Q(K_Y + \mathbf{M}_Y) \\ &\leq \frac{1}{N}\operatorname{mult}_Q \operatorname{div}_Y(a_i) - \operatorname{mult}_Q(K_Y + \mathbf{M}_Y) = \operatorname{mult}_Q B^+_{i,Y} \leq 1.\end{aligned}$$

Let $Q_1, \dots, Q_m$ be the components of Ξ whose coefficients are > 1. Since $\operatorname{mult}_Q \Xi \leq 1$ for any prime divisor Q on Y such that $s_i \in (g \circ \mu)(Q)$ for some i, we have $s_i \notin (g \circ \mu)(Q_k)$ for any i and k. Since $g \circ \mu$ is projective, $(g \circ \mu)(Q_k)$ is closed in S. Let

$$S^0 := S \setminus \bigcup_{k=1}^m (g \circ \mu)(Q_k) \quad \text{and} \quad Y^0 := (g \circ \mu)^{-1}(S^0).$$

Then S^0 is an open neighborhood of $\{s_1, \dots, s_r\}$, $\operatorname{Supp}\Xi|_{Y^0}$ is a simple normal crossing divisor, and the coefficients of $\Xi|_{Y^0}$ are ≤ 1.

Step 5. In this step, we take a general element $a \in V_{\mathfrak{a}}$ and construct B^+.

By [Kle97, Theorem 4.1], on any variety, a general member of a linear system without fixed components has no singular points outside the base locus of the linear system and the singular locus of the variety. We use [Kle97, Theorem 4.1] for the restriction of the base point free linear system $\{L_a \mid 0 \neq a \in V_{\mathfrak{a}}\}$ to the smooth variety Y^0, and for the restrictions to the strata of $\operatorname{Supp}\Xi|_{Y^0}$, that is, the irreducible components of the intersections of components of $\operatorname{Supp}\Xi|_{Y^0}$. There are finitely many strata, any stratum is smooth, and any restricted linear system is base point free. A stratum W is contained in $\operatorname{Supp} L_a$ only if a is in the kernel of the restriction map to W, which is a proper linear subspace of $V_{\mathfrak{a}}$ since the linear system has no base point on W. Since $\mathbb{C}$ is infinite, there exists a non-zero $a \in V_{\mathfrak{a}}$ satisfying finitely many non-empty open conditions, such that $L_a|_{Y^0}$ is smooth, L_a contains no stratum, and the restriction of L_a to any stratum of

positive dimension is smooth. Then $L_a|_{Y^0}$ is reduced, no component of $L_a|_{Y^0}$ is a component of $\Xi|_{Y^0}$, and $\mathrm{Supp}(\Xi+L_a)|_{Y^0}$ is a simple normal crossing divisor. Since the coefficients of $\Xi|_{Y^0}$ are ≤ 1 and $\frac{1}{N}\leq 1$, $(Y^0,(\Xi+\frac{1}{N}L_a)|_{Y^0})$ is sub-lc by [KM98, Corollary 2.31(3)].

Recall from **Step 3** that $\mathcal{G}=\mathcal{O}_X(-NK_X-\lceil NB\rceil-N\mathbf{M}_X)$ and $a'_1,\dots,a'_t$ are global sections of $\mathcal{G}$, so a is a global section of $\mathcal{G}$. Let $X^0:=g^{-1}(S^0)$ and

$$B^+:=\frac{1}{N}\operatorname{div}_X(a)-K_X-\mathbf{M}_X \text{ on } X^0.$$

Since a is a global section of $\mathcal{G}$,

$$NB^+\geq\lceil NB\rceil\geq NB\geq 0.$$

Thus $B^+\geq B$, NB^+ is a divisor, and $N(K_X+B^++\mathbf{M}_X)=\operatorname{div}_X(a)\sim 0$ on X^0. Moreover, on Y^0,

$$\mu^*(K_X+B^++\mathbf{M}_X)=\frac{1}{N}\operatorname{div}_Y(a)=\frac{1}{N}(F+L_a)=K_Y+\Xi+\frac{1}{N}L_a+\mathbf{M}_Y.$$

Since $\mathbf{M}$ descends to Y, $a(E,X^0,B^+,\mathbf{M})=a(E,Y^0,(\Xi+\frac{1}{N}L_a)|_{Y^0})\geq 0$ for any prime divisor E over X^0. Thus $(X^0,B^+,\mathbf{M})$ is generalized lc.

Let b be a coefficient of B. If $b<1$, then by Lemma 2.13(1),

$$\lceil Nb\rceil\geq\lfloor(N+1)b\rfloor=N\lfloor b\rfloor+\lfloor(N+1)\{b\}\rfloor.$$

If $b=1$, then

$$\lceil Nb\rceil=N=N\lfloor b\rfloor+\lfloor(N+1)\{b\}\rfloor.$$

Since $NB^+\geq\lceil NB\rceil$, we have $NB^+\geq N\lfloor B\rfloor+\lfloor(N+1)\{B\}\rfloor$ on X^0.

Let $U^0:=U\backslash h(S\backslash S^0)$. Since h is finite, $h(S\backslash S^0)$ is closed in U. Since S^0 contains $h^{-1}(u)=\{s_1,\dots,s_r\}$, U^0 is an open neighborhood of u. Moreover, $h^{-1}(U^0)\subset S^0$, so $\pi^{-1}(U^0)\subset X^0$. The restriction of B^+ to $\pi^{-1}(U^0)$ satisfies the conclusion of the proposition. □

4.3. **Coefficients in the unit interval.** In this subsection, we prove the boundedness of complements of generalized pairs of Fano type with coefficients in $[0,1]$, with indices divisible by given integers.

When the coefficients of B are in $[0,1]$, complements are in general not monotonic, and a non-monotonic n-complement is not necessarily a kn-complement for a positive integer k, since the rounding inequality in Definition 2.11(2) may fail for kn. When we lift complements in Sections 10 and 11, we need $p\mid n$ and $l\mid n$ for the index n, and the conditions $p\mid n$ and $l\mid n$ have to be built into the index before taking the complement. [CHHX25, Theorem 8.1] allows this.

Proposition 4.3. *Let d, p, and l be positive integers. Then there exists a finite set $\mathcal{J}(d,p,l)$ of positive integers divisible by p and l satisfying the following. Let $(X/U\ni u,B,\mathbf{M})$ be a klt $\mathbb{R}$-complementary generalized pair such that $X\to U$ is a contraction, $\dim X\leq d$, $B\in[0,1]$, and $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier. Assume that X is of Fano type over a neighborhood of u, or that $-(K_X+B+\mathbf{M}_X)$ is nef and big over a neighborhood of u. Then $(X/U\ni u,B,\mathbf{M})$ has an n-complement for some $n\in\mathcal{J}(d,p,l)$.*

Proof. Let $I:=\mathrm{lcm}(p,l)$ and $\Phi:=\Phi(\{0,1\})$. We use [CHHX25, Theorem 8.1], see [CHHX25, Definitions 2.6, 2.8, and 2.17] for generalized pairs, Fano type, and complements there. For any positive integer e, [CHHX25, Theorem 8.1] with the parameters e, p, I, and Φ states the following. There exists a finite set $\mathcal{N}_e$ of positive integers, depending only on e, p, I, and Φ, such that $I\mid n$ for any $n\in\mathcal{N}_e$, satisfying the following. Let $(Z/T\ni t,B_Z,\mathbf{M})$ be a generalized pair such that $Z\to T$ is a contraction, $\dim Z=e$, $B_Z\in[0,1]$, $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier, Z is of Fano type over T, and $(Z/T\ni t,(B_Z)_{\Gamma(\mathcal{N}_e,\Phi)},\mathbf{M})$ is klt $\mathbb{R}$-complementary, where $(B_Z)_{\Gamma(\mathcal{N}_e,\Phi)}$ is defined by Lemma 2.14(1) and Definition 2.12. Then $(Z/T\ni t,B_Z,\mathbf{M})$ has an n-complement for some $n\in\mathcal{N}_e$. Let

$$\mathcal{J}(d,p,l):=\{I\}\cup\bigcup_{e=1}^{d}\mathcal{N}_e.$$

Then $\mathcal{J}(d,p,l)$ is a finite set depending only on d, p, and l, and any element of $\mathcal{J}(d,p,l)$ is divisible by I, hence by p and l.

Let $(X/U\ni u, B, \mathbf{M})$ be as in the proposition, $\pi: X\to U$ the contraction, and $e:=\dim X$. If $e=0$, then X is a point, $B=0$, and $K_X+\mathbf{M}_X=0$, so $(X/U\ni u, 0, \mathbf{M})$ is an I-complement of $(X/U\ni u, B, \mathbf{M})$, where $I\mathbf{M}$ is $\boldsymbol{b}$-Cartier since $p\mid I$. Thus we may assume that $1\le e\le d$.

Let $(X/U\ni u, B^+, \mathbf{M})$ be a klt $\mathbb{R}$-complement of $(X/U\ni u, B, \mathbf{M})$. Over a neighborhood of u,

$$B^+-B=(K_X+B^++\mathbf{M}_X)-(K_X+B+\mathbf{M}_X)$$

is $\mathbb{R}$-Cartier and $B^+-B\ge 0$, so $a(E,X,B,\mathbf{M})\ge a(E,X,B^+,\mathbf{M})>0$ for any prime divisor E over X. Thus $(X,B,\mathbf{M})$ is generalized klt over a neighborhood of u. If $-(K_X+B+\mathbf{M}_X)$ is nef and big over a neighborhood of u, then by Lemma 4.1, applied over an affine open neighborhood of u, X is of Fano type over a neighborhood of u. Thus in either case, there exists an open neighborhood U^1 of u such that $X^1:=\pi^{-1}(U^1)$ is of Fano type over U^1, $(X^1,B^+,\mathbf{M})$ is generalized klt, and $K_X+B^++\mathbf{M}_X\sim_{\mathbb{R}}0$ on X^1.

If u is not a closed point, then we pick a closed point u_0 of U^1 in the closure of u. Since any open neighborhood of u_0 contains u, if $(X/U\ni u_0, B^+, \mathbf{M})$ is an n-complement of $(X/U\ni u_0, B, \mathbf{M})$, then $(X/U\ni u, B^+, \mathbf{M})$ is an n-complement of $(X/U\ni u, B, \mathbf{M})$. Thus, possibly replacing u with u_0 and shrinking U^1, we may assume that u is a closed point and U^1 is affine.

Since X^1 is of Fano type over U^1, there exists $\Delta\ge 0$ on X^1 such that (X^1,Δ) is klt and $-(K_{X^1}+\Delta)$ is nef and big over U^1. By [BCHM10, Corollary 1.4.3], there exists a small $\mathbb{Q}$-factorialization $\alpha: X'\to X^1$ (α is projective, being constructed by an MMP over X^1 in [BCHM10, Corollary 1.4.3]). Let $\pi':=\pi\circ\alpha: X'\to U^1$, and let B', B'^+, and Δ' be the strict transforms of B, B^+, and Δ on X'. Since α is small, for each of the equalities

$$\begin{aligned}K_{X'}+B'+\mathbf{M}_{X'}&=\alpha^*(K_X+B+\mathbf{M}_X),\\ K_{X'}+B'^++\mathbf{M}_{X'}&=\alpha^*(K_X+B^++\mathbf{M}_X),\\ K_{X'}+\Delta'&=\alpha^*(K_{X^1}+\Delta),\end{aligned}$$

the coefficients of the left-hand side and of the right-hand side at any prime divisor on X' are equal, so the equalities hold. Thus (X',Δ') is klt and $-(K_{X'}+\Delta')$ is nef and big over U^1, since the pullback of a divisor big over U^1 under a birational morphism is big over U^1 (cf. [KMM87, Corollary 0-3-4]). Hence X' is of Fano type over U^1. Moreover, π' is a contraction, $\dim X'=e$, $B'\in[0,1]$, $(X',B'^+,\mathbf{M})$ is generalized klt, and $K_{X'}+B'^++\mathbf{M}_{X'}\sim_{\mathbb{R}}0$. Since X' is $\mathbb{Q}$-factorial, $K_{X'}+B'_{\Gamma(\mathcal{N}_e,\Phi)}+\mathbf{M}_{X'}$ is $\mathbb{R}$-Cartier, so $(X'/U^1\ni u, B'_{\Gamma(\mathcal{N}_e,\Phi)},\mathbf{M})$ is a generalized pair. By Definition 2.12, $B'_{\Gamma(\mathcal{N}_e,\Phi)}\le B'\le B'^+$, so $(X'/U^1\ni u, B'^+,\mathbf{M})$ is a klt $\mathbb{R}$-complement of $(X'/U^1\ni u, B'_{\Gamma(\mathcal{N}_e,\Phi)},\mathbf{M})$. Thus all conditions of [CHHX25, Theorem 8.1] hold for $(X'/U^1\ni u, B',\mathbf{M})$, with the parameters e, p, I, and Φ.

By [CHHX25, Theorem 8.1], there exist $n\in\mathcal{N}_e$, an open neighborhood $U^0\subset U^1$ of u, and a $\mathbb{Q}$-divisor Θ' on $X'^0:=\pi'^{-1}(U^0)$ such that $(X'^0,\Theta',\mathbf{M})$ is generalized lc, $n(K_{X'}+\Theta'+\mathbf{M}_{X'})\sim 0$ on X'^0, and

$$n\Theta'\ge n\lfloor B'\rfloor+\lfloor(n+1)\{B'\}\rfloor.$$

Write $n(K_{X'}+\Theta'+\mathbf{M}_{X'})=\operatorname{div}_{X'}(a)$ on X'^0 for a non-zero rational function a on X'. Let $X^0:=\pi^{-1}(U^0)$ and $\Theta^0:=\alpha_*\Theta'$. Pushing forward by α, we obtain

$$n(K_X+\Theta^0+\mathbf{M}_X)=\operatorname{div}_X(a)\text{ on }X^0.$$

Thus

$$\alpha^*(K_X+\Theta^0+\mathbf{M}_X)=\frac{1}{n}\operatorname{div}_{X'}(a)=K_{X'}+\Theta'+\mathbf{M}_{X'}\text{ on }X'^0,$$

so $a(E,X^0,\Theta^0,\mathbf{M})=a(E,X'^0,\Theta',\mathbf{M})\ge 0$ for any prime divisor E over X^0, and $(X^0,\Theta^0,\mathbf{M})$ is generalized lc. Since α is small, the prime divisors on X'^0 and on X^0 correspond to each other

via α, and $B = \alpha_* B'$. Thus

$$n\Theta^0 \geq n\lfloor B \rfloor + \lfloor (n+1)\{B\} \rfloor \text{ on } X^0.$$

Since $p \mid n$, $n\mathbf{M}$ is $\boldsymbol{b}$-Cartier. Possibly shrinking U to a neighborhood of u, we may assume that $U = U^0$. Then $(X/U \ni u, \Theta^0, \mathbf{M})$ is an n-complement of $(X/U \ni u, B, \mathbf{M})$, and $n \in \mathcal{N}_e \subset \mathcal{J}(d, p, l)$. □

5. Semi-ample models via the minimal model program

In this section, we prove Theorem 5.1, Lemma 5.2, and Proposition 5.3. Theorem 5.1 provides the termination of an MMP with scaling and the semi-ampleness of the log canonical divisor where it terminates, for both main theorems, Theorems 1.3 and 1.6. Lemma 5.2 replaces an $\mathbb{R}$-complement with a $\mathbb{Q}$-complement without changing the non-klt locus. Proposition 5.3 reduces Theorem 1.3 to the case when $-(K_X + B)$ is the pullback of a divisor on a variety T over Z which is ample over Z. The constructions in Sections 6, 7, and 8 take place on T.

We state Theorem 5.1 with a proof rather than cite it: [OAI26a, Theorem 11.1] gives good minimal models of projective lc pairs with $\mathbb{R}$-boundary and pseudo-effective log canonical divisor, and [OAI26a, Corollary 14.1] gives relative semi-ampleness in the complex analytic setting for lc pairs with $\mathbb{Q}$-boundary and log canonical divisor nef over the base. Proposition 5.3, Proposition 9.3, and Lemma 11.3 need the algebraic relative form over a quasi-projective base: termination of the $(K_X + B)$-MMP with scaling of an ample divisor, and semi-ampleness over the base of the log canonical divisor of the pair where it terminates. Termination follows from [OAI26a, Theorem 11.1] through [FG14a, Conjecture 5.1, Lemma 5.3] and [Bir11, Theorems 1.4 and 1.5], and semi-ampleness follows from [OAI26a, Corollary 14.1] by relative GAGA.

Theorem 5.1. *Let $X \to U$ be a projective morphism to a normal quasi-projective variety, and (X, B) a $\mathbb{Q}$-factorial dlt pair such that B is a $\mathbb{Q}$-divisor. Assume that $K_X + B \sim_{\mathbb{Q},U} G$ for some $\mathbb{Q}$-divisor $G \geq 0$. Then there exists a $(K_X + B)$-MMP over U with scaling of an ample divisor which terminates with a $\mathbb{Q}$-factorial dlt pair (X', B') such that $K_{X'} + B'$ is semi-ample over U.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we show that for any projective lc pair (X_1, B_1) such that $\dim X_1 \leq \dim X$ and $K_{X_1} + B_1$ is pseudo-effective, $K_{X_1} + B_1$ is $\mathbb{R}$-linearly equivalent to an effective $\mathbb{R}$-divisor. In particular, [FG14a, Conjecture 5.1], which asks for numerical equivalence to an $\mathbb{R}$-divisor ≥ 0, holds in dimension $\leq \dim X$.

Let (X_1, B_1) be a projective lc pair such that $\dim X_1 \leq \dim X$ and $K_{X_1} + B_1$ is pseudo-effective. By [OAI26a, Theorem 11.1, Lemma 2.2], (X_1, B_1) has a good log minimal model (X_1', B_1'), and for a common resolution $p_1 : W_1 \to X_1$ and $q_1 : W_1 \to X_1'$,

$$p_1^*(K_{X_1} + B_1) = q_1^*(K_{X_1'} + B_1') + E_1$$

for some $\mathbb{R}$-divisor $E_1 \geq 0$. Since (X_1', B_1') is a good log minimal model, $K_{X_1'} + B_1'$ is semi-ample, which for $\mathbb{R}$-divisors means, as defined at the beginning of [OAI26a, Section 2], that there exist a contraction $g_1 : X_1' \to T_1$ and an ample $\mathbb{R}$-divisor A_1 on T_1 such that $K_{X_1'} + B_1' \sim_{\mathbb{R}} g_1^* A_1$, where $A_1 = 0$ if T_1 is a point. As A_1 is a positive real linear combination of ample Cartier divisors [Laz04, Definition 1.3.11, Proposition 1.3.13], and a positive multiple of an ample Cartier divisor is linearly equivalent to an effective divisor, $A_1 \sim_{\mathbb{R}} A_1'$ for some $\mathbb{R}$-divisor $A_1' \geq 0$. Thus

$$K_{X_1} + B_1 = p_{1*}p_1^*(K_{X_1} + B_1) \sim_{\mathbb{R}} p_{1*}\left(q_1^* g_1^* A_1' + E_1\right) \geq 0.$$

Step 2. In this step, we run the MMP and show that it terminates with a $\mathbb{Q}$-factorial dlt pair (X', B') such that $K_{X'} + B'$ is nef over U.

By **Step 1** and [FG14a, Remark 5.2, Lemma 5.3], for any projective morphism $X_1 \to Z_1$ and any lc pair (X_1, B_1) such that $\dim X_1 \leq \dim X$ and $K_{X_1} + B_1$ is pseudo-effective over Z_1, $K_{X_1} + B_1$ is $\mathbb{R}$-linearly equivalent over Z_1 to an $\mathbb{R}$-divisor $D_1 \geq 0$. In particular, [Bir11,

Conjecture 1.3] holds in dimension $\leq \dim X$ (see the paragraph after [FG14a, Remark 5.2]): for any $\mathbb{Q}$-factorial dlt pair (X_1, B_1) with a projective morphism $X_1 \to Z_1$ such that $\dim X_1 \leq \dim X$ and $K_{X_1} + B_1$ is pseudo-effective over Z_1, $K_{X_1} + B_1 \equiv M_1$ over Z_1 for some $\mathbb{R}$-divisor $M_1 \geq 0$. Hence, by [Bir11, Theorem 1.4], any lc pair (X_1, B_1) with a projective morphism $X_1 \to Z_1$ such that $\dim X_1 \leq \dim X$ and $K_{X_1} + B_1$ is pseudo-effective over Z_1 has a log minimal model over Z_1, which is the hypothesis of [Bir11, Theorem 1.5] in dimension $\leq \dim X$.

Let $A \geq 0$ be a $\mathbb{Q}$-divisor on X which is ample over U such that $(X, B + A)$ is lc and $K_X + B + A$ is nef over U. For example, let A_0 be an ample Cartier divisor on X such that $K_X + B + A_0$ is nef over U (a sufficiently large multiple of any ample Cartier divisor on X is such an A_0, since $K_X + B$ is $\mathbb{Q}$-Cartier [Har77, Exercise II.7.5(b)]), $k \geq 2$ an integer such that kA_0 is very ample, H a general member of the free linear system $|kA_0|$, and $A := \frac{1}{k}H$. By [KM98, Lemma 5.17(2)] applied to the lc pair (X, B), $(X, B + H)$ is lc, so $(X, B + A)$ is lc, and $K_X + B + A \sim_{\mathbb{Q}} K_X + B + A_0$ is nef over U. By [FG14a, 2.20], we can run a $(K_X + B)$-MMP over U with scaling of A,

$$X = X_0 \dashrightarrow X_1 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots,$$

where each $X_i \dashrightarrow X_{i+1}$ is a divisorial contraction or a flip over U. Let B_i be the pushforward of B on X_i. By [KM98, 3.31, Propositions 3.36, 3.37, Corollary 3.44], each (X_i, B_i) is a $\mathbb{Q}$-factorial dlt pair. In [Bir11, Theorem 1.5], C denotes the divisor by which the MMP is scaled. With $C = A$, the pair $(X, B + A)$ is $\mathbb{Q}$-factorial and lc, $K_X + B + A$ is nef over U, $B, A \geq 0$, and (X, B) is dlt, and A is itself an ample divisor over U with $A \leq C$. Hence [Bir11, Theorem 1.5] (cf. the last paragraph of [FG14a, 2.20]) implies that the MMP terminates with a pair $(X', B') := (X_l, B_l)$ for some l. Thus either $K_{X'} + B'$ is nef over U, or there exists a $(K_{X'} + B')$-Mori fiber space $X' \to T$ over U.

Suppose that $X' \to T$ is a $(K_{X'} + B')$-Mori fiber space over U. Let G' be the pushforward of G on X'. Since the MMP is over U and $X \dashrightarrow X'$ extracts no divisor, $K_{X'} + B' \sim_{\mathbb{Q},U} G' \geq 0$. Since $\dim T < \dim X'$, any fiber of $X' \to T$ has positive dimension. Let $x \in X' \setminus \operatorname{Supp} G'$ be a closed point and C_0 a curve such that $x \in C_0$ and C_0 is contained in a fiber of $X' \to T$. Then $C_0 \not\subset \operatorname{Supp} G'$, and C_0 is contracted over U, so

$$((K_{X'} + B') \cdot C_0) = (G' \cdot C_0) \geq 0.$$

On the other hand, $-(K_{X'} + B')$ is ample over T, so $((K_{X'} + B') \cdot C_0) < 0$, a contradiction. Thus $K_{X'} + B'$ is nef over U.

Step 3. In this step, we reduce the semi-ampleness of $K_{X'} + B'$ over U to the case when $X' \to U$ is a contraction.

Let $X' \to U' \to U$ be the Stein factorization of $X' \to U$. Then U' is normal and quasi-projective, $U' \to U$ is finite, and $X' \to U'$ is a contraction. Since $U' \to U$ is finite, a curve on X' is contracted to a point of U if and only if it is contracted to a point of U'. Hence $K_{X'} + B'$ is nef over U'. Assume that $K_{X'} + B'$ is semi-ample over U', and let m be a positive integer such that $m(K_{X'} + B')$ is Cartier and $\mathcal{O}_{X'}(m(K_{X'} + B'))$ is generated over U'. Since $U' \to U$ is finite, it is affine, so the preimage U_0' of any affine open subset $U_0 \subset U$ is affine. As the preimages of U_0' and U_0 in X' coincide, $\mathcal{O}_{X'}(m(K_{X'} + B'))$ is generated by global sections on the preimage of U_0. Thus $K_{X'} + B'$ is semi-ample over U. Possibly replacing U with U', we may assume that $h : X' \to U$ is a contraction. We use the surjectivity of h only for [OAI26a, Corollary 14.1] in **Step 4**.

Step 4. In this step, we show that $K_{X'} + B'$ is semi-ample over U.

Since X' is quasi-compact, there exists a positive integer m_0 such that $L := m_0(K_{X'} + B')$ is Cartier. Let

$$h^{\mathrm{an}} : X'^{\mathrm{an}} \to U^{\mathrm{an}}$$

be the analytification of h. Then X'^{an} and U^{an} are normal [Gro71, Exp. XII, Proposition 2.1] and irreducible [Gro71, Exp. XII, Proposition 2.4], and h^{an} is projective and surjective. The

analytification of a log resolution of (X', B') is a log resolution of $(X'^{\mathrm{an}}, B'^{\mathrm{an}})$, and the log discrepancy of each exceptional prime divisor is that of the corresponding algebraic divisor, so $(X'^{\mathrm{an}}, B'^{\mathrm{an}})$ is lc. For any point $u \in U^{\mathrm{an}}$, the fiber $(h^{\mathrm{an}})^{-1}(u)$ is the analytification of the projective variety $h^{-1}(u)$. By Chow's theorem [Cho49], any compact analytic curve C in $(h^{\mathrm{an}})^{-1}(u)$ is algebraic, so $((K_{X'} + B') \cdot C) \geq 0$. Hence $(K_{X'} + B')^{\mathrm{an}}$ is nef over U^{an}.

Fix a closed point $s \in U$. By [OAI26a, Corollary 14.1], applied to the compact subset $W = \{s\}$ of U^{an}, there exist an open neighborhood N_s of s in U^{an} and a positive integer b_s such that $b_s(K_{X'} + B')^{\mathrm{an}}$ is Cartier on $(h^{\mathrm{an}})^{-1}(N_s)$ and the line bundle $\mathcal{L}_s$ associated to $b_s(K_{X'} + B')^{\mathrm{an}}$ is generated over N_s with respect to h^{an}. Since

$$\mathcal{O}_{X'}(b_s L)^{\mathrm{an}} = \mathcal{L}_s^{\otimes m_0} \text{ on } (h^{\mathrm{an}})^{-1}(N_s),$$

$\mathcal{O}_{X'}(b_s L)^{\mathrm{an}}$ is also generated over N_s with respect to h^{an}, that is, the evaluation map

$$(h^{\mathrm{an}})^* h_*^{\mathrm{an}} \mathcal{O}_{X'}(b_s L)^{\mathrm{an}} \to \mathcal{O}_{X'}(b_s L)^{\mathrm{an}}$$

is surjective over N_s. Let $\mathcal{Q}_s$ be the cokernel of the evaluation map $h^* h_* \mathcal{O}_{X'}(b_s L) \to \mathcal{O}_{X'}(b_s L)$. Since h is proper, $\mathcal{Q}_s$ is coherent. By relative GAGA [Gro71, Exp. XII, Théorème 4.2], the natural map

$$(h_* \mathcal{O}_{X'}(b_s L))^{\mathrm{an}} \to h_*^{\mathrm{an}} \mathcal{O}_{X'}(b_s L)^{\mathrm{an}}$$

is an isomorphism compatible with the evaluation maps. Since analytification is exact, $\mathcal{Q}_s^{\mathrm{an}}$ is the cokernel of the analytic evaluation map, hence $\mathcal{Q}_s^{\mathrm{an}} = 0$ on $(h^{\mathrm{an}})^{-1}(N_s)$. Let $x \in h^{-1}(s)$ be a closed point. Then

$$\mathcal{Q}_{s,x} \otimes_{\mathcal{O}_{X',x}} \mathcal{O}_{X'^{\mathrm{an}},x} = 0.$$

Tensoring with the residue field $\mathbb{C}$ of $\mathcal{O}_{X'^{\mathrm{an}},x}$, we obtain $\mathcal{Q}_{s,x}/\mathfrak{m}_x \mathcal{Q}_{s,x} = 0$, where $\mathfrak{m}_x$ is the maximal ideal of $\mathcal{O}_{X',x}$. By Nakayama's lemma, $\mathcal{Q}_{s,x} = 0$. Thus the closed subset $\operatorname{Supp} \mathcal{Q}_s \cap h^{-1}(s)$ of X' contains no closed point, so $\operatorname{Supp} \mathcal{Q}_s \cap h^{-1}(s) = \emptyset$. Since h is proper, $U_s := U \backslash h(\operatorname{Supp} \mathcal{Q}_s)$ is an open neighborhood of s, and $\mathcal{O}_{X'}(b_s L)$ is generated over U_s.

The complement of $\bigcup_s U_s$ in U is closed and contains no closed point, so $U = \bigcup_s U_s$. Since U is quasi-compact, $U = U_{s_1} \cup \cdots \cup U_{s_e}$ for some closed points $s_1, \ldots, s_e$. Let $m := \operatorname{lcm}(b_{s_1}, \ldots, b_{s_e})$. Since the tensor powers of a line bundle generated over U_{s_i} are generated over U_{s_i}, $\mathcal{O}_{X'}(mL)$ is generated over U_{s_i} for any i, hence over U. Thus $K_{X'} + B'$ is semi-ample over U. □

Lemma 5.2. *Let $(X/Z \ni z, B)$ be a $\mathbb{Q}$-pair and $(X/Z \ni z, B^+)$ an $\mathbb{R}$-complement of $(X/Z \ni z, B)$. Then there exists a $\mathbb{Q}$-complement $(X/Z \ni z, B_0^+)$ of $(X/Z \ni z, B)$ such that $\operatorname{Nklt}(X, B_0^+) = \operatorname{Nklt}(X, B^+)$. In particular, if $(X/Z \ni z, B^+)$ is a klt $\mathbb{R}$-complement of $(X/Z \ni z, B)$, then $(X/Z \ni z, B_0^+)$ is klt.*

Proof. Possibly shrinking Z to a neighborhood of z, we may assume that $(X/Z, B^+)$ is an $\mathbb{R}$-complement of $(X/Z, B)$. Write $B^+ = \sum_{i=1}^m b_i B_i$, $\boldsymbol{b} := (b_1, \ldots, b_m)$, $V \ni \boldsymbol{b}$ the rational envelope of $\boldsymbol{b}$ in $\mathbb{R}^m$, and $B(\boldsymbol{v}) := \sum_{i=1}^m v_i B_i$ for any $\boldsymbol{v} = (v_1, \ldots, v_m) \in \mathbb{R}^m$. By [HLS26, Lemma 5.3 and Theorem 5.6], there exists an open neighborhood $U \ni \boldsymbol{b}$ in V, such that for any $\boldsymbol{v} \in U$, $K_X + B(\boldsymbol{v}) \sim_{\mathbb{R},Z} 0$ and $\operatorname{Nklt}(X, B(\boldsymbol{v})) = \operatorname{Nklt}(X, B(\boldsymbol{b}))$. Since B is a $\mathbb{Q}$-divisor, possibly shrinking U, we may assume that $B(\boldsymbol{v}) \geq B$ for any $\boldsymbol{v} \in U$. We may let $B_0^+ := B(\boldsymbol{v})$ for some $\boldsymbol{v} \in U \cap \mathbb{Q}^m$. □

Without the assumption that X is of Fano type over Z, we cannot run an arbitrary MMP on $-(K_X + B)$. Instead, after raising to 1 the coefficients of the boundary along the exceptional prime divisors of a dlt modification and along the components of $\lfloor B^+ \rfloor$ for an $\mathbb{R}$-complement $(X/Z \ni z, B^+)$ of $(X/Z \ni z, B)$ (cf. [HLS19, Lemma 5.19(4)]), the negative of the log canonical divisor is, over Z, $\mathbb{Q}$-linearly equivalent to a positive rational multiple of the log canonical divisor of a dlt pair, and we may use Theorem 5.1. The price is a larger boundary with more components of coefficient 1, and Lemma 2.15 carries a monotonic n-complement on the new model back to a monotonic n-complement of $(X/Z \ni z, B)$.

Proposition 5.3. *Let $\Gamma \subset [0,1] \cap \mathbb{Q}$ be a finite set and $(X/Z \ni z, B)$ an $\mathbb{R}$-complementary pair such that $B \in \Gamma$. Then, after replacing Z with an affine neighborhood of z, there exist a $\mathbb{Q}$-factorial dlt pair (Y, B_Y) with $B_Y \in \Gamma \cup \{1\}$, a birational map $X \dashrightarrow Y$ over Z, and contractions $g : Y \to T$ and $T \to Z$, satisfying the following.*

(1) $-(K_Y + B_Y) \sim_{\mathbb{Q}} g^*H$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H on T which is ample over Z.
(2) $q^*(K_Y + B_Y) \geq p^*(K_X + B)$ for a common resolution $p : W \to X$ and $q : W \to Y$.

In particular, for any positive integer n, if $(Y/Z \ni z, B_Y)$ has a monotonic n-complement, then $(X/Z \ni z, B)$ has a monotonic n-complement.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct a crepant $\mathbb{Q}$-factorial dlt modification $v : V \to X$ and $\mathbb{Q}$-divisors $B_V^+ \geq B_V$ on V such that $K_V + B_V \geq v^*(K_X + B)$.

Let $(X/Z \ni z, B^+)$ be an $\mathbb{R}$-complement of $(X/Z \ni z, B)$. Since $\Gamma \subset \mathbb{Q}$, B is a $\mathbb{Q}$-divisor, so $K_X + B$ is $\mathbb{Q}$-Cartier over a neighborhood of z, since an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor is $\mathbb{Q}$-Cartier (cf. [HLS26, Lemma 5.3]). By Lemma 5.2, there exists a $\mathbb{Q}$-complement $(X/Z \ni z, B_0^+)$ of $(X/Z \ni z, B)$. Since B_0^+ is a $\mathbb{Q}$-divisor and $K_X + B_0^+ \sim_{\mathbb{R}} 0$ over some neighborhood of z, $K_X + B_0^+ \sim_{\mathbb{Q}} 0$ over that neighborhood. Possibly replacing Z with an affine neighborhood of z, we may assume that $K_X + B$ is $\mathbb{Q}$-Cartier, (X, B_0^+) is lc, and

$$K_X + B_0^+ = \sum_{i=1}^{r} \lambda_i \operatorname{div}_X(\varphi_i)$$

for some rational numbers λ_i and rational functions φ_i on X. Since $B \leq B_0^+$, (X, B) is lc.

By [KK10, Theorem 3.1, Definitions and Notation 1.9], there exists a crepant $\mathbb{Q}$-factorial dlt modification $v : V \to X$ of (X, B_0^+), and v is projective: [KK10, Theorem 3.1] constructs v as a log terminal model over X [BCHM10] (cf. [Kol13, Theorem 1.34]). Thus V is $\mathbb{Q}$-factorial, $K_V + B_V^+ = v^*(K_X + B_0^+)$ for a $\mathbb{Q}$-divisor B_V^+ such that (V, B_V^+) is dlt, and any v-exceptional prime divisor has coefficient 1 in B_V^+. Let B_V be the $\mathbb{Q}$-divisor on V such that, for any prime divisor P on V, $\operatorname{mult}_P B_V = 1$ if P is v-exceptional or a component of $\lfloor B_V^+ \rfloor$, and $\operatorname{mult}_P B_V = \operatorname{mult}_P v_*^{-1} B$ otherwise. Since $B \in \Gamma$ and $B \leq B_0^+$, we have $B_V \in \Gamma \cup \{1\}$, $0 \leq B_V \leq B_V^+$, and $\lfloor B_V \rfloor = \lfloor B_V^+ \rfloor$. Let

$$G := B_V^+ - B_V \geq 0.$$

Then no component of $\lfloor B_V^+ \rfloor$ is a component of G.

Let B_V' be the $\mathbb{Q}$-divisor such that $K_V + B_V' = v^*(K_X + B)$. For any prime divisor P on V which is not v-exceptional, $\operatorname{mult}_P B_V' = \operatorname{mult}_P v_*^{-1} B \leq \operatorname{mult}_P B_V$. For any v-exceptional prime divisor E, $\operatorname{mult}_E B_V' = 1 - a(E, X, B) \leq 1 = \operatorname{mult}_E B_V$, since (X, B) is lc. Hence

$$K_V + B_V - v^*(K_X + B) = B_V - B_V' \geq 0. \tag{5.1}$$

Step 2. In this step, we run an MMP over Z on $K_V + \Delta_V$ for a dlt pair (V, Δ_V), and construct (Y, B_Y) satisfying (2).

By [KM98, Theorem 2.44], there exists a log resolution $h : W' \to V$ of (V, B_V^+) such that $a(E, V, B_V^+) > 0$ for any h-exceptional prime divisor E. For a rational number $t > 0$, let

$$\Delta_V := (1 + t)B_V^+ - tB_V = B_V^+ + tG.$$

Since $\operatorname{Supp} G \subset \operatorname{Supp} B_V^+$, $\operatorname{Supp} \Delta_V = \operatorname{Supp} B_V^+$, so h is a log resolution of (V, Δ_V). Since V is $\mathbb{Q}$-factorial, G is $\mathbb{Q}$-Cartier, and $a(E, V, \Delta_V) = a(E, V, B_V^+) - t \operatorname{mult}_E G$ for any h-exceptional prime divisor E. Since no component of $\lfloor B_V^+ \rfloor$ is a component of G and the set of h-exceptional prime divisors is finite, we may fix t sufficiently small such that any coefficient of Δ_V is at most 1 and $a(E, V, \Delta_V) > 0$ for any h-exceptional prime divisor E. By [KM98, Theorem 2.44], (V, Δ_V)

is a $\mathbb{Q}$-factorial dlt pair, and Δ_V is a $\mathbb{Q}$-divisor. Since

$$K_V + B_V^+ = v^*(K_X + B_0^+) = \sum_i \lambda_i \operatorname{div}_V(\varphi_i),$$

we have

$$K_V + \Delta_V = (1+t)(K_V + B_V^+) - t(K_V + B_V) \sim_{\mathbb{Q},Z} tG \sim_{\mathbb{Q},Z} -t(K_V + B_V).$$

In particular, $K_V + \Delta_V \sim_{\mathbb{Q},Z} tG$ and $tG \geq 0$.

Since v and $X \to Z$ are projective, $V \to Z$ is projective, so by Theorem 5.1, there exists a $(K_V + \Delta_V)$-MMP $\psi : V \dashrightarrow Y$ over Z which terminates with a $\mathbb{Q}$-factorial dlt pair (Y, Δ_Y), where $\Delta_Y := \psi_* \Delta_V$, such that $K_Y + \Delta_Y$ is semi-ample over Z. Let $B_Y^+ := \psi_* B_V^+$ and $B_Y := \psi_* B_V$. Since ψ extracts no divisor, $B_Y \in \Gamma \cup \{1\}$,

$$\Delta_Y = (1+t)B_Y^+ - tB_Y \geq B_Y, \quad \text{and} \quad K_Y + B_Y^+ = \sum_{i=1}^{r} \lambda_i \operatorname{div}_Y(\varphi_i).$$

Since Y is $\mathbb{Q}$-factorial and (Y, Δ_Y) is dlt, (Y, B_Y) is dlt [KM98, Corollary 2.39(1)]. The birational map $X \dashrightarrow Y$ over Z is $\psi \circ v^{-1}$.

Let $p : W \to X$ and $q : W \to Y$ be a common resolution such that $p = v \circ p_V$ for a morphism $p_V : W \to V$ and $q = \psi \circ p_V$. Since ψ is a composition of $(K_V + \Delta_V)$-negative divisorial contractions and flips over Z, [KM98, Lemma 3.38], applied to each of them, implies that $a(E, V, \Delta_V) \leq a(E, Y, \Delta_Y)$ for any prime divisor E over V. Hence

$$p_V^*(K_V + \Delta_V) - q^*(K_Y + \Delta_Y) \geq 0.$$

Moreover,

$$p_V^*(K_V + B_V^+) = \sum_i \lambda_i \operatorname{div}_W(\varphi_i) = q^*(K_Y + B_Y^+).$$

Since $\Delta_V = (1+t)B_V^+ - tB_V$ and $\Delta_Y = (1+t)B_Y^+ - tB_Y$, we obtain

$$t\big(q^*(K_Y + B_Y) - p_V^*(K_V + B_V)\big) = p_V^*(K_V + \Delta_V) - q^*(K_Y + \Delta_Y) \geq 0.$$

By (5.1),

$$q^*(K_Y + B_Y) \geq p_V^*(K_V + B_V) \geq p_V^* v^*(K_X + B) = p^*(K_X + B).$$

This proves (2).

Step 3. In this step, we construct $g : Y \to T$ and H satisfying (1), and show that g and $T \to Z$ are contractions.

Since $K_Y + B_Y^+ \sim_{\mathbb{Q}} 0$,

$$K_Y + \Delta_Y = (1+t)(K_Y + B_Y^+) - t(K_Y + B_Y) \sim_{\mathbb{Q}} -t(K_Y + B_Y).$$

Since $K_Y + \Delta_Y$ is semi-ample over Z and Z is affine, there exists a positive integer m such that $-m(K_Y + B_Y)$ is Cartier and $\mathcal{O}_Y(-m(K_Y + B_Y))$ is generated by global sections. Since $Y \to Z$ is projective, finitely many global sections $s_0, \ldots, s_M$ generate $\mathcal{O}_Y(-m(K_Y + B_Y))$. They define a morphism $Y \to \mathbb{P}_Z^M$ over Z, and we let J be the image. Let $Y \to T \to J$ be the Stein factorization and $g : Y \to T$ the induced morphism. Then $g_* \mathcal{O}_Y = \mathcal{O}_T$, T is normal as Y is normal, $T \to J$ is finite, and g and $T \to Z$ are projective. The pullback of $\mathcal{O}_J(1)$ to T is ample over Z since $T \to J$ is finite. Let H_m be a Cartier divisor on T whose associated line bundle is the pullback of $\mathcal{O}_J(1)$, and let $H := \frac{1}{m} H_m$. Then $-m(K_Y + B_Y) \sim g^* H_m$, so $-(K_Y + B_Y) \sim_{\mathbb{Q}} g^* H$, and H is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor which is ample over Z (cf. [Laz04, Theorem 2.1.27]). This proves (1).

The morphism $Y \to Z$ is projective, and it is surjective since Y is birational to X over Z. Let $Y \to Z' \to Z$ be the Stein factorization of $Y \to Z$. Then $Z' \to Z$ is finite, and the contraction $X \to Z$ is the composition of the dominant rational map $X \dashrightarrow Y \to Z'$ and the finite morphism $Z' \to Z$. By Lemma 2.17(4), applied to the contraction $X \to Z$, $Z' \to Z$ is finite and birational. Since Z is normal, $Z' \to Z$ is an isomorphism, so $Y \to Z$ is a contraction. Since $g_* \mathcal{O}_Y = \mathcal{O}_T$,

the pushforward of $\mathcal{O}_T$ to Z is the pushforward of $\mathcal{O}_Y$ to Z, which is $\mathcal{O}_Z$. As T is projective over Z, $T \to Z$ is a contraction. Since Y is projective over Z, so is g, and g is a contraction.

Step 4. In this step, we prove the last statement.

Let n be a positive integer and $(Y/Z \ni z, D_Y)$ a monotonic n-complement of $(Y/Z \ni z, B_Y)$. Possibly replacing Z with a neighborhood of z, we may assume that (Y, D_Y) is lc, $D_Y \geq B_Y$, and $n(K_Y + D_Y) = \mathrm{div}_Y(\varphi)$ for some rational function φ on Y. By (2), $p_*\big(q^*(K_Y + B_Y) - p^*(K_X + B)\big) \geq 0$. Since $X \to Z$ and $Y \to Z$ are contractions (**Step 3**), Lemma 2.15, applied to $\psi \circ v^{-1} : X \dashrightarrow Y$, the common resolution p, q, the zero divisor on Z, φ, and D_Y, implies that $(X/Z \ni z, D)$ is a monotonic n-complement of $(X/Z \ni z, B)$, where D is the $\mathbb{Q}$-divisor on X such that $K_X + D = p_*q^*(K_Y + D_Y)$. □

6. Galois quotients of canonical bundle formulas

In this section, we prove Theorem 6.4. For a minimal lc center W over a minimal image V of the lc centers (step (3) of the sketch of the proof in Section 1), with Stein factorization $W \to V' \to V$, the morphism $V' \to V$ is the quotient of V' by a finite group, and we descend the canonical bundle formula of $W \to V'$ to V.

Set-up 6.1. Let d, m, $h : X \to V'$, (X, B), ψ, G, and $V \to U$ be as follows.

(1) $d \geq 2$ and $m \geq 1$ are integers.
(2) $h : X \to V'$ is a contraction such that $0 < \dim V' \leq \dim X \leq d$.
(3) (X, B) is an lc pair such that B is a $\mathbb{Q}$-divisor and (X, B) is klt over the generic point of V'.
(4) $K_X + B + \frac{1}{m}\,\mathrm{div}(\psi) = h^*D_{V'}$ is a trivialization of degree m of $K_X + B$ over V', with induced generalized sub-pair $(V', B_{V'}, \mathbf{M})$.
(5) G is a finite group acting faithfully on V', and $\rho : V' \to V := V'/G$ is the quotient morphism (the quotient exists by [Gro71, Exp. V, Proposition 1.8], and it is a normal quasi-projective variety since V' is).
(6) $V \to U$ is a projective morphism to a normal quasi-projective variety U.
(7) For any $\sigma \in G$, there exists a birational map $\phi : X \dashrightarrow X$ which is crepant for (X, B) and satisfies $h \circ \phi = \sigma \circ h$.

Fix a rational 1-canonical form ω_V on V. For any normal variety V_1 projective and birational over V and any normal variety V_1' projective and birational over V', for instance the normalization of V_1 in V' (Definition 2.7), let $K_{V_1} := \mathrm{div}_{V_1}(\omega_V)$ and $K_{V_1'} := \mathrm{div}_{V_1'}(\rho^*\omega_V)$. The moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ in (4), and the $\mathbb{Q}$-divisors B_V and B_{V_1} in Theorem 6.4, are computed with the canonical divisors K_{V_1} and $K_{V_1'}$ above.

Condition (7) of Set-up 6.1 only asks that any element of G lifts to a birational map $X \dashrightarrow X$ which is crepant for (X, B), and it does not ask that the lifts form a group acting on X. Condition (7) holds over a minimal image of lc centers, where the contraction $h : X \to V'$ corresponds to $W \to V'$ (Lemma 7.5 in Section 7).

Theorem 6.2 ([OAI26b, Theorem 6.1])**.** *For integers $r \geq 0$ and $m \geq 1$ there exists a positive integer $\ell(r, m)$ satisfying the following. Let (X, B) be a projective klt pair of dimension r such that $m(K_X + B) \sim 0$. Let ω be a rational m-canonical form on X with $\mathrm{div}(\omega) = -mB$. Then for any birational map $\phi : X \dashrightarrow X$ which is crepant for (X, B), we have $\phi^*\omega = c\omega$ for some $c \in \mathbb{C}$ with $c^{\ell(r,m)} = 1$.*

Remark 6.3. For $r = 0$ we may take $\ell(0, m) = 1$, since then X is a point and $\phi = \mathrm{id}_X$.

Theorem 6.4. *Notation and conditions as in Set-up 6.1. Let*

$$b := m \cdot \mathrm{lcm}\{\ell(j, m) \mid 0 \leq j \leq d-1\},$$
$$a := 2b \cdot \mathrm{lcm}\{q(j, b) \mid 0 \leq j \leq d-1\},$$
$$p := a \cdot \mathrm{lcm}\{\ell(j, a) \mid 0 \leq j \leq d-1\},$$

where $q(\cdot,\cdot)$ and $\ell(\cdot,\cdot)$ are the integers of Theorems 3.5 and 6.2. Then there exist a rational function χ on V', a rational b-canonical form θ on X, a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D_V on V, a $\mathbb{Q}$-divisor $B_V \geq 0$ on V, and a $\boldsymbol{b}$-$\mathbb{Q}$-Cartier $\mathbb{Q}$-$\boldsymbol{b}$-divisor $\mathbf{N}$ over V, satisfying the following.

(1) $\rho^*D_V = D_{V'} + \frac{1}{b}\operatorname{div}(\chi)$, $\operatorname{div}(\theta) = bh^*\rho^*D_V - bB$, and $\phi^*\theta = \theta$ for any birational map $\phi : X \dashrightarrow X$ which is crepant for (X,B) and satisfies $h \circ \phi = \sigma \circ h$ for some $\sigma \in G$.

(2) $\rho^*\mathbf{N} = \mathbf{M} + \frac{1}{b}\operatorname{div}(\chi)$ and $D_V = K_V + B_V + \mathbf{N}_V$. Moreover, for any normal variety V_1 projective and birational over V, the normalization V_1' of V_1 in V', and the $\mathbb{Q}$-divisor B_{V_1} with $K_{V_1} + B_{V_1} + \mathbf{N}_{V_1}$ equal to the pullback of D_V, the pullback of $K_{V_1} + B_{V_1}$ to V_1' is $K_{V_1'} + B_{V_1'}$, where $B_{V_1'}$ is the trace on V_1' of the discriminant $\boldsymbol{b}$-divisor of (X,B) over V'.

(3) $(V, B_V, \mathbf{N})/U$ is generalized lc, and $\mathbf{N}$ is nef over U.

(4) $p\mathbf{N}_{V_2}$ is a Cartier divisor for any smooth variety V_2 projective and birational over V.

We prove Theorem 6.4 at the end of Subsection 6.2.

6.1. **Invariant pluricanonical forms.** In this subsection, we construct a pluricanonical form on X which is fixed by all lifts of elements of G, and we descend invariant divisors and $\boldsymbol{b}$-divisors along ρ.

The lifts of the identity of V' act on the pluricanonical forms of the general fibers of h, which are klt log Calabi–Yau pairs, by characters of crepant birational self-maps (cf. [OAI26b, Lemma 2.1]), and Theorem 6.2 bounds the orders of such characters in terms of the dimension and the degree. Thus, in the degree b of Lemma 6.5, which depends only on d and m, the action of the lifts on rational b-canonical forms factors through G, and Hilbert's Theorem 90 yields an invariant form (see part (a) of the sketch of the proof in Section 1).

Lemma 6.5. *Notation and conditions as in Set-up 6.1. Let $b := m \cdot \operatorname{lcm}\{\ell(j,m) \mid 0 \leq j \leq d-1\}$. Then there exist a rational function χ on V' and a rational b-canonical form θ on X such that*

$$\operatorname{div}(\theta) = bh^*D_{V'} + h^*\operatorname{div}(\chi) - bB$$

and $\phi^*\theta = \theta$ for any birational map $\phi : X \dashrightarrow X$ which is crepant for (X,B) and satisfies $h \circ \phi = \sigma \circ h$ for some $\sigma \in G$.

Proof. The proof is long, so we divide it into several steps.

Let $s := b/m$ and $r := \dim X - \dim V'$. Since $\dim V' > 0$, we have $0 \leq r \leq d-1$, so $\ell(r,m)$ divides s. For $\sigma \in G$, we call a birational map $\phi : X \dashrightarrow X$ a *lift* of σ if ϕ is crepant for (X,B) and $h \circ \phi = \sigma \circ h$. Let ω_X be a rational 1-canonical form on X with $\operatorname{div}(\omega_X) = K_X$, and let $\Theta := \psi\omega_X^{\otimes m}$. By Set-up 6.1(4),

$$\operatorname{div}(\Theta) = \operatorname{div}(\psi) + mK_X = mh^*D_{V'} - mB.$$

In particular, $K_X + B$ is $\mathbb{Q}$-Cartier.

Step 1. In this step we show that for any $\sigma \in G$ and any lift ϕ of σ, there exists a non-zero rational function c_ϕ on V' such that

$$\phi^*\Theta = h^*c_\phi \cdot \Theta \quad \text{and} \quad \operatorname{div}(c_\phi) = m\sigma^*D_{V'} - mD_{V'}.$$

Since ϕ is crepant for (X,B), there exist a normal variety Y and projective birational morphisms $p : Y \to X$ and $q : Y \to X$ such that $q = \phi \circ p$ and

$$p^*(K_X + B) - \operatorname{div}(p^*\omega_X) = q^*(K_X + B) - \operatorname{div}(q^*\omega_X)$$

(Definition 2.4). Since $\operatorname{div}(\psi) = mh^*D_{V'} - m(K_X + B)$, we have

$$\operatorname{div}(p^*\Theta) = mp^*h^*D_{V'} - m\big(p^*(K_X + B) - \operatorname{div}(p^*\omega_X)\big),$$

and

$$\operatorname{div}(q^*\Theta) = mq^*h^*D_{V'} - m\big(q^*(K_X + B) - \operatorname{div}(q^*\omega_X)\big).$$

Since $q^*\Theta = p^*\phi^*\Theta$ and $h \circ q = \sigma \circ h \circ p$, we obtain

$$\operatorname{div}\left(p^*\frac{\phi^*\Theta}{\Theta}\right) = \operatorname{div}(q^*\Theta) - \operatorname{div}(p^*\Theta) = (h \circ p)^*(m\sigma^*D_{V'} - mD_{V'}).$$

The morphism $h \circ p : Y \to V'$ is a contraction and $m\sigma^* D_{V'} - mD_{V'}$ is $\mathbb{Q}$-Cartier. By Lemma 2.17(2), there exists a non-zero rational function c_ϕ on V' such that $p^*(\phi^*\Theta/\Theta) = (h\circ p)^* c_\phi$ and $\operatorname{div}(c_\phi) = m\sigma^* D_{V'} - mD_{V'}$. Thus $\phi^*\Theta = h^* c_\phi \cdot \Theta$.

Step 2. In this step we show that $c_\phi^s = 1$ for any lift ϕ of $\mathrm{id}_{V'}$.

By **Step 1**, $\operatorname{div}(c_\phi) = 0$. Since V' is normal, c_ϕ is a regular function on V' without zeros. If $r = 0$, then h is birational, so $h \circ \phi = h$ implies $\phi = \mathrm{id}_X$, and $c_\phi = 1$. Thus we may assume that $r > 0$.

Let $\omega_{V'} := \rho^* \omega_V$, where ω_V is the form fixed in Set-up 6.1. Let $X^\circ \subset h^{-1}(V'_{\mathrm{reg}})$ be the open subset on which h is smooth, which is non-empty by generic smoothness, and $\omega_{X^\circ/V'}$ the top exterior power of $\Omega_{X^\circ/V'}$. By the cotangent sequence of the smooth morphism $X^\circ \to V'_{\mathrm{reg}}$,

$$\omega_{X^\circ} = \omega_{X^\circ/V'} \otimes h^* \omega_{V'_{\mathrm{reg}}}.$$

For a rational j-canonical form Ξ on X, the quotient $\Xi/h^*\omega_{V'}^{\otimes j}$ is the rational section of $\omega_{X^\circ/V'}^{\otimes j}$ whose wedge product with $h^*\omega_{V'}^{\otimes j}$ is Ξ.

Let p and q be the morphisms of **Step 1** for ϕ. There exists a non-empty open subset $V'^\circ \subset V'$ satisfying the following: for any closed point $t \in V'^\circ$, the fiber $X_t := h^{-1}(t)$ is a normal projective variety of dimension r, the pair (X_t, B_t) with $B_t := B|_{X_t}$ is klt, ϕ restricts to a birational map $\phi_t : X_t \dashrightarrow X_t$ which is crepant for (X_t, B_t), and the quotient $\Theta/h^*\omega_{V'}^{\otimes m}$ restricts to a rational m-canonical form Θ_t on X_t with $\operatorname{div}(\Theta_t) = -mB_t$. The open subset V'° exists since (X, B) is klt over the generic point of V' and since

$$\operatorname{div}(\Theta) + mB - mh^* \operatorname{div}(\omega_{V'})$$

is the pullback of a divisor on V'. Over V'°, p and q restrict to projective birational morphisms $p_t, q_t : Y_t \to X_t$ with $q_t = \phi_t \circ p_t$, where Y_t, the fiber of $h \circ p$ over t, is a normal variety. Since $h \circ q = h \circ p$, we may divide $p^*\omega_X$ and $q^*\omega_X$ by $(h \circ p)^*\omega_{V'}$ and restrict them to Y_t. Then the equality of Definition 2.4 for ϕ restricts to the equality for ϕ_t and the restriction ω_t of $\omega_X/h^*\omega_{V'}$ to X_t, hence ϕ_t is crepant for (X_t, B_t). The remaining properties follow from generic smoothness [Har77, Corollary III.10.7], applied to a log resolution of (X, B) and to the restriction of $h \circ p$ to the regular locus of Y, from Lemma 2.17(4), applied to h and to $h \circ p$, from generic flatness [Gro65, Théorème 6.9.1], and from [Har77, Corollary III.9.6]. For example, by generic flatness and [Har77, Corollary III.9.6], the closed subsets of X over which p or q is not an isomorphism intersect X_t in dimension less than r, so p_t and q_t are birational. We omit the verification of the other properties.

Let $t \in V'^\circ$ be a closed point. Since $h \circ \phi = h$, we have $\phi^* h^* \omega_{V'} = h^* \omega_{V'}$, so

$$\phi_t^* \Theta_t = c_\phi(t) \Theta_t.$$

Since $\operatorname{div}(\Theta_t) = -mB_t$, we have $m(K_{X_t} + B_t) \sim 0$. By Theorem 6.2, $c_\phi(t)^{\ell(r,m)} = 1$. Since closed points of V'° are dense in V', $c_\phi^{\ell(r,m)} = 1$. Since $\ell(r, m)$ divides s, $c_\phi^s = 1$.

Step 3. In this step we construct χ and θ.

Let $\sigma \in G$, and let ϕ_1 and ϕ_2 be two lifts of σ. Since compositions and inverses of birational maps which are crepant for (X, B) are crepant for (X, B), which one sees by passing to a common resolution (cf. [KM98, Lemma 2.30]), $\phi_0 := \phi_1 \circ \phi_2^{-1}$ is a lift of $\mathrm{id}_{V'}$. Since $\phi_1 = \phi_0 \circ \phi_2$ and $h \circ \phi_2 = \sigma \circ h$,

$$h^* c_{\phi_1} \cdot \Theta = \phi_1^* \Theta = \phi_2^*(h^* c_{\phi_0} \cdot \Theta) = h^*(\sigma^* c_{\phi_0} \cdot c_{\phi_2}) \cdot \Theta.$$

Thus $c_{\phi_1} = \sigma^* c_{\phi_0} \cdot c_{\phi_2}$, and **Step 2** implies $c_{\phi_1}^s = c_{\phi_2}^s$. By Set-up 6.1(7), any $\sigma \in G$ has a lift ϕ, and we let $c_\sigma := c_\phi^s$, which depends only on σ.

Let $\sigma, \tau \in G$, and let ϕ and ϕ' be lifts of σ and τ respectively. Then $h \circ \phi \circ \phi' = \sigma \circ \tau \circ h$, so $\phi \circ \phi'$ is a lift of $\sigma\tau$, and

$$(\phi \circ \phi')^* \Theta = \phi'^*(h^* c_\phi \cdot \Theta) = h^*(\tau^* c_\phi \cdot c_{\phi'}) \cdot \Theta.$$

Taking s-th powers, we obtain

$$c_{\sigma\tau} = \tau^* c_\sigma \cdot c_\tau \quad \text{for any } \sigma, \tau \in G.$$

Since $V = V'/G$, a rational function on V' is G-invariant if and only if it is the pullback of a rational function on V. Since G acts faithfully on the separated variety V', the identity is the only element $\sigma \in G$ such that $\sigma^*\varphi = \varphi$ for any rational function φ on V'. Thus the function field of V' is a Galois extension of the function field of V with group G, acting by pullback. Pullback is a right action, and $\sigma \to (\sigma^{-1})^*$ is a left action of G on the non-zero rational functions on V'. For $a_\sigma := (\sigma^{-1})^* c_\sigma$, the identity $c_{\sigma\tau} = \tau^* c_\sigma \cdot c_\tau$ implies $a_{\sigma\tau} = a_\sigma \cdot (\sigma^{-1})^* a_\tau$, so $\sigma \to a_\sigma$ is a 1-cocycle for the left action. By Hilbert's Theorem 90 [Ser79, Ch. X, §1, Proposition 2], there exists a non-zero rational function χ on V' such that $a_\sigma = (\sigma^{-1})^*\chi/\chi$ for any $\sigma \in G$. Applying σ^*, we obtain

$$c_\sigma = \frac{\chi}{\sigma^*\chi} \quad \text{for any } \sigma \in G.$$

Let $\theta := h^*\chi \cdot \Theta^{\otimes s}$. Then θ is a rational b-canonical form on X, and

$$\operatorname{div}(\theta) = h^* \operatorname{div}(\chi) + s \operatorname{div}(\Theta) = bh^* D_{V'} + h^* \operatorname{div}(\chi) - bB.$$

Let ϕ be a lift of $\sigma \in G$. By **Step 1**,

$$\phi^*\theta = (h \circ \phi)^*\chi \cdot h^* c_\phi^s \cdot \Theta^{\otimes s} = h^*(\sigma^*\chi \cdot c_\sigma) \cdot \Theta^{\otimes s} = h^*\chi \cdot \Theta^{\otimes s} = \theta. \quad \square$$

Lemma 6.6. *Let V' be a normal variety, G a finite group acting faithfully on V', $V := V'/G$ the quotient, $\rho : V' \to V$ the quotient morphism, and $V \to U$ a projective morphism to a normal quasi-projective variety U.*

(1) For any G-invariant $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D' on V', there exists a unique $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on V such that $\rho^*D = D'$. If D' is nef over U, then D is nef over U.

(2) For any G-invariant $\boldsymbol{b}$-$\mathbb{Q}$-Cartier $\mathbb{Q}$-$\boldsymbol{b}$-divisor $\mathbf{M}'$ over V', there exists a $\boldsymbol{b}$-$\mathbb{Q}$-Cartier $\mathbb{Q}$-$\boldsymbol{b}$-divisor $\mathbf{N}$ over V such that $\rho^*\mathbf{N} = \mathbf{M}'$. If $\mathbf{M}'$ is nef over U, then $\mathbf{N}$ is nef over U.

Proof. Since $V \to U$ is projective and U is quasi-projective, V is quasi-projective. Since ρ is finite, V' is quasi-projective. The variety V is normal since $\mathcal{O}_V = (\rho_*\mathcal{O}_{V'})^G$. Thus V and V' are normal quasi-projective varieties, as in Definition 2.7. As in **Step 3** of the proof of Lemma 6.5, a G-invariant rational function on V' is the pullback of a rational function on V, and the function field of V' is a Galois extension of the function field of V with group G. For a non-zero rational function φ on V', the rational function $\prod_{\sigma\in G} \sigma^*\varphi$ is G-invariant, and we let $\operatorname{Nm}(\varphi)$ be the non-zero rational function on V such that

$$\rho^* \operatorname{Nm}(\varphi) = \prod_{\sigma \in G} \sigma^*\varphi.$$

Then $\operatorname{Nm}(\varphi)$ is the norm of φ for the finite extension of function fields given by ρ.

(1) Let $D := \frac{1}{|G|}\rho_* D'$, and let k be a positive integer such that kD' is Cartier. Let $v \in V$ be a closed point. Since V' is quasi-projective, kD' is principal on an open neighborhood of the finite set $\rho^{-1}(v)$: a finite subset of a quasi-projective variety is contained in an affine open subset, and an invertible module over a semilocal ring is free. Since ρ is finite, hence closed, there exist an open neighborhood W of v and a non-zero rational function φ on V' such that $kD'|_{\rho^{-1}(W)} = \operatorname{div}(\varphi)$. By [Ful98, Proposition 1.4(b)] applied to the finite surjective morphism $\rho^{-1}(W) \to W$,

$$k|G|D|_W = \rho_* \operatorname{div}(\varphi) = \operatorname{div}(\operatorname{Nm}(\varphi)).$$

Thus $k|G|D$ is Cartier, so D is $\mathbb{Q}$-Cartier. Moreover, since $\rho^{-1}(W)$ is G-stable and D' is G-invariant,

$$k|G|\rho^*D|_{\rho^{-1}(W)} = \operatorname{div}(\rho^* \operatorname{Nm}(\varphi)) = \sum_{\sigma\in G} \operatorname{div}(\sigma^*\varphi) = \sum_{\sigma\in G} \sigma^*(kD')|_{\rho^{-1}(W)} = k|G|D'|_{\rho^{-1}(W)}.$$

Since v is arbitrary, $\rho^*D = D'$.

For uniqueness, let D_0 be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on V such that $\rho^* D_0 = 0$. For any prime divisor F on V, there exists a prime divisor P on V' with $\rho(P) = F$, and $\operatorname{mult}_P D_0 = e \operatorname{mult}_F D_0$, where $e \geq 1$ is the ramification index of ρ along P. Thus $D_0 = 0$.

Assume that D' is nef over U. Let $C \subset V$ be a curve which is contracted over U. Since ρ is finite and surjective, there exists an irreducible component C' of $\rho^{-1}(C)$ such that $\rho(C') = C$. Then C' is contracted over U. By the projection formula [Ful98, Proposition 2.5(c)] applied to the Cartier divisor $k|G|D$,

$$(D' \cdot C') = (\rho^* D \cdot C') = \deg(C' \to C)(D \cdot C).$$

Since $(D' \cdot C') \geq 0$ and $\deg(C' \to C) > 0$, we have $(D \cdot C) \geq 0$. Thus D is nef over U (cf. [Laz04, Example 1.4.4]).

(2) Since $V' \to U$ is projective, by Definition 2.7 there exists a normal variety P'' with a projective birational morphism $g : P'' \to V'$ such that $\mathbf{M}'$ descends to P'', and if $\mathbf{M}'$ is nef over U, we may moreover assume that $\mathbf{M}'_{P''}$ is nef over U. For $\sigma \in G$, let $g_\sigma := \sigma \circ g : P'' \to V'$. Let P' be the normalization of the unique irreducible component of the fiber product of the morphisms g_σ, $\sigma \in G$, over V' which dominates V'. Then $P' \to V'$ is projective and birational, and P' dominates P'' via the factor $g_{\mathrm{id}_{V'}} = g$. For $\tau \in G$, the isomorphism of the fiber product sending the factor indexed by σ to the factor indexed by $\tau\sigma$ covers the automorphism τ of V', since $\tau \circ g_\sigma = g_{\tau\sigma}$. Thus G acts on P', and $P' \to V'$ is G-equivariant. Since $\mathbf{M}'$ descends to P'', $\mathbf{M}'$ descends to P', and the trace $\mathbf{M}'_{P'}$ is the pullback of $\mathbf{M}'_{P''}$. Since $\mathbf{M}'$ is G-invariant, $\sigma^* \mathbf{M}'_{P'} = \mathbf{M}'_{P'}$ for any $\sigma \in G$ (Definition 2.7). If $\mathbf{M}'_{P''}$ is nef over U, then $\mathbf{M}'_{P'}$ is nef over U by the projection formula.

The group G acts on P' by automorphisms over V, and P' is projective over V. Since P' is quasi-projective, the quotient P'/G exists [Gro71, Exp. V, Proposition 1.8], and it is proper over V, since $P' \to V$ is proper and $P' \to P'/G$ is surjective. The morphism $P' \to V'$ is an isomorphism over the complement of a proper closed subset $Z' \subset V'$. Let $V^0 := V \backslash \rho(Z')$, which is a non-empty open subset of V since ρ is finite. Then $P' \to V'$ is a G-equivariant isomorphism over $\rho^{-1}(V^0)$. Since the structure sheaves of the quotients P'/G and V are the G-invariant parts of the pushforwards of the structure sheaves of P' and V', the morphism $P'/G \to V$ is an isomorphism over V^0, so P'/G is birational over V.

By Chow's lemma [Har77, Exercise II.4.10] followed by normalization, there exists a normal variety V'' projective and birational over V which dominates P'/G. Let P'_1 be the normalization of V'' in V' (Definition 2.7). Then P'_1 is projective and birational over V', P'_1 dominates P', $\mathbf{M}'$ descends to P'_1 with trace the pullback of $\mathbf{M}'_{P'}$, and G acts on P'_1 with quotient V''. Here G acts on P'_1 through the action of G on the factor V' of $V'' \times_V V'$, which preserves the unique irreducible component dominating V'. Since P' is normal, finite over P'/G, and birational over V', P' is the normalization of P'/G in V', and since V'' dominates P'/G, P'_1 dominates P' by the universal property of normalization. The quotient P'_1/G is finite over V'', and birational over V'' as for P'/G above, so $P'_1/G = V''$ since V'' is normal. The trace of $\mathbf{M}'$ on P'_1 is G-invariant, and nef over U if $\mathbf{M}'_{P'}$ is, as for P' above.

Replacing P' with P'_1, we may assume that $V'' = P'/G$. The group G acts faithfully on P', since $P' \to V'$ is G-equivariant and birational and G acts faithfully on V'. Let $\rho'' : P' \to V''$ be the quotient morphism. Applying (1) to ρ'', the G-invariant $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\mathbf{M}'_{P'}$, and the projective morphism $V'' \to U$, we obtain a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor N'' on V'' such that $\rho''^* N'' = \mathbf{M}'_{P'}$, and N'' is nef over U if $\mathbf{M}'_{P'}$ is nef over U. Let $\mathbf{N}$ be the $\mathbb{Q}$-$\boldsymbol{b}$-divisor over V which descends to V'' with trace $\mathbf{N}_{V''} = N''$. Then $\mathbf{N}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, and $\mathbf{N}$ is nef over U if $\mathbf{M}'$ is nef over U.

The variety P' is normal, finite over V'', and birational over V'. Thus P' is the normalization of V'' in V'. By Definition 2.7, $\rho^* \mathbf{N}$ descends to P' with trace $\rho''^* N'' = \mathbf{M}'_{P'}$. Since $\mathbf{M}'$ also descends to P' with trace $\mathbf{M}'_{P'}$, we have $\rho^* \mathbf{N} = \mathbf{M}'$. □

6.2. **Descent to the quotient.** In this subsection, we prove Theorem 6.4. The main input is Lemma 6.9, which controls the weight of an invariant form at a fixed point of an automorphism of

a curve, uniformly in the order of the automorphism. We only use the projectivity of morphisms, and we do not assume that X, V', or V is a projective variety.

We first compute the value group and the residue field of a valuation whose values on a regular system of parameters are linearly independent over $\mathbb{Q}$.

Lemma 6.7. *Let $(R, \mathfrak{m})$ be a regular local ring with a regular system of parameters $s_1, \dots, s_k$, and μ a valuation of the fraction field of R with values in $\mathbb{R}$ such that $\mu \geq 0$ on R, $\mu > 0$ on $\mathfrak{m}$, and $\mu(s_1), \dots, \mu(s_k)$ are linearly independent over $\mathbb{Q}$. Then the value group of μ is $\bigoplus_{i=1}^k \mathbb{Z}\mu(s_i)$, and the natural map from the residue field of R to the residue field of μ is an isomorphism.*

Proof. For $I = (I_1, \dots, I_k) \in \mathbb{Z}^k_{\geq 0}$, let $s^I := s_1^{I_1} \cdots s_k^{I_k}$ and $|I| := I_1 + \cdots + I_k$. Since $\mu(s_1), \dots, \mu(s_k)$ are linearly independent over $\mathbb{Q}$, $\mu(s^I) \neq \mu(s^J)$ for $I \neq J$. Let $f \in R \backslash \{0\}$, and l a positive integer such that $l \min_i \mu(s_i) > \mu(f)$. Since $\mathfrak{m}^j$ is generated by the monomials s^I with $|I| = j$ for any $j \geq 0$, there exist elements $\lambda_I \in R$ for $|I| < l$, each a unit of R or zero, such that $f - \sum_{|I|<l} \lambda_I s^I \in \mathfrak{m}^l$. Units of R have value 0, and elements of $\mathfrak{m}^l$ have value at least $l \min_i \mu(s_i) > \mu(f)$. Thus

$$\mu(f) = \mu\Big(\sum_{|I|<l} \lambda_I s^I \Big) = \min\{\mu(s^I) \mid \lambda_I \neq 0\},$$

and the minimum is attained at exactly one index $I(f)$. Hence $\mu(f) \in \bigoplus_i \mathbb{Z}\mu(s_i)$, which proves the statement on the value group. Let $f, h \in R \backslash \{0\}$ with $\mu(f) = \mu(h)$. Then $I(f) = I(h) =: I$. Let λ_I and λ'_I be the coefficients of s^I in the expansions of f and h above. In the expansion of $\lambda'_I f - \lambda_I h$, the coefficient of s^I is zero, and any monomial s^J with $J \neq I$ and non-zero coefficient satisfies $\mu(s^J) > \mu(f)$. Thus $\mu(\lambda'_I f - \lambda_I h) > \mu(\lambda'_I f)$, and the residue of f/h is the residue of $\lambda_I / \lambda'_I \in R$. This proves the statement on the residue fields. □

The following lemma is a local version of [OAI26b, Lemma 7.2].

Lemma 6.8. *Let $f : N \to C$ be a projective contraction from a normal variety N to a smooth curve C, $c \in C$ a closed point, and σ an automorphism of C such that $\sigma(c) = c$. Let $\Delta \geq 0$ be a $\mathbb{Q}$-divisor on N, a a positive even integer, and η a rational a-canonical form on N. Assume that over a neighborhood of c, (N, Δ) is dlt and not klt, any lc center of (N, Δ) is contained in $f^{-1}(c)$, and $\operatorname{div}(\eta) = -a\Delta$. Let $\gamma : N \dashrightarrow N$ be a birational map such that $f \circ \gamma = \sigma \circ f$ and $\gamma^* \eta = f^* v \cdot \eta$ for a rational function v on C which is regular and non-zero at c. Then there exist minimal lc centers Z and Z' of (N, Δ) contained in $f^{-1}(c)$ and a birational map $\gamma_Z : Z \dashrightarrow Z'$ which is crepant for (Z, Δ_Z) and $(Z', \Delta_{Z'})$, such that $\gamma_Z^* \eta_{Z'} = v(c) \eta_Z$, where Δ_Z and $\Delta_{Z'}$ are given by dlt adjunction (Lemma 2.2), and η_Z and $\eta_{Z'}$ are the Poincaré residues of η on Z and Z' (see [Kol16, Definition 13]).*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we shrink C and resolve γ.

Let $C^1 \subset C$ be an open neighborhood of c such that the assumptions of the lemma hold over C^1, and v is regular and nowhere zero on C^1. Let $C^0 := C^1 \cap \sigma^{-1}(C^1)$, $N^0 := f^{-1}(C^0)$, and $N'^0 := f^{-1}(\sigma(C^0))$. Since $f \circ \gamma = \sigma \circ f$, γ restricts to a birational map $N^0 \dashrightarrow N'^0$. Since $\sigma(c) = c$, both N^0 and N'^0 are open subsets of $f^{-1}(C^1)$ containing $f^{-1}(c)$. Since any lc center of $(f^{-1}(C^1), \Delta)$ is contained in $f^{-1}(c)$, the lc centers of (N^0, Δ) and of (N'^0, Δ) are exactly the lc centers of $(f^{-1}(C^1), \Delta)$. Let $n := \dim N$, and k the largest codimension of an lc center of $(f^{-1}(C^1), \Delta)$. Since $(f^{-1}(C^1), \Delta)$ is not klt, $k \geq 1$.

Let $p : Y \to N^0$ be a log resolution of (N^0, Δ) such that $q := \gamma \circ p : Y \to N'^0$ is a morphism and

$$\Sigma := \operatorname{Exc}(p) \cup p^{-1}(\operatorname{Supp} \Delta) \cup \operatorname{Exc}(q) \cup q^{-1}(\operatorname{Supp} \Delta)$$

is an snc divisor [KM98, Theorem 0.2]. Since $f \circ q = \sigma \circ f \circ p$ and $f \circ p : Y \to C^0$ is proper, q is proper. Let $\Delta_Y := -\frac{1}{a} \operatorname{div}(p^* \eta)$. Since $\operatorname{div}(\eta) = -a\Delta$ on N^0, we have $K_Y + \Delta_Y = p^*(K_N + \Delta)$

over N^0. Since $(f\circ p)^*v$ is regular and nowhere zero on Y, and

$$q^*\eta = p^*\gamma^*\eta = (f\circ p)^*v\cdot p^*\eta, \tag{6.1}$$

we have $\operatorname{div}(q^*\eta) = -a\Delta_Y$. Since $\operatorname{div}(\eta) = -a\Delta$ on N'^0, we have $K_Y+\Delta_Y = q^*(K_N+\Delta)$ over N'^0. Thus $a(F,N^0,\Delta) = a(F,Y,\Delta_Y) = a(F,N'^0,\Delta)$ for any prime divisor F over Y.

Step 2. In this step we construct a valuation μ on the non-zero rational functions on N and a subvariety $S\subset Y$.

Let Z be an lc center of (N^0,Δ) with $\operatorname{codim}_N Z = k$, and ξ_Z the generic point of Z. Any lc center properly contained in Z has codimension larger than k, so Z is a minimal lc center, and $Z\subset f^{-1}(c)$. By [KM98, Definition 2.37] and [Kol13, Theorem 4.16(1)], (N^0,Δ) is snc at ξ_Z, and Z is a component of $D_1\cap\cdots\cap D_k$ for components $D_1,\dots,D_k$ of Δ with coefficient 1. Since $\operatorname{Supp}\Delta$ is snc at ξ_Z and $\operatorname{codim}_N Z = k$, no other component of $\operatorname{Supp}\Delta$ contains Z. Let $R := \mathcal{O}_{N,\xi_Z}$, $x_i\in R$ a local equation of D_i for $1\le i\le k$, and $y_1,\dots,y_{n-k}\in R$ elements whose restrictions to Z define a dominant rational map $Z\dashrightarrow \mathbb{A}^{n-k}$. Then $dx_1,\dots,dx_k,dy_1,\dots,dy_{n-k}$ form a basis of the module of differentials Ω_R of R over $\mathbb{C}$, by the conormal sequence

$$0\to \mathfrak{m}/\mathfrak{m}^2\to \Omega_R\otimes\kappa\to\Omega_\kappa\to 0,$$

where $\mathfrak{m}$ and κ are the maximal ideal and the residue field of R. The conormal sequence is exact since $\kappa/\mathbb{C}$ is separable [Mat89, Theorems 25.2 and 26.9]. Since $\operatorname{div}(\eta) = -a(D_1+\cdots+D_k)$ near ξ_Z, there exists a unit g of R such that

$$\eta = g\Big(\frac{dx_1}{x_1}\wedge\cdots\wedge\frac{dx_k}{x_k}\wedge dy_1\wedge\cdots\wedge dy_{n-k}\Big)^{\otimes a},$$

and

$$\eta_Z = g|_Z(dy_1|_Z\wedge\cdots\wedge dy_{n-k}|_Z)^{\otimes a}$$

by [Kol16, Definition 13].

Let $\alpha_1,\dots,\alpha_k$ be positive real numbers which are linearly independent over $\mathbb{Q}$. By [Mat89, Theorems 28.3 and 29.7], the completion of R is isomorphic to $\kappa[[x_1,\dots,x_k]]$, where κ is the residue field of R. Let μ be the restriction to the non-zero elements of the fraction field of R, that is, to the non-zero rational functions on N, of the monomial valuation of the fraction field of $\kappa[[x_1,\dots,x_k]]$ with $\mu(x_i)=\alpha_i$ for $1\le i\le k$. Then $\mu\ge 0$ on R and $\mu>0$ on the maximal ideal of R. By Lemma 6.7, the value group of μ is $\bigoplus_i\mathbb{Z}\alpha_i$, and the natural map $\kappa(\xi_Z)\to\kappa(\mu)$ of residue fields is an isomorphism. Since p and q are birational, μ is also a valuation on the non-zero rational functions on Y and on those on N'^0.

Since p is proper, μ has a center ξ on Y with $p(\xi)=\xi_Z$ [Har77, Theorem II.4.7]. Let S be the closure of ξ in Y. The natural maps $\kappa(\xi_Z)\to\kappa(\xi)\to\kappa(\mu)$ compose to an isomorphism. Thus $\kappa(\xi)=\kappa(\xi_Z)$, that is, $p|_S: S\to Z$ is birational, and $\operatorname{codim}_Y S = k$.

Let $Y_1,\dots,Y_s$ be the components of Σ containing S, and $w_l\in\mathcal{O}_{Y,\xi}$ a local equation of Y_l for $1\le l\le s$. Since Σ is snc and $\operatorname{codim}_Y S = k$, $s\le k$. For $1\le i\le k$, since $\operatorname{div}(x_i) = D_i$ near ξ_Z, any prime divisor on Y containing S and contained in $\operatorname{Supp}\operatorname{div}(p^*x_i)$ is p-exceptional or the strict transform of D_i, and hence is one of $Y_1,\dots,Y_s$. Thus

$$p^*x_i = u_i\prod_{l=1}^{s} w_l^{m_{il}}$$

for a unit u_i of $\mathcal{O}_{Y,\xi}$ and integers $m_{il}\ge 0$. Let $\beta_l := \mu(w_l)>0$. Then $\alpha_i = \sum_l m_{il}\beta_l$ for $1\le i\le k$. Since $\alpha_1,\dots,\alpha_k$ are linearly independent over $\mathbb{Q}$, $s\ge k$. Thus $s=k$, $w_1,\dots,w_k$ is a regular system of parameters of $\mathcal{O}_{Y,\xi}$, $\det(m_{il})\ne 0$, and $\beta_1,\dots,\beta_k$ are linearly independent over $\mathbb{Q}$. By Lemma 6.7 applied to $\mathcal{O}_{Y,\xi}$, the value group of μ is also $\bigoplus_l\mathbb{Z}\beta_l$. Since $\alpha_i=\sum_l m_{il}\beta_l$ and $\bigoplus_i\mathbb{Z}\alpha_i = \bigoplus_l\mathbb{Z}\beta_l$, we have $\det(m_{il})=\pm 1$.

Step 3. In this step we compute $p^*\eta$ near ξ.

Let $t_j := p^*y_j$ for $1\le j\le n-k$. Since $p|_S$ is birational, the restrictions of $t_1,\dots,t_{n-k}$ to S define a dominant rational map $S\dashrightarrow\mathbb{A}^{n-k}$. Since $w_1,\dots,w_k$ is a regular system of parameters

of $\mathcal{O}_{Y,\xi}$, by the conormal sequence as in **Step 2**, the forms $dw_1,\dots,dw_k,dt_1,\dots,dt_{n-k}$ form a basis of the module of differentials of $\mathcal{O}_{Y,\xi}$ over $\mathbb{C}$. Let

$$\omega_Y := \frac{dw_1}{w_1}\wedge\cdots\wedge\frac{dw_k}{w_k}\wedge dt_1\wedge\cdots\wedge dt_{n-k},$$

and write

$$du_i = \sum_l \lambda_{il}dw_l + \sum_j \nu_{ij}dt_j$$

with $\lambda_{il},\nu_{ij}\in\mathcal{O}_{Y,\xi}$. Since

$$p^*\frac{dx_i}{x_i} = \frac{du_i}{u_i} + \sum_l m_{il}\frac{dw_l}{w_l}$$

and $p^*dy_j = dt_j$,

$$p^*\Big(\frac{dx_1}{x_1}\wedge\cdots\wedge\frac{dx_k}{x_k}\wedge dy_1\wedge\cdots\wedge dy_{n-k}\Big) = \delta\omega_Y,\qquad \delta := \det\Big(m_{il}+\frac{\lambda_{il}w_l}{u_i}\Big)_{1\le i,l\le k}.$$

Thus $\delta\in\mathcal{O}_{Y,\xi}$, $\delta|_S = \det(m_{il}) = \pm 1$, and

$$p^*\eta = (g\circ p)\delta^a\omega_Y^{\otimes a}, \tag{6.2}$$

where $(g\circ p)\delta^a$ is a unit of $\mathcal{O}_{Y,\xi}$. Since $\Delta_Y = -\frac{1}{a}\operatorname{div}(p^*\eta)$, we have $\Delta_Y = Y_1+\cdots+Y_k$ near ξ.

Step 4. In this step we construct Z' and compute $q^*\eta$ near ξ.

Let $Y^\circ\subset Y$ be an open neighborhood of ξ such that $S\cap Y^\circ$ is smooth and $\Delta_Y|_{Y^\circ} = (Y_1+\cdots+Y_k)|_{Y^\circ}$. Let F be the exceptional divisor of the blow-up of Y° along $S\cap Y^\circ$. By **Step 3** and [KM98, Lemma 2.29], $a(F,Y,\Delta_Y) = k-k = 0$. By **Step 1**, $a(F,N'^0,\Delta) = 0$, so $Z' := q(S)$ is an lc center of (N'^0,Δ). By **Step 1**, $Z'\subset f^{-1}(c)$ and $\operatorname{codim}_N Z'\le k$. Since $\dim Z'\le\dim S = n-k$, we have $\operatorname{codim}_N Z' = k$. As in the case of Z, Z' is a minimal lc center.

Let $\xi_{Z'} = q(\xi)$ be the generic point of Z', and $R' := \mathcal{O}_{N,\xi_{Z'}}$. As in **Step 2**, there exist components $D'_1,\dots,D'_k$ of Δ with coefficient 1 containing Z', local equations $x'_i\in R'$ of D'_i, elements $y'_1,\dots,y'_{n-k}\in R'$ whose restrictions to Z' define a dominant rational map $Z'\dashrightarrow\mathbb{A}^{n-k}$, and a unit g' of R' such that

$$\eta = g'\Big(\frac{dx'_1}{x'_1}\wedge\cdots\wedge\frac{dx'_k}{x'_k}\wedge dy'_1\wedge\cdots\wedge dy'_{n-k}\Big)^{\otimes a},\qquad \eta_{Z'} = g'|_{Z'}(dy'_1|_{Z'}\wedge\cdots\wedge dy'_{n-k}|_{Z'})^{\otimes a}.$$

Since the components of Σ containing S are $Y_1,\dots,Y_k$, the argument of **Step 2** with q in place of p implies that

$$q^*x'_i = u'_i\prod_{l=1}^k w_l^{m'_{il}}$$

for a unit u'_i of $\mathcal{O}_{Y,\xi}$ and integers $m'_{il}\ge 0$. Write

$$du'_i = \sum_l \lambda'_{il}dw_l + \sum_{j'}\nu'_{ij'}dt_{j'},\qquad d(q^*y'_j) = \sum_l e'_{jl}dw_l + \sum_{j'}e_{jj'}dt_{j'}$$

with coefficients in $\mathcal{O}_{Y,\xi}$. As in **Step 3**,

$$q^*\Big(\frac{dx'_1}{x'_1}\wedge\cdots\wedge\frac{dx'_k}{x'_k}\wedge dy'_1\wedge\cdots\wedge dy'_{n-k}\Big) = \delta'\omega_Y,$$

where

$$\delta' := \det\begin{pmatrix}\big(m'_{il}+\lambda'_{il}w_l/u'_i\big)_{i,l} & \big(\nu'_{ij'}/u'_i\big)_{i,j'}\\ \big(e'_{jl}w_l\big)_{j,l} & \big(e_{jj'}\big)_{j,j'}\end{pmatrix},$$

and $\delta'|_S = \det(m'_{il})\det(e_{jj'}|_S)$. By (6.1) and (6.2),

$$(g'\circ q)\delta'^a = (f\circ p)^*v\cdot(g\circ p)\delta^a, \tag{6.3}$$

which is a unit of $\mathcal{O}_{Y,\xi}$. Thus $\delta'|_S \neq 0$, and hence $\det(m'_{il}) \neq 0$. Since $\mu(x'_i) = \sum_l m'_{il}\beta_l$ and $\beta_1, \dots, \beta_k$ are linearly independent over $\mathbb{Q}$, the values $\mu(x'_1), \dots, \mu(x'_k)$ are linearly independent over $\mathbb{Q}$. Since $q(\xi) = \xi_{Z'}$, $\mu \geq 0$ on R' and $\mu > 0$ on the maximal ideal of R'. By Lemma 6.7 applied to R', the value group of μ is $\bigoplus_i \mathbb{Z}\mu(x'_i)$, and $\kappa(\xi_{Z'}) \to \kappa(\mu)$ is an isomorphism. As in **Step 2**, $\det(m'_{il}) = \pm 1$, and $q|_S : S \to Z'$ is birational.

Step 5. In this step we construct γ_Z.

Since $f \circ p$ maps S to c, the restriction of $(f \circ p)^* v$ to S is the constant $v(c)$. Since a is even, $\det(m_{il})^a = \det(m'_{il})^a = 1$. Thus restricting (6.3) to S implies

$$(g'|_{Z'} \circ q|_S) \det(e_{jj'}|_S)^a = v(c)(g|_Z \circ p|_S).$$

Since

$$(q|_S)^*(dy'_1|_{Z'} \wedge \cdots \wedge dy'_{n-k}|_{Z'}) = \det(e_{jj'}|_S) dt_1|_S \wedge \cdots \wedge dt_{n-k}|_S,$$
$$(p|_S)^*(dy_1|_Z \wedge \cdots \wedge dy_{n-k}|_Z) = dt_1|_S \wedge \cdots \wedge dt_{n-k}|_S,$$

we obtain

$$(q|_S)^* \eta_{Z'} = v(c)(p|_S)^* \eta_Z.$$

Let $\gamma_Z := q|_S \circ (p|_S)^{-1} : Z \dashrightarrow Z'$. Then $\gamma_Z^* \eta_{Z'} = v(c)\eta_Z$. By [Kol16, Definition 13], $\operatorname{div}(\eta_Z) = -a\Delta_Z$ and $\operatorname{div}(\eta_{Z'}) = -a\Delta_{Z'}$. Since $v(c) \neq 0$ and the log discrepancies of (Z, Δ_Z) and $(Z', \Delta_{Z'})$ are determined by η_Z and $\eta_{Z'}$, γ_Z is crepant for (Z, Δ_Z) and $(Z', \Delta_{Z'})$. □

Lemma 6.9 below is the key input for Theorem 6.4(4). At a prime divisor over V' with ramification index e over V, the coefficient of the moduli part on V is the coefficient w of $\mathbf{M} + \frac{1}{b}\operatorname{div}(\chi)$ divided by e, and w is the weight of the invariant form on a general transverse curve (Lemma 6.10). The integrality of pw alone only shows that pe times the coefficient on V is an integer. Lemma 6.9 shows that e divides pw, and this is the reason why the integer p in Theorem 6.4(4) is independent of $|G|$ and of the ramification indices. Because of the factor 2 in the definition of a in Lemma 6.9, the integer a is even. Both Lemma 6.8 and the $\mathbb{P}^1$-links of [Kol16, Theorem 10, Proposition 14] use this: on a $\mathbb{P}^1$-link with fiber coordinate x, the residues of dx/x along the zero section and the polar section of x are 1 and -1, so the residues of an m-canonical form along the zero section and the polar section of x differ by $(-1)^m$.

For a variety X over $\mathbb{C}(C)$ and a birational map $\gamma : X \dashrightarrow X$ over the automorphism of $\mathbb{C}(C)$ induced by an automorphism σ of C, crepancy is understood as in Definition 2.4, where Y is a normal variety over $\mathbb{C}(C)$, $p_1 : Y \to X$ is projective and birational over $\mathbb{C}(C)$, and $p_2 := \gamma \circ p_1$, which play the roles of p and q in Definition 2.4. For a rational relative pluricanonical form β on X, $\gamma^*\beta$ denotes the semilinear pullback of β along γ. By [OAI26c, Lemma 4.3(i)], the weight $w_c(\varphi)$ of Definition 3.4 does not depend on the choice of z.

Lemma 6.9. *Let $d \geq 2$ and $b \geq 1$ be integers, $a := 2b \cdot \operatorname{lcm}\{q(j,b) \mid 0 \leq j \leq d-1\}$, and $p := a \cdot \operatorname{lcm}\{\ell(j,a) \mid 0 \leq j \leq d-1\}$, where $q(\cdot,\cdot)$ and $\ell(\cdot,\cdot)$ are the integers of Theorems 3.5 and 6.2. Let C be a smooth projective curve, $c \in C$ a closed point, and σ an automorphism of C of finite order such that $\sigma(c) = c$ and σ acts on the tangent space of C at c by a primitive e-th root of unity. Let X be a normal geometrically integral projective variety over $\mathbb{C}(C)$ of dimension $r \leq d-1$, and $B \geq 0$ a $\mathbb{Q}$-divisor on X such that (X, B) is geometrically klt. Let β be a rational relative b-canonical form on X such that $\operatorname{div}(\beta) + bB = 0$, and assume that $\gamma^*\beta = \beta$ for a birational map $\gamma : X \dashrightarrow X$ over the automorphism of $\mathbb{C}(C)$ induced by σ which is crepant for (X, B). Then $pw_c(\beta)$ is an integer divisible by e.*

Proof. Let n be the order of σ, ζ the primitive e-th root of unity by which σ acts on the tangent space of C at c, and $\mathfrak{m}_c$ the maximal ideal of $\mathcal{O}_{C,c}$. Then σ^* acts on $\mathfrak{m}_c/\mathfrak{m}_c^2$ by ζ, and $\zeta^n = 1$ since $\sigma^n = \mathrm{id}_C$. Let z be a rational function on C with $\operatorname{mult}_c \operatorname{div}(z) = 1$, and

$$u := \frac{1}{n} \sum_{j=0}^{n-1} \zeta^{-j} (\sigma^j)^* z.$$

Since $(\sigma^j)^*z/z$ is regular at c with value ζ^j, the function u/z is regular at c with value 1. Thus $\mathrm{mult}_c \operatorname{div}(u) = 1$. Since $\sigma^n = \mathrm{id}_C$ and $\zeta^n = 1$, we have $\sigma^* u = \zeta u$. Since the weight does not depend on the choice of z, we may compute weights at c with u in place of z in Definition 3.4.

Suppose first that $r = 0$. Then X is the generic point of C, $B = 0$, β is a non-zero rational function on C, and $\gamma^*\beta = \sigma^*\beta$. By Definition 3.4, the only prime divisor in the definition of $w_c(\beta)$ is the point c of C, and

$$\mathrm{mult}_c \operatorname{div}\left(\beta \cdot (du/u)^{\otimes b}\right) = \mathrm{mult}_c \operatorname{div}(\beta) - b,$$

so $w_c(\beta) = \mathrm{mult}_c \operatorname{div}(\beta)/b$. Let $k := \mathrm{mult}_c \operatorname{div}(\beta)$ and write $\beta = u^k g$, where g is a rational function on C which is regular and non-zero at c. Since $\sigma^*\beta = \beta$ and $\sigma^* u = \zeta u$,

$$\zeta^{-k} g = \frac{\beta}{(\zeta u)^k} = \frac{\sigma^*\beta}{(\sigma^* u)^k} = \sigma^* g.$$

Evaluating at c and using $(\sigma^* g)(c) = g(\sigma(c)) = g(c) \neq 0$, we obtain $\zeta^{-k} = 1$, that is, $e \mid k$. Since $b \mid a \mid p$, the number $p w_c(\beta) = (p/b)k$ is an integer divisible by e.

From now on, assume that $r \geq 1$. Let $w := w_c(\beta)$. By Theorem 3.5, $q(r,b)w \in \mathbb{Z}$. Since $r \leq d-1$, $q(r,b)$ divides a, and hence $aw \in \mathbb{Z}$. Let $\Omega := \beta \wedge (du/u)^{\otimes b}$. By [OAI26c, Proposition 4.5], applied with $m = b$, $\varphi = \beta$, and $z = u$, there exist a projective contraction $f : N \to C$ and a horizontal $\mathbb{Q}$-divisor $H_N \geq 0$ on N such that, with $T := (f^*c)_{\mathrm{red}}$, the following hold:

(i) N is $\mathbb{Q}$-factorial, the generic fiber N_ξ of f is geometrically integral and birational to X over $\mathbb{C}(C)$, and $(N, T + H_N)$ is dlt;
(ii) $K_N + T + H_N \sim_{\mathbb{Q},C} 0$;
(iii) the equality $\frac{1}{b}\operatorname{div}_N(\Omega) + T + H_N = w f^* c$ holds on an open neighborhood of T, and $\operatorname{div}(\beta) + b H_{N,\xi} = 0$ on N_ξ, where $H_{N,\xi}$ is the restriction of H_N to N_ξ.

Since $H_{N,\xi} = -\frac{1}{b}\operatorname{div}(\beta)$ on N_ξ and $B = -\frac{1}{b}\operatorname{div}(\beta)$ on X, the birational map between N_ξ and X over $\mathbb{C}(C)$ is crepant for $(N_\xi, H_{N,\xi})$ and (X, B). Since (X, B) is geometrically klt and $H_{N,\xi} \geq 0$, $(N_\xi, H_{N,\xi})$ is geometrically klt. Since H_N is horizontal and $(N_\xi, H_{N,\xi})$ is klt, the coefficients of H_N are less than 1. Thus the lc centers of $(N, T + H_N)$ over a neighborhood of c are the strata of T [Kol13, Theorem 4.16], and they are contained in $f^{-1}(c)$.

Since f is proper, there exists an open neighborhood $C^\circ \subset C$ of c such that $\operatorname{div}(u)|_{C^\circ} = c$ and the equality in (iii) holds on $N^\circ := f^{-1}(C^\circ)$. Let $\eta := u^{-aw}\Omega^{\otimes a/b}$, a rational a-canonical form on N. On N°,

$$\operatorname{div}(\eta) = -awf^*c + \frac{a}{b}\operatorname{div}(\Omega) = -awf^*c + a\big(wf^*c - T - H_N\big) = -a(T + H_N). \tag{6.4}$$

The birational map γ induces a birational map $N \dashrightarrow N$, which we also denote by γ, such that $f \circ \gamma = \sigma \circ f$. Since $\gamma^*\beta = \beta$ and $\sigma^* u = \zeta u$, we have $\gamma^*(du/u) = du/u$ and $\gamma^*\Omega = \Omega$. Thus

$$\gamma^*\eta = (\zeta u)^{-aw}\Omega^{\otimes a/b} = \zeta^{-aw}\eta.$$

By (6.4), Lemma 6.8 applied to f, c, σ, $\Delta := T + H_N$, a, η, γ, and the constant function $v := \zeta^{-aw}$ implies that there exist minimal lc centers Z and Z' of $(N, T + H_N)$ contained in $f^{-1}(c)$ and a birational map $\gamma_Z : Z \dashrightarrow Z'$ such that

$$\gamma_Z^* \eta_{Z'} = \zeta^{-aw}\eta_Z,$$

where η_Z and $\eta_{Z'}$ are the Poincaré residues of η.

We now compare Z and Z' by [Kol16]. Let $f^\circ := f|_{N^\circ} : N^\circ \to C^\circ$. Then f° is projective, hence proper, and f° is a contraction, hence surjective with connected fibers. Moreover, $(N^\circ, T + H_N)$ is dlt, and by (6.4), η induces an isomorphism $\omega_{N^\circ}^{[a]}(a(T + H_N)) \cong \mathcal{O}_{N^\circ}$. Thus f° is a dlt crepant log structure in the sense of [Kol16, Definition 2], and $f^\circ(Z) = f^\circ(Z') = \{c\}$. Moreover, a is even by the definition of a. By [Kol16, Theorem 10, Proposition 14], there exists a birational map $\psi : Z' \dashrightarrow Z$ such that $\psi^*\eta_Z = \eta_{Z'}$. Let $\gamma_1 := \psi \circ \gamma_Z : Z \dashrightarrow Z$. Then

$$\gamma_1^*\eta_Z = \gamma_Z^*\psi^*\eta_Z = \gamma_Z^*\eta_{Z'} = \zeta^{-aw}\eta_Z.$$

Let Δ_Z be given by dlt adjunction (Lemma 2.2). Since Z is a minimal lc center of the dlt pair $(N^\circ, T + H_N)$, Lemma 2.2(1) and (3) show that Z is normal, $\Delta_Z \geq 0$, (Z, Δ_Z) is dlt, and any lc center of (Z, Δ_Z) is an lc center of $(N^\circ, T + H_N)$ properly contained in Z. By the minimality of Z, (Z, Δ_Z) has no lc center, so (Z, Δ_Z) is klt. Since Z is closed in $f^{-1}(c)$ and f is projective, Z is a projective variety. By (6.4) and [Kol16, Definition 13], $\operatorname{div}(\eta_Z) = -a\Delta_Z$, and hence $a(K_Z + \Delta_Z) \sim 0$. Since the log discrepancies of (Z, Δ_Z) are determined by η_Z, the identity $\gamma_1^* \eta_Z = \zeta^{-aw} \eta_Z$ implies that γ_1 is crepant for (Z, Δ_Z). Let $j := \dim Z$. Then $j \leq \dim f^{-1}(c) = r$. By Theorem 6.2 applied to (Z, Δ_Z), η_Z, and γ_1,

$$\zeta^{-aw\ell(j,a)} = 1.$$

Since $j \leq r \leq d - 1$, $\ell(j, a)$ divides p/a, and hence $\zeta^{-pw} = 1$. Since ζ is a primitive e-th root of unity and $pw = (p/a) \cdot aw \in \mathbb{Z}$, the integer pw is divisible by e. □

The next lemma realizes the coefficients of $\mathbf{M} + \frac{1}{b} \operatorname{div}(\chi)$ as weights of an invariant form on curves on which an inertia group acts.

Lemma 6.10. *Notation and conditions as in Set-up 6.1. Let b be a positive integer, χ a rational function on V', and θ a rational b-canonical form on X such that*

$$\operatorname{div}(\theta) = bh^* D_{V'} + h^* \operatorname{div}(\chi) - bB$$

and $\phi^* \theta = \theta$ for any birational map $\phi : X \dashrightarrow X$ which is crepant for (X, B) and satisfies $h \circ \phi = \sigma' \circ h$ for some $\sigma' \in G$. Let V_1 be a smooth variety projective and birational over V, V_1' the normalization of V_1 in V', Q a prime divisor on V_1', e the ramification index of $V_1' \to V_1$ along Q, and $\sigma \in G$ a generator of the inertia group of Q. Then there exist the following data.

1. *A smooth projective curve C, a closed point $c \in C$, and an automorphism σ_C of C of finite order such that $\sigma_C(c) = c$ and σ_C acts on the tangent space of C at c by a primitive e-th root of unity.*
2. *A normal geometrically integral projective variety X_C over $\mathbb{C}(C)$ of dimension $\dim X - \dim V'$, and a $\mathbb{Q}$-divisor $B_C \geq 0$ on X_C such that (X_C, B_C) is geometrically klt.*
3. *A rational relative b-canonical form β_C on X_C such that $\operatorname{div}(\beta_C) + bB_C = 0$.*
4. *A birational map $\gamma : X_C \dashrightarrow X_C$ over the automorphism of $\mathbb{C}(C)$ induced by σ_C which is crepant for (X_C, B_C) and satisfies $\gamma^* \beta_C = \beta_C$.*

Moreover, the data in (1)–(4) satisfy

$$w_c(\beta_C) = \operatorname{mult}_Q \left(\mathbf{M}_{V_1'} + \frac{1}{b} \operatorname{div}_{V_1'}(\chi) \right).$$

Proof. **Step 1.** In this step we fix notation and construct a σ-eigenfunction u with $\operatorname{mult}_Q \operatorname{div}(u) = 1$ and σ-invariant functions $t_2, \ldots, t_k$.

Let $g' : V_1' \to V'$ be the induced morphism, $D_\beta := D_{V'} + \frac{1}{b} \operatorname{div}(\chi)$, $k := \dim V'$, and $r := \dim X - \dim V'$. Let $\omega_{V'} := \rho^* \omega_V$, where ω_V is the form fixed in Set-up 6.1, so that $K_{V_1'} = \operatorname{div}_{V_1'}(\omega_{V'})$. Since $\rho \circ \sigma' = \rho$, we have $\sigma'^* \omega_{V'} = \omega_{V'}$ for any $\sigma' \in G$. Let $\beta := \theta / h^* \omega_{V'}^{\otimes b}$ be the rational section of $\omega_{X^\circ/V'}^{\otimes b}$ such that $\theta = \beta \wedge (h^* \omega_{V'})^{\otimes b}$, where $X^\circ \subset X$ and the quotient are as in **Step 2** of the proof of Lemma 6.5.

By [Ser79, Ch. I, §7, Proposition 21 and its Corollary], [Ser79, Ch. IV, §1, Remark 2], and [Ser79, Ch. IV, §2, Proposition 7, Corollaries 1 and 2], the inertia group of Q is cyclic of order e, and there exists a primitive e-th root of unity ζ such that for any rational function z on V_1' with $\operatorname{mult}_Q \operatorname{div}(z) = 1$, the function $\sigma^* z / z$ is regular at the generic point of Q with restriction ζ to Q. Fix such z, and let

$$u := \frac{1}{e} \sum_{j=0}^{e-1} \zeta^{-j} (\sigma^j)^* z.$$

Since σ acts trivially on Q, the restriction of $(\sigma^j)^* z / z$ to Q is ζ^j, so the restriction of u/z to Q is 1. Thus $\operatorname{mult}_Q \operatorname{div}(u) = 1$. Since $\sigma^e = \mathrm{id}_{V_1'}$, we have $\sigma^* u = \zeta u$. Let $y_2, \ldots, y_k$ be rational

functions on Q which define a dominant rational map $Q \dashrightarrow \mathbb{A}^{k-1}$, $\tilde{y}_i$ a rational function on V_1' which is regular at the generic point of Q with restriction y_i to Q, and

$$t_i := \frac{1}{e}\sum_{j=0}^{e-1}(\sigma^j)^*\tilde{y}_i$$

for $2 \le i \le k$. Then $\sigma^* t_i = t_i$, and the restriction of t_i to Q is y_i. Hence there exists a σ-stable open subset $W \subset V_1'$ with $W \cap Q \neq \emptyset$ such that W is smooth, $u, t_2, \dots, t_k$ are regular on W, $Q \cap W$ is the zero locus of u, and $du \wedge dt_2 \wedge \cdots \wedge dt_k$ vanishes nowhere on W. Let $t := (t_2, \dots, t_k) : W \to \mathbb{A}^{k-1}$, $\omega_0 := du/u \wedge dt_2 \wedge \cdots \wedge dt_k$, and φ_0 the non-zero rational function on V_1' such that $g'^*\omega_{V'} = \varphi_0\omega_0$. Then $\mathrm{mult}_Q \operatorname{div}(\omega_0) = -1$.

Step 2. In this step we express the lc threshold t_Q of Q on a log resolution.

Let Y be a smooth variety with a projective birational morphism $p_Y : Y \to X$ such that $h \circ p_Y = g' \circ g_Y$ for a morphism $g_Y : Y \to V_1'$, and such that $\operatorname{Supp} B_Y \cup \operatorname{Exc}(p_Y) \cup \operatorname{Supp} \operatorname{div}(g_Y^* u)$ is an snc divisor, where $K_Y + B_Y = p_Y^*(K_X + B)$ (such Y exists by [KM98, Theorem 0.2], applied to the normalization of the closure of the graph of the rational map $X \dashrightarrow V_1'$). Since $Q \cap W = \operatorname{div}(u)|_W$, on $g_Y^{-1}(W)$ the set $g_Y^{-1}(Q)$ is a union of components of $\operatorname{Supp} \operatorname{div}(g_Y^* u)$, so $\operatorname{Supp} B_Y \cup \operatorname{Exc}(p_Y) \cup g_Y^{-1}(Q)$ is an snc divisor on $g_Y^{-1}(W)$. By Set-up 6.1(7), there exists a birational map $\phi : X \dashrightarrow X$ which is crepant for (X, B) and satisfies $h \circ \phi = \sigma \circ h$. Let $p_1, p_2 : Y_\phi \to X$ be the morphisms p and q of Definition 2.4 for ϕ. For any component E of $g_Y^{-1}(Q)$ with $g_Y(E) = Q$, E intersects $g_Y^{-1}(W)$, so E is a prime divisor on Y. For such E, let $n_E := \mathrm{mult}_E Q$, computed over the generic point of Q, where Q is Cartier, that is, $n_E = \mathrm{mult}_E \operatorname{div}(g_Y^* u)$, and let $b_E := \mathrm{mult}_E B_Y$. By [Kol13, Notation 2.6, 2.20 and Corollary 2.11], the lc threshold t_Q of Q (Definition 3.1(2)) satisfies

$$t_Q = \min_E \frac{1 - b_E}{n_E}. \tag{6.5}$$

Step 3. In this step we construct C, c, and σ_C.

Let $c \in Q \cap W$ be a general closed point, and $\lambda := t(c)$. Since t is smooth at c, there exists a unique component C° of $t^{-1}(\lambda)$ containing c. Shrinking C°, we may assume that C° is a smooth curve. Since σ acts trivially on Q and $\sigma^* t_i = t_i$, we have $\sigma(c) = c$ and $\sigma(C^\circ) = C^\circ$. Since $(u, t_2 - t_2(c), \dots, t_k - t_k(c))$ is a system of local parameters at c, the restriction of u to C° vanishes to order 1 at c. Let C be the smooth projective curve containing C° as an open subset, and σ_C the automorphism of C extending $\sigma|_{C^\circ}$. Then $\sigma_C(c) = c$, $\sigma_C^e = \mathrm{id}_C$, and $\sigma_C^* u = \zeta u$, where u also denotes the restriction of u to C. Thus σ_C acts on the tangent space of C at c by ζ.

Since c is general in Q and the restriction of t to Q is dominant, λ is general in $\mathbb{A}^{k-1}$, and C° is not contained in any given proper closed subset of V_1'. Thus we may assume the following.

(a) The generic point ξ of C° is contained in an open subset $V_1'^0 \subset V_1'$ such that $g'|_{V_1'^0}$ is an open immersion, h is flat over $g'(V_1'^0)$ with geometrically normal and geometrically integral fibers, g_Y and the strata of $\operatorname{Supp} B_Y \cup \operatorname{Exc}(p_Y)$ are smooth over $V_1'^0$, the components of B_Y intersecting $g_Y^{-1}(V_1'^0)$ have coefficients less than 1, and $g'(V_1'^0)$ is disjoint from $\operatorname{Supp} D_\beta$, from $\operatorname{Supp} \operatorname{div}(\omega_{V'})$, and from the images of the components of B not dominating V'. Moreover, p_1 and p_2 restrict to birational morphisms from the fiber of $h \circ p_1$ over $g'(\xi)$ to the fiber of h over $g'(\xi)$, and the fiber of $h \circ p_1$ over $g'(\xi)$ is normal and geometrically integral, by the references given below.

(b) For $S = Y$ and for any stratum S of $\operatorname{Supp} B_Y \cup \operatorname{Exc}(p_Y) \cup g_Y^{-1}(Q)$, the restriction of $t \circ g_Y$ to $S \cap g_Y^{-1}(W)$ is smooth over a neighborhood of λ.

(c) The point c is not contained in $g_Y(E)$ for any component E of $g_Y^{-1}(Q)$ with $g_Y(E) \neq Q$, nor in any component of $\operatorname{Supp} g'^* D_\beta \cup \operatorname{Supp} \operatorname{div}(\varphi_0)$ other than Q.

The open subset in (a) exists by generic flatness [Gro65, Théorème 6.9.1], by generic smoothness [Har77, Corollary III.10.7], applied to g_Y and to the restrictions of g_Y to the strata of $\operatorname{Supp} B_Y \cup$

$\mathrm{Exc}(p_Y)$, by Lemma 2.17(4), applied to the contractions h and $h\circ p_1$, and by the klt property of (X,B) over the generic point of V'. By generic flatness and [Har77, Corollary III.9.6], applied also at the non-closed point $g'(\xi)$, the fiber of h over $g'(\xi)$ is of dimension r, and the closed subsets of X over which p_1 or p_2 is not an isomorphism intersect the fiber of h over $g'(\xi)$ in dimension less than r, so the restrictions of p_1 and p_2 in (a) are birational. Property (b) follows from generic smoothness [Har77, Corollary III.10.7] applied to $t\circ g_Y$ on Y and on the strata. Property (c) holds since c is general.

Step 4. In this step we construct X_C, B_C, β_C, and γ.

Let X_C be the fiber of h over $g'(\xi)$, and $B_C := B|_{X_C}$. By (a), X_C is a normal geometrically integral projective variety over $\mathbb{C}(C)$ with $\dim X_C = r$. By (a), the fiber Y_ξ of g_Y over ξ is a log resolution of (X_C, B_C), also after base change to an algebraic closure of $\mathbb{C}(C)$, the restriction of B_Y to Y_ξ has coefficients less than 1, and $K_{Y_\xi}+B_Y|_{Y_\xi}$ is the pullback of $K_{X_C}+B_C$. Thus (X_C,B_C) is geometrically klt [KM98, Corollary 2.31(3), Lemma 2.30]. Since $h\circ\phi=\sigma\circ h$ and $\sigma(\xi)=\xi$, by (a) the map ϕ restricts to a birational map $\gamma: X_C \dashrightarrow X_C$ over the automorphism of $\mathbb{C}(C)$ induced by σ_C, and p_1 and p_2 restrict to birational morphisms showing that γ is crepant for (X_C,B_C). By (a), the restriction β_C of β to X_C is defined, and since $\frac{1}{b}\operatorname{div}(\theta)+B = h^*D_\beta$, we have $\operatorname{div}(\beta_C)+bB_C=0$. Since $\phi^*\theta=\theta$ and $\phi^*h^*\omega_{V'} = h^*\sigma^*\omega_{V'} = h^*\omega_{V'}$, we have $\phi^*\beta=\beta$, and hence $\gamma^*\beta_C=\beta_C$.

Step 5. In this step we compute $w_c(\beta_C)$.

Let Y_C be the closure of Y_ξ in $g_Y^{-1}(C^\circ)$, and $g_{Y_C}: Y_C\to C^\circ$ the restriction of g_Y. By (b), $g_Y^{-1}(C^\circ)$ is smooth of pure dimension $r+1$ near $g_Y^{-1}(c)$, and by (c), $\dim g_Y^{-1}(c)=r$. Thus Y_C coincides with $g_Y^{-1}(C^\circ)$ near $g_Y^{-1}(c)$. By (b) and (c), near $g_Y^{-1}(c)$, the variety Y_C is smooth, the restriction of $\operatorname{Supp} B_Y\cup\mathrm{Exc}(p_Y)\cup g_Y^{-1}(Q)$ to Y_C is an snc divisor, and the components of $g_{Y_C}^*c$ are the components E_C of $E\cap Y_C$ for the components E of $g_Y^{-1}(Q)$ with $g_Y(E)=Q$, with

$$\operatorname{mult}_{E_C} c = n_E,\quad \operatorname{mult}_{E_C}\left(B_Y|_{Y_C}\right)=b_E.$$

Let $\Omega_C := \beta_C\wedge(du/u)^{\otimes b}$, and

$$\delta := \operatorname{mult}_Q\left(g'^*D_\beta - \operatorname{div}(\varphi_0)\right).$$

Since

$$p_Y^*\theta = g_Y^*\varphi_0^b\cdot\beta\wedge\left(\frac{du}{u}\right)^{\otimes b}\wedge(dt_2\wedge\cdots\wedge dt_k)^{\otimes b},$$

and $t\circ g_Y$ is smooth near $g_Y^{-1}(c)$ by (b), we have, near $g_Y^{-1}(c)$ on Y_C,

$$\begin{aligned}\frac{1}{b}\operatorname{div}(\Omega_C)+B_Y|_{Y_C} &= \left(\frac{1}{b}\operatorname{div}_Y(p_Y^*\theta)+B_Y-g_Y^*\operatorname{div}(\varphi_0)\right)\Big|_{Y_C}\\ &= g_Y^*\left(g'^*D_\beta-\operatorname{div}(\varphi_0)\right)\big|_{Y_C}\\ &= \delta g_{Y_C}^*c.\end{aligned}\tag{6.6}$$

Here the second equality holds since

$$\frac{1}{b}\operatorname{div}_Y(p_Y^*\theta)+B_Y = p_Y^*\left(\frac{1}{b}\operatorname{div}(\theta)+B\right) = g_Y^*g'^*D_\beta,$$

and the third equality holds by (c), since $Q\cap W$ is the zero locus of u and the restriction of u to C° vanishes to order 1 at c.

Since g_Y is projective, $g_{Y_C}: Y_C\to C^\circ$ is projective, and since Y_C is smooth near $g_{Y_C}^{-1}(c)$, after shrinking C° around c we may assume that Y_C is smooth. The generic fiber Y_ξ of g_{Y_C} is birational to X_C via p_Y (**Step 4**). Thus we may take $W=Y_C$ and $\pi=g_{Y_C}$ in Definition 3.4, with u in place of z. Let F be a prime divisor over Y_C whose center on Y_C is contained in $g_{Y_C}^{-1}(c)$, that is, one of the prime divisors in Definition 3.4 with $W=Y_C$ and $\pi=g_{Y_C}$. Then (6.6) implies

$$\frac{\operatorname{mult}_F\operatorname{div}(\Omega_C)+b}{b\operatorname{mult}_F\operatorname{div}(g_{Y_C}^*u)} = \delta+\frac{a(F,Y_C,B_Y|_{Y_C})}{\operatorname{mult}_F\operatorname{div}(g_{Y_C}^*u)}.$$

By [KM98, Corollary 2.31(3), Lemma 2.30], near $g_{Y_C}^{-1}(c)$ the sub-pair $(Y_C, B_Y|_{Y_C} + sg_{Y_C}^*c)$ is sub-lc for $s := \min_E(1-b_E)/n_E$, so

$$a(F, Y_C, B_Y|_{Y_C}) \geq s \operatorname{mult}_F \operatorname{div}(g_{Y_C}^* u),$$

with equality for $F = E_C$ when E attains the minimum. By (6.5), $w_c(\beta_C) = \delta + t_Q$. Finally, $\operatorname{mult}_Q \operatorname{div}(\omega_0) = -1$, $\operatorname{mult}_Q B_{V_1'} = 1 - t_Q$ by Definition 3.1(2), and

$$K_{V_1'} = \operatorname{div}(\varphi_0) + \operatorname{div}(\omega_0),$$

so

$$\operatorname{mult}_Q \left(\mathbf{M}_{V_1'} + \frac{1}{b} \operatorname{div}_{V_1'}(\chi) \right) = \operatorname{mult}_Q \left(g'^* D_\beta - K_{V_1'} - B_{V_1'} \right) = \delta + 1 - (1 - t_Q) = w_c(\beta_C).$$

□

Proof of Theorem 6.4. The proof is long, so we divide it into several steps.

Step 1. In this step we construct χ, θ, and D_V, and prove (1).

By Lemma 6.5, there exist a rational function χ on V' and a rational b-canonical form θ on X such that

$$\operatorname{div}(\theta) = bh^* D_{V'} + h^* \operatorname{div}(\chi) - bB$$

and $\phi^*\theta = \theta$ for any birational map ϕ as in (1). Let $D_\beta := D_{V'} + \frac{1}{b}\operatorname{div}(\chi)$. Then

$$\frac{1}{b} \operatorname{div}(\theta) + B = h^* D_\beta. \tag{6.7}$$

We show that D_β is G-invariant. Let $\sigma \in G$. By Set-up 6.1(7), there exists a birational map $\phi : X \dashrightarrow X$ which is crepant for (X, B) and satisfies $h \circ \phi = \sigma \circ h$. Let $p_1, p_2 : Y \to X$ be the morphisms p and q of Definition 2.4 for ϕ, and B_Y the $\mathbb{Q}$-divisor on Y such that

$$\frac{1}{b} \operatorname{div}_Y(p_1^*\theta) + B_Y = p_1^* \Big(\frac{1}{b} \operatorname{div}(\theta) + B \Big).$$

Since $\phi^*\theta = \theta$, we have $p_1^*\theta = p_2^*\theta$. Since ϕ is crepant for (X, B), we also have

$$\frac{1}{b} \operatorname{div}_Y(p_2^*\theta) + B_Y = p_2^* \Big(\frac{1}{b} \operatorname{div}(\theta) + B \Big).$$

Since $h \circ p_2 = \sigma \circ h \circ p_1$, (6.7) implies

$$p_1^* h^* D_\beta = \frac{1}{b} \operatorname{div}_Y(p_1^*\theta) + B_Y = \frac{1}{b} \operatorname{div}_Y(p_2^*\theta) + B_Y = p_2^* h^* D_\beta = p_1^* h^* \sigma^* D_\beta.$$

Since $h \circ p_1$ is surjective and the pullback of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors under a surjective morphism between normal varieties is injective, $\sigma^* D_\beta = D_\beta$. Since $\sigma \in G$ is arbitrary, D_β is G-invariant. By Lemma 6.6(1), there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D_V on V such that $\rho^* D_V = D_\beta$. Then (1) follows from (6.7).

Step 2. In this step we construct $\mathbf{N}$ and B_V, and prove (2).

We first show that $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U. Let Γ be the set of coefficients of B. Then Γ is a finite subset of $[0,1] \cap \mathbb{Q}$, so Γ satisfies the DCC. Moreover, $K_X + B \sim_{\mathbb{Q}} h^* D_{V'}$. Let m' be the integer m of Theorem 3.3 for d and Γ. By Theorem 3.3(1), there exists a trivialization $K_X + B + \frac{1}{m'} \operatorname{div}(\psi') = h^* D'$ of degree m' of $K_X + B$ over V'. Let $\lambda := \psi^{m'} / \psi'^m$. Then

$$\operatorname{div}(\lambda) = m' \operatorname{div}(\psi) - m \operatorname{div}(\psi') = mm' h^* (D_{V'} - D').$$

By Lemma 2.17(2), applied to h and the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $mm'(D_{V'} - D')$, the function λ is the pullback of a rational function on V', which we also denote by λ, and $D_{V'} - D' = \frac{1}{mm'} \operatorname{div}(\lambda)$. The discriminant $\boldsymbol{b}$-divisor of (X, B) over V' does not depend on the trivialization (Definition 3.1(2)), so by Definition 3.1(3), the moduli $\boldsymbol{b}$-divisor $\mathbf{M}'$ of the trivialization of degree m' satisfies $\mathbf{M} = \mathbf{M}' + \frac{1}{mm'} \operatorname{div}(\lambda)$. Since $V' \to V$ is finite and $V \to U$ is projective, $V' \to U$ is projective. By Theorem 3.3(2), $\mathbf{M}'$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U. Thus $\mathbf{M}$ and $\mathbf{M} + \frac{1}{b} \operatorname{div}(\chi)$ are $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U. We do not use the integer p of Theorem 3.3, which depends on Γ.

Next, we show that $\mathbf{M}+\frac{1}{b}\operatorname{div}(\chi)$ is G-invariant. Let $\omega_{V'}:=\rho^*\omega_V$, where ω_V is the form fixed in Set-up 6.1. Since $\rho\circ\sigma=\rho$, we have $\sigma^*\omega_{V'}=\omega_{V'}$ for any $\sigma\in G$. Let $\sigma\in G$ and ϕ be as in **Step 1**. Let $g_1':V_1'\to V'$ be a projective birational morphism from a normal variety and P a prime divisor on V_1'. Let $\sigma(P)$ be the prime divisor P on V_1' regarded over V' through $\sigma\circ g_1'$, so that, for a $\mathbb{Q}$-$\boldsymbol{b}$-divisor $\mathbf{L}$ over V', the coefficient of P in the trace of $\sigma^*\mathbf{L}$ on V_1' is the coefficient of $\sigma(P)$ in $\mathbf{L}$. Let $p_1,p_2:Y'\to X$ be the morphisms p and q of Definition 2.4 for ϕ. Replacing Y' with a normal variety projective and birational over Y', which keeps the property of Definition 2.4, we may assume that $h\circ p_1=g_1'\circ g_{Y'}$ for a morphism $g_{Y'}:Y'\to V_1'$. Then

$$h\circ p_2=\sigma\circ h\circ p_1=(\sigma\circ g_1')\circ g_{Y'}.$$

Let ω_X be a rational 1-canonical form on X with $\operatorname{div}(\omega_X)=K_X$. Since ϕ is crepant for (X,B),

$$\Delta_{Y'}:=p_1^*(K_X+B)-\operatorname{div}(p_1^*\omega_X)=p_2^*(K_X+B)-\operatorname{div}(p_2^*\omega_X).$$

By Definition 3.1(2), applied on Y' with p_1 and $g_{Y'}$ for P, and with p_2 and $g_{Y'}$ for $\sigma(P)$, both t_P and $t_{\sigma(P)}$ are the supremum of the real numbers t such that $(Y',\Delta_{Y'}+tg_{Y'}^*P)$ is sub-lc over a neighborhood of the generic point of P on which P is Cartier. Thus $t_P=t_{\sigma(P)}$, and hence the discriminant $\boldsymbol{b}$-divisor of (X,B) over V' is G-invariant. For any normal variety V_1' with a projective birational morphism $g_1':V_1'\to V'$, the trace of $\mathbf{M}+\frac{1}{b}\operatorname{div}(\chi)$ on V_1' is

$$g_1'^*D_\beta-\operatorname{div}_{V_1'}(\omega_{V'})-B_{V_1'}.$$

Since D_β, $\omega_{V'}$, and the discriminant $\boldsymbol{b}$-divisor are G-invariant, $\mathbf{M}+\frac{1}{b}\operatorname{div}(\chi)$ is G-invariant. By Lemma 6.6(2), there exists a $\boldsymbol{b}$-$\mathbb{Q}$-Cartier $\mathbb{Q}$-$\boldsymbol{b}$-divisor $\mathbf{N}$ over V such that $\rho^*\mathbf{N}=\mathbf{M}+\frac{1}{b}\operatorname{div}(\chi)$, and $\mathbf{N}$ is nef over U.

Let $B_V:=D_V-K_V-\mathbf{N}_V$. Let V_1, V_1', and B_{V_1} be as in (2), $\rho_1:V_1'\to V_1$ the induced finite morphism, and $g_1:V_1\to V$ and $g_1':V_1'\to V'$ the induced morphisms. The trace of $\rho^*\mathbf{N}$ on V_1' is $\rho_1^*\mathbf{N}_{V_1}$: indeed, over the generic point of any prime divisor on V_1', both coincide with the corresponding traces on the normalization in V' of a determining model of $\mathbf{N}$ which is projective and birational over V_1 (Definition 2.7). Since $\rho_1^*g_1^*D_V=g_1'^*\rho^*D_V=g_1'^*D_\beta$, Definition 3.1(3) implies

$$\rho_1^*(K_{V_1}+B_{V_1})=\rho_1^*\big(g_1^*D_V-\mathbf{N}_{V_1}\big)=g_1'^*D_\beta-\mathbf{M}_{V_1'}-\frac{1}{b}\operatorname{div}_{V_1'}(\chi)=g_1'^*D_{V'}-\mathbf{M}_{V_1'}=K_{V_1'}+B_{V_1'}.$$

This proves (2).

Step 3. In this step we prove (3).

Let V_1, V_1', ρ_1, and B_{V_1} be as in **Step 2**, R a prime divisor on V_1, Q a prime divisor on V_1' with $\rho_1(Q)=R$, and e the ramification index of ρ_1 along Q. By (2) and the Hurwitz formula [Kol13, 2.41] at the generic point of Q (the coefficient of Q in the ramification divisor of ρ_1 is $e-1$, since the ramification is tame in characteristic zero),

$$1-\operatorname{mult}_Q B_{V_1'}=e\big(1-\operatorname{mult}_R B_{V_1}\big).\tag{6.8}$$

Since (X,B) is lc, Lemma 3.8(2) applied with $P=0$ implies that $\operatorname{mult}_Q B_{V_1'}\le 1$. Thus $\operatorname{mult}_R B_{V_1}\le 1$ by (6.8). For $V_1=V$, Theorem 3.3(2) applied to the trivialization of degree m' in **Step 2** implies that $B_{V'}\ge 0$. Thus $0\le\operatorname{mult}_Q B_{V'}\le 1$, and by (6.8),

$$\operatorname{mult}_R B_V=1-\frac{1-\operatorname{mult}_Q B_{V'}}{e}\in[0,1].$$

Hence $B_V\ge 0$. Since $\mathbf{N}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U (**Step 2**), and $K_V+B_V+\mathbf{N}_V=D_V$ is $\mathbb{Q}$-Cartier, $(V,B_V,\mathbf{N})/U$ is a generalized pair. Since the coefficients of B_{V_1} are at most 1 for any V_1 as in (2), $(V,B_V,\mathbf{N})/U$ is generalized lc. This proves (3).

Step 4. In this step we prove (4).

Let V_2 be a smooth variety projective and birational over V, R a prime divisor on V_2, V_2' the normalization of V_2 in V', Q a prime divisor on V_2' over R, e the ramification index of $V_2' \to V_2$ along Q, and $\sigma \in G$ a generator of the inertia group of Q. By **Step 2** and (2),

$$e \operatorname{mult}_R \mathbf{N}_{V_2} = \operatorname{mult}_Q \left(\mathbf{M}_{V_2'} + \frac{1}{b} \operatorname{div}_{V_2'}(\chi) \right). \tag{6.9}$$

By Lemma 6.10, applied with $V_1 := V_2$ and with b, χ, and θ as in **Step 1**, there exist C, c, σ_C, X_C, B_C, β_C, and γ satisfying the conclusion of Lemma 6.10, with $w_c(\beta_C)$ equal to the right-hand side of (6.9). Since $\dim X_C = \dim X - \dim V' \leq d-1$, and the integers a and p of Lemma 6.9 for d and b are the integers a and p of the theorem, Lemma 6.9 with $X := X_C$, $B := B_C$, $\beta := \beta_C$, $\sigma := \sigma_C$, the integers d, b, and e of this step, and the map $\gamma : X_C \dashrightarrow X_C$ of Lemma 6.10 shows that $pw_c(\beta_C)$ is an integer divisible by e. By (6.9),

$$p \operatorname{mult}_R \mathbf{N}_{V_2} = \frac{pw_c(\beta_C)}{e} \in \mathbb{Z}.$$

Since R is arbitrary, the coefficients of $p\mathbf{N}_{V_2}$ are integers. Since V_2 is smooth, $p\mathbf{N}_{V_2}$ is a Cartier divisor. This proves (4). □

7. Sections over a minimal log canonical image

In this section, we prove Theorem 7.13, which is used in step (3) of the sketch of the proof in Section 1. It determines which sections on a minimal lc center over a subvariety minimal among the images of the lc centers are Poincaré residues of restrictions of sections on a log resolution: they are exactly the sections invariant under a Galois group.

7.1. **The setting.** In this subsection, we fix the setting of this section and the constant N_0.

Set-up 7.1. Let d, Γ, $X \xrightarrow{g} T \to U$, (X, B), H, and V be as follows.

(1) $d \geq 2$ is an integer and $\Gamma \subset [0,1] \cap \mathbb{Q}$ is a finite set.
(2) $X \xrightarrow{g} T \to U$ are contractions such that U is affine and $\dim X \leq d$.
(3) (X, B) is a $\mathbb{Q}$-factorial dlt pair such that $B \in \Gamma \cup \{1\}$.
(4) $-(K_X + B) \sim_{\mathbb{Q}} g^*H$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H on T which is ample over U.
(5) $V \subsetneq T$ is a minimal element, with respect to inclusion, of the set of images on T of the lc centers of (X, B).

We call V as in Set-up 7.1(5) a *minimal lc image* of (X, B) over T.

Condition (5) ensures that the minimal lc centers of (X, B) with image V are klt for the pairs given by dlt adjunction, and that any lc center of (X, B) with image contained in V, and any irreducible component of an intersection of such lc centers, dominates V (Lemma 7.5). Thus sections on different lc centers with image V can be related over the generic point of V, and the divisors added to the boundaries on such lc centers are pulled back from the bases of their Stein factorizations (Lemma 7.6).

Construction 7.2. Notation and conditions as in Set-up 7.1.

(1) Let Σ_V be the union of the lc centers of (X, B) whose images on T are equal to V.
(2) Let $W \subset \Sigma_V$ be an lc center of (X, B) which is minimal with respect to inclusion among the lc centers with image V, and B_W the $\mathbb{Q}$-divisor on W given by dlt adjunction $K_W + B_W = (K_X + B)|_W$ (Lemma 2.2).
(3) Let $W \xrightarrow{h} V' \xrightarrow{\rho} V$ be the Stein factorization of $W \to V$, and G the group of automorphisms of V' over V.
(4) Let $\mu : \tilde{X} \to X$ be a log resolution of (X, B) which is an isomorphism over the generic point of any lc center of (X, B), and write $K_{\tilde{X}} + \tilde{S} + E = \mu^*(K_X + B)$, where $\tilde{S} := \mu_*^{-1} \lfloor B \rfloor$.
(5) Let $\tilde{\Sigma}_V$ be the union of the strata of $\tilde{S}$, that is, the irreducible components of intersections of components of $\tilde{S}$, whose images on T are equal to V, and $\tilde{W} := \mu_*^{-1} W$.

(6) For a positive integer N, let $L_N := -NK_{\tilde{X}} - N\tilde{S} - \lfloor (N+1)E \rfloor$. We regard $\mathcal{O}_{\tilde{X}}(L_N)$ as the sheaf of rational $(-N)$-canonical forms τ on $\tilde{X}$ such that $\operatorname{div}(\tau) \geq N\tilde{S} + \lfloor (N+1)E \rfloor$, together with 0.
(7) For a positive integer N and an lc center $W' \subset \Sigma_V$ of (X, B) with dlt adjunction $K_{W'} + B_{W'} = (K_X + B)|_{W'}$, let $\mathcal{J}_{N,W'}$ be the sheaf of rational $(-N)$-canonical forms t on W' such that $\operatorname{div}(t) \geq \lceil NB_{W'} \rceil$, together with 0, so that $\mathcal{J}_{N,W'} \cong \mathcal{O}_{W'}(-NK_{W'} - \lceil NB_{W'} \rceil)$. For a rational $(-N)$-canonical form t on W', let $B_t := \frac{1}{N} \operatorname{div}(t)$.

The general points of the lc centers of (X, B) are contained in the open subset of X where (X, B) is snc [KM98, Definition 2.37]. Thus there exists a log resolution of (X, B) which is an isomorphism over that open subset [Kol13, Theorem 10.45(2)], and in particular μ exists.

Construction 7.3. Let $d \geq 2$ be an integer and $\Gamma \subset [0, 1] \cap \mathbb{Q}$ a finite set. Let $\mathfrak{R} \subset [0, 1] \cap \mathbb{Q}$ be a finite set with $0, 1 \in \mathfrak{R}$, depending only on d and Γ, such that $\Gamma \cup \{1\} \subset \Phi(\mathfrak{R})$, and such that for any dlt pair (X, B) of dimension at most d with $B \in \Gamma \cup \{1\}$ and any lc center W' of (X, B), the $\mathbb{Q}$-divisor $B_{W'}$ given by dlt adjunction $K_{W'} + B_{W'} = (K_X + B)|_{W'}$ satisfies $B_{W'} \in \Phi(\mathfrak{R})$. Let $b_{\mathfrak{R}}$ be a positive integer such that $b_{\mathfrak{R}} r \in \mathbb{Z}$ for any $r \in \mathfrak{R}$, m the integer of Theorem 3.3 for $d - 1$ and $\Phi(\mathfrak{R})$,

$$m_0 := \operatorname{lcm}\{m, I(1, \Phi(\mathfrak{R})), \dots, I(d-1, \Phi(\mathfrak{R}))\},$$

where $I(j, \Phi(\mathfrak{R}))$ is the integer of Theorem 2.21, $a := 2 \operatorname{lcm}\{b_{\mathfrak{R}}, m_0\}$, and

$$N_0 := a \cdot \operatorname{lcm}\{\ell(j, a) \mid 0 \leq j \leq d-1\},$$

where $\ell(j, a)$ is the integer of Theorem 6.2.

Let $\mathfrak{R}_0 := \{1 - b \mid b \in \Gamma\} \cup \{0, 1\}$, so that $\Gamma \cup \{1\} \subset \Phi(\mathfrak{R}_0)$. By [Bir19, Lemma 3.3], for $i \geq 0$ there exists a finite set $\mathfrak{R}_{i+1} \subset [0, 1] \cap \mathbb{Q}$ containing $\mathfrak{R}_i$, depending only on $\mathfrak{R}_i$, such that for any lc pair (X, B) with $B \in \Phi(\mathfrak{R}_i)$ and any normal component S of $\lfloor B \rfloor$, the different B_S with $K_S + B_S = (K_X + B)|_S$ satisfies $B_S \in \Phi(\mathfrak{R}_{i+1})$. An lc center of codimension $k \leq d$ of a dlt pair is reached by k such adjunctions ([Kol13, Theorem 4.16] and Lemma 2.2), so we may take $\mathfrak{R} := \mathfrak{R}_d$, and since $\mathfrak{R}$ is finite, $\Phi(\mathfrak{R})$ satisfies the DCC [Bir19, 2.2] and Theorems 3.3 and 2.21 apply to $\Phi(\mathfrak{R})$.

Since $b_{\mathfrak{R}} \mid a$, for any positive integer N with $N_0 \mid N$, the roundings in Construction 7.2(6) and (7) are compatible (Lemma 7.6(1)). The factor 2 in a gives that N_0 and N are even, because for odd N the Poincaré residue of $(-N)$-canonical forms along the strata of $\tilde{S}$ is defined only up to sign [Kol16, Definition 13], and [Kol16, Proposition 14], used in Lemma 7.11, requires an even power (the residues of dx/x at 0 and ∞ on the fibers $\mathbb{P}^1$ of a $\mathbb{P}^1$-link are 1 and -1).

7.2. **Log canonical centers and residues.** In this subsection, we study the lc centers of (X, B) contained in Σ_V, and the restrictions of $\mathcal{O}_{\tilde{X}}(L_N)$ to their strict transforms on $\tilde{X}$.

Lemma 7.4. *Notation and conditions as in Set-up 7.1 and Construction 7.2. Then for any stratum $\tilde{Y}$ of $\tilde{S}$, the variety $Y := \mu(\tilde{Y})$ is an lc center of (X, B), μ is an isomorphism over the generic point of Y, and $\tilde{Y} = \mu_*^{-1} Y$. Conversely, $\mu_*^{-1} Y$ is a stratum of $\tilde{S}$ for any lc center Y of (X, B).*

Proof. Since μ is a log resolution of (X, B), $\operatorname{Exc}(\mu)$ is a divisor and $\operatorname{Exc}(\mu) \cup \operatorname{Supp} \tilde{S}$ is a simple normal crossing divisor. Let $\tilde{Y}$ be an irreducible component of $\tilde{S}_1 \cap \cdots \cap \tilde{S}_k$ for distinct components $\tilde{S}_1, \dots, \tilde{S}_k$ of $\tilde{S}$, so that $\operatorname{codim} \tilde{Y} = k$. If $\tilde{Y} \subset \operatorname{Exc}(\mu)$, then $\tilde{Y} \subset P$ for an exceptional prime divisor P, and $\tilde{Y}$ is contained in the intersection of the $k + 1$ components $\tilde{S}_1, \dots, \tilde{S}_k, P$ of a simple normal crossing divisor, which has codimension $k + 1$, a contradiction. Thus μ is an isomorphism over a neighborhood of the generic point of $Y := \mu(\tilde{Y})$ (Zariski's main theorem [Har77, Corollary III.11.4], as X is normal), so Y is an irreducible component of $\mu(\tilde{S}_1) \cap \cdots \cap \mu(\tilde{S}_k)$ and $\tilde{Y} = \mu_*^{-1} Y$. Since (X, B) is dlt, the irreducible components of intersections of components of $\lfloor B \rfloor$ are exactly the lc centers of (X, B) [Kol13, Theorem 4.16]. Hence Y is an lc center of

(X, B). Conversely, let Y be an lc center of (X, B). Then Y is an irreducible component of an intersection of components of $\lfloor B \rfloor$ [Kol13, Theorem 4.16], and μ is an isomorphism over the generic point of Y (Construction 7.2(4)), so $\mu_*^{-1}Y$ is a stratum of $\tilde{S}$. □

Lemma 7.5. *Notation and conditions as in Set-up 7.1 and Construction 7.2. Then V is normal, (W, B_W) is klt, V' is normal, G is finite, ρ induces an isomorphism $V'/G \to V$, and any element of G is induced by a birational map $W \dashrightarrow W$ over V which is crepant for (W, B_W). Moreover, the coefficients of E are less than 1, and any lc center of (X, B) contained in Σ_V, and any irreducible component of a non-empty intersection of such lc centers, has image V on T.*

Proof. We first show that the coefficients of E are less than 1. Since $\mu_* E = B - \lfloor B \rfloor$, the coefficient of E along any prime divisor on $\tilde{X}$ which is not μ-exceptional is less than 1. Since (X, B) is lc, the coefficients of E are at most 1. Suppose that a μ-exceptional prime divisor P on $\tilde{X}$ has coefficient 1 in E. Then $a(P, X, B) = 0$, so $\mu(P)$ is an lc center of (X, B). Since μ is an isomorphism over the generic point of $\mu(P)$, no μ-exceptional prime divisor has center $\mu(P)$ on X, a contradiction.

Next we show that V is normal. We have

$$K_{\tilde{X}} + \tilde{S} + E = \mu^*(K_X + B) \sim_{\mathbb{Q}} -(g\mu)^* H.$$

Since $\mu_* E \geq 0$ and the coefficients of E are less than 1, $\lceil -E \rceil \geq 0$ is a μ-exceptional divisor. Thus $\mu_* \mathcal{O}_{\tilde{X}}(\lceil -E \rceil) = \mathcal{O}_X$. Since $g_* \mathcal{O}_X = \mathcal{O}_T$, the natural map $\mathcal{O}_T \to (g\mu)_* \mathcal{O}_{\tilde{X}}(\lceil -E \rceil)$ is an isomorphism. Hence $g\mu : (\tilde{X}, \tilde{S} + E) \to T$ defines a quasi-log structure on $[T, -H]$ [Fuj14, Definition 3.2] (here $(\tilde{X}, \tilde{S} + E)$ is a globally embedded simple normal crossing pair, viewing $\tilde{X}$ as the divisor $\tilde{X} \times \{0\}$ of $\tilde{X} \times \mathbb{A}^1$ [Fuj14, Definition 3.1]). Since $\tilde{S} + E$ has snc support and the coefficients of E are less than 1, the lc centers of $(\tilde{X}, \tilde{S} + E)$ are the strata of $\tilde{S}$. By Lemma 7.4, the images on X of the strata of $\tilde{S}$ are exactly the lc centers of (X, B). Thus the qlc centers of $[T, -H]$ are T and the images on T of the lc centers of (X, B). Let $v \in V$ be a point. By [Fuj14, Theorem 3.4(ii)], there exists a unique minimal qlc center T_v of $[T, -H]$ containing v, and T_v is normal at v. Since V is a qlc center containing v, we have $T_v \subset V \subsetneq T$, so T_v is the image on T of an lc center of (X, B). By Set-up 7.1(5), $T_v = V$. Thus V is normal at v. Since $v \in V$ is arbitrary, V is normal.

Next we show that the lc centers contained in Σ_V, and the irreducible components of their intersections, have image V on T. Let C be an lc center of (X, B) contained in Σ_V. Then $g(C) \subset g(\Sigma_V) = V$, so $g(C) = V$ by Set-up 7.1(5). Since (X, B) is dlt, the lc centers of (X, B) are exactly the irreducible components of the intersections of components of $\lfloor B \rfloor$ [Kol13, Theorem 4.16]. Thus any irreducible component R of a non-empty intersection of lc centers of (X, B) contained in Σ_V is an lc center of (X, B) (cf. [Kol13, Corollary 4.41(2)]) contained in Σ_V, and so $g(R) = V$.

Next we show that (W, B_W) is klt. By Lemma 2.2(1)(3), the pair (W, B_W) is dlt, and any lc center of (W, B_W) is an lc center of (X, B). Suppose that C_W is an lc center of (W, B_W). Then $C_W \subsetneq W$ is an lc center of (X, B) contained in Σ_V, so $g(C_W) = V$ by the previous paragraph. This contradicts the minimality of W in Construction 7.2(2). Thus (W, B_W) has no lc center, and since (W, B_W) is lc, (W, B_W) is klt.

Finally we show the conclusions on V' and G. Since W is normal (Lemma 2.2(1)) and $W \xrightarrow{h} V' \xrightarrow{\rho} V$ is the Stein factorization of $W \to V$, V' is normal and ρ is finite. Since $K_X + B \sim_{\mathbb{Q}} -g^* H$, $g : (X, B) \to T$ is a crepant log structure [Kol16, Definition 2]. By the minimality of W in Construction 7.2(2), (W, B_W) is a source of V, and V' is the spring of V, with respect to g [Kol16, Definition 18]. By [Kol16, Proposition 19 and Lemma 20], the function field of V' is a Galois extension of the function field of V, and any automorphism of the function field of V' over the function field of V is induced by a birational map $W \dashrightarrow W$ over V which is crepant for (W, B_W). Since V' is normal and finite over V, V' is the normalization of V in the function field of V'. Thus any such automorphism of function fields extends uniquely to an automorphism

of V' over V, and any automorphism of V' over V is determined by the pullback of rational functions on V' under it. Hence G is the Galois group of the function field of V' over the function field of V, and in particular G is finite.

Since V' is quasi-projective, the quotient V'/G exists [Gro71, Exp. V, Proposition 1.8], it is a normal variety since V' is, and ρ induces a finite morphism $V'/G \to V$. The rational functions on V'/G are the G-invariant rational functions on V', which are the pullbacks of the rational functions on V since the extension is Galois with group G. Thus $V'/G \to V$ is finite and birational, and it is an isomorphism since V is normal (Zariski's main theorem [Har77, Corollary III.11.4]). □

Lemma 7.6. *Notation and conditions as in Set-up 7.1, Construction 7.2, and Construction 7.3. Let N be a positive integer with $N_0 \mid N$, $W' \subset \Sigma_V$ an lc center of (X, B), $\tilde{W}' := \mu_*^{-1}W'$, $\mu_{W'} : \tilde{W}' \to W'$ the induced morphism, $P_{\tilde{W}'}$ the restriction to $\tilde{W}'$ of the components of $\tilde{S}$ not containing $\tilde{W}'$, and $W' \xrightarrow{f_{W'}} V_{W'} \to V$ the Stein factorization of $W' \to V$. Then:*

(1) the Poincaré residue along the components of $\tilde{S}$ containing $\tilde{W}'$ induces an isomorphism from $\mu_{W'*}\big(\mathcal{O}_{\tilde{X}}(L_N)|_{\tilde{W}'}\big)$ onto $\mathcal{J}_{N,W'}$, under which a section corresponds to a rational $(-N)$-canonical form β on $\tilde{W}'$ such that

$$\operatorname{div}(\beta) \geq NP_{\tilde{W}'} + \lfloor (N+1)E|_{\tilde{W}'} \rfloor;$$

(2) for any open subset $V^0 \subset V$ and any non-zero section t of $\mathcal{J}_{N,W'}$ over the inverse image of V^0, over V^0 we have $B_t \geq B_{W'}$ and $B_t = B_{W'} + f_{W'}^* E_t$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $E_t \geq 0$ on the inverse image of V^0 in $V_{W'}$.

Proof. Since $N_0 \mid N$ and $2b_{\mathfrak{R}} \mid N_0$, N is even and $Nr \in \mathbb{Z}$ for any $r \in \mathfrak{R}$. By Construction 7.3, $B_{W'} \in \Phi(\mathfrak{R})$.

(1) By Lemma 7.4, $\tilde{W}'$ is a stratum of $\tilde{S}$ and $\mu_{W'}$ is birational. Let $\tilde{S}_1, \dots, \tilde{S}_k$ be the components of $\tilde{S}$ containing $\tilde{W}'$. Then $\tilde{W}'$ is an irreducible component of $\tilde{S}_1 \cap \cdots \cap \tilde{S}_k$, and $\tilde{W}'$ is smooth. Since $\operatorname{Supp}(\tilde{S} + E)$ is snc, no component of $\tilde{S} - \sum_{i=1}^k \tilde{S}_i$ or of E contains $\tilde{W}'$, and distinct components of $\tilde{S} + E$ restrict to divisors on $\tilde{W}'$ without common components. In particular,

$$\lfloor (N+1)E \rfloor|_{\tilde{W}'} = \lfloor (N+1)E|_{\tilde{W}'} \rfloor,$$

and no prime divisor on $\tilde{W}'$ is contained in both $\operatorname{Supp} P_{\tilde{W}'}$ and $\operatorname{Supp} E|_{\tilde{W}'}$. By adjunction for snc pairs and since $K_{W'} + B_{W'} = (K_X + B)|_{W'}$, we have

$$K_{\tilde{W}'} + P_{\tilde{W}'} + E|_{\tilde{W}'} = (K_{\tilde{X}} + \tilde{S} + E)|_{\tilde{W}'} = \mu_{W'}^*(K_{W'} + B_{W'}). \tag{7.1}$$

In particular, $\mu_{W'*}(P_{\tilde{W}'} + E|_{\tilde{W}'}) = B_{W'}$.

Since N is even, the Poincaré residue along $\tilde{S}_1, \dots, \tilde{S}_k$ does not depend on the order of $\tilde{S}_1, \dots, \tilde{S}_k$ and induces an isomorphism

$$\mathcal{O}_{\tilde{X}}\big(-N(K_{\tilde{X}} + \tilde{S}_1 + \cdots + \tilde{S}_k)\big)|_{\tilde{W}'} \cong \mathcal{O}_{\tilde{W}'}(-NK_{\tilde{W}'})$$

[Kol16, Definition 13]. Since

$$L_N = -N(K_{\tilde{X}} + \tilde{S}_1 + \cdots + \tilde{S}_k) - N\Big(\tilde{S} - \sum_{i=1}^k \tilde{S}_i\Big) - \lfloor (N+1)E \rfloor$$

and the restrictions of the last two divisors to $\tilde{W}'$ are $NP_{\tilde{W}'}$ and $\lfloor (N+1)E|_{\tilde{W}'} \rfloor$, the residue induces an isomorphism from $\mathcal{O}_{\tilde{X}}(L_N)|_{\tilde{W}'}$ onto the sheaf of rational $(-N)$-canonical forms β on $\tilde{W}'$ such that

$$\operatorname{div}(\beta) \geq NP_{\tilde{W}'} + \lfloor (N+1)E|_{\tilde{W}'} \rfloor,$$

together with 0.

Since $\mu_{W'}$ is birational, the rational $(-N)$-canonical forms on $\tilde{W}'$ are the rational $(-N)$-canonical forms on W', and

$$\operatorname{div}_{W'}(\beta) = \mu_{W'*} \operatorname{div}_{\tilde{W}'}(\beta)$$

for any such form β. Thus it suffices to show the following: for any open subset $W'^0 \subset W'$ and any rational $(-N)$-canonical form β on W',

$$\operatorname{div}_{\tilde{W}'}(\beta) \geq NP_{\tilde{W}'} + \lfloor (N+1)E|_{\tilde{W}'} \rfloor \text{ over } \mu_{W'}^{-1}(W'^0) \tag{7.2}$$

holds if and only if

$$\operatorname{div}_{W'}(\beta) \geq \lceil NB_{W'} \rceil \text{ over } W'^0. \tag{7.3}$$

Let Q be a prime divisor on W' which intersects W'^0, $\tilde{Q} := \mu_{W'*}^{-1}Q$, and $b := \operatorname{mult}_Q B_{W'}$. Then $b \in \Phi(\mathfrak{R})$, and by (7.1), b is the coefficient of $\tilde{Q}$ in $P_{\tilde{W}'} + E|_{\tilde{W}'}$. Since the coefficients of E are less than 1 (Lemma 7.5), $b = 1$ if and only if $\tilde{Q} \subset \operatorname{Supp} P_{\tilde{W}'}$. Thus the coefficient of $\tilde{Q}$ in $NP_{\tilde{W}'} + \lfloor (N+1)E|_{\tilde{W}'} \rfloor$ is

$$N\lfloor b \rfloor + \lfloor (N+1)\{b\} \rfloor = \lceil Nb \rceil,$$

where the equality follows from Lemma 2.13(2). Hence

$$\mu_{W'*}\big(NP_{\tilde{W}'} + \lfloor (N+1)E|_{\tilde{W}'} \rfloor\big) = \lceil NB_{W'} \rceil.$$

Pushing forward (7.2) by $\mu_{W'}$, we see that (7.2) implies (7.3).

Conversely, assume (7.3). Then

$$NB_\beta = \operatorname{div}_{W'}(\beta) \geq \lceil NB_{W'} \rceil \geq NB_{W'} \text{ over } W'^0,$$

so $B_\beta - B_{W'} \geq 0$ over W'^0. Since $N(K_{W'} + B_\beta) \sim 0$ and $K_{W'} + B_{W'} = (K_X + B)|_{W'}$, the $\mathbb{Q}$-divisor $B_\beta - B_{W'} = (K_{W'} + B_\beta) - (K_{W'} + B_{W'})$ is $\mathbb{Q}$-Cartier. Since $N(K_{W'} + B_\beta)$ is the principal divisor whose pullback to $\tilde{W}'$ is $NK_{\tilde{W}'} + \operatorname{div}_{\tilde{W}'}(\beta)$, we have

$$K_{\tilde{W}'} + \frac{1}{N}\operatorname{div}_{\tilde{W}'}(\beta) = \mu_{W'}^*(K_{W'} + B_\beta).$$

By (7.1),

$$\frac{1}{N}\operatorname{div}_{\tilde{W}'}(\beta) - \big(P_{\tilde{W}'} + E|_{\tilde{W}'}\big) = \mu_{W'}^*(B_\beta - B_{W'}) \geq 0 \text{ over } \mu_{W'}^{-1}(W'^0).$$

Let $\tilde{Q}$ be a prime divisor on $\tilde{W}'$ which intersects $\mu_{W'}^{-1}(W'^0)$, e the coefficient of $\tilde{Q}$ in $P_{\tilde{W}'} + E|_{\tilde{W}'}$, and e^+ the coefficient of $\tilde{Q}$ in $\frac{1}{N}\operatorname{div}_{\tilde{W}'}(\beta)$. Then $e^+ \geq e$ and $Ne^+ \in \mathbb{Z}$. If $\tilde{Q} \subset \operatorname{Supp} P_{\tilde{W}'}$, then $e = 1$, $\tilde{Q} \not\subset \operatorname{Supp} E|_{\tilde{W}'}$, and the coefficient of $\tilde{Q}$ in $NP_{\tilde{W}'} + \lfloor (N+1)E|_{\tilde{W}'} \rfloor$ is $N \leq Ne^+$. Otherwise, e is a coefficient of E, so $e < 1$ by Lemma 7.5, and e may be negative. For any real number $e < 1$, we have $(N+1)e < Ne + 1 \leq \lceil Ne \rceil + 1$, so

$$\lfloor (N+1)e \rfloor \leq \lceil Ne \rceil \leq Ne^+,$$

where the second inequality holds since Ne^+ is an integer and $Ne^+ \geq Ne$. Thus (7.2) holds. This proves (1).

(2) Let $V_{W'}^0$ be the inverse image of V^0 in $V_{W'}$. Then $f_{W'}^{-1}(V_{W'}^0)$ is the inverse image of V^0 in W', and $f_{W'}^{-1}(V_{W'}^0) \to V_{W'}^0$ is a contraction. Over $V_{W'}^0$, we have

$$NB_t = \operatorname{div}(t) \geq \lceil NB_{W'} \rceil \geq NB_{W'},$$

so $B_t \geq B_{W'}$. Since $N(K_{W'} + B_t) \sim 0$ and $K_{W'} + B_{W'} = (K_X + B)|_{W'}$, the $\mathbb{Q}$-divisor $B_t - B_{W'} = (K_{W'} + B_t) - (K_{W'} + B_{W'})$ is $\mathbb{Q}$-Cartier. Let $H_{W'}$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on $V_{W'}$ which is $\mathbb{Q}$-linearly equivalent to the pullback of H under $V_{W'} \to V \subset T$. Then

$$B_t - B_{W'} \sim_{\mathbb{Q}} -(K_{W'} + B_{W'}) = -(K_X + B)|_{W'} \sim_{\mathbb{Q}} (g^*H)|_{W'} \sim_{\mathbb{Q}} f_{W'}^* H_{W'}$$

over $V_{W'}^0$. By Lemma 2.17(3), there exists a unique $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $E_t \geq 0$ on $V_{W'}^0$ such that $B_t - B_{W'} = f_{W'}^* E_t$ over $V_{W'}^0$. □

7.3. **Invariant sections.** In this subsection, we prove Theorem 7.13 and Proposition 7.14. As in part (b) of the sketch of the proof in Section 1, Lemma 7.11(2), proved with Lemma 7.10, shows that the image of the restriction map in Theorem 7.13(2) consists of G-invariant sections, and Lemma 7.12 proves the converse. We first fix a rational a-canonical form $\theta_{W'}$ on any lc center W' of (X, B) contained in Σ_V (Construction-Proposition 7.7) and define residue maps between such lc centers (Construction-Proposition 7.8).

Construction-Proposition 7.7. Notation and conditions as in Set-up 7.1, Construction 7.2, and Construction 7.3. Let N be a positive integer with $N_0 \mid N$, $W' \subset \Sigma_V$ an lc center of (X, B), and $W' \xrightarrow{f_{W'}} V_{W'} \to V$ the Stein factorization of $W' \to V$. Then there exist a rational a-canonical form $\theta_{W'}$ on W' and a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D_{W'}$ on $V_{W'}$ such that

$$\operatorname{div}(\theta_{W'}) = -aB_{W'} + af_{W'}^*D_{W'}.$$

We fix such $\theta_{W'}$ and $D_{W'}$. Let $M_{N,W'}$ be the set of rational $(-N)$-canonical forms t on W' such that $f_{W'}(\operatorname{Supp}(B_t - B_{W'})) \neq V_{W'}$, together with 0. Then:

(1) any $t \in M_{N,W'}\backslash\{0\}$ can be written uniquely as $t = f_{W'}^*\chi \cdot \theta_{W'}^{-N/a}$ with χ a non-zero rational function on $V_{W'}$, and

$$B_t = B_{W'} + f_{W'}^*E_t, \qquad E_t := -D_{W'} + \frac{1}{N}\operatorname{div}(\chi);$$

(2) for any non-empty open subset $V^0 \subset V$, any non-zero section of $\mathcal{J}_{N,W'}$ over the inverse image of V^0 belongs to $M_{N,W'}$, and an element $t \in M_{N,W'}\backslash\{0\}$ is a section of $\mathcal{J}_{N,W'}$ over the inverse image of V^0 if and only if $E_t \geq 0$ over V^0;

(3) if t is a section of $\mathcal{J}_{N,W'}$ over the inverse image of an open subset $V^0 \subset V$, then over V^0 the $\mathbb{Q}$-divisor E_t of (1) is the $\mathbb{Q}$-divisor E_t of Lemma 7.6(2).

Proof. Since $V \neq T$, W' is contained in $\lfloor B \rfloor$, so $\dim W' \leq d-1$. By Lemma 2.2 and Set-up 7.1(4), $K_{W'} + B_{W'} \sim_{\mathbb{Q}} -(g|_{W'})^*H$, and $-(g|_{W'})^*H$ is the pullback by $f_{W'}$ of the pullback of $-H$ to $V_{W'}$. Moreover, $B_{W'} \in \Phi(\mathfrak{R})$ by Construction 7.3. Fix a non-zero rational 1-canonical form ω on W' and let $K_{W'} := \operatorname{div}(\omega)$.

Suppose that $\dim V > 0$. Then $0 < \dim V_{W'} \leq \dim W' \leq d-1$. By Theorem 3.3(1), applied to $f_{W'}$ and $(W', B_{W'})$, there exist a rational function ψ on W' and a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D_{W'}$ on $V_{W'}$ such that

$$K_{W'} + B_{W'} + \frac{1}{m}\operatorname{div}(\psi) = f_{W'}^*D_{W'}.$$

Since $m \mid a$, we let $\theta_{W'} := \omega^a\psi^{a/m}$. Suppose that $\dim V = 0$. Then V and $V_{W'}$ are points and we let $D_{W'} := 0$. If $\dim W' = 0$, then $B_{W'} = 0$ and we let $\theta_{W'} := 1$. Otherwise, W' is projective and $K_{W'} + B_{W'} \sim_{\mathbb{Q}} 0$, so Theorem 2.21 and $I(\dim W', \Phi(\mathfrak{R})) \mid a$ imply that $a(K_{W'} + B_{W'}) = \operatorname{div}(\lambda)$ for a rational function λ on W', and we let $\theta_{W'} := \omega^a\lambda^{-1}$. In both cases a direct calculation shows $\operatorname{div}(\theta_{W'}) = -aB_{W'} + af_{W'}^*D_{W'}$.

(1) Let $\varphi := t \cdot \theta_{W'}^{N/a}$, a rational function on W'. Then

$$\operatorname{div}(\varphi) = N(B_t - B_{W'}) + Nf_{W'}^*D_{W'}.$$

By the definition of $M_{N,W'}$, the closed subset $f_{W'}(\operatorname{Supp}(B_t - B_{W'}))$ is not $V_{W'}$, and no component of $f_{W'}^*D_{W'}$ dominates $V_{W'}$. Thus no component of $\operatorname{div}(\varphi)$ dominates $V_{W'}$, and by Lemma 2.17(4), $\varphi = f_{W'}^*\chi$ for a non-zero rational function χ on $V_{W'}$, which is unique since $f_{W'}$ is surjective. The formula for $\operatorname{div}(\varphi)$ then implies $B_t = B_{W'} + f_{W'}^*E_t$.

(2) The first statement follows from Lemma 7.6(2). An element $t \in M_{N,W'}\backslash\{0\}$ is a section of $\mathcal{J}_{N,W'}$ over an open subset of W' if and only if $\operatorname{div}(t) \geq \lceil NB_{W'} \rceil$ there. Since $\operatorname{div}(t)$ is a divisor, the inequality holds if and only if $B_t \geq B_{W'}$ there, that is, $f_{W'}^*E_t \geq 0$ there. Hence the second statement follows from Lemma 2.17(1).

(3) Over V^0, the pullbacks by $f_{W'}$ of the $\mathbb{Q}$-divisor E_t of (1) and of the $\mathbb{Q}$-divisor E_t of Lemma 7.6(2) are both equal to $B_t - B_{W'}$. Since $f_{W'}^*$ is injective on $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors (Lemma 2.17(1)), the $\mathbb{Q}$-divisor E_t of (1) is the $\mathbb{Q}$-divisor E_t of Lemma 7.6(2) over V^0. □

Since rational $(-N)$-canonical forms on W' which coincide over a non-empty open subset of V are equal, the residue maps below are defined on all of $M_{N,W'}$ without a choice of an open subset of V. For an lc center $W'' \subset W'$ of (X, B) with $W'' \subset \Sigma_V$, the variety $\tilde{W}''$ is contained in $\tilde{W}'$, since μ is an isomorphism over the generic point of W''. The Poincaré residue of [Kol16, Definition 13] then induces maps between the sets $M_{N,W'}$ as follows.

Construction-Proposition 7.8. Notation and conditions as in Construction-Proposition 7.7. Let $W'' \subset W'$ be lc centers of (X, B) contained in Σ_V. For $t \in M_{N,W'} \backslash \{0\}$, let $V^0 \subset V$ be a non-empty open subset over which $E_t \geq 0$ (so that t is a section of $\mathcal{J}_{N,W'}$ over the inverse image of V^0 by Construction-Proposition 7.7(2)), and let $\mathrm{Res}_{W',W''}(t)$ be the image of t under the composition of the isomorphism of Lemma 7.6(1) for W', the restriction from $\tilde{W}'$ to $\tilde{W}''$ over V^0, and the isomorphism of Lemma 7.6(1) for W'', regarded as a rational $(-N)$-canonical form on W''. Let $\mathrm{Res}_{W',W''}(0) := 0$. Thus $\mathrm{Res}_{W',W''}$ takes rational $(-N)$-canonical forms on W' to rational $(-N)$-canonical forms on W''. For a union $\tilde{Z}$ of strata of $\tilde{S}$, we write $\mathcal{O}_{\tilde{Z}}(L_N) := \mathcal{O}_{\tilde{X}}(L_N)|_{\tilde{Z}}$. For a union $\tilde{Z}$ of strata of $\tilde{S}$ containing $\tilde{W}'$, a non-empty open subset $V^0 \subset V$, and a section s of $\mathcal{O}_{\tilde{Z}}(L_N)$ over the inverse image of V^0, let $s_{W'}$ be the image of $s|_{\tilde{W}'}$ under the isomorphism of Lemma 7.6(1). Then:

(1) $\mathrm{Res}_{W',W''}(t)$ does not depend on V^0, $\mathrm{Res}_{W',W''}(t) \in M_{N,W''}$, $s_{W'} \in M_{N,W'}$, and

$$s_{W''} = \mathrm{Res}_{W',W''}(s_{W'}), \qquad \mathrm{Res}_{W'',W'''} \circ \mathrm{Res}_{W',W''} = \mathrm{Res}_{W',W'''}$$

for any lc center $W''' \subset W''$ of (X, B);

(2) for any rational function χ on $V_{W'}$ and any $t \in M_{N,W'}$,

$$\mathrm{Res}_{W',W''}(f_{W'}^*\chi \cdot t) = (f_{W'}|_{W''})^*\chi \cdot \mathrm{Res}_{W',W''}(t);$$

(3) for any $t \in M_{N,W'} \backslash \{0\}$, $\mathrm{Res}_{W',W''}(t) \neq 0$ and

$$B_{\mathrm{Res}_{W',W''}(t)} = B_{W''} + (f_{W'}|_{W''})^* E_t.$$

Proof. The map $W'' \to V_{W'}$ is surjective: $f_{W'}(W'')$ is closed and irreducible, and it is finite over V with image V, so $\dim f_{W'}(W'') = \dim V = \dim V_{W'}$, and $f_{W'}(W'') = V_{W'}$ since $V_{W'}$ is irreducible. Hence $(f_{W'}|_{W''})^*\chi$ is defined, and the sections over smaller open subsets of V are restrictions of the sections over larger ones. Thus $\mathrm{Res}_{W',W''}(t)$ does not depend on V^0.

(2) Since Lemma 7.6(1) is an isomorphism of $\mathcal{O}_{W'}$-modules and restriction is $\mathcal{O}$-linear, (2) holds for χ regular over V^0, and the general case follows by writing χ as a quotient of two such functions.

(3) If $W'' = W'$, then $\mathrm{Res}_{W',W'}$ is the identity and (3) is Construction-Proposition 7.7(1). Suppose that $W'' \subsetneq W'$. By (2) and Construction-Proposition 7.7(1), we may assume that $t = \theta_{W'}^{-N/a}$, so that $E_t = -D_{W'}$. Let q be a positive integer such that $qaD_{W'}$ is Cartier. Locally on $V_{W'}$, write $qaD_{W'} = \mathrm{div}(\lambda)$. Then $f_{W'}^*\lambda^{-1} \cdot \theta_{W'}^q$ is a local generator of $\omega_{W'}^{[qa]}(qaB_{W'})$, viewed as the sheaf of rational qa-canonical forms η with $\mathrm{div}(\eta) \geq -qaB_{W'}$. Since $(W', B_{W'})$ is dlt, qa is even, and $B_{W''}$ is the different of $(W', B_{W'})$ on W'' (Lemma 2.2(1)(2), applied to $W'' \subsetneq W'$), the Poincaré residue of $f_{W'}^*\lambda^{-1} \cdot \theta_{W'}^q$ along W'' is a local generator of $\omega_{W''}^{[qa]}(qaB_{W''})$ [Kol13, Theorem 4.19(2)]. Since the Poincaré residue is multiplicative, as both residues are the snc residue at the generic point of W'', where μ is an isomorphism [KM98, Definition 2.37] (cf. [Kol13, (4.18.2)]), we obtain

$$\mathrm{div}\left(\mathrm{Res}_{W',W''}(\theta_{W'}^{-N/a})\right) = NB_{W''} - N(f_{W'}|_{W''})^* D_{W'},$$

which is (3) for $t = \theta_{W'}^{-N/a}$, and in particular $\mathrm{Res}_{W',W''}(t) \neq 0$.

(1) The isomorphisms of Lemma 7.6(1) are given by Poincaré residues, which are transitive along strata [Kol16, Definition 13]. Hence $s_{W''} = \mathrm{Res}_{W',W''}(s_{W'})$ and the transitivity in (1) hold. The relation $s_{W'} \in M_{N,W'}$ follows from Construction-Proposition 7.7(2), and $\mathrm{Res}_{W',W''}(t) \in M_{N,W''}$ follows from (3). □

Lemma 7.9. *Notation and conditions as in Construction-Proposition 7.7. Let $W', W'' \subset \Sigma_V$ be lc centers of (X, B) and $\phi : W'' \dashrightarrow W'$ a birational map over V which is crepant for $(W'', B_{W''})$ and $(W', B_{W'})$. Then ϕ induces an isomorphism $\kappa : V_{W''} \to V_{W'}$ over V with $f_{W'} \circ \phi = \kappa \circ f_{W''}$, and for any $t \in M_{N,W'} \backslash \{0\}$, we have $\phi^* t \in M_{N,W''}$, $E_{\phi^* t} = \kappa^* E_t$, and ϕ is crepant for $(W'', B_{\phi^* t})$ and (W', B_t). In particular, for any non-empty open subset $V^0 \subset V$, ϕ^* maps sections of $\mathcal{J}_{N,W'}$ over the inverse image of V^0 to sections of $\mathcal{J}_{N,W''}$ over the inverse image of V^0.*

Proof. Let $p : Y \to W''$ and $q : Y \to W'$ be projective birational morphisms from a normal variety with $q = \phi \circ p$ as in Definition 2.4. Since ϕ is over V, the composite $\pi_Y : Y \to V$ of q with $W' \to V$ is also the composite of p with $W'' \to V$. Since W' and W'' are normal, $q_* \mathcal{O}_Y = \mathcal{O}_{W'}$ and $p_* \mathcal{O}_Y = \mathcal{O}_{W''}$, so $f_{W'} \circ q$ and $f_{W''} \circ p$ are contractions. Since $V_{W'} \to V$ is finite, hence affine, $V_{W'}$ is the relative spectrum over V of $\pi_{Y*} \mathcal{O}_Y$, and so is $V_{W''}$ (cf. [Har77, Corollary III.11.5]). Hence there exists a unique isomorphism $\kappa : V_{W''} \to V_{W'}$ over V with $f_{W'} \circ q = \kappa \circ f_{W''} \circ p$, and $f_{W'} \circ \phi = \kappa \circ f_{W''}$ as rational maps.

Fix a non-zero rational 1-canonical form ω on W', and let $K_{W'} := \mathrm{div}(\omega)$, $K_Y := \mathrm{div}(q^* \omega)$, and $K_{W''} := p_* K_Y$. Writing $t = \psi \omega^{-N}$ with a non-zero rational function ψ on W', we have $N(K_{W'} + B_t) = \mathrm{div}(\psi)$, and hence

$$K_Y + \frac{1}{N} \mathrm{div}_Y(q^* t) = q^*(K_{W'} + B_t) = q^*(K_{W'} + B_{W'}) + p^* f_{W''}^* \kappa^* E_t.$$

Since ϕ is crepant for $(W'', B_{W''})$ and $(W', B_{W'})$, $q^*(K_{W'} + B_{W'}) = p^*(K_{W''} + B_{W''})$. Since $q^* t = p^* \phi^* t$, pushing forward by p implies $B_{\phi^* t} = B_{W''} + f_{W''}^* \kappa^* E_t$. Thus $\phi^* t \in M_{N,W''}$, $E_{\phi^* t} = \kappa^* E_t$, and $q^*(K_{W'} + B_t) = p^*(K_{W''} + B_{\phi^* t})$, so ϕ is crepant for $(W'', B_{\phi^* t})$ and (W', B_t). The last statement follows from Construction-Proposition 7.7(2). □

Lemma 7.10. *Notation and conditions as in Construction-Proposition 7.7. Let $W' \subset \Sigma_V$ be an lc center of (X, B) which is minimal with respect to inclusion among the lc centers of (X, B) with image V, $\mathbb{K}$ a finite extension of $\mathbb{C}(V_{W'})$, $F := W' \times_{V_{W'}} \mathrm{Spec}\, \mathbb{K}$, and $\alpha : F \dashrightarrow F$ a birational map over $\mathbb{K}$ which is crepant for $(F, B_{W'}|_F)$. Then $\alpha^* t = t$ for any $t \in M_{N,W'}$, where t is regarded as a rational $(-N)$-canonical form on F over $\mathbb{C}$.*

Proof. Since $f_{W'}$ is a contraction, F is the base change to $\mathbb{K}$ of a geometrically normal and geometrically integral projective variety, namely the fiber of $f_{W'}$ over the generic point of $V_{W'}$ (Lemma 2.17(4)). Thus F is a geometrically integral normal projective variety over $\mathbb{K}$. By Lemma 7.5, applied with W' in place of W, $(W', B_{W'})$ is klt, so (F, B_F) is klt, where $B_F := B_{W'}|_F$. Let $r := \dim W' - \dim V \leq d - 1$. The exact sequence of Kähler differentials of $\mathbb{C} \subset \mathbb{K} \subset \mathbb{C}(F)$ induces an isomorphism

$$\bigwedge^{\dim W'} \Omega_{\mathbb{C}(F)/\mathbb{C}} \cong \bigwedge^{r} \Omega_{\mathbb{C}(F)/\mathbb{K}} \otimes_{\mathbb{K}} \bigwedge^{\dim V} \Omega_{\mathbb{K}/\mathbb{C}}, \tag{7.4}$$

which is compatible with α^*, and α^* is the identity on $\bigwedge^{\dim V} \Omega_{\mathbb{K}/\mathbb{C}}$. Write $\theta_{W'} = \theta_F \otimes \xi$ under the isomorphism (7.4), where θ_F is a rational a-canonical form on F over $\mathbb{K}$. Since $f_{W'}^* D_{W'}$ is zero on F, $\mathrm{div}(\theta_F) = -aB_F$.

Let $\bar{\mathbb{K}}$ be an algebraic closure of $\mathbb{K}$. Since $\mathbb{K}$ is finitely generated over $\mathbb{C}$, $\bar{\mathbb{K}}$ is an algebraically closed field of characteristic zero with the cardinality of $\mathbb{C}$, so $\bar{\mathbb{K}} \cong \mathbb{C}$ by Steinitz's theorem. Projectivity, the klt property, the equality $\mathrm{div}(\theta_F) = -aB_F$, and crepancy of α are preserved by the base change from $\mathbb{K}$ to $\bar{\mathbb{K}}$, since the base change of a log resolution is a log resolution and the discrepancies of corresponding divisors are equal. Thus, after an isomorphism $\bar{\mathbb{K}} \cong \mathbb{C}$, by Theorem 6.2 applied to the base change of (F, B_F), θ_F, and α, we have $\alpha^* \theta_F = c\theta_F$ for some

$c \in \bar{\mathbb{K}}$ with $c^{\ell(r,a)} = 1$, and $c \in \mathbb{K}$ since $c^{\ell(r,a)} = 1$ and the $\ell(r,a)$-th roots of unity in $\bar{\mathbb{K}}$ belong to $\mathbb{C} \subset \mathbb{K}$. Hence $\alpha^*\theta_{W'} = c\theta_{W'}$.

Let $t \in M_{N,W'} \backslash \{0\}$. By Construction-Proposition 7.7(1), $t = f_{W'}^*\chi \cdot \theta_{W'}^{-N/a}$ with χ a non-zero rational function on $V_{W'}$. On F, $f_{W'}^*\chi$ is the pullback of χ by $F \to \operatorname{Spec} \mathbb{K} \to V_{W'}$. Since α is over $\mathbb{K}$, $\alpha^*t = c^{-N/a}t$. By Construction 7.3, $\ell(r,a)$ divides N_0/a, hence N/a. Thus $\alpha^*t = t$. □

Minimal lc centers with image V are compared by $\mathbb{P}^1$-links [Kol16, Definition 9, Theorem 10]. The next lemma shows that crepant birational maps between minimal lc centers with image V carry the restriction of a section on the strata to one such center onto the restriction of the section to the other.

Lemma 7.11. *Notation and conditions as in Construction-Proposition 7.7. Let $(Z, B_Z, \tilde{Z}, V_Z)$ be one of the following:*

(i) $(X, B, \tilde{\Sigma}_V, T)$;

(ii) $(W_1, B_{W_1}, \tilde{W}_1, V_{W_1})$ for an lc center $W_1 \subset \Sigma_V$ of (X, B).

Let $W', W'' \subset Z$ be lc centers of (X, B) which are minimal with respect to inclusion among the lc centers of (X, B) with image V. In case (i), a birational map $W'' \dashrightarrow W'$ is over $V_Z = T$ if and only if it is over V, since W' and W'' have image V on T. Then:

(1) there exists a birational map $W'' \dashrightarrow W'$ over V_Z which is crepant for $(W'', B_{W''})$ and $(W', B_{W'})$;

(2) for any birational map $\phi : W'' \dashrightarrow W'$ over V_Z which is crepant for $(W'', B_{W''})$ and $(W', B_{W'})$, any non-empty open subset $V^0 \subset V$, and any section s of $\mathcal{O}_{\tilde{Z}}(L_N)$ over the inverse image of V^0, we have $\phi^*(s_{W'}) = s_{W''}$.

Proof. If we are in case (ii) and W_1 itself is minimal with respect to inclusion among the lc centers of (X, B) with image V, then $W' = W'' = W_1$, and (1) holds with the identity map. Let $f_1 := f_{W_1}$. We prove (2) in this case using the general fibers of f_1, that is, the fibers over the closed points of a non-empty open subset of V_{W_1}. General fibers suffice, since each property of the fibers used below holds for the fibers over all closed points of one non-empty open subset of V_{W_1}, and on each such fiber we use Theorem 6.2 for the single integer $\ell(r,a)$.

Let $\phi : W_1 \dashrightarrow W_1$ be a birational map over V_{W_1} which is crepant for (W_1, B_{W_1}). Let s be a section as in (2). By Construction-Proposition 7.8(1), $s_{W_1} \in M_{N,W_1}$, and we may assume that $s_{W_1} \neq 0$. Let $\theta := \theta_{W_1}$, so that $\operatorname{div}(\theta) = -aB_{W_1} + af_1^*D_{W_1}$. By Construction-Proposition 7.7(1), $s_{W_1} = f_1^*\chi \cdot \theta^{-N/a}$ for a non-zero rational function χ on V_{W_1}. We apply the computation of **Step 1** of the proof of Lemma 6.5 with f_1, θ, a, D_{W_1}, and the identity of V_{W_1} in place of h, Θ, m, $D_{V'}$, and σ. Since ϕ is crepant for (W_1, B_{W_1}) and $f_1 \circ \phi = f_1$, the computation shows that $\phi^*\theta = f_1^*c \cdot \theta$ for a regular function c without zeros on the normal variety V_{W_1}.

Let $r := \dim W_1 - \dim V$. Since $V \neq T$, W_1 is contained in $\lfloor B \rfloor$, so $r \leq d-1$. By Lemma 7.5, applied with W_1 in place of W, (W_1, B_{W_1}) is klt. If $\dim V = 0$, then V_{W_1} is a point, $D_{W_1} = 0$, W_1 is projective, and Theorem 6.2 applied to (W_1, B_{W_1}), θ, and ϕ implies $c^{\ell(r,a)} = 1$. If $\dim V > 0$, let ω_1 be a rational 1-canonical form on V_{W_1}. As in **Step 2** of the proof of Lemma 6.5, applied to f_1, (W_1, B_{W_1}), θ, ϕ, and ω_1, there exists a non-empty open subset $V_1^\circ \subset V_{W_1}$ satisfying the following: for any closed point $v \in V_1^\circ$, the fiber $W_v := f_1^{-1}(v)$ is a normal projective variety of dimension r, the pair (W_v, B_v) with $B_v := B_{W_1}|_{W_v}$ is klt, ϕ restricts to a birational map $\phi_v : W_v \dashrightarrow W_v$ which is crepant for (W_v, B_v), and $\theta/f_1^*\omega_1^{\otimes a}$ restricts to a rational a-canonical form θ_v on W_v with $\operatorname{div}(\theta_v) = -aB_v$. Let $v \in V_1^\circ$ be a closed point. Since $f_1 \circ \phi = f_1$, we have $\phi_v^*\theta_v = c(v)\theta_v$, and Theorem 6.2 implies $c(v)^{\ell(r,a)} = 1$. Since the closed points of V_1° are dense in V_{W_1}, $c^{\ell(r,a)} = 1$.

In both cases, $c^{\ell(r,a)} = 1$, and $\ell(r,a)$ divides N_0/a, hence N/a (Construction 7.3). Since $f_1 \circ \phi = f_1$, we obtain

$$\phi^*(s_{W_1}) = f_1^*\chi \cdot f_1^*c^{-N/a} \cdot \theta^{-N/a} = s_{W_1}.$$

Thus we may assume that W' and W'' are properly contained in Z.

In case (i), $Z \to V_Z$ is $g : X \to T$, and v_Z is the generic point of V. In case (ii), let $Z \to V_Z$ denote f_{W_1}, and v_Z the generic point of V_{W_1}. In both cases $Z \to V_Z$ is a contraction, (Z, B_Z) is dlt, and $K_Z + B_Z$ is $\mathbb{Q}$-linearly equivalent to a pullback from V_Z, by Set-up 7.1 and Lemma 2.2(1). By Lemma 2.2(2)(3), the lc centers of (Z, B_Z) are the lc centers of (X, B) properly contained in Z. An lc center $Y \subset Z$ of (X, B) has image V on T if and only if the image of Y on V_Z is the closure of v_Z, in case (ii) because $V_{W_1} \to V$ is finite. Thus W' and W'' are minimal lc centers of (Z, B_Z) over v_Z, and (1) follows from [Kol16, Definition 9, Theorem 10, Definition 18].

(2) Let $\mathbb{K}$ be a finite Galois extension of $\mathbb{C}(v_Z)$ such that $\mathbb{C}(V_{W'})$ and $\mathbb{C}(V_{W''})$ admit embeddings into $\mathbb{K}$ over $\mathbb{C}(v_Z)$. Let $\mathcal{O}_{V_Z, v_Z}$ be the local ring of V_Z at v_Z, $p \in \mathcal{O}_{V_Z, v_Z}[x]$ a monic polynomial whose reduction is the minimal polynomial of a primitive element of $\mathbb{K}/\mathbb{C}(v_Z)$, and T' the spectrum of the localization of $\mathcal{O}_{V_Z, v_Z}[x]/(p)$ at $\partial p/\partial x$. Then $T' \to \operatorname{Spec} \mathcal{O}_{V_Z, v_Z}$ is étale, and the fiber over v_Z is one point v' with residue field $\mathbb{K}$. Let $(Z', B_{Z'})$ be the base change of (Z, B_Z) to T'. It is dlt, $K_{Z'} + B_{Z'} \sim_{\mathbb{Q}} 0$ over T', the fiber of Z' over v' is connected since $Z \to V_Z$ is a contraction, and the lc centers of $(Z', B_{Z'})$ are the irreducible components of the inverse images of the lc centers of (Z, B_Z), since log discrepancies are étale local. By Lemma 2.17(4), applied to the contraction $W' \to V_{W'}$, the fiber of $W' \to V_{W'}$ over the generic point of $V_{W'}$ is geometrically integral. Hence the fiber of W' over v' is the disjoint union of the varieties $W'^e := W' \times_{V_{W'}} \operatorname{Spec} \mathbb{K}$, where e runs over the embeddings $\mathbb{C}(V_{W'}) \to \mathbb{K}$ over $\mathbb{C}(v_Z)$. Similarly, the fiber of W'' over v' is the disjoint union of the varieties $W''^e := W'' \times_{V_{W''}} \operatorname{Spec} \mathbb{K}$, where e runs over the embeddings $\mathbb{C}(V_{W''}) \to \mathbb{K}$ over $\mathbb{C}(v_Z)$.

Fix an embedding e''. Since ϕ is over V_Z, the base change of ϕ restricts to a birational map $\phi^{e''} : W''^{e''} \dashrightarrow W'^{e'}$ over $\mathbb{K}$ for some embedding e', crepant for the restrictions of $(W'', B_{W''})$ and $(W', B_{W'})$. By [Kol16, Theorem 10], applied to $(Z', B_{Z'}) \to T'$ at v', the closures of $W''^{e''}$ and $W'^{e'}$ are connected by a chain of $\mathbb{P}^1$-links, and each link is either an lc center C' of $(Z', B_{Z'})$ whose image on T' is the closure of v', or, in case (ii), $C' = Z'$ [Kol16, Definition 9] (cf. [Kol16, pp. 36–37]). Thus C' is a component of the inverse image of an lc center $C \subset Z$ of (X, B) with image V, where $C = Z$ if $C' = Z'$, so $\tilde{C} \subset \tilde{Z}$. The link C' joins two minimal lc centers of $(Z', B_{Z'})$ in the chain, which are components of the inverse images of lc centers $Y, Y' \subset C$ of (X, B) with image V, and $s_Y = \operatorname{Res}_{C,Y}(s_C)$ and $s_{Y'} = \operatorname{Res}_{C,Y'}(s_C)$ (Construction-Proposition 7.8(1)). Since $\operatorname{Res}_{C,Y}(0) = 0$ and $\operatorname{Res}_{C,Y}$ maps non-zero elements to non-zero elements (Construction-Proposition 7.8(3)), $s_Y = 0$ if and only if $s_C = 0$, if and only if $s_{Y'} = 0$. Going along the chain, if $s_{W''} = 0$, then $s_C = 0$ for every link and $s_{W'} = 0$, so $\phi^*(s_{W'}) = 0 = s_{W''}$. Otherwise $s_C \neq 0$ for every link C' of the chain, and we argue as follows.

Since $s_C \in M_{N,C}$ (Construction-Proposition 7.8(1)), $B_{s_C} = B_C$ over the generic point of V, so the form s_C^{-1} restricted to the fiber $C'_{v'}$ of C' over v' is, via the decomposition (7.4) for C, a nowhere vanishing section of $\omega_{C'_{v'}}^{[N]}(NB_C|_{C'_{v'}})$, and N is even. Hence the projective dlt pair on $C'_{v'}$ satisfies the hypotheses of [Kol16, Proposition 14], and we compose the birational maps given by the links. By Construction-Proposition 7.8(1), for each link C' the Poincaré residues of s_C along the minimal lc centers joined by C' are s_Y and $s_{Y'}$, and the minimal lc centers at the ends of the chain are components of the inverse images of W'' and W'. Thus we obtain a birational map $\psi : W''^{e''} \dashrightarrow W'^{e'}$ over $\mathbb{K}$ such that $\psi^*(s_{W'}) = s_{W''}$ on $W''^{e''}$.

We show that ψ is crepant for the restrictions of $(W'', B_{W''})$ and $(W', B_{W'})$. Under the decomposition (7.4) for W', which ψ^* preserves, write $s_{W'} = \lambda \omega^{-N} \otimes \xi$, where ω is a non-zero rational 1-canonical form on $W'^{e'}$ over $\mathbb{K}$, λ is a non-zero rational function on $W'^{e'}$, and

$$\xi \in (\textstyle\bigwedge^{\dim V} \Omega_{\mathbb{K}/\mathbb{C}})^{\otimes(-N)}.$$

Let $K_{W'^{e'}} := \operatorname{div}(\omega)$ and $K_{W''^{e''}} := \operatorname{div}(\psi^*\omega)$. Since $B_{s_{W'}} = B_{W'}$ on $W'^{e'}$ and $B_{s_{W''}} = B_{W''}$ on $W''^{e''}$ (Construction-Proposition 7.7(1)), and $s_{W''} = \psi^*\lambda \cdot (\psi^*\omega)^{-N} \otimes \xi$, we have

$$N(K_{W'^{e'}} + B_{W'}) = \operatorname{div}(\lambda), \qquad N(K_{W''^{e''}} + B_{W''}) = \operatorname{div}(\psi^*\lambda)$$

on $W'^{e'}$ and $W''^{e''}$. The pullbacks of $\mathrm{div}(\lambda)$ and $\mathrm{div}(\psi^*\lambda)$ to a common resolution of $W''^{e''}$ and $W'^{e'}$ are equal, so ψ is crepant for the restrictions of $(W'', B_{W''})$ and $(W', B_{W'})$.

Let $\alpha := \phi^{e''} \circ \psi^{-1}$, a birational map $W'^{e'} \dashrightarrow W'^{e'}$ over $\mathbb{K}$ which is crepant for the restriction of $(W', B_{W'})$ (compositions and inverses of crepant birational maps are crepant, as one checks on a common resolution). By Lemma 7.10, applied to W', the field $\mathbb{K}$, which is a finite extension of $\mathbb{C}(V_{W'})$ via e', and α, we have $\alpha^*(s_{W'}) = s_{W'}$ on $W'^{e'}$. Hence

$$(\phi^{e''})^*(s_{W'}) = \psi^*\alpha^*(s_{W'}) = \psi^*(s_{W'}) = s_{W''}$$

on $W''^{e''}$. Since the projection $W''^{e''} \to W''$ is dominant, the rational $(-N)$-canonical forms on W'' embed into those on $W''^{e''}$ by pullback, and we obtain $\phi^*(s_{W'}) = s_{W''}$. □

The following lemma constructs a section on any lc center in Σ_V from an invariant form on W.

Lemma 7.12. *Notation and conditions as in Construction-Proposition 7.7. Let $W_1 \subset \Sigma_V$ be an lc center of (X, B). Let $\beta \in H^0(W, \mathcal{J}_{N,W})$ be a section such that $\phi^*\beta = \beta$ for any birational map $\phi : W \dashrightarrow W$ over V which is crepant for (W, B_W). Then there exists a unique $t_1 \in M_{N,W_1}$ such that*

$$\mathrm{Res}_{W_1,W'}(t_1) = \psi^*\beta$$

for any lc center $W' \subset W_1$ of (X, B) which is minimal with respect to inclusion among the lc centers of (X, B) with image V, and any birational map $\psi : W' \dashrightarrow W$ over V which is crepant for $(W', B_{W'})$ and (W, B_W). Moreover, $t_1 \in H^0(W_1, \mathcal{J}_{N,W_1})$, and if $\beta \neq 0$, then $t_1 = f_{W_1}^*\chi \cdot \theta_{W_1}^{-N/a}$ and $E_{t_1} = -D_{W_1} + \frac{1}{N}\mathrm{div}(\chi) \geq 0$ for some non-zero rational function χ on V_{W_1}.

Proof. Let $W_2 \subset W_1$ be an lc center of (X, B) which is minimal with respect to inclusion among the lc centers of (X, B) with image V, and $\phi_2 : W_2 \dashrightarrow W$ a birational map over V which is crepant for (W_2, B_{W_2}) and (W, B_W) (Lemma 7.11(1) in case (i)). Uniqueness of t_1 follows from the injectivity of Res_{W_1,W_2} (Construction-Proposition 7.8(2)(3) and Construction-Proposition 7.7(1)). If $\beta = 0$, then $t_1 := 0$. Suppose that $\beta \neq 0$ and let $\beta_2 := \phi_2^*\beta$, a non-zero element of $H^0(W_2, \mathcal{J}_{N,W_2})$ by Lemma 7.9. Let $f_2 := f_{W_2}$ and $\kappa : V_{W_2} \to V_{W_1}$ the finite morphism with $\kappa \circ f_2 = f_{W_1}|_{W_2}$. Let

$$u := \mathrm{Res}_{W_1,W_2}\left(\theta_{W_1}^{-N/a}\right) \in M_{N,W_2}\backslash\{0\}.$$

By Construction-Proposition 7.7(1) applied to W_2, $\beta_2 = f_2^*\chi \cdot u$ for some non-zero rational function χ on V_{W_2}.

We show that $\chi = \kappa^*\chi_1$ for a non-zero rational function χ_1 on V_{W_1}. If W_1 itself is minimal with respect to inclusion among the lc centers of (X, B) with image V, then $W_2 = W_1$ and $\kappa = \mathrm{id}$, and we let $\chi_1 := \chi$. Moreover, any W' as in the statement is W_1, and below we take $\phi_3 = \mathrm{id}$. Suppose that $W_2 \neq W_1$. Let G_2 be the group of automorphisms of V_{W_2} over V_{W_1}. By Lemmas 7.9 and 7.5, V_{W_2} is isomorphic to V' over V, and ρ is the quotient morphism of V' by the finite group G. Hence the composite $V_{W_2} \xrightarrow{\kappa} V_{W_1} \to V$ is the quotient morphism of V_{W_2} by a finite group. Since κ and $V_{W_1} \to V$ are finite and surjective and V_{W_1} is normal, the Galois correspondence for the extensions of function fields given by κ and $V_{W_1} \to V$ implies that κ is the quotient morphism of V_{W_2} by G_2.

The variety W_2 is a minimal lc center of (W_1, B_{W_1}) over the generic point of V_{W_1}, and $W_2 \to V_{W_2} \to V_{W_1}$ is the Stein factorization of $W_2 \to V_{W_1}$. By [Kol16, Definition 18, Proposition 19], applied to $f_{W_1} : (W_1, B_{W_1}) \to V_{W_1}$, any automorphism of the function field of V_{W_2} over the function field of V_{W_1} is induced by a birational map $W_2 \dashrightarrow W_2$ over V_{W_1} which is crepant for (W_2, B_{W_2}). Since V_{W_2} is normal and finite over V_{W_1}, every such automorphism of function fields is the pullback of rational functions by a unique element of G_2, and every element of G_2 arises in this way. Thus any $\gamma \in G_2$ is induced by a birational map $\alpha : W_2 \dashrightarrow W_2$ over V_{W_1} which is crepant for (W_2, B_{W_2}), that is, $f_2 \circ \alpha = \gamma \circ f_2$. Since $\phi_2 \circ \alpha \circ \phi_2^{-1}$ is a birational map $W \dashrightarrow W$

over V which is crepant for (W, B_W),

$$\alpha^*\beta_2 = \phi_2^*(\phi_2 \circ \alpha \circ \phi_2^{-1})^*\beta = \phi_2^*\beta = \beta_2.$$

Let $V^0 \subset V$ be a non-empty open subset over which $D_{W_1} = 0$. By Construction-Proposition 7.7(2), $\theta_{W_1}^{-N/a}$ is a section of $\mathcal{J}_{N,W_1}$ over the inverse image of V^0, and it corresponds to a section s of $\mathcal{O}_{\tilde{W}_1}(L_N)$ over the inverse image of V^0 with $s_{W_2} = u$. By Lemma 7.11(2) in case (ii), $\alpha^*u = u$. Hence $f_2^*\gamma^*\chi = \alpha^* f_2^*\chi = f_2^*\chi$, so $\gamma^*\chi = \chi$ since f_2 is surjective. Thus χ is a G_2-invariant rational function on V_{W_2}, and since κ is the quotient morphism by G_2, $\chi = \kappa^*\chi_1$ for a non-zero rational function χ_1 on V_{W_1}.

Let $t_1 := f_{W_1}^*\chi_1 \cdot \theta_{W_1}^{-N/a} \in M_{N,W_1}$. By Construction-Proposition 7.7(1), $E_{t_1} = -D_{W_1} + \frac{1}{N}\operatorname{div}(\chi_1)$. By Construction-Proposition 7.8(2),

$$\operatorname{Res}_{W_1,W_2}(t_1) = (f_{W_1}|_{W_2})^*\chi_1 \cdot u = f_2^*\kappa^*\chi_1 \cdot u = f_2^*\chi \cdot u = \beta_2.$$

By Construction-Proposition 7.8(3) and Construction-Proposition 7.7(2) for W_2 with $V^0 = V$,

$$f_2^*\kappa^* E_{t_1} = B_{\beta_2} - B_{W_2} = f_2^* E_{\beta_2}, \qquad E_{\beta_2} \geq 0.$$

By Lemma 2.17(1), $\kappa^* E_{t_1} = E_{\beta_2} \geq 0$. Since κ is a finite surjective morphism between normal varieties, any prime divisor P on V_{W_1} is dominated by a prime divisor on V_{W_2} along which the coefficient of $\kappa^* E_{t_1}$ is a positive multiple of the coefficient of E_{t_1} along P. Thus $E_{t_1} \geq 0$, and $t_1 \in H^0(W_1, \mathcal{J}_{N,W_1})$ by Construction-Proposition 7.7(2).

Finally, let W' and ψ be as in the statement. By Lemma 7.11(1) in case (ii), there exists a birational map $\phi_3 : W' \dashrightarrow W_2$ over V_{W_1} which is crepant for $(W', B_{W'})$ and (W_2, B_{W_2}). By Lemma 7.11(2) in case (ii), applied to the section of $\mathcal{O}_{\tilde{W}_1}(L_N)$ corresponding to t_1, and by Construction-Proposition 7.8(1),

$$\operatorname{Res}_{W_1,W'}(t_1) = \phi_3^* \operatorname{Res}_{W_1,W_2}(t_1) = \phi_3^*\phi_2^*\beta = (\phi_2 \circ \phi_3)^*\beta.$$

Since $\psi \circ (\phi_2 \circ \phi_3)^{-1}$ is a birational map $W \dashrightarrow W$ over V which is crepant for (W, B_W), $(\phi_2 \circ \phi_3)^*\beta = \psi^*\beta$. □

We now prove the main result of this section.

Theorem 7.13. *Notation and conditions as in Set-up 7.1, Construction 7.2, and Construction 7.3, and let N be a positive integer with $N_0 \mid N$. Then:*

(1) any birational map $\phi : W \dashrightarrow W$ over V which is crepant for (W, B_W) induces an automorphism $\kappa_\phi \in G$ of V' over V with $h \circ \phi = \kappa_\phi \circ h$, and the pullback ϕ^* of rational $(-N)$-canonical forms maps $H^0(W, \mathcal{J}_{N,W})$ into itself and depends only on κ_ϕ; since any element of G is κ_ϕ for some such ϕ (Lemma 7.5), $\sigma \cdot t := (\phi^{-1})^*t$ for $\kappa_\phi = \sigma$ defines an action of G on $H^0(W, \mathcal{J}_{N,W})$;

(2) restriction to $\tilde{W}$, followed by the isomorphism of Lemma 7.6(1) for $W' = W$, induces an injective map

$$H^0(\tilde{\Sigma}_V, \mathcal{O}_{\tilde{\Sigma}_V}(L_N)) \to H^0(W, \mathcal{J}_{N,W})$$

whose image is $H^0(W, \mathcal{J}_{N,W})^G$, and the restriction map

$$H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(L_N)) \to H^0(\tilde{\Sigma}_V, \mathcal{O}_{\tilde{\Sigma}_V}(L_N))$$

is surjective.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove (1). Let $\operatorname{Bir}_V^c(W)$ be the group of birational maps $W \dashrightarrow W$ over V which are crepant for (W, B_W). By Lemma 7.9 with $W' = W'' = W$ and $V^0 = V$, any $\phi \in \operatorname{Bir}_V^c(W)$ induces an automorphism $\kappa_\phi \in G$ of V' over V, and ϕ^* maps $H^0(W, \mathcal{J}_{N,W})$ into itself. We have $\kappa_{\phi\circ\phi'} = \kappa_\phi \circ \kappa_{\phi'}$ and $(\phi \circ \phi')^* = \phi'^* \circ \phi^*$ for $\phi, \phi' \in \operatorname{Bir}_V^c(W)$. Suppose that $\kappa_\phi = \mathrm{id}_{V'}$. Then ϕ is a birational map over V'. For $W' = W$, the Stein factorization

$W \xrightarrow{f_W} V_W \to V$ in Construction-Proposition 7.7 is $W \xrightarrow{h} V' \xrightarrow{\rho} V$. Let $t \in H^0(W, \mathcal{J}_{N,W})$, and s the global section of $\mathcal{O}_{\tilde{W}}(L_N)$ corresponding to t under the isomorphism of Lemma 7.6(1) for $W' = W$, so that $s_W = t$. By Lemma 7.11(2) in case (ii) with $W_1 = W' = W'' = W$ and $V^0 = V$, we have $\phi^* t = t$. Thus $\phi^* t$ depends only on κ_ϕ. This proves (1).

Step 2. In this step we describe the lc centers of $(\tilde{X}, \Delta_N)$, where $\Delta_N := \tilde{S} + \{(N+1)E\}$.

By Lemma 7.4, the strata of $\tilde{S}$ are the varieties $\mu_*^{-1}Y$ for the lc centers Y of (X, B), and μ is an isomorphism over the generic point of any lc center of (X, B). Since the coefficients of $\{(N+1)E\}$ are less than 1 and $\operatorname{Supp} \Delta_N$ is a simple normal crossing divisor, $(\tilde{X}, \Delta_N)$ is lc and the lc centers of $(\tilde{X}, \Delta_N)$ are the strata of $\tilde{S}$.

In particular, $\tilde{\Sigma}_V$ is the union of the varieties $\tilde{W}'$ for the lc centers $W' \subset \Sigma_V$ of (X, B). Let $W^{(1)}, \dots, W^{(n)}$ be the irreducible components of Σ_V. Then the irreducible components of $\tilde{\Sigma}_V$ are $\tilde{W}^{(1)}, \dots, \tilde{W}^{(n)}$, and for $i \neq j$, the scheme $\tilde{W}^{(i)} \cap \tilde{W}^{(j)}$ is reduced, and any irreducible component of it is a stratum of $\tilde{S}$, hence equal to $\tilde{R}$ for an lc center $R \subset W^{(i)} \cap W^{(j)}$ of (X, B), and R has image V (Lemma 7.5).

Step 3. In this step we show that the map $s \to s_W$ in (2) is injective with image contained in $H^0(W, \mathcal{J}_{N,W})^G$. Let $s \in H^0(\tilde{\Sigma}_V, \mathcal{O}_{\tilde{\Sigma}_V}(L_N))$. Then $s_W \in H^0(W, \mathcal{J}_{N,W})$ by Lemma 7.6(1). Let $\sigma \in G$ and $\phi \in \operatorname{Bir}^c_V(W)$ with $\kappa_\phi = \sigma$. By Lemma 7.11(2) in case (i) with $W' = W'' = W$, $\phi^* s_W = s_W$. Hence s_W is G-invariant by **Step 1**.

Suppose that $s_W = 0$. Let $1 \leq i \leq n$ and $W' \subset W^{(i)}$ an lc center of (X, B) which is minimal with respect to inclusion among the lc centers of (X, B) with image V. By Lemma 7.11(1) and (2) in case (i), $s_{W'} = \phi'^* s_W = 0$ for a birational map $\phi' : W' \dashrightarrow W$ over V which is crepant for $(W', B_{W'})$ and (W, B_W). By Construction-Proposition 7.8(1) and (3), $s_{W'} = \operatorname{Res}_{W^{(i)},W'}(s_{W^{(i)}})$ and $\operatorname{Res}_{W^{(i)},W'}$ maps non-zero elements to non-zero elements, so $s_{W^{(i)}} = 0$, that is, $s|_{\tilde{W}^{(i)}} = 0$. Since $\tilde{\Sigma}_V$ is the reduced union of $\tilde{W}^{(1)}, \dots, \tilde{W}^{(n)}$ (**Step 2**), $s = 0$.

Step 4. In this step we show that any $\beta \in H^0(W, \mathcal{J}_{N,W})^G$ is of the form s_W for a section s of $\mathcal{O}_{\tilde{\Sigma}_V}(L_N)$ over $\tilde{\Sigma}_V$. By **Step 1**, $\phi^* \beta = \beta$ for any $\phi \in \operatorname{Bir}^c_V(W)$. For $1 \leq i \leq n$, let $t_{W^{(i)}} \in H^0(W^{(i)}, \mathcal{J}_{N,W^{(i)}})$ be the element of Lemma 7.12 for $W_1 := W^{(i)}$, and $\tilde{s}_{W^{(i)}}$ the global section of $\mathcal{O}_{\tilde{W}^{(i)}}(L_N)$ corresponding to $t_{W^{(i)}}$ under Lemma 7.6(1).

Let $1 \leq i < j \leq n$ and $\tilde{R}$ an irreducible component of $\tilde{W}^{(i)} \cap \tilde{W}^{(j)}$, where R is an lc center of (X, B) with image V (**Step 2**). Let $W' \subset R$ be an lc center of (X, B) which is minimal with respect to inclusion among the lc centers of (X, B) with image V, and $\psi : W' \dashrightarrow W$ a birational map over V which is crepant for $(W', B_{W'})$ and (W, B_W) (Lemma 7.11(1)). By Construction-Proposition 7.8(1) and Lemma 7.12,

$$\operatorname{Res}_{R,W'}\left(\operatorname{Res}_{W^{(i)},R}(t_{W^{(i)}})\right) = \operatorname{Res}_{W^{(i)},W'}(t_{W^{(i)}}) = \psi^*\beta = \operatorname{Res}_{R,W'}\left(\operatorname{Res}_{W^{(j)},R}(t_{W^{(j)}})\right).$$

By Construction-Proposition 7.8(2)(3) and Construction-Proposition 7.7(1), $\operatorname{Res}_{R,W'}$ is injective, so

$$\operatorname{Res}_{W^{(i)},R}(t_{W^{(i)}}) = \operatorname{Res}_{W^{(j)},R}(t_{W^{(j)}}).$$

Thus $\tilde{s}_{W^{(i)}}|_{\tilde{R}}$ and $\tilde{s}_{W^{(j)}}|_{\tilde{R}}$ coincide over a non-empty open subset of V, and hence on $\tilde{R}$, since they are sections of an invertible sheaf on an integral scheme. Since $\tilde{W}^{(i)} \cap \tilde{W}^{(j)}$ is reduced, $\tilde{s}_{W^{(i)}}$ and $\tilde{s}_{W^{(j)}}$ coincide on $\tilde{W}^{(i)} \cap \tilde{W}^{(j)}$.

We glue the sections $\tilde{s}_{W^{(i)}}$. Let $x \in \tilde{X}$. There exist local coordinates of $\tilde{X}$ at x, that is, an étale morphism from a neighborhood of x to an affine space, such that the components of $\tilde{S}$ through x are coordinate hyperplanes. Then near x, each $\tilde{W}^{(i)}$ is defined by an ideal I_i generated by a subset of the coordinates. For ideals $I_1, \dots, I_n$ of a polynomial ring generated by subsets of the variables, the sequence

$$0 \to \mathcal{O} \Big/ \bigcap_i I_i \to \bigoplus_i \mathcal{O}/I_i \to \bigoplus_{i<j} \mathcal{O}/(I_i + I_j)$$

is exact, since a monomial is non-zero in $\mathcal{O}/I_i$ if and only if it involves none of the variables in I_i, so a family in the kernel of the second map assigns equal coefficients to a monomial in any two $\mathcal{O}/I_i$ in which it is non-zero, and the polynomial with the coefficients so obtained maps to the family. Exactness is preserved by flat base change to the local ring of $\tilde{X}$ at x and by tensoring with $\mathcal{O}_{\tilde{X}}(L_N)$. Since $\bigcap_i I_i$ is the ideal of the reduced union $\tilde{\Sigma}_V$ and $I_i + I_j$ is the ideal of $\tilde{W}^{(i)} \cap \tilde{W}^{(j)}$, the sections $\tilde{s}_{W^{(i)}}$ glue to a section $s \in H^0(\tilde{\Sigma}_V, \mathcal{O}_{\tilde{\Sigma}_V}(L_N))$. Finally, $W \subset W^{(i)}$ for some i, and by Construction-Proposition 7.8(1) and Lemma 7.12 with $W' = W$ and $\psi = \mathrm{id}_W$, we have $s_W = \mathrm{Res}_{W^{(i)},W}(t_{W^{(i)}}) = \beta$.

Step 5. In this step we prove the surjectivity of

$$H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(L_N)) \to H^0(\tilde{\Sigma}_V, \mathcal{O}_{\tilde{\Sigma}_V}(L_N)).$$

Since $-(K_X + B)$ is in general not big, we use [Fuj17, Theorem 6.4], which needs only semi-ampleness, in place of the Kawamata–Viehweg vanishing theorem. Since U is affine and H is ample over U, H is ample. Since (X, B) has finitely many lc centers, there exist a positive integer k such that kH is Cartier and $\mathcal{I}_V \otimes \mathcal{O}_T(kH)$ is globally generated, where $\mathcal{I}_V$ is the ideal sheaf of V, and a Cartier divisor $D_T \geq 0$ with $D_T \sim kH$ such that $V \subset \operatorname{Supp} D_T$ and $g(Y) \not\subset \operatorname{Supp} D_T$ for any lc center Y of (X, B) with $g(Y) \not\subset V$. Let $D := (g\mu)^* D_T$. Then

$$L_N - (K_{\tilde{X}} + \Delta_N) = -(N+1)(K_{\tilde{X}} + \tilde{S} + E) = -(N+1)\mu^*(K_X + B) \sim_{\mathbb{Q}} (N+1)(g\mu)^* H$$

is semi-ample over U, and

$$D \sim_{\mathbb{Q}} \frac{k}{N+1}\big(L_N - (K_{\tilde{X}} + \Delta_N)\big).$$

By **Step 2**, the lc centers of $(\tilde{X}, \Delta_N)$ are the varieties $\tilde{Y}$ for the lc centers Y of (X, B), and $\tilde{Y} \subset \operatorname{Supp} D$ if and only if $g(Y) \subset \operatorname{Supp} D_T$, if and only if $g(Y) \subset V$, if and only if $g(Y) = V$ by the minimality of V. Thus the union of the lc centers of $(\tilde{X}, \Delta_N)$ contained in $\operatorname{Supp} D$ is $\tilde{\Sigma}_V$. By [Fuj17, Theorem 6.4], applied to $(\tilde{X}, \Delta_N)$, the composite $\pi : \tilde{X} \xrightarrow{g\mu} T \to U$, the Cartier divisor L_N, the $\mathbb{Q}$-divisor $L_N - (K_{\tilde{X}} + \Delta_N)$ which is semi-ample over U, and D, the restriction map

$$\pi_* \mathcal{O}_{\tilde{X}}(L_N) \to \pi_* \mathcal{O}_{\tilde{\Sigma}_V}(L_N)$$

is surjective. Since U is affine, the restriction map on global sections is surjective. □

Proposition 7.14. *Notation and conditions as in Set-up 7.1, Construction 7.2, and Construction 7.3. Let N be a positive integer with $N_0 \mid N$, $\tau \in H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(L_N))$ a section such that $s := \tau|_{\tilde{\Sigma}_V} \neq 0$, $t \in H^0(W, \mathcal{J}_{N,W})$ the image of s under the map of Theorem 7.13(2), and $B^+ := \frac{1}{N}\operatorname{div}_X(\tau)$, where τ is regarded as a rational $(-N)$-canonical form on X. Let W_1 be an lc center of (X, B) which is maximal with respect to inclusion among the lc centers of (X, B) with image V on T, and $W_1 \xrightarrow{f_1} V_1 \to V$ the Stein factorization of $W_1 \to V$. Then:*

(1) $B^+ \geq B$ and $N(K_X + B^+) \sim 0$;
(2) $W_1 \not\subset \operatorname{Supp}(B^+ - B)$, and $(B^+ - B)|_{W_1} = f_1^* E_1$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $E_1 \geq 0$ on V_1;
(3) there exist an lc center $W_2 \subset W_1$ of (X, B) which is minimal with respect to inclusion among the lc centers of (X, B) with image V, and a birational map $\phi : W_2 \dashrightarrow W$ over V which is crepant for $(W_2, B_{W_2} + (f_1|_{W_2})^* E_1)$ and (W, B_t). In particular, if (W, B_t) is lc, then $(W_2, B_{W_2} + (f_1|_{W_2})^* E_1)$ is lc.

Proof. (1) Since $\tau \in H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(L_N))$,

$$\operatorname{div}_{\tilde{X}}(\tau) \geq N\tilde{S} + \lfloor (N+1)E \rfloor.$$

Since $\mu_* \tilde{S} = \lfloor B \rfloor$ and $\mu_* E = \{B\}$, pushing forward by μ implies

$$\operatorname{div}_X(\tau) \geq N\lfloor B \rfloor + \lfloor (N+1)\{B\} \rfloor = \lceil NB \rceil \geq NB$$

by Lemma 2.13(2), since $B \in \Gamma \cup \{1\} \subset \Phi(\mathfrak{R})$ and $b_{\mathfrak{R}} \mid N$. Thus $B^+ \geq B$. Writing $\tau = \psi\omega^{-N}$ for a non-zero rational 1-canonical form ω on X with $K_X = \operatorname{div}(\omega)$ and a non-zero rational function ψ on X, we have $N(K_X + B^+) = \operatorname{div}(\psi) \sim 0$.

(2) We have $f_1 = f_{W_1}$ and $V_1 = V_{W_1}$. By Theorem 7.13(2), $t = s_W \neq 0$. Let $W_2 \subset W_1$ be an lc center of (X, B) which is minimal with respect to inclusion among the lc centers of (X, B) with image V, and $\phi : W_2 \dashrightarrow W$ a birational map over V which is crepant for (W_2, B_{W_2}) and (W, B_W) (Lemma 7.11(1) in case (i)). By Lemma 7.11(2) in case (i) and Construction-Proposition 7.8(1),

$$\operatorname{Res}_{W_1,W_2}(s_{W_1}) = s_{W_2} = \phi^* t \neq 0,$$

so $t_1 := s_{W_1} \neq 0$.

Let ξ be the generic point of W_1 and $k := \operatorname{codim} W_1$. Since (X, B) is dlt, μ is an isomorphism over a neighborhood of ξ, and there exist local coordinates $z_1, \dots, z_n$ of X at ξ such that $B = \sum_{i=1}^{k}(z_i = 0)$ near ξ [KM98, Definition 2.37] and [Kol13, Theorem 4.16]. Near ξ, the sheaf $\mathcal{O}_{\tilde{X}}(L_N)$ is generated by $e := (z_1 \cdots z_k)^N (dz_1 \wedge \cdots \wedge dz_n)^{-N}$. Write $\tau = ve$ with $v \in \mathcal{O}_{X,\xi}$. Since $t_1 \neq 0$, $\tau|_{\tilde{W}_1} \neq 0$, so v is not in the maximal ideal of $\mathcal{O}_{X,\xi}$, that is, v is a unit at ξ. Near ξ we obtain

$$NB^+ = \operatorname{div}(\tau) = NB + \operatorname{div}(v), \qquad \xi \notin \operatorname{Supp} \operatorname{div}(v).$$

Thus $W_1 \not\subset \operatorname{Supp}(B^+ - B)$.

Since X is $\mathbb{Q}$-factorial, there exists a positive integer q such that $q(B^+ - B)$ is Cartier. Locally on X, write $q(B^+ - B) = \operatorname{div}(\lambda)$, so that $\lambda|_{W_1}$ is a non-zero rational function. Then $\lambda^N \tau^{-q}$ is a local generator of $\omega_X^{[qN]}(qNB)$, and qN is even since a is even and $a \mid N_0 \mid N$ (Construction 7.3). By [Kol13, Theorem 4.19(2)], the Poincaré residue of $\lambda^N \tau^{-q}$ along W_1 is a local generator of $\omega_{W_1}^{[qN]}(qNB_{W_1})$. Since the Poincaré residue is multiplicative, as both residues are the snc residue at the generic point of W_1, where μ is an isomorphism [KM98, Definition 2.37] (cf. [Kol13, (4.18.2)]), and t_1 is the Poincaré residue of τ along W_1 (Lemma 7.6(1)), we obtain $\operatorname{div}(t_1) = NB_{W_1} + N(B^+ - B)|_{W_1}$, that is,

$$B_{t_1} = B_{W_1} + (B^+ - B)|_{W_1}.$$

By Lemma 7.6(2) with $V^0 = V$, $(B^+ - B)|_{W_1} = f_1^* E_1$ for the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $E_1 := E_{t_1} \geq 0$ on V_1.

(3) We use the lc center W_2 and the birational map ϕ from (2). By (2), $\phi^* t = s_{W_2} = \operatorname{Res}_{W_1,W_2}(t_1)$ and E_1, the $\mathbb{Q}$-divisor E_{t_1} of Lemma 7.6(2), is the $\mathbb{Q}$-divisor E_{t_1} of Construction-Proposition 7.7(1) by Construction-Proposition 7.7(3). By Construction-Proposition 7.8(3), $B_{\phi^* t} = B_{W_2} + (f_1|_{W_2})^* E_1$. By Lemma 7.9, ϕ is crepant for $(W_2, B_{\phi^* t})$ and (W, B_t), that is, for $(W_2, B_{W_2} + (f_1|_{W_2})^* E_1)$ and (W, B_t). Since log discrepancies are preserved by crepant birational maps, if (W, B_t) is lc, then $(W_2, B_{W_2} + (f_1|_{W_2})^* E_1)$ is lc. □

8. Complements of pairs with semi-ample anti-log canonical divisor

In this section, we prove Theorem 8.3. On the models obtained in Proposition 5.3, we construct monotonic N-complements with N depending only on the dimension and the coefficients. We distinguish two cases: any lc center dominates T (Proposition 8.4), or some lc center does not dominate T (Proposition 8.8 in Subsection 8.4).

8.1. **The setting and the constants.**

In this subsection, we fix the setting of this section and define the integer $N(d, \Gamma)$.

Set-up 8.1. Let d, Γ, $(X/U \ni u, B)$, $X \xrightarrow{g} T \to U$, and H be as follows.

(1) d is a positive integer and $\Gamma \subset [0, 1] \cap \mathbb{Q}$ is a finite set.
(2) $(X/U \ni u, B)$ is a pair such that U is affine, $\dim X \leq d$, X is $\mathbb{Q}$-factorial, (X, B) is dlt, and $B \in \Gamma \cup \{1\}$.
(3) $X \xrightarrow{g} T \to U$ are contractions whose composite is the contraction $X \to U$.
(4) $-(K_X + B) \sim_{\mathbb{Q}} g^* H$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H on T which is ample over U.

(5) The image on T of any lc center of (X, B) intersects the fiber T_u of $T \to U$ over u.

Condition (5) of Set-up 8.1 ensures that the minimal lc image V in Proposition 8.5 intersects T_u, so that the lc property proved near Σ_V extends by connectedness to the fiber of $X \to U$ over u (see Proposition 8.8).

Construction 8.2. Let d be a positive integer, $\Gamma \subset [0,1] \cap \mathbb{Q}$ a finite set, and $d' := \max\{d, 2\}$. Let $\mathfrak{R}$ and N_0 be as in Construction 7.3 for d' and Γ. Let $m \mid p$ be the integers of Theorem 3.3 for d' and $\Phi(\mathfrak{R})$, and $\Lambda := \Phi(\mathfrak{R}[p])$. Let b and p' be the integers b and p of Theorem 6.4 for d' and m. Let n_1 be the integer N of Proposition 4.2 for d', p', $\operatorname{lcm}\{N_0, b, p, p'\}$, and Λ, and n_2 the integer N of Proposition 4.2 for d', p, 1, and Λ. Let δ be a positive integer such that $\delta\Gamma \subset \mathbb{Z}$. We define

$$N(d, \Gamma) := \operatorname{lcm}\{N_0, n_1, n_2, p, p', b, \delta, I(1, \Phi(\mathfrak{R})), \dots, I(d', \Phi(\mathfrak{R}))\}.$$

The integers in the definition of $N(d, \Gamma)$ depend only on d and Γ. In particular, they do not depend on the degree or the ramification of the finite morphism $V' \to V$ in Proposition 8.5, nor on the number of points over u of the Stein factorization of $V \to U$. We use d' in place of d since Construction 7.3 requires the dimension to be at least 2, while Set-up 8.1 and Set-up 7.1 only require $\dim X$ to be at most the given dimension. The integer n_2 is the index of the complement on T in Proposition 8.4, where $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier by Theorem 3.3, and n_1 is the index of the complement on the minimal lc image V in Proposition 8.5, where $p'\mathbf{N}$ is $\boldsymbol{b}$-Cartier by Theorem 6.4(4). The other integers are used in the remaining steps: $N_0 \mid N(d, \Gamma)$ is used for Theorem 7.13, $b \mid N(d, \Gamma)$ lets the invariant form θ of Theorem 6.4(1) enter a section as $\theta^{-N(d,\Gamma)/b}$, $\delta \mid N(d, \Gamma)$ implies that the elements of $N(d, \Gamma)\Gamma$ are integers, and $I(j, \Phi(\mathfrak{R})) \mid N(d, \Gamma)$ treats the case when T or V is a point (Theorem 2.21).

Theorem 8.3. *Notation and conditions as in Set-up 8.1, and let $N := N(d, \Gamma)$ be the integer of Construction 8.2. Then $(X/U \ni u, B)$ has a monotonic N-complement.*

Theorem 8.3 follows from Propositions 8.4 and 8.8, and we prove it at the end of Subsection 8.4.

8.2. **Log canonical centers dominating the base.** In this subsection, we treat the case when any lc center of (X, B) dominates T. In this case, the generalized pair induced on T by the canonical bundle formula is generalized klt, and we lift a complement of it from T to X.

Proposition 8.4. *Notation and conditions as in Set-up 8.1, and let $N := N(d, \Gamma)$ be the integer of Construction 8.2. Assume that any lc center of (X, B) dominates T. Then $(X/U \ni u, B)$ has a monotonic N-complement.*

Proof. We use the notation of Construction 8.2. Since (X, B) is dlt and $B \in \Gamma \cup \{1\} \subset \Phi(\mathfrak{R})$ (Construction 7.3), (X, B) is an lc pair with $B \in \Phi(\mathfrak{R})$.

First assume that $\dim T = 0$. Since $T \to U$ is a contraction, U is a point, X is projective, and $H = 0$. Thus $K_X + B \sim_{\mathbb{Q}} 0$. Let $n := I(\dim X, \Phi(\mathfrak{R}))$ if $\dim X \geq 1$, and $n := 1$ if $\dim X = 0$. If $\dim X \geq 1$, then $n(K_X + B) \sim 0$ by Theorem 2.21, as $\dim X \leq d \leq d'$. If $\dim X = 0$, then $K_X + B = 0$. Since $n \mid N$, $(X/U \ni u, B)$ is a monotonic N-complement of $(X/U \ni u, B)$.

From now on, assume that $\dim T > 0$. Since $0 < \dim T \leq \dim X \leq d'$ and $K_X + B \sim_{\mathbb{Q}} g^*(-H)$, Theorem 3.3(1) implies a trivialization of degree m

$$K_X + B + \frac{1}{m}\operatorname{div}(\psi) = g^* D_T, \qquad D_T \sim_{\mathbb{Q}} -H.$$

Let $(T, B_T, \mathbf{M})$ be the induced generalized sub-pair, so that $D_T = K_T + B_T + \mathbf{M}_T$. By Theorem 3.3(2), $B_T \geq 0$, $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U, and $p\mathbf{M}_{T'}$ is Cartier for any smooth variety T' projective and birational over T. Since $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, it descends to some smooth variety projective and birational over T, and therefore $p\mathbf{M}$ is $\boldsymbol{b}$-Cartier. Since $\mathbf{M}_T$ is the pushforward of $\mathbf{M}_{T'}$ for a resolution $T' \to T$, $p\mathbf{M}_T$ has integer coefficients. As $B \in \Phi(\mathfrak{R})$ and $m \mid p$, Lemma 3.7 implies that $B_T \in \Lambda$.

We show that $(T, B_T, \mathbf{M})/U$ is generalized klt. Let E be a prime divisor over T. There exists a log resolution $T'' \to T$ of $(T, \operatorname{Supp} B_T)$ such that $\mathbf{M}$ descends to T'' and E is a divisor on T'' (cf. [KM98, Lemma 2.45]). Since $K_{T''} + B_{T''} + \mathbf{M}_{T''}$ is the pullback of D_T (Definition 3.1(3)), we have $a(E, T, B_T, \mathbf{M}) = 1 - \operatorname{mult}_E B_{T''}$. Since (X, B) is lc and any lc center of (X, B) dominates T, the coefficients of $B_{T''}$ are less than 1 by Lemma 3.8(3). Thus $a(E, T, B_T, \mathbf{M}) > 0$, and $(T, B_T, \mathbf{M})/U$ is generalized klt.

Since $-(K_T + B_T + \mathbf{M}_T) = -D_T \sim_{\mathbb{Q}} H$ is ample over U and U is affine, T is of Fano type over U by Lemma 4.1. Since $\mathfrak{R}[p]$ is a finite subset of $[0,1] \cap \mathbb{Q}$ containing 0 and 1 (Lemma 3.7), Λ satisfies the DCC and the only accumulation point of Λ is 1. The Stein factorization of $T \to U$ is $T \to U \xrightarrow{\mathrm{id}_U} U$, since $T \to U$ is a contraction, and u belongs to the image of T. By Proposition 4.2, applied with parameters d', p, 1, and Λ to $T \to U$ and $(T, B_T, \mathbf{M})/U$, there exist an open neighborhood U^0 of u and a $\mathbb{Q}$-divisor $B_T^+ \geq B_T$ on the inverse image T^0 of U^0 such that $(T^0, B_T^+, \mathbf{M})$ is generalized lc and

$$n_2(K_{T^0} + B_T^+ + \mathbf{M}_{T^0}) \sim 0.$$

In particular, $B_T^+ - B_T|_{T^0} = (K_{T^0} + B_T^+ + \mathbf{M}_{T^0}) - D_T|_{T^0}$ is $\mathbb{Q}$-Cartier.

Possibly shrinking U to an affine open neighborhood of u in U^0 and replacing X, T with their inverse images, we may assume that $U^0 = U$. Let $P := B_T^+ - B_T$ and $B^+ := B + g^*P$. Then $P \geq 0$ is $\mathbb{Q}$-Cartier, $B_T + P \geq B_T$, and $B^+ \geq B$. Since $p \mid n_2$ (Proposition 4.2), $n_2\mathbf{M}$ is $\boldsymbol{b}$-Cartier. Since $n_2 \mid N$, $(T/U \ni u, B_T + P, \mathbf{M})$ is a monotonic N-complement of $(T/U \ni u, B_T, \mathbf{M})$. Since $p \mid N$, Proposition 3.9(2), applied to the trivialization above, implies that $(X/U \ni u, B^+)$ is a monotonic N-complement of $(X/U \ni u, B)$. □

8.3. **A minimal lc image.** In this subsection, we construct an invariant section on a minimal lc center over a minimal lc image (Set-up 7.1(5)), whose divisor defines an lc boundary.

If some lc center of (X, B) does not dominate T, then an lc place of that lc center defines, on some higher model of T, a prime divisor whose coefficient in the discriminant $\boldsymbol{b}$-divisor is equal to 1. Thus the generalized pair induced on T by the canonical bundle formula is not generalized klt, and the propositions of Section 4 cannot be used on T. Instead, as in step (3) of the sketch of the proof in Section 1, we use Proposition 4.2 on a minimal lc image V.

Proposition 8.5. *Notation and conditions as in Set-up 8.1, and let $N := N(d, \Gamma)$ be the integer of Construction 8.2. Let $V \subsetneq T$, W, B_W, $h : W \to V'$, $\rho : V' \to V$, and G be as in Set-up 7.1 and Construction 7.2 for the integer d' of Construction 8.2 and Γ. Then, after replacing U with an affine open neighborhood of u and X, T, V, V', W with their inverse images, there exists a non-zero G-invariant section $t \in H^0(W, \mathcal{J}_{N,W})$ such that (W, B_t) is lc.*

Proof. The proof is long, so we divide it into several steps. We use the notation of Construction 8.2. The conditions of Set-up 7.1 hold for d' and Γ by Set-up 8.1. By Lemma 7.5, V is normal, (W, B_W) is klt, V' is normal, G is finite, ρ induces an isomorphism $V'/G \to V$, and any element of G is induced by a birational map $W \dashrightarrow W$ over V which is crepant for (W, B_W). By Construction 7.3, $B_W \in \Phi(\mathfrak{R})$.

Step 1. In this step we treat the case when $\dim V = 0$.

We take U itself as the neighborhood of u. Since V is a normal point and V' is normal, connected, and finite over V, we have $V' = V$ and G is trivial. Since $W \to V$ is projective, W is projective. Since $g(W) = V$ is a point, $(g^*H)|_W \sim_{\mathbb{Q}} 0$, and hence

$$K_W + B_W = (K_X + B)|_W \sim_{\mathbb{Q}} 0.$$

If $\dim W = 0$, then $B_W = 0$, $\mathcal{J}_{N,W} = \mathcal{O}_W$, and $t := 1$ satisfies the conclusion. Assume that $\dim W \geq 1$. Since (W, B_W) is klt, $B_W \in \Phi(\mathfrak{R})$, and $\dim W \leq d'$, Theorem 2.21 implies that $I(\dim W, \Phi(\mathfrak{R}))(K_W + B_W) \sim 0$. Since $I(\dim W, \Phi(\mathfrak{R})) \mid N$, we have $N(K_W + B_W) \sim 0$, so there exists a rational $(-N)$-canonical form t on W with $\operatorname{div}(t) = NB_W$. Thus NB_W has integer

coefficients, $\operatorname{div}(t) = \lceil NB_W \rceil$, and t is a non-zero section of $\mathcal{J}_{N,W}$, which is G-invariant as G is trivial. Moreover, $B_t = B_W$, so (W, B_t) is lc.

Step 2. In this step we reduce to the case when $V \not\subset \operatorname{Supp} H$, and we show that $V = V'/G$.

From now on, assume that $\dim V > 0$. Since U is affine and H is ample over U, there exists a positive integer k such that kH is Cartier and very ample over U. Let H_1 be a general member of $|kH|$. Then $V \not\subset \operatorname{Supp} H_1$, $\frac{1}{k}H_1$ is ample over U, and $-(K_X + B) \sim_{\mathbb{Q}} g^*(\frac{1}{k}H_1)$, so condition (4) of Set-up 8.1 holds with H replaced by $\frac{1}{k}H_1$. Possibly replacing H with $\frac{1}{k}H_1$, we may assume that $V \not\subset \operatorname{Supp} H$. Then $H|_V$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on V which is ample over U.

By definition, G acts faithfully on V' by automorphisms over V. Since G is finite and ρ induces an isomorphism $V'/G \to V$ (Lemma 7.5), we have $V = V'/G$, and ρ is the quotient morphism. Moreover, for any $\sigma \in G$, there exists a birational map $\phi : W \dashrightarrow W$ which is crepant for (W, B_W) and satisfies $h \circ \phi = \sigma \circ h$.

Step 3. In this step we descend the canonical bundle formula of h to V.

Since $g|_W = \rho \circ h$, we have

$$K_W + B_W = (K_X + B)|_W \sim_{\mathbb{Q}} -h^*\rho^*(H|_V).$$

Since h is a contraction with $0 < \dim V' = \dim V \le \dim W \le d'$, (W, B_W) is lc, and $B_W \in \Phi(\mathfrak{R})$, Theorem 3.3(1) implies a trivialization of degree m

$$K_W + B_W + \frac{1}{m}\operatorname{div}(\psi) = h^*D_{V'}, \qquad D_{V'} \sim_{\mathbb{Q}} -\rho^*(H|_V).$$

Let $(V', B_{V'}, \mathbf{M})$ be the induced generalized sub-pair. Since $V' \to U$ is projective, Theorem 3.3(2) implies that $B_{V'} \ge 0$, $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over U, and $p\mathbf{M}$ is Cartier on any smooth variety projective and birational over V'. Since $\mathbf{M}_{V'}$ is the pushforward of the trace of $\mathbf{M}$ on a resolution of V', $p\mathbf{M}_{V'}$ has integer coefficients. As $B_W \in \Phi(\mathfrak{R})$ and $m \mid p$, Lemma 3.7 implies that $B_{V'} \in \Lambda$.

The conditions of Set-up 6.1 hold for d', m, $h : W \to V'$, (W, B_W), ψ, G, and $V \to U$: condition (3) holds since (W, B_W) is klt, conditions (5) and (7) hold by **Step 2**, and condition (6) holds since V is a closed subvariety of T. Here $\mathbf{M}$ is computed with $K_{V'} = \operatorname{div}(\rho^*\omega_V)$, where ω_V is the rational 1-canonical form on V fixed in Set-up 6.1, and the conclusions of Theorem 3.3(2) hold for this $\mathbf{M}$. By Theorem 6.4, with b and p' the integers of Construction 8.2, there exist a rational function χ on V', a rational b-canonical form θ on W, a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D_V on V, a $\mathbb{Q}$-divisor $B_V \ge 0$ on V, and a $\boldsymbol{b}$-$\mathbb{Q}$-Cartier $\mathbb{Q}$-$\boldsymbol{b}$-divisor $\mathbf{N}$ over V, satisfying (1)–(4) of Theorem 6.4. In particular,

$$\rho^*D_V = D_{V'} + \frac{1}{b}\operatorname{div}(\chi), \qquad \operatorname{div}(\theta) = bh^*\rho^*D_V - bB_W, \qquad D_V = K_V + B_V + \mathbf{N}_V,$$

and $\mathbf{N}$ is nef over U.

Step 4. In this step we show that $(V, B_V, \mathbf{N})/U$ is generalized klt, $B_V \in \Lambda$, and $D_V \sim_{\mathbb{Q}} -H|_V$.

Let E be a prime divisor over V. Let $V_1 \to V$ be a log resolution of $(V, \operatorname{Supp} B_V)$ such that $\mathbf{N}$ descends to V_1 and E is a divisor on V_1 (cf. [KM98, Lemma 2.45]), V_1' the normalization of V_1 in V' (Definition 2.7), $\rho_1 : V_1' \to V_1$ the induced finite morphism, and B_{V_1} the $\mathbb{Q}$-divisor with $K_{V_1} + B_{V_1} + \mathbf{N}_{V_1}$ equal to the pullback of D_V. Then $a(E, V, B_V, \mathbf{N}) = 1 - \operatorname{mult}_E B_{V_1}$. By Theorem 6.4(2), $\rho_1^*(K_{V_1} + B_{V_1}) = K_{V_1'} + B_{V_1'}$, where $B_{V_1'}$ is the trace on V_1' of the discriminant $\boldsymbol{b}$-divisor of (W, B_W) over V'. Let Q be a prime divisor on V_1' over E and e the ramification index of ρ_1 along Q. By the Hurwitz formula [Kol13, 2.41] (the coefficient of Q in the ramification divisor of ρ_1 is $e - 1$, since the characteristic is zero),

$$1 - \operatorname{mult}_Q B_{V_1'} = e\,(1 - \operatorname{mult}_E B_{V_1})\,.$$

Since (W, B_W) is klt and $V_1' \to V'$ is projective and birational (since V_1' is the normalization of the irreducible component of $V_1 \times_V V'$ dominating V', and $V_1 \to V$ is projective and birational), $\operatorname{mult}_Q B_{V_1'} < 1$ by Lemma 3.8(3). Thus $a(E, V, B_V, \mathbf{N}) > 0$, and $(V, B_V, \mathbf{N})/U$ is generalized klt.

Taking $V_1 = V$ in Theorem 6.4(2), we have $\rho^*(K_V + B_V) = K_{V'} + B_{V'}$. Let E be a prime divisor on V, Q a prime divisor on V' over E, and e the ramification index of ρ along Q. Since $B_{V'} \in \Lambda$, we may write $\operatorname{mult}_Q B_{V'} = 1 - \frac{r}{j}$ with $r \in \mathfrak{R}[p]$ and j a positive integer. By the Hurwitz formula, $\operatorname{mult}_E B_V = 1 - \frac{r}{je} \in \Lambda$. Thus $B_V \in \Lambda$.

Let $D := D_V + H|_V$. Then

$$\rho^* D = D_{V'} + \rho^*(H|_V) + \frac{1}{b}\operatorname{div}(\chi) \sim_{\mathbb{Q}} 0,$$

so there exist a positive integer k' and a rational function ξ on V' such that $\rho^* D = \frac{1}{k'}\operatorname{div}(\xi)$. For a non-zero rational function η on V', let $V^\circ \subset V$ be a non-empty open subset such that $\rho_*\mathcal{O}_{V'}$ is locally free of rank $\deg\rho$ on V° and η is regular on $\rho^{-1}(V^\circ)$, and let $\operatorname{Nm}(\eta)$ be the determinant of the $\mathcal{O}_{V^\circ}$-linear endomorphism of $\rho_*\mathcal{O}_{V'}|_{V^\circ}$ given by multiplication by η. Then $\operatorname{Nm}(\eta)$ is a non-zero rational function on V, which does not depend on V°. By [Ful98, Proposition 1.4(b)],

$$\rho_* \operatorname{div}(\eta) = \operatorname{div}(\operatorname{Nm}(\eta)).$$

Applying the last equality to the pullbacks $\rho^* f$ of local equations f of a Cartier multiple of D, for which $\operatorname{Nm}(\rho^* f) = f^{\deg\rho}$, we obtain $\rho_*\rho^* D = (\deg\rho)D$. Thus

$$(\deg\rho)k' D = k'\rho_*\rho^* D = \operatorname{div}(\operatorname{Nm}(\xi)),$$

and $D_V \sim_{\mathbb{Q}} -H|_V$.

Step 5. In this step we construct a complement of $(V, B_V, \mathbf{N})$ over a neighborhood of u, and we shrink U.

Let $V \to U_V \to U$ be the Stein factorization of $V \to U$. Since $U_V \to U$ is finite and U is affine, U_V is affine [Har77, Exercise II.3.4]. Since $\mathbf{N}$ is nef over U and $H|_V$ is ample over U, $\mathbf{N}$ is nef over U_V and $-(K_V + B_V + \mathbf{N}_V) = -D_V \sim_{\mathbb{Q}} H|_V$ is ample over U_V. By **Step 4** and Lemma 4.1, V is of Fano type over U_V.

Since $\mathbf{N}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, it descends to a smooth variety projective and birational over V, on which $p'\mathbf{N}$ is Cartier by Theorem 6.4(4). Thus $p'\mathbf{N}$ is $\boldsymbol{b}$-Cartier. Since $\mathfrak{R}[p]$ is a finite subset of $[0,1] \cap \mathbb{Q}$ containing 0 and 1 (Lemma 3.7), Λ satisfies the DCC and the only accumulation point of Λ is 1. By Set-up 8.1(5), V intersects T_u, so u belongs to the image of V in U. Moreover, $1 \le \dim V \le d'$ and $B_V \in \Lambda$ (**Step 4**). By Proposition 4.2, applied with parameters d', p', $\operatorname{lcm}\{N_0, b, p, p'\}$, and Λ to $V \to U$ and $(V, B_V, \mathbf{N})/U_V$, there exist an open neighborhood U^0 of u and a $\mathbb{Q}$-divisor $B_V^+ \ge B_V$ on the inverse image V^0 of U^0 such that $(V^0, B_V^+, \mathbf{N})$ is generalized lc and

$$n_1(K_{V^0} + B_V^+ + \mathbf{N}_{V^0}) \sim 0.$$

Possibly shrinking U to an affine open neighborhood of u in U^0 and replacing X, T, V, V', W with their inverse images, we may assume that $U^0 = U$.

Let $E_V := B_V^+ - B_V \ge 0$. Since $K_V + B_V^+ + \mathbf{N}_V = D_V + E_V$ and D_V is $\mathbb{Q}$-Cartier, E_V is $\mathbb{Q}$-Cartier, and $(V, B_V + E_V, \mathbf{N})/U$ is generalized lc. Since $n_1 \mid N$, there exists a rational function λ on V such that

$$N(D_V + E_V) = \operatorname{div}(\lambda).$$

Step 6. In this step we define t and show that t is a non-zero G-invariant section of $\mathcal{J}_{N,W}$.

Recall that θ is the rational b-canonical form on W of **Step 3** with $\operatorname{div}(\theta) = bh^*\rho^* D_V - bB_W$. Since $b \mid N$, we may define the rational $(-N)$-canonical form

$$t := (h^*\rho^*\lambda)\theta^{-N/b}$$

on W. Then

$$\operatorname{div}(t) = Nh^*\rho^*(D_V + E_V) - \frac{N}{b}(bh^*\rho^* D_V - bB_W) = NB_W + Nh^*\rho^* E_V,$$

so $B_t = B_W + h^*\rho^* E_V$. Since $\operatorname{div}(t)$ has integer coefficients and $\operatorname{div}(t) \ge NB_W$, we have $\operatorname{div}(t) \ge \lceil NB_W \rceil$. Thus t is a non-zero element of $H^0(W, \mathcal{J}_{N,W})$.

Let $\phi : W \dashrightarrow W$ be a birational map which is crepant for (W, B_W) and satisfies $h \circ \phi = \sigma \circ h$ for some $\sigma \in G$. By Theorem 6.4(1), $\phi^*\theta = \theta$. Since $\rho \circ \sigma = \rho$, we have

$$\phi^* h^* \rho^* \lambda = h^* \sigma^* \rho^* \lambda = h^* \rho^* \lambda.$$

Thus $\phi^* t = t$. Since $N_0 \mid N$, G acts on $H^0(W, \mathcal{J}_{N,W})$ through such birational maps ϕ (Theorem 7.13(1) and Lemma 7.5). Therefore t is G-invariant.

Step 7. In this step we show that (W, B_t) is lc.

We use the trivialization of degree m of **Step 3**, whose moduli $\boldsymbol{b}$-divisor is $\mathbf{M}$, and we let $P := \rho^* E_V$. The discriminant $\boldsymbol{b}$-divisor depends only on (W, B_W) (Definition 3.1(2)), so the trace of the discriminant $\boldsymbol{b}$-divisor on V_1' is the $\mathbb{Q}$-divisor $B_{V_1'}$ in Theorem 6.4(2) for any V_1. By Lemma 3.8(2), it suffices to show that, for any normal variety V_2' with a projective birational morphism $\tau_2 : V_2' \to V'$, the coefficients of $B_{V_2'} + \tau_2^* P$ are at most 1.

Let $V_1 \to V$ be a log resolution of $(V, \mathrm{Supp}(B_V + E_V))$ such that $\mathbf{N}$ descends to V_1 (cf. [KM98, Theorem 0.2]), V_1' the normalization of V_1 in V' (Definition 2.7), $\rho_1 : V_1' \to V_1$ the induced finite morphism, E_1 the pullback of E_V to V_1, and B_{V_1} the $\mathbb{Q}$-divisor with $K_{V_1} + B_{V_1} + \mathbf{N}_{V_1}$ equal to the pullback of D_V. Then $K_{V_1} + B_{V_1} + E_1 + \mathbf{N}_{V_1}$ is the pullback of $K_V + B_V + E_V + \mathbf{N}_V$. Since $\mathbf{N}$ descends to V_1 and $(V, B_V + E_V, \mathbf{N})/U$ is generalized lc, the sub-pair $(V_1, B_{V_1} + E_1)$ is sub-lc. By Theorem 6.4(2),

$$\rho_1^*(K_{V_1} + B_{V_1} + E_1) = K_{V_1'} + B_{V_1'} + \rho_1^* E_1,$$

so $(V_1', B_{V_1'} + \rho_1^* E_1)$ is sub-lc by [KM98, Proposition 5.20]. Since $P = \rho^* E_V$, E_1 is the pullback of E_V to V_1, and ρ_1 is compatible with ρ, $\rho_1^* E_1$ is the pullback of P to V_1'.

By Definition 2.7, $\rho^*\mathbf{N}$ descends to V_1'. Since $\mathbf{M} = \rho^*\mathbf{N} - \frac{1}{b}\mathrm{div}(\chi)$ (Theorem 6.4(2)), $\mathbf{M}$ also descends to V_1'. Let V_3' be a normal variety with a projective birational morphism $\tau : V_3' \to V_1'$, and P_3 the pullback of P to V_3'. Since $K_{V_3'} + B_{V_3'} + \mathbf{M}_{V_3'}$ and $K_{V_1'} + B_{V_1'} + \mathbf{M}_{V_1'}$ are the pullbacks of $D_{V'}$ (Definition 3.1(3)) and $\mathbf{M}_{V_3'} = \tau^* \mathbf{M}_{V_1'}$, we have

$$K_{V_3'} + B_{V_3'} + P_3 = \tau^* \left(K_{V_1'} + B_{V_1'} + \rho_1^* E_1 \right).$$

Thus the coefficients of $B_{V_3'} + P_3$ are at most 1. For any normal variety V_2' with a projective birational morphism $\tau_2 : V_2' \to V'$, we choose V_3' projective and birational over both V_1' and V_2'. Then $B_{V_2'} + \tau_2^* P$ is the pushforward of $B_{V_3'} + P_3$, so the coefficients of $B_{V_2'} + \tau_2^* P$ are at most 1.

By Lemma 3.8(2), $(W, B_W + h^*\rho^* E_V)$ is sub-lc. Since $B_t = B_W + h^*\rho^* E_V \geq 0$, (W, B_t) is lc. □

8.4. **Extension of invariant sections.** In this subsection, we treat the case when some lc center of (X, B) does not dominate T (Proposition 8.8), and then we prove Theorem 8.3.

Lemma 8.6. *Let (X, B) be a dlt pair such that B is a $\mathbb{Q}$-divisor, S an lc center of (X, B), and $c := \mathrm{codim}_X S$. Let $S_0 := X$ and $B_{S_0} := B$. Then there exist lc centers $S_1 \supsetneq \cdots \supsetneq S_c = S$ of (X, B) with $\mathrm{codim}_X S_i = i$ for $1 \leq i \leq c$ such that, with B_{S_i} the different of (X, B) on S_i for $1 \leq i \leq c$, for any $0 \leq i < c$ the pair (S_i, B_{S_i}) is dlt, B_{S_i} is a $\mathbb{Q}$-divisor, S_{i+1} is a component of $\lfloor B_{S_i} \rfloor$, and $B_{S_{i+1}}$ is the different of (S_i, B_{S_i}) on S_{i+1}.*

Proof. By [Kol13, Theorem 4.16], there exist components $S_1', \ldots, S_c'$ of $\lfloor B \rfloor$ such that S is an irreducible component of $S_1' \cap \cdots \cap S_c'$. For $1 \leq i \leq c$, let S_i be the irreducible component of $S_1' \cap \cdots \cap S_i'$ containing S (it is unique, since (X, B) is snc at the generic point of S [KM98, Definition 2.37], hence $S_{i+1} \subset S_i$ for $1 \leq i < c$, and $S_c = S$), which is an lc center of (X, B) with $\mathrm{codim}_X S_i = i$ [Kol13, Theorem 4.16]. For $1 \leq i < c$, Lemma 2.2(1)(2) and [Kol13, Theorem 4.19(1)], applied to the lc centers $S_{i+1} \subsetneq S_i$ of (X, B), show that (S_i, B_{S_i}) is dlt, B_{S_i} is a $\mathbb{Q}$-divisor, S_{i+1} is an lc center of (S_i, B_{S_i}) of codimension one in S_i, that is, a component of $\lfloor B_{S_i} \rfloor$, and $B_{S_{i+1}}$ is the different of (S_i, B_{S_i}) on S_{i+1}. For $i = 0$, the pair $(S_0, B_{S_0}) = (X, B)$ is dlt, B is a $\mathbb{Q}$-divisor, $S_1 = S_1'$ is a component of $\lfloor B \rfloor$, and B_{S_1} is the different of (X, B) on S_1 by definition. □

By Proposition 8.5, there exists a G-invariant section t on W, and log canonicity is only known for the pair (W, B_t). After extending t to X, we have to prove log canonicity along any lc center W_1 of (X, B) which is maximal among the lc centers with image V, and W_1 need not be birational to W. Proposition 7.14(3) provides an lc center contained in W_1 which is crepant birational to (W, B_t) and dominates the base of the Stein factorization of $W_1 \to V$. The following lemma shows that such a horizontal lc center detects the lc threshold of any pullback from the base.

Lemma 8.7. *Let $f : X \to Z$ be a contraction, (X, B) a dlt pair, S an lc center of (X, B), and B_S the divisor on S given by dlt adjunction $K_S + B_S = (K_X + B)|_S$. Assume that $\dim Z > 0$, B is a $\mathbb{Q}$-divisor, $K_X + B \sim_{\mathbb{Q},Z} 0$, and S dominates Z. Then for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D \geq 0$ on Z,*

$$\operatorname{lct}(X, B; f^*D) = \operatorname{lct}(S, B_S; (f|_S)^*D).$$

In particular, $(X, B + f^*D)$ is lc if and only if $(S, B_S + (f|_S)^*D)$ is lc.

Proof. **Step 1.** In this step we prove the lemma when S is a component of $\lfloor B \rfloor$.

Let $F := f^*D$ and $F_S := (f|_S)^*D$. Since S dominates Z and $\operatorname{Supp} D \neq Z$, we have $S \not\subset \operatorname{Supp} F$. If $D = 0$, then both thresholds are $+\infty$. Assume that $D \neq 0$. Since f and $f|_S$ are surjective, $F \neq 0$ and $F_S \neq 0$, so both thresholds are finite. Let $t := \operatorname{lct}(X, B; F)$. Let $\varphi : X' \to X$ be a log resolution of $(X, B + F)$. By [Kol13, Notation 2.6, 2.20 and Corollary 2.11], t is the minimum of the numbers $a(P, X, B)/\operatorname{mult}_P F$ over the finitely many prime divisors P on X' with $\operatorname{mult}_P F > 0$, and $a(P, X, B)/\operatorname{mult}_P F \in \mathbb{Q}$ for any such P since B and D are $\mathbb{Q}$-divisors. Hence $t \in \mathbb{Q}$, $(X, B + tF)$ is lc, and a prime divisor P on X' attaining the minimum satisfies

$$a(P, X, B + tF) = 0 \quad \text{and} \quad \operatorname{mult}_P F > 0.$$

Since $(X, B + tF)$ is lc, by [KK10, Theorem 3.1 and Definitions and Notation 1.9], there exists a projective birational morphism $\mu : Y \to X$ such that Y is $\mathbb{Q}$-factorial, (Y, B_Y) is dlt, and

$$K_Y + B_Y = \mu^*(K_X + B + tF),$$

where B_Y is the sum of $\mu_*^{-1}(B + tF)$ and the μ-exceptional prime divisors. Here μ is projective, since [KK10, Theorem 3.1] constructs μ as a log terminal model over X [BCHM10] (cf. [Kol13, Theorem 1.34]). Since $S \not\subset \operatorname{Supp} F$, the coefficient of S in $B + tF$ is 1, so the strict transform S_Y of S is a component of $\lfloor B_Y \rfloor$. Let $q := f \circ \mu$. Since $\mu_* \mathcal{O}_Y = \mathcal{O}_X$, we have $q_* \mathcal{O}_Y = \mathcal{O}_Z$, and $K_Y + B_Y \sim_{\mathbb{Q},Z} 0$ because $K_X + B + tF \sim_{\mathbb{Q},Z} 0$. Thus $q : (Y, B_Y) \to Z$ is a dlt crepant log structure [Kol16, Definition 2].

Let $\mu_S : S_Y \to S$ be the birational morphism induced by μ, and B_{S_Y} the divisor on S_Y given by dlt adjunction $K_{S_Y} + B_{S_Y} = (K_Y + B_Y)|_{S_Y}$. Since F is $\mathbb{Q}$-Cartier and $S \not\subset \operatorname{Supp} F$, the $\mathbb{Q}$-divisor $B_S + tF_S$ is the different of $(X, B + tF)$ on S. Since differents are compatible with crepant birational morphisms [Kol13, Proposition 4.6],

$$K_{S_Y} + B_{S_Y} = \mu_S^*(K_S + B_S + tF_S). \tag{8.1}$$

By Lemma 2.2(1), (S_Y, B_{S_Y}) is dlt. Thus $(S, B_S + tF_S)$ is lc, and $\operatorname{lct}(S, B_S; F_S) \geq t$.

For the converse inequality, let C_Y be the center of P on Y. Then

$$a(P, Y, B_Y) = a(P, X, B + tF) = 0,$$

so the variety C_Y is an lc center of (Y, B_Y). Let $C := q(C_Y)$. Since $\operatorname{mult}_P F > 0$, we have $C_Y \subset \operatorname{Supp} \mu^* F = q^{-1}(\operatorname{Supp} D)$, hence $C \subset \operatorname{Supp} D$. Let $S_Y \xrightarrow{h_S} Z_S \xrightarrow{\rho_S} Z$ be the Stein factorization of $q|_{S_Y}$. Since S_Y dominates Z, ρ_S is finite and surjective, so $\rho_S^{-1}(C) \neq \emptyset$. Let C' be an irreducible component of $\rho_S^{-1}(C)$. Since C is an lc center of the crepant log structure q, by [Kol16, Corollary 11(3)], $C' = h_S(R)$ for an lc center R of (S_Y, B_{S_Y}). Let P' be a prime divisor over S_Y with center R such that $a(P', S_Y, B_{S_Y}) = 0$. Since $q(R) \subset C \subset \operatorname{Supp} D$, we have $R \subset \operatorname{Supp}(q|_{S_Y})^* D$, hence $\operatorname{mult}_{P'} D > 0$. Since $(q|_{S_Y})^* D = \mu_S^* F_S$, (8.1) implies that

$$a(P', S, B_S + tF_S) = 0 \quad \text{and} \quad \operatorname{mult}_{P'} F_S > 0.$$

Therefore, for any real number $\delta > 0$,

$$a(P', S, B_S + (t+\delta)F_S) = -\delta \operatorname{mult}_{P'} F_S < 0,$$

so $(S, B_S + (t+\delta)F_S)$ is not lc. Thus $\operatorname{lct}(S, B_S; F_S) = t$.

Step 2. In this step we prove the lemma for an arbitrary lc center S dominating Z.

Let $c := \operatorname{codim}_X S$. By Lemma 8.6, there exist lc centers $S_1 \supsetneq \cdots \supsetneq S_c = S$ of (X, B) with $\operatorname{codim}_X S_i = i$ such that, with $S_0 := X$, $B_{S_0} := B$, and B_{S_i} given by dlt adjunction $K_{S_i} + B_{S_i} = (K_X + B)|_{S_i}$, for any $0 \le i < c$ the pair (S_i, B_{S_i}) is dlt, B_{S_i} is a $\mathbb{Q}$-divisor, S_{i+1} is a component of $\lfloor B_{S_i} \rfloor$, and $B_{S_{i+1}}$ is the different of (S_i, B_{S_i}) on S_{i+1}.

Fix $0 \le i < c$, and let $S_i \xrightarrow{f_i} Z_i \xrightarrow{\tau_i} Z$ be the Stein factorization of $f|_{S_i}$, where $Z_0 = Z$ and $f_0 = f$. Then f_i is a contraction, τ_i is finite, and $\dim Z_i = \dim Z > 0$. Since $K_{S_i} + B_{S_i} = (K_X + B)|_{S_i} \sim_{\mathbb{Q}, Z} 0$, we have $K_{S_i} + B_{S_i} \sim_{\mathbb{Q}, Z_i} 0$. Since S_{i+1} dominates Z and τ_i is finite, $\dim f_i(S_{i+1}) = \dim Z = \dim Z_i$, so S_{i+1} dominates Z_i. The $\mathbb{Q}$-divisor $\tau_i^* D \ge 0$ is $\mathbb{Q}$-Cartier, and $f_i^* \tau_i^* D = (f|_{S_i})^* D$. **Step 1** for f_i, (S_i, B_{S_i}), S_{i+1}, and $\tau_i^* D$ implies that

$$\operatorname{lct}(S_i, B_{S_i}; (f|_{S_i})^* D) = \operatorname{lct}(S_{i+1}, B_{S_{i+1}}; (f|_{S_{i+1}})^* D). \tag{8.2}$$

Combining (8.2) for $0 \le i < c$, we obtain $\operatorname{lct}(X, B; f^* D) = \operatorname{lct}(S, B_S; (f|_S)^* D)$.

Finally, when the lc threshold of a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D' \ge 0$ with respect to an lc pair is finite, it is attained, since the description of t in **Step 1** holds for any lc pair whose boundary is a $\mathbb{Q}$-divisor and any non-zero $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $D' \ge 0$. Hence $(X, B + f^* D)$ is lc if and only if $\operatorname{lct}(X, B; f^* D) \ge 1$, and $(S, B_S + (f|_S)^* D)$ is lc if and only if $\operatorname{lct}(S, B_S; (f|_S)^* D) \ge 1$. This proves the last statement. □

Proposition 8.8. *Notation and conditions as in Set-up 8.1, and let $N := N(d, \Gamma)$ be the integer of Construction 8.2. Assume that some lc center of (X, B) does not dominate T. Then $(X/U \ni u, B)$ has a monotonic N-complement.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we choose a minimal lc image V and fix notation.

Let d' be the integer of Construction 8.2. By [Kol13, Theorem 4.16], (X, B) has finitely many lc centers. Since g is proper, the image on T of an lc center of (X, B) which does not dominate T is a proper closed subset of T. Thus there exists a minimal element V, with respect to inclusion, of the set of images on T of the lc centers of (X, B) such that $V \subsetneq T$. Since $\dim X \le d \le d'$, the conditions of Set-up 7.1 hold for d', Γ, $X \xrightarrow{g} T \to U$, (X, B), H, and V. By Set-up 8.1(5), $V \cap T_u \ne \emptyset$. Let Σ_V, W, B_W, $h : W \to V'$, $\rho : V' \to V$, G, $\mu : \tilde{X} \to X$, $\tilde{\Sigma}_V$, L_N, and $\mathcal{J}_{N,W}$ be as in Construction 7.2, and N_0 the integer of Construction 7.3, for d' and Γ. By Construction 8.2, $N_0 \mid N$.

Step 2. In this step we extend an invariant section from W to $\tilde{X}$ and construct the $\mathbb{Q}$-divisor B^+.

By Proposition 8.5, possibly replacing U with an affine open neighborhood of u and X, T, V, V', W, and the objects of Construction 7.2 with their inverse images, we may assume that there exists a non-zero G-invariant section $t \in H^0(W, \mathcal{J}_{N,W})$ such that (W, B_t) is lc.

Since $N_0 \mid N$ and t is G-invariant, by Theorem 7.13(2), t is the image of a section $s \in H^0(\tilde{\Sigma}_V, \mathcal{O}_{\tilde{\Sigma}_V}(L_N))$ under the injective map of Theorem 7.13(2). Since $t \ne 0$, the section s is non-zero. Since U is affine, by Theorem 7.13(2) again, there exists $\tau \in H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(L_N))$ such that $\tau|_{\tilde{\Sigma}_V} = s$. Regard τ as a rational $(-N)$-canonical form on X, and let $B^+ := \frac{1}{N} \operatorname{div}_X(\tau)$. By Proposition 7.14(1),

$$B^+ \ge B \quad \text{and} \quad N(K_X + B^+) \sim 0.$$

Since X is $\mathbb{Q}$-factorial, $B^+ - B \ge 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor.

Step 3. In this step we prove log canonicity of the pair induced by B^+ on any lc center which is maximal among the lc centers with image V.

Let W_1 be an lc center of (X, B) which is maximal with respect to inclusion among the lc centers of (X, B) with image V on T, $W_1 \xrightarrow{f_1} V_1 \to V$ the Stein factorization of $W_1 \to V$, and B_{W_1} the divisor on W_1 given by dlt adjunction $K_{W_1} + B_{W_1} = (K_X + B)|_{W_1}$. By Proposition 7.14(2), $W_1 \not\subset \operatorname{Supp}(B^+ - B)$ and

$$(B^+ - B)|_{W_1} = f_1^* E_1$$

for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $E_1 \geq 0$ on V_1. Since (W, B_t) is lc, by Proposition 7.14(3), there exists an lc center $W_2 \subset W_1$ of (X, B) with image V on T such that $(W_2, B_{W_2} + (f_1|_{W_2})^* E_1)$ is lc, where $K_{W_2} + B_{W_2} = (K_X + B)|_{W_2}$ is given by dlt adjunction. We show that $(W_1, B_{W_1} + f_1^* E_1)$ is lc.

Suppose first that $\dim V > 0$. If $W_2 = W_1$, then $(W_1, B_{W_1} + f_1^* E_1) = (W_2, B_{W_2} + (f_1|_{W_2})^* E_1)$ is lc, so we may assume $W_2 \subsetneq W_1$. By Lemma 2.2(1)(2) and [Kol13, Theorem 4.19(1)], (W_1, B_{W_1}) is dlt, B_{W_1} is a $\mathbb{Q}$-divisor, W_2 is an lc center of (W_1, B_{W_1}), and the different of (W_1, B_{W_1}) on W_2 is B_{W_2}. The morphism f_1 is a contraction, and $\dim V_1 = \dim V > 0$. We have

$$K_{W_1} + B_{W_1} = (K_X + B)|_{W_1} \sim_{\mathbb{Q}} -(g|_{W_1})^* H,$$

and $g|_{W_1} : W_1 \to T$ factors through f_1, so $K_{W_1} + B_{W_1} \sim_{\mathbb{Q}, V_1} 0$. The set $f_1(W_2) \subset V_1$ is closed. Since $V_1 \to V$ is finite and $W_2 \to V$ is surjective, $\dim f_1(W_2) = \dim V = \dim V_1$. Since V_1 is irreducible, W_2 dominates V_1. Thus Lemma 8.7 for $f_1 : W_1 \to V_1$, (W_1, B_{W_1}), $S := W_2$, and $D := E_1$ implies that $(W_1, B_{W_1} + f_1^* E_1)$ is lc.

Suppose that $\dim V = 0$. Then V and V_1 are points, so $E_1 = 0$. Since (W_1, B_{W_1}) is dlt (Lemma 2.2(1)), $(W_1, B_{W_1} + f_1^* E_1) = (W_1, B_{W_1})$ is lc.

Step 4. In this step we prove that (X, B^+) is lc on a neighborhood of Σ_V.

Let W_1 be an lc center of (X, B) which is maximal with respect to inclusion among the lc centers with image V, as in **Step 3**. Let $c := \operatorname{codim}_X W_1$. By Lemma 8.6 for the lc center W_1, there exist lc centers $S_1 \supsetneq \cdots \supsetneq S_c = W_1$ of (X, B) such that, with $S_0 := X$, $B_{S_0} := B$, and $K_{S_i} + B_{S_i} = (K_X + B)|_{S_i}$ given by dlt adjunction, for any $0 \leq i < c$ the variety S_{i+1} is a component of $\lfloor B_{S_i} \rfloor$ and $B_{S_{i+1}}$ is the different of (S_i, B_{S_i}) on S_{i+1}. Since $W_1 \subset S_i$ and $W_1 \not\subset \operatorname{Supp}(B^+ - B)$, we have $S_i \not\subset \operatorname{Supp}(B^+ - B)$. Let

$$B_i^+ := B_{S_i} + (B^+ - B)|_{S_i}$$

for $0 \leq i \leq c$. Then $B_0^+ = B^+$, $K_{S_i} + B_i^+ = (K_X + B^+)|_{S_i}$ is $\mathbb{Q}$-Cartier, and $B_c^+ = B_{W_1} + f_1^* E_1$ by **Step 3**. For $0 \leq i < c$, since $(B^+ - B)|_{S_i} \geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor and $S_{i+1} \not\subset \operatorname{Supp}(B^+ - B)$, the coefficient of S_{i+1} in B_i^+ is 1, and B_{i+1}^+ is the different of (S_i, B_i^+) on S_{i+1}.

We show by descending induction on i that there exists an open subset $O_i \subset X$ containing W_1 such that $(S_i \cap O_i, B_i^+)$ is lc. For $i = c$, let $O_c := X$. By **Step 3**, $(W_1, B_{W_1} + f_1^* E_1)$ is lc. Assume that O_{i+1} exists for some $0 \leq i < c$. By [Kaw07, Theorem] applied to the pair $(S_i \cap O_{i+1}, B_i^+)$ and the reduced divisor $S_{i+1} \cap O_{i+1}$, which is normal (Lemma 2.2(1)), the pair $(S_i \cap O_{i+1}, B_i^+)$ is lc on an open neighborhood of $S_{i+1} \cap O_{i+1}$ in $S_i \cap O_{i+1}$. Since S_i is closed in X, there exists an open subset $O_i \subset O_{i+1}$ containing $S_{i+1} \cap O_{i+1} \supset W_1$ such that $(S_i \cap O_i, B_i^+)$ is lc. For $i = 0$, we obtain that (X, B^+) is lc on the open neighborhood O_0 of W_1.

Any lc center of (X, B) with image V is contained in an lc center which is maximal among the lc centers with image V. Hence Σ_V is the union of the maximal ones, and (X, B^+) is lc on a neighborhood of Σ_V.

Step 5. In this step we prove that (X, B^+) is lc over a neighborhood of u, and conclude the proof.

Let $\pi : X \to U$ be the composite of g and $T \to U$, and $\operatorname{Nlc}(X, B^+)$ the set of points of X near which (X, B^+) is not lc. Then $\operatorname{Nlc}(X, B^+)$ is closed, and since π is proper, $\pi(\operatorname{Nlc}(X, B^+))$ is closed in U. Let u' be a closed point of the closure of $\{u\}$ in U. Suppose that $u \in \pi(\operatorname{Nlc}(X, B^+))$. Then $u' \in \pi(\operatorname{Nlc}(X, B^+))$.

Let W_1 be a maximal lc center as in **Step 3**. Since $g(W_1) = V$ and $V \cap T_u \neq \emptyset$, the closed subset $\pi(W_1) \subset U$ contains u, hence contains u'. Let P be a prime divisor over X with center

W_1 such that $a(P, X, B) = 0$. Since $B^+ - B$ is $\mathbb{Q}$-Cartier and $W_1 \not\subset \operatorname{Supp}(B^+ - B)$, we have $\operatorname{mult}_P(B^+ - B) = 0$, so

$$a(P, X, B^+) = a(P, X, B) - \operatorname{mult}_P(B^+ - B) = 0.$$

By **Step 4**, (X, B^+) is lc on a neighborhood of W_1. Thus W_1 is an lc center of (X, B^+). Since $N(K_X + B^+) \sim 0$, $-(K_X + B^+)$ is semi-ample over U. Moreover, $\pi_*\mathcal{O}_X = \mathcal{O}_U$ and $B^+ \geq B \geq 0$. Since $\operatorname{Nlc}(X, B^+) \cap \pi^{-1}(u') \neq \emptyset$ and $W_1 \cap \pi^{-1}(u') \neq \emptyset$, by [Fuj17, Theorem 6.3], $\operatorname{Nlc}(X, B^+)$ intersects $W_1 \cap \pi^{-1}(u')$. This contradicts the log canonicity of (X, B^+) on a neighborhood of W_1. Hence $u \notin \pi(\operatorname{Nlc}(X, B^+))$.

Therefore $U \backslash \pi(\operatorname{Nlc}(X, B^+))$ is an open neighborhood of u over which (X, B^+) is lc and $N(K_X + B^+) \sim 0$. Since $B^+ \geq B$ is a $\mathbb{Q}$-divisor, $(X/U \ni u, B^+)$ is a monotonic N-complement of $(X/U \ni u, B)$. □

Proof of Theorem 8.3. If any lc center of (X, B) dominates T, then Theorem 8.3 follows from Proposition 8.4. Otherwise, some lc center of (X, B) does not dominate T, and Theorem 8.3 follows from Proposition 8.8. □

9. A common base for two truncations

This section and Sections 10 and 11 prepare the proof of Theorem 12.2. As in the two-model construction in [CHX24, Section 6], the proof compares the relative Iitaka dimensions of the differences between a klt reference boundary and the truncations of the reference boundary. In this section, we treat two consecutive truncations B_1 and B_2 whose differences from the reference boundary have equal relative Iitaka dimensions. We prove that, after a D_2-MMP followed by a D_1-MMP, the base of the second MMP is a common base for B_1 and B_2 (Lemma 9.2), and that the canonical bundle formulas of B_1 and B_2 share the moduli $\boldsymbol{b}$-divisor over the common base, whose denominator depends only on $\dim X$ and the coefficients of B_1 (Proposition 9.3).

9.1. The geometry of two MMPs.

In this subsection, we introduce the set-up of this section and prove Lemma 9.2.

Set-up 9.1. Let $\pi : X \to Z$, (X, B), B_1, B_2, and k be as follows.

(1) $\pi : X \to Z$ is a contraction such that Z is affine and X is $\mathbb{Q}$-factorial.
(2) (X, B) is an lc pair such that B is a $\mathbb{Q}$-divisor and $K_X + B \sim_{\mathbb{Q},Z} 0$.
(3) B_1 and B_2 are $\mathbb{Q}$-divisors such that $0 \leq B_1 \leq B_2 \leq B$, and $D_i := B - B_i$ for $i = 1, 2$.
(4) $\kappa(X/Z, D_1) = \kappa(X/Z, D_2) = k$ and $k + \dim Z > 0$.

For a $\mathbb{Q}$-divisor D on X and a birational map $X \dashrightarrow X_j$, we write D_{X_j} for the strict transform of D on X_j.

By condition (4), the bases of the MMPs in Lemma 9.2 are birational, so that B_1 and B_2 share the moduli $\boldsymbol{b}$-divisor over the common base, and the integers m and p attached to B_2 depend only on $\dim X$ and the coefficient set of B_1 (Proposition 9.3). The case excluded by the condition $k + \dim Z > 0$, where Z is a point and D_2 has Iitaka dimension zero, is treated in Lemma 11.3. We run the D_2-MMP first because Section 11 takes complements of the generalized pair induced on the common base by B_2.

Lemma 9.2. *Notation and conditions as in Set-up 9.1. Assume that there exist a D_2-MMP $X \dashrightarrow X_1$ over Z such that D_{2,X_1} is semi-ample over Z and defines a contraction $f_1 : X_1 \to T_1$ over Z, and a D_{1,X_1}-MMP $X_1 \dashrightarrow X_2$ over T_1 such that D_{1,X_2} is semi-ample over T_1 and defines a contraction $f_2 : X_2 \to T_2$ over T_1. Let $\tau : T_2 \to T_1$ be the induced morphism. Then:*

(1) τ is birational and $\dim T_1 = \dim T_2 = \dim Z + k$;
(2) $D_{2,X_1} = f_1^* A_1$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $A_1 \geq 0$ on T_1 which is ample over Z, and $D_{2,X_2} = f_2^* \tau^* A_1$;
(3) $B_{2,X_2} - B_{1,X_2} = f_2^* G$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ on T_2;
(4) $h'^*(K_{X_2} + B_{2,X_2}) \geq h^*(K_X + B_2)$ for a common resolution $h : W \to X$ and $h' : W \to X_2$;

(5) if

$$K_{X_2} + B_{1,X_2} + \frac{1}{m}\operatorname{div}(\psi) = f_2^*(K_{T_2} + \Delta_1 + \mathbf{M}_{T_2})$$

is a trivialization of degree m with discriminant Δ_1 and moduli $\boldsymbol{b}$-divisor $\mathbf{M}$, then

$$K_{X_2} + B_{2,X_2} + \frac{1}{m}\operatorname{div}(\psi) = f_2^*(K_{T_2} + \Delta_1 + G + \mathbf{M}_{T_2})$$

is a trivialization of degree m whose discriminant is $\Delta_1 + G$ and whose moduli $\boldsymbol{b}$-divisor is $\mathbf{M}$.

Proof. We have the commutative diagram

$$\begin{array}{ccccc} X & \dashrightarrow & X_1 & \dashrightarrow & X_2 \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle f_1} & & \downarrow{\scriptstyle f_2} \\ Z & \longleftarrow & T_1 & \xleftarrow{\tau} & T_2 \end{array}$$

where $T_1 \to Z$ is the structure morphism of T_1 over Z. Let $g := \tau \circ f_2 : X_2 \to T_1$. By Definition 2.18, X_1, X_2, and the varieties between the steps of both MMPs are $\mathbb{Q}$-factorial, so $\mathbb{Q}$-divisors on them are $\mathbb{Q}$-Cartier. In particular, the D_{1,X_1}-MMP over T_1 is defined. Since $D_1 \geq D_2 \geq 0$, we have $D_{1,X_j} \geq D_{2,X_j} \geq 0$ for $j = 1, 2$. Since both MMPs consist of divisorial contractions and flips, any prime divisor on X_1 or X_2 is the strict transform of a prime divisor on X. We prove (2) and (3) first, since the proof of (1) uses (2) and the $\mathbb{Q}$-divisor A_2 constructed in **Step 2**.

Step 1. In this step we prove (2).

Since D_{2,X_1} is semi-ample over Z and defines f_1, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H_1 on T_1 which is ample over Z such that $D_{2,X_1} \sim_{\mathbb{Q}} f_1^* H_1$. Since $D_{2,X_1} \geq 0$, by Lemma 2.17(3), $D_{2,X_1} = f_1^* A_1$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $A_1 \geq 0$ on T_1 such that $A_1 \sim_{\mathbb{Q}} H_1$. In particular, A_1 is ample over Z.

Let $h_2 : W_2 \to X_1$ and $h_2' : W_2 \to X_2$ be a common resolution. Since any prime divisor on X_2 is the strict transform of a prime divisor on X_1, any h_2-exceptional divisor is h_2'-exceptional, so $D_{2,X_2} = h_{2*}' h_2^* D_{2,X_1}$. Since $X_1 \dashrightarrow X_2$ is over T_1, $f_1 \circ h_2 = g \circ h_2'$. Thus

$$D_{2,X_2} = h_{2*}' h_2^* f_1^* A_1 = h_{2*}' h_2'^* g^* A_1 = g^* A_1 = f_2^* \tau^* A_1.$$

Step 2. In this step we prove (3).

Since D_{1,X_2} is semi-ample over T_1 and defines f_2, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H_2 on T_2 which is ample over T_1 such that $D_{1,X_2} \sim_{\mathbb{Q}} f_2^* H_2$. Since $D_{1,X_2} \geq 0$, by Lemma 2.17(3), $D_{1,X_2} = f_2^* A_2$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $A_2 \geq 0$ on T_2 such that $A_2 \sim_{\mathbb{Q}} H_2$. In particular, A_2 is ample over T_1. Let $G := A_2 - \tau^* A_1$. By (2),

$$B_{2,X_2} - B_{1,X_2} = D_{1,X_2} - D_{2,X_2} = f_2^* A_2 - f_2^* \tau^* A_1 = f_2^* G.$$

Since $B_2 \geq B_1$, $f_2^* G \geq 0$. By Lemma 2.17(1), $G \geq 0$.

Step 3. In this step we prove (1).

Let $f : X_j \to T$ be a contraction over Z for some $j \in \{1, 2\}$, and $A \geq 0$ a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T which is ample over Z. Let $\pi_j : X_j \to Z$ and $t : T \to Z$ be the induced morphisms, so that $\pi_j = t \circ f$. Since X_j is normal and birational to X over Z, pushing forward the structure sheaf of a common resolution of X and X_j shows that $\pi_{j*}\mathcal{O}_{X_j} = \pi_* \mathcal{O}_X = \mathcal{O}_Z$. Thus π_j is a contraction, and so is t, since $t_* \mathcal{O}_T = t_* f_* \mathcal{O}_{X_j} = \mathcal{O}_Z$. Let r be a positive integer such that rA is Cartier. Since $f_* \mathcal{O}_{X_j} = \mathcal{O}_T$, the projection formula implies

$$\pi_{j*}\mathcal{O}_{X_j}(lrf^*A) = t_* \mathcal{O}_T(lrA)$$

for any positive integer l. By Lemma 2.19(2), $\kappa(X_j/Z, f^*A) = \kappa(T/Z, A)$. Let F_T be a very general fiber of t. Then F_T is a normal projective variety of dimension $\dim T - \dim Z$

(Lemma 2.19(1)), and $A|_{F_T}$ is ample since A is ample over Z. Thus $\kappa(T/Z, A) = \kappa(F_T, A|_{F_T}) = \dim F_T$, and

$$\kappa(X_j/Z, f^*A) = \dim T - \dim Z. \tag{9.1}$$

By Lemma 2.20(2) applied to $D_1 \geq D_2$ and the D_2-MMP $X \dashrightarrow X_1$, we have $\kappa(X_1/Z, D_{1,X_1}) = \kappa(X_1/Z, D_{2,X_1}) = k$. By (2) and (9.1), $k = \kappa(X_1/Z, f_1^*A_1) = \dim T_1 - \dim Z$.

For any variety Y projective over T_1, $\overline{\mathrm{NE}}(Y/T_1)$ is an extremal face of $\overline{\mathrm{NE}}(Y/Z)$, since $\overline{\mathrm{NE}}(Y/T_1) = \overline{\mathrm{NE}}(Y/Z) \cap (q^*H)^\perp$ for a Cartier divisor H on T_1 ample over Z, where $q : Y \to T_1$ is the structure morphism. Thus any step of $X_1 \dashrightarrow X_2$ is a D_{1,X_1}-negative divisorial contraction or flip over Z, and $X_1 \dashrightarrow X_2$ is a D_{1,X_1}-MMP over Z. By Lemma 2.20(2) applied to $D_{1,X_1} \geq D_{1,X_1}$, $\kappa(X_2/Z, D_{1,X_2}) = \kappa(X_1/Z, D_{1,X_1}) = k$.

Since A_1 is ample over Z and A_2 is ample over T_1, by [Sta, Tag 0C4K] applied to a positive integer multiple of A_1 and A_2 which is Cartier, there exists a positive integer a such that $A_2 + a\tau^*A_1$ is ample over Z. Let $\epsilon := 1/a$. Then $\tau^*A_1 + \epsilon A_2 = \epsilon(A_2 + a\tau^*A_1)$ is ample over Z. By (2) and **Step 2**,

$$D_{2,X_2} + \epsilon D_{1,X_2} = f_2^*(\tau^*A_1 + \epsilon A_2).$$

Since $\tau^*A_1 + \epsilon A_2 \geq 0$, (9.1) implies that $\kappa(X_2/Z, D_{2,X_2} + \epsilon D_{1,X_2}) = \dim T_2 - \dim Z$. On the other hand, since $D_{1,X_2} \geq D_{2,X_2} \geq 0$,

$$(1+\epsilon)D_{1,X_2} \geq D_{2,X_2} + \epsilon D_{1,X_2} \geq 0,$$

and both divisors have support $\operatorname{Supp} D_{1,X_2}$. By Lemma 2.20(1),

$$\kappa(X_2/Z, D_{2,X_2} + \epsilon D_{1,X_2}) = \kappa(X_2/Z, (1+\epsilon)D_{1,X_2}) = \kappa(X_2/Z, D_{1,X_2}) = k,$$

where the second equality holds because the Iitaka dimension does not change when a divisor is multiplied by a positive rational number. Thus $\dim T_2 = \dim Z + k = \dim T_1$.

Since $f_1 \circ h_2 = g \circ h_2'$ and $f_1 \circ h_2$ is surjective, $g = \tau \circ f_2$ is surjective, so τ is surjective. Since $\dim T_1 = \dim T_2$, the morphism τ is generically finite. Let $\alpha : X_1 \dashrightarrow T_2$ be the composite of $X_1 \dashrightarrow X_2$ and f_2, which is a dominant rational map. Since $f_1 \circ h_2 = \tau \circ f_2 \circ h_2'$, we have $f_1 = \tau \circ \alpha$. By Lemma 2.17(4) applied to the contraction f_1, τ is birational.

Step 4. In this step we prove (4).

Since $K_X + B \sim_{\mathbb{Q},Z} 0$, there exist a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L_0 on Z, a positive integer r, and a non-zero rational function φ_0 on X such that

$$K_X + B = \pi^*L_0 + \frac{1}{r}\operatorname{div}(\varphi_0). \tag{9.2}$$

Let X' be X_1, X_2, or a variety between two steps of the MMPs. Since $X \dashrightarrow X'$ is over Z and any prime divisor on X' is the strict transform of a prime divisor on X, pushing forward (9.2) to X' implies that $K_{X'} + B_{X'}$ is the sum of the pullback of L_0 to X' and $\frac{1}{r}\operatorname{div}(\varphi_0)$. Thus the pullbacks of $K_X + B$ and $K_{X'} + B_{X'}$ to any common resolution of X and X' are both equal to the sum of the pullback of L_0 and $\frac{1}{r}\operatorname{div}(\varphi_0)$. In particular, $(X', B_{X'})$ is crepant to (X, B), and

$$h^*(K_X + B) = h'^*(K_{X_2} + B_{X_2}). \tag{9.3}$$

Fix a rational number $\epsilon > 0$ and let $\Theta := B + \epsilon D_2$. Then $K_{X'} + \Theta_{X'} \sim_{\mathbb{Q},Z} \epsilon D_{2,X'}$ for X' equal to X_1, X_2, or a variety between two steps of the MMPs. Write the D_2-MMP as $X = Y_0 \dashrightarrow Y_1 \dashrightarrow \cdots \dashrightarrow Y_l = X_1$. Then any step $Y_i \dashrightarrow Y_{i+1}$ is a $(K_{Y_i} + \Theta_{Y_i})$-negative divisorial contraction or flip over Z. Let $f : Y_i \to Y_i'$ be the contraction of the extremal ray. If $Y_i \dashrightarrow Y_{i+1}$ is a flip, let $f' : Y_{i+1} \to Y_i'$ be the flipped contraction. If $Y_i \dashrightarrow Y_{i+1}$ is a divisorial contraction, let $Y_i' = Y_{i+1}$ and $f' = \mathrm{id}_{Y_{i+1}}$. Then $f_*\Theta_{Y_i} = f'_*\Theta_{Y_{i+1}}$, $-(K_{Y_i} + \Theta_{Y_i})$ is f-ample, and $K_{Y_{i+1}} + \Theta_{Y_{i+1}}$ is $\mathbb{Q}$-Cartier and f'-nef. By [KM98, Lemma 3.38], $a(E, Y_i, \Theta_{Y_i}) \leq a(E, Y_{i+1}, \Theta_{Y_{i+1}})$ for any prime divisor E exceptional over Y_i'. If E is a prime divisor on Y_i', then the centers of E on Y_i and Y_{i+1} are the strict transforms of E, so $a(E, Y_i, \Theta_{Y_i}) = a(E, Y_{i+1}, \Theta_{Y_{i+1}})$. Thus

$a(E,X,\Theta)\le a(E,X_1,\Theta_{X_1})$ for any prime divisor E over X. Hence, for any common resolution $h_1:W_1\to X$ and $h_1':W_1\to X_1$,

$$h_1^*(K_X+\Theta)\ge h_1'^*(K_{X_1}+\Theta_{X_1}).$$

Since $h_1^*(K_X+B)=h_1'^*(K_{X_1}+B_{X_1})$ and $\Theta-B=\epsilon D_2$, we deduce that

$$h_1^*D_2\ge h_1'^*D_{2,X_1}. \tag{9.4}$$

If (4) holds on a common resolution of X and X_2 which is higher than W, then (4) holds on W by pushing forward. Possibly replacing W with a higher resolution, we may assume that W is also a common resolution of X and X_1, and we take $W_1=W$ and $h_1=h$ in (9.4). Since $X_1\dashrightarrow X_2$ is over T_1, $f_1\circ h_1'=g\circ h'$. By **Step 1**,

$$h_1'^*D_{2,X_1}=h_1'^*f_1^*A_1=h'^*g^*A_1=h'^*D_{2,X_2}.$$

By (9.4), $h^*D_2\ge h'^*D_{2,X_2}$. By (9.3),

$$h'^*(K_{X_2}+B_{2,X_2})=h'^*(K_{X_2}+B_{X_2})-h'^*D_{2,X_2}\ge h^*(K_X+B)-h^*D_2=h^*(K_X+B_2).$$

Since the common resolution W in (4) is arbitrary, (4) holds for any common resolution of X and X_2.

Step 5. In this step we prove (5).

Since $K_X+B\sim_{\mathbb{Q},Z}0$ and both MMPs are over Z, by **Step 4**, (X_2,B_{X_2}) is crepant to (X,B), hence lc. Since $0\le B_{1,X_2}\le B_{2,X_2}\le B_{X_2}$, (X_2,B_{1,X_2}) and (X_2,B_{2,X_2}) are lc.

The $\mathbb{Q}$-divisor Δ_1 is the trace on T_2 of the discriminant $\boldsymbol{b}$-divisor of (X_2,B_{1,X_2}) over T_2, that is, $B_{Z'}$ in Definition 3.1 with $Z'=T_2$. By Lemma 3.8(1) applied to $P=G$,

$$K_{X_2}+B_{1,X_2}+f_2^*G+\frac{1}{m}\operatorname{div}(\psi)=f_2^*(K_{T_2}+\Delta_1+G+\mathbf{M}_{T_2})$$

is a trivialization of degree m whose discriminant $\boldsymbol{b}$-divisor has trace Δ_1+G on T_2 and whose moduli $\boldsymbol{b}$-divisor is $\mathbf{M}$. By (3), $B_{1,X_2}+f_2^*G=B_{2,X_2}$. In particular, $K_{X_2}+B_{2,X_2}$ is $\mathbb{Q}$-linearly equivalent to the pullback of a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T_2. Since (X_2,B_{2,X_2}) is lc, it satisfies the conditions of Definition 3.1, and (5) follows (cf. [CHX24, Lemma 3.2]). □

9.2. **Existence and denominators.** In this subsection, we construct the MMPs of Lemma 9.2 when (X,B) is klt, and show that $K_{X_2}+B_{1,X_2}$ has a trivialization over T_2 whose degree and the denominators of whose moduli $\boldsymbol{b}$-divisor depend only on $\dim X$ and the coefficient set of B_1.

Proposition 9.3. *Notation and conditions as in Set-up 9.1. Let d be a positive integer and $\mathfrak{R}_1\subset[0,1]\cap\mathbb{Q}$ a finite set with $0,1\in\mathfrak{R}_1$. Assume that $\dim X\le d$, (X,B) is klt, and $B_1\in\Phi(\mathfrak{R}_1)$, and let $m\mid p$ be the integers of Theorem 3.3 for d and $\Phi(\mathfrak{R}_1)$. Then there exist a D_2-MMP $X\dashrightarrow X_1$ over Z such that D_{2,X_1} is semi-ample over Z and defines a contraction $f_1:X_1\to T_1$ over Z, and a D_{1,X_1}-MMP $X_1\dashrightarrow X_2$ over T_1 such that D_{1,X_2} is semi-ample over T_1 and defines a contraction $f_2:X_2\to T_2$ over T_1. Moreover, X_1 and X_2 are $\mathbb{Q}$-factorial, (X_1,B_{X_1}) and (X_2,B_{X_2}) are klt, and $K_{X_2}+B_{1,X_2}$ has a trivialization of degree m over T_2 with moduli $\boldsymbol{b}$-divisor $\mathbf{M}$ such that $\mathbf{M}$ is nef over Z and $p\mathbf{M}_{T'}$ is Cartier for any smooth variety T' projective and birational over T_2.*

Proof. For any $r\in\mathfrak{R}_1$, since $0\le r\le 1$, we have $1-\frac{r}{j}\in[0,1]\cap\mathbb{Q}$ for any positive integer j, and the sequence $\{1-\frac{r}{j}\}_{j\ge1}$ is increasing in j. Since $\Phi(\mathfrak{R}_1)$ is the union of the finitely many sequences $\{1-\frac{r}{j}\}_{j\ge1}$ with $r\in\mathfrak{R}_1$, $\Phi(\mathfrak{R}_1)\subset[0,1]\cap\mathbb{Q}$ satisfies the DCC. Thus Theorem 3.3 can be applied to d and $\Phi(\mathfrak{R}_1)$, and m and p are defined.

Since $K_X+B\sim_{\mathbb{Q},Z}0$, there exist a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L_0 on Z, a positive integer r, and a non-zero rational function φ_0 on X such that

$$K_X+B=\pi^*L_0+\frac{1}{r}\operatorname{div}(\varphi_0). \tag{9.5}$$

Let $X \dashrightarrow X'$ be a composite of divisorial contractions and flips over Z. Pushing forward (9.5) to X', the pullbacks of $K_X + B$ and $K_{X'} + B_{X'}$ to any common resolution of X and X' are both equal to the sum of the pullback of L_0 and $\frac{1}{r}\operatorname{div}(\varphi_0)$. Thus $(X', B_{X'})$ is crepant to (X, B), hence klt.

We construct the first MMP. Since (X, B) is klt and $D_2 \geq 0$ is $\mathbb{Q}$-Cartier, there exists a rational number $\epsilon_2 > 0$ such that $(X, B + \epsilon_2 D_2)$ is klt [KM98, Corollary 2.35(2)]. Then $(X, B + \epsilon_2 D_2)$ is a $\mathbb{Q}$-factorial dlt pair, and

$$K_X + B + \epsilon_2 D_2 \sim_{\mathbb{Q},Z} \epsilon_2 D_2 \geq 0.$$

Since Z is affine, by Theorem 5.1, there exists a $(K_X + B + \epsilon_2 D_2)$-MMP $X \dashrightarrow X_1$ over Z which terminates with a $\mathbb{Q}$-factorial dlt pair $(X_1, B_{X_1} + \epsilon_2 D_{2,X_1})$ such that $K_{X_1} + B_{X_1} + \epsilon_2 D_{2,X_1}$ is semi-ample over Z. The varieties between the steps of $X \dashrightarrow X_1$ are $\mathbb{Q}$-factorial [KM98, 3.31, Propositions 3.36, 3.37 and Corollary 3.44]. On X_1 and on any variety X' between two steps of $X \dashrightarrow X_1$,

$$K_{X'} + B_{X'} + \epsilon_2 D_{2,X'} \sim_{\mathbb{Q},Z} \epsilon_2 D_{2,X'},$$

so any step of $X \dashrightarrow X_1$ is a D_2-negative divisorial contraction or D_2-flip over Z. Thus $X \dashrightarrow X_1$ is a D_2-MMP over Z in the sense of Definition 2.18. Since

$$\epsilon_2 D_{2,X_1} \sim_{\mathbb{Q},Z} K_{X_1} + B_{X_1} + \epsilon_2 D_{2,X_1},$$

D_{2,X_1} is semi-ample over Z. Let $f_1 : X_1 \to T_1$ be the contraction over Z defined by D_{2,X_1}. By the previous paragraph, (X_1, B_{X_1}) is klt.

We construct the second MMP. Since f_1 is projective and Z is affine, T_1 is a normal quasi-projective variety. Since (X_1, B_{X_1}) is klt and $D_{1,X_1} \geq 0$ is $\mathbb{Q}$-Cartier, there exists a rational number $\epsilon_1 > 0$ such that $(X_1, B_{X_1} + \epsilon_1 D_{1,X_1})$ is klt [KM98, Corollary 2.35(2)], hence $\mathbb{Q}$-factorial dlt. Since $K_{X_1} + B_{X_1} \sim_{\mathbb{Q},Z} 0$, we have

$$K_{X_1} + B_{X_1} + \epsilon_1 D_{1,X_1} \sim_{\mathbb{Q},T_1} \epsilon_1 D_{1,X_1} \geq 0.$$

By Theorem 5.1 over T_1, there exists a $(K_{X_1} + B_{X_1} + \epsilon_1 D_{1,X_1})$-MMP $X_1 \dashrightarrow X_2$ over T_1 which terminates with a $\mathbb{Q}$-factorial dlt pair $(X_2, B_{X_2} + \epsilon_1 D_{1,X_2})$ such that $K_{X_2} + B_{X_2} + \epsilon_1 D_{1,X_2}$ is semi-ample over T_1. As for the first MMP, $X_1 \dashrightarrow X_2$ is a D_{1,X_1}-MMP over T_1 in the sense of Definition 2.18, and D_{1,X_2} is semi-ample over T_1. Let $f_2 : X_2 \to T_2$ be the contraction over T_1 defined by D_{1,X_2}. Since $X \dashrightarrow X_2$ is a composite of divisorial contractions and flips over Z, (X_2, B_{X_2}) is klt.

We construct the trivialization. Since D_{1,X_2} defines f_2, there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H_2 on T_2 such that $D_{1,X_2} \sim_{\mathbb{Q}} f_2^* H_2$. Let L_2 be the pullback of L_0 to T_2. Pushing forward (9.5) to X_2, we have

$$K_{X_2} + B_{X_2} = f_2^* L_2 + \frac{1}{r}\operatorname{div}(\varphi_0),$$

so

$$K_{X_2} + B_{1,X_2} = K_{X_2} + B_{X_2} - D_{1,X_2} \sim_{\mathbb{Q}} f_2^*(L_2 - H_2).$$

Since $0 \leq B_{1,X_2} \leq B_{X_2}$ and (X_2, B_{X_2}) is klt, (X_2, B_{1,X_2}) is lc. Since the coefficients of B_{1,X_2} are coefficients of B_1, $B_{1,X_2} \in \Phi(\mathfrak{R}_1)$. By Lemma 9.2(1) and Set-up 9.1(4), $\dim T_2 = \dim Z + k$ and $\dim Z + k > 0$. Moreover, $\dim T_2 \leq \dim X_2 = \dim X \leq d$. By Theorem 3.3(1) applied to f_2 and (X_2, B_{1,X_2}), $K_{X_2} + B_{1,X_2}$ has a trivialization of degree m over T_2 with moduli $\boldsymbol{b}$-divisor $\mathbf{M}$. Since $T_2 \to Z$ is a projective morphism to a normal quasi-projective variety, by Theorem 3.3(2), $\mathbf{M}$ is nef over Z and $p\mathbf{M}_{T'}$ is a Cartier divisor for any smooth variety T' projective and birational over T_2. □

10. Exposing models and lifting complements

The goal of this section is to prove Propositions 10.3 and 10.4, which are used in case (3) of Lemma 11.4 (see the sketch of the proof in Section 1). Proposition 10.3 provides the higher model of the base on which a complement of the induced generalized pair (Set-up 10.2) is taken, and Proposition 10.4 pulls the complement back to X.

10.1. **Exposing models.** In this subsection, we introduce exposing models and construct, for klt pairs, exposing models on which the trace of the discriminant $\boldsymbol{b}$-divisor has non-negative coefficients.

Definition 10.1. Let $\alpha : X \dashrightarrow Y$ be a birational map between normal varieties and $g : Y \to T$ a contraction. A projective birational morphism $\sigma : T' \to T$ from a normal variety T' is an *exposing model* for α if the induced rational map $X \dashrightarrow T'$ maps every prime divisor P on X which does not dominate T onto a prime divisor Q_P on T'. Here a prime divisor P on X *dominates* T if the rational map $g \circ \alpha : X \dashrightarrow T$ is defined at the generic point of P and maps P onto a dense subset of T. For any prime divisor P on X which does not dominate T, the rational map $X \dashrightarrow T'$ is then defined at the generic point of P, and Q_P is Cartier on a neighborhood of the generic point of Q_P since T' is normal. We let e_P be the coefficient of P in the pullback of Q_P under $X \dashrightarrow T'$, which is a positive integer.

The prime divisors that an exposing model has to expose are the prime divisors P on X which do not dominate T and whose image in T has codimension at least 2. Such a P may have coefficient zero in the boundary, and α may contract P. When a complement on the base which is not monotonic is pulled back to X, the coefficient at such a P is not determined by the coefficient at a single prime divisor on the base, and neither effectivity nor rounding can be checked at P. On an exposing model, any such P corresponds to the prime divisor Q_P, and the coefficient at P is controlled by the coefficient at Q_P alone.

Set-up 10.2. Let X, Y, Z, T, α, A, A_Y, g, p, ψ, Δ_T, and $\mathbf{M}$ be as follows.

(1) $X \to Z$ and $Y \to Z$ are contractions, and $\alpha : X \dashrightarrow Y$ is a birational map over Z.
(2) $A \geq 0$ is a $\mathbb{Q}$-divisor on X such that $K_X + A$ and $K_Y + A_Y$ are $\mathbb{Q}$-Cartier, where $A_Y := \alpha_* A$.
(3) $h'^*(K_Y + A_Y) \geq h^*(K_X + A)$ for a common resolution $h : W \to X$ and $h' : W \to Y$ of α.
(4) $g : Y \to T$ is a contraction over Z such that $\dim T > 0$.
(5) The equality
$$K_Y + A_Y + \frac{1}{p}\operatorname{div}(\psi) = g^*(K_T + \Delta_T + \mathbf{M}_T)$$
is a trivialization of degree p of $K_Y + A_Y$ over T, with induced generalized sub-pair $(T, \Delta_T, \mathbf{M})$.
(6) For any projective birational morphism $\sigma : T' \to T$ from a normal variety T', let $\Delta_{T'}$ be the trace on T' of the discriminant $\boldsymbol{b}$-divisor of (Y, A_Y) over T, so that $K_{T'} + \Delta_{T'} + \mathbf{M}_{T'} = \sigma^*(K_T + \Delta_T + \mathbf{M}_T)$.

Proposition 4.3 requires a boundary with non-negative coefficients, while on a smooth higher model of T to which $\mathbf{M}$ descends the discriminant may have negative coefficients. Condition (3) of Set-up 10.2 guarantees that extracting the prime divisors Q_P over T onto which prime divisors P on X are mapped does not make the discriminant negative. Only finitely many of the prime divisors Q_P are exceptional over T. The complement is taken on the $\mathbb{Q}$-factorial exposing model of Proposition 10.3. When it is pulled back to X, one passes to a smooth higher model on which $\mathbf{M}$ descends, and the rounding is checked only at the prime divisors Q_P.

Proposition 10.3. *Notation and conditions as in Set-up 10.2. Assume that (Y, A_Y) is klt and $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over Z. Then there exists an exposing model $\sigma : T' \to T$ for α such that T' is $\mathbb{Q}$-factorial, $\Delta_{T'} \geq 0$, and $(T', \Delta_{T'}, \mathbf{M})/Z$ is generalized klt.*

Proof. The proof is long, so we divide it into several steps. Let $h : W \to X$ and $h' : W \to Y$ be the common resolution in Set-up 10.2(3), let $f := g \circ h' : W \to T$, and let C_W be the $\mathbb{Q}$-divisor on W such that $K_W + C_W = h'^*(K_Y + A_Y)$. For any prime divisor P on X, let P_W be the strict transform of P on W. Since h is an isomorphism over the generic point of P and $g \circ \alpha = f \circ h^{-1}$, the rational map $g \circ \alpha : X \dashrightarrow T$ is defined at the generic point of P and maps P onto a dense subset of the closed subset $f(P_W)$ of T. Thus P dominates T if and only if $f(P_W) = T$.

Step 1. In this step we show that $c_P \in [\mathrm{mult}_P A, 1)$ for any prime divisor P on X, where $c_P := 1 - a(P, Y, A_Y)$.

By [KM98, Lemma 2.30], $a(P, Y, A_Y) = a(P_W, W, C_W) = 1 - \mathrm{mult}_{P_W} C_W$, so $c_P = \mathrm{mult}_{P_W} C_W$. Since h is an isomorphism over the generic point of P, $\mathrm{mult}_{P_W}(h^*(K_X + A) - K_W) = \mathrm{mult}_P A$. Thus Set-up 10.2(3) implies

$$c_P - \mathrm{mult}_P A = \mathrm{mult}_{P_W}\left(h'^*(K_Y + A_Y) - h^*(K_X + A)\right) \geq 0.$$

Since (Y, A_Y) is klt, $a(P, Y, A_Y) > 0$, that is, $c_P < 1$.

Step 2. In this step we construct a projective birational morphism $\tau : T_1 \to T$ from a smooth variety T_1 such that the rational map $X \dashrightarrow T_1$ maps every prime divisor P on X which does not dominate T onto a prime divisor Q_P on T_1, and we show that only finitely many of the prime divisors Q_P are exceptional over T.

Since g is a contraction and h' is birational, f is a contraction. By [ACSS21, Theorem 2.2] (whose proof is [AK00, Theorem 2.1 and Proposition 4.4]), applied to the pair $(W, 0)$ and f, there exist projective birational morphisms $\beta : W' \to W$ and $\tau : T_1 \to T$, and an equidimensional toroidal morphism $f_1 : W' \to T_1$ with $\tau \circ f_1 = f \circ \beta$, where W' is a normal variety with quotient toric singularities and T_1 is smooth. Since $\tau \circ f_1 = f \circ \beta$ is proper and τ is separated, f_1 is proper [Har77, Corollary II.4.8(e)]. Since $f \circ \beta$ is surjective and τ is birational, the image of f_1 is dense, so f_1 is surjective. Since f_1 is equidimensional, any fiber F of f_1 satisfies $\dim F = d$, where $d := \dim W - \dim T$.

Let P be a prime divisor on X which does not dominate T. Since W is normal, β is an isomorphism over the generic point of P_W, so the strict transform $P_{W'}$ of P_W on W' is a prime divisor. Let $Q_P := f_1(P_{W'})$, which is an irreducible closed subset of T_1. Since the fibers of $f_1|_{P_{W'}} : P_{W'} \to Q_P$ are contained in fibers of f_1, $\dim P_{W'} \leq \dim Q_P + d$, that is, $\dim Q_P \geq \dim T - 1$. Since $\tau(Q_P) = f(\beta(P_{W'})) = f(P_W) \neq T$, we have $Q_P \neq T_1$. Thus Q_P is a prime divisor on T_1. Since $h \circ \beta$ is an isomorphism over the generic point of P and $\tau \circ f_1 = g \circ h' \circ \beta$, the rational map $X \dashrightarrow T_1$ induced by $g \circ \alpha$ is $f_1 \circ (h \circ \beta)^{-1}$, which is defined at the generic point of P and maps P onto a dense subset of Q_P. Let $e_P := \mathrm{mult}_{P_{W'}} Q_P$, which is the coefficient of P in the pullback of Q_P under $X \dashrightarrow T_1$. Since T_1 is smooth, Q_P is Cartier, and since $f_1(P_{W'}) = Q_P$, e_P is a positive integer.

If $f(P_W)$ is a prime divisor on T, then τ is an isomorphism over the generic point of $f(P_W)$ since T is normal, and Q_P, which is a prime divisor mapped onto $f(P_W)$ by τ, is the strict transform of $f(P_W)$ on T_1. If $f(P_W)$ is not a prime divisor on T, then $\tau(Q_P) = f(P_W)$ has codimension at least 2 in T, so Q_P is exceptional over T. Since τ has only finitely many exceptional prime divisors, only finitely many of the prime divisors Q_P are exceptional over T. Let $S_1, \ldots, S_r$ be the distinct prime divisors on T_1 among the Q_P which are exceptional over T, where P runs through the prime divisors on X which do not dominate T, and $r = 0$ if no Q_P is exceptional over T.

Step 3. In this step we show that $(T, \Delta_T, \mathbf{M})/Z$ is a generalized klt pair, and $a(S_i, T, \Delta_T, \mathbf{M}) \in (0, 1]$ for any i.

Since g is surjective and $Y \to Z$ is projective, $T \to Z$ is proper, hence projective as T is quasi-projective. By Set-up 10.2(5) and Definition 3.1(3), $K_T + \Delta_T + \mathbf{M}_T$ is $\mathbb{Q}$-Cartier, and $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over Z by assumption.

We show that $\Delta_T \geq 0$. Let Q be a prime divisor on T and t_Q the lc threshold of Q (Definition 3.1(2)), so that $\mathrm{mult}_Q \Delta_T = 1 - t_Q$. We compute t_Q on Y itself, over a neighborhood of the generic point of Q on which Q is Cartier. Since g is a contraction, g^*Q has a component G which dominates Q. Then $\mathrm{mult}_G Q$ is a positive integer and $\mathrm{mult}_G A_Y \geq 0$. For any real number t such that $(Y, A_Y + tg^*Q)$ is sub-lc over a neighborhood of the generic point of Q, we have $\mathrm{mult}_G A_Y + t \,\mathrm{mult}_G Q \leq 1$, that is,

$$t \leq \frac{1 - \mathrm{mult}_G A_Y}{\mathrm{mult}_G Q} \leq 1.$$

Thus $t_Q \leq 1$, and $\operatorname{mult}_Q \Delta_T = 1 - t_Q \geq 0$. Hence $(T, \Delta_T, \mathbf{M})/Z$ is a generalized pair.

Let E be a prime divisor over T and $\rho : T'' \to T$ a log resolution of $(T, \operatorname{Supp} \Delta_T)$ such that $\mathbf{M}$ descends to T'' and E is a prime divisor on T''. By Set-up 10.2(6), $K_{T''} + \Delta_{T''} + \mathbf{M}_{T''} = \rho^*(K_T + \Delta_T + \mathbf{M}_T)$, so $a(E, T, \Delta_T, \mathbf{M}) = 1 - \operatorname{mult}_E \Delta_{T''} = t_E$, where t_E is the lc threshold of E. Since (Y, A_Y) is klt, $\operatorname{mult}_E \Delta_{T''} < 1$ by Lemma 3.8(3). Thus $a(E, T, \Delta_T, \mathbf{M}) > 0$, and $(T, \Delta_T, \mathbf{M})/Z$ is generalized klt.

Now fix i, and let P be a prime divisor on X which does not dominate T such that $S_i = Q_P$. By the previous paragraph, applied with $E = S_i$, $a(S_i, T, \Delta_T, \mathbf{M}) = t_{S_i}$, the lc threshold of S_i, which depends only on S_i as a prime divisor over T (Definition 3.1(2)). We compute t_{S_i} on T_1, with $h' \circ \beta : W' \to Y$ and $f_1 : W' \to T_1$ in place of h_X and f' in Definition 3.1(2). Write

$$K_{W'} + C_{W'} = \beta^*(K_W + C_W) = (h' \circ \beta)^*(K_Y + A_Y).$$

Since β is an isomorphism over the generic point of P_W, $\operatorname{mult}_{P_{W'}} C_{W'} = \operatorname{mult}_{P_W} C_W = c_P$ by **Step 1**, and $\operatorname{mult}_{P_{W'}} S_i = e_P$ by **Step 2**. By Definition 3.1(2), t_{S_i} is the supremum of the real numbers t such that $(W', C_{W'} + t f_1^* S_i)$ is sub-lc over a neighborhood of the generic point of S_i. Since $f_1(P_{W'}) = S_i$, for any such t, $c_P + t e_P \leq 1$. Since $c_P \geq \operatorname{mult}_P A \geq 0$ by **Step 1** and $e_P \geq 1$, we have

$$a(S_i, T, \Delta_T, \mathbf{M}) = t_{S_i} \leq \frac{1 - c_P}{e_P} \leq 1.$$

Combined with the previous paragraph, $a(S_i, T, \Delta_T, \mathbf{M}) \in (0, 1]$.

Step 4. In this step we construct σ and conclude.

By **Step 3**, $(T, \Delta_T, \mathbf{M})/Z$ is a generalized klt pair, and $S_1, \dots, S_r$ are prime divisors exceptional over T whose generalized log discrepancies with respect to $(T, \Delta_T, \mathbf{M})$ are at most 1. The triple $(T, \Delta_T, \mathbf{M})$ with data $\tilde{T} \to T \to Z$, where $\tilde{T}$ is a resolution of a determining model of $\mathbf{M}$ on which the trace of $\mathbf{M}$ is nef over Z (**Step 3** and Definition 2.7), is a generalized polarized pair in the sense of [BZ16, Definition 1.4]: $T \to Z$ is projective, Δ_T is a boundary ($\Delta_T \geq 0$ by **Step 3**, and the coefficients of Δ_T are less than 1 since $(T, \Delta_T, \mathbf{M})/Z$ is generalized klt) and $\mathbf{M}_{\tilde{T}}$ is nef over Z, as the pullback of a $\mathbb{Q}$-divisor which is nef over Z, and $K_T + \Delta_T + \mathbf{M}_T$ is $\mathbb{R}$-Cartier. Here T need not be $\mathbb{Q}$-factorial. By [BZ16, Lemma 4.5], there exists a projective birational morphism $\sigma : T' \to T$ from a $\mathbb{Q}$-factorial normal variety T' whose exceptional divisors are exactly $S_1, \dots, S_r$.

We show that $\Delta_{T'} \geq 0$. Let Q be a prime divisor on T'. By Definition 3.1(2) and Set-up 10.2(6), $\operatorname{mult}_Q \Delta_{T'} = 1 - t_Q$, and t_Q depends only on Q as a prime divisor over T. If Q is not exceptional over T, then $\sigma(Q)$ is a prime divisor on T and σ is an isomorphism over the generic point of $\sigma(Q)$ since T is normal, so $t_Q = t_{\sigma(Q)}$ and $\operatorname{mult}_Q \Delta_{T'} = \operatorname{mult}_{\sigma(Q)} \Delta_T \geq 0$ by **Step 3**. If $Q = S_i$ for some i, then $\operatorname{mult}_Q \Delta_{T'} = 1 - a(S_i, T, \Delta_T, \mathbf{M}) \in [0, 1)$ by **Step 3**. Thus $\Delta_{T'} \geq 0$.

We show that $(T', \Delta_{T'}, \mathbf{M})/Z$ is generalized klt. By Set-up 10.2(6),

$$K_{T'} + \Delta_{T'} + \mathbf{M}_{T'} = \sigma^*(K_T + \Delta_T + \mathbf{M}_T)$$

is $\mathbb{Q}$-Cartier, and T' is projective over Z. Thus $(T', \Delta_{T'}, \mathbf{M})/Z$ is a generalized pair. Let E be a prime divisor over T' and $\rho' : T'' \to T'$ a log resolution of $(T', \operatorname{Supp} \Delta_{T'})$ such that $\mathbf{M}$ descends to T'' and E is a prime divisor on T''. By Set-up 10.2(6) applied to $\sigma \circ \rho'$,

$$K_{T''} + \Delta_{T''} + \mathbf{M}_{T''} = \rho'^*(K_{T'} + \Delta_{T'} + \mathbf{M}_{T'}).$$

Thus $a(E, T', \Delta_{T'}, \mathbf{M}) = 1 - \operatorname{mult}_E \Delta_{T''}$, which is positive by Lemma 3.8(3).

Finally, we show that σ is an exposing model for α. Let P be a prime divisor on X which does not dominate T. Suppose first that $f(P_W)$ is a prime divisor on T. Since T is normal, σ is an isomorphism over the generic point of $f(P_W)$. Since $g \circ \alpha$ is defined at the generic point of P and maps P onto a dense subset of $f(P_W)$, the rational map $X \dashrightarrow T'$, which is $\sigma^{-1} \circ g \circ \alpha$, maps P onto the strict transform of $f(P_W)$ on T', which is a prime divisor on T'. Suppose now that $f(P_W)$ is not a prime divisor on T. By **Step 2**, $Q_P = S_i$ for some i, and the rational map $X \dashrightarrow T_1$ maps P onto S_i. Since T_1 and T' are normal and S_i is a prime divisor on both of them,

the birational map $T_1 \dashrightarrow T'$ over T restricts to an isomorphism between neighborhoods of the generic points of S_i. Thus $X \dashrightarrow T'$ maps P onto the prime divisor S_i on T'. In both cases, $T_1 \dashrightarrow T'$ is an isomorphism near the generic point of the prime divisor Q_P on T_1 of **Step 2** and maps it onto the prime divisor Q_P on T' of Definition 10.1, so the integer e_P of Definition 10.1 equals $\mathrm{mult}_{P_{W'}} Q_P$, the integer e_P of **Step 2**. Thus σ is an exposing model for α. □

10.2. **Lifting with a fixed index.** In this subsection, we lift a complement on an exposing model to X without changing the index (cf. [CHX24, Proposition 3.5]).

Proposition 10.4. *Notation and conditions as in Set-up 10.2, and let $z \in Z$. Assume that α^{-1} does not contract any divisor and (Y, A_Y) is lc. Let $\sigma : T' \to T$ be an exposing model for α such that T' is smooth and $\mathbf{M}$ descends to T'. Let n be a positive integer with $p \mid n$, and $\Delta^+_{T'}$ a $\mathbb{Q}$-divisor on T' such that:*

(1) $(T', \Delta^+_{T'})$ is sub-lc;
(2) $n(K_{T'} + \Delta^+_{T'} + \mathbf{M}_{T'}) \sim 0$ over a neighborhood of z;
(3) for any prime divisor P on X which does not dominate T, with Q_P as in Definition 10.1 and $\delta_P := \mathrm{mult}_{Q_P} \Delta_{T'}$,

$$n \,\mathrm{mult}_{Q_P} \Delta^+_{T'} \geq n\lfloor \delta_P \rfloor + \lfloor (n+1)\{\delta_P\} \rfloor.$$

Then $(X/Z \ni z, A)$ has an n-complement.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct a $\mathbb{Q}$-divisor B^+ on X and a $\mathbb{Q}$-divisor C^+_W on a common resolution W such that $n(K_X + B^+)$ is principal and $K_W + C^+_W$ is the pullback of $K_X + B^+$.

Possibly shrinking Z to a neighborhood of z, we may assume that

$$n(K_{T'} + \Delta^+_{T'} + \mathbf{M}_{T'}) = \mathrm{div}_{T'}(u)$$

for a non-zero rational function u on T'. The inequality in Set-up 10.2(3) still holds after pulling back to any higher resolution of W. Thus, possibly replacing W with a resolution of the main component of $W \times_T T'$, we may assume that there exists a morphism $q : W \to T'$ such that $\sigma \circ q = g \circ h'$. Let $q = q_2 \circ q_1$ be the Stein factorization of q, with $q_1 : W \to V$ a contraction and $q_2 : V \to T'$ finite. Then $g = (\sigma \circ q_2) \circ (q_1 \circ h'^{-1})$, where $q_1 \circ h'^{-1} : Y \dashrightarrow V$ is a dominant rational map and $\sigma \circ q_2 : V \to T$ is generically finite. Since g is a contraction, Lemma 2.17(4) implies that $\sigma \circ q_2$ is birational. Thus q_2 is a finite birational morphism onto the normal variety T', hence an isomorphism. Thus q is a contraction.

Let C_W be the $\mathbb{Q}$-divisor on W such that $K_W + C_W = h'^*(K_Y + A_Y)$. Since (Y, A_Y) is lc, (W, C_W) is sub-lc. Pulling back the trivialization in Set-up 10.2(5) by h' and applying Set-up 10.2(6), we obtain

$$K_W + C_W + \frac{1}{p}\mathrm{div}_W(\psi \circ h') = q^*\sigma^*(K_T + \Delta_T + \mathbf{M}_T) = q^*(K_{T'} + \Delta_{T'} + \mathbf{M}_{T'}), \tag{10.1}$$

which is a trivialization of degree p of $K_W + C_W$ over T'. By Definition 3.1(2), the lc threshold of a prime divisor over T' can be computed on any normal variety projective and birational over W which dominates the corresponding higher model of T', and (W, C_W) is crepant to (Y, A_Y). Thus the discriminant $\boldsymbol{b}$-divisor of (W, C_W) over T' has trace $\Delta_{T''}$ on any normal variety T'' projective and birational over T'. By Definition 3.1(3), the moduli $\boldsymbol{b}$-divisor of the trivialization (10.1) is $\mathbf{M}$. Hence the induced generalized sub-pair is $(T', \Delta_{T'}, \mathbf{M})$.

Let $F := \Delta^+_{T'} - \Delta_{T'}$, which is $\mathbb{Q}$-Cartier since T' is smooth. Let $C^+_W := C_W + q^*F$ and $B^+ := h_*C^+_W$. Since $p \mid n$, n/p is an integer. Let

$$\varphi := \frac{u \circ q}{(\psi \circ h')^{n/p}},$$

which is a non-zero rational function on W. Then

$$n(K_W + C_W^+) = q^*\left(n(K_{T'} + \Delta_{T'}^+ + \mathbf{M}_{T'})\right) - \frac{n}{p}\operatorname{div}_W(\psi \circ h') = \operatorname{div}_W(\varphi).$$

Pushing forward by h, we obtain

$$n(K_X + B^+) = \operatorname{div}_X(\varphi). \tag{10.2}$$

In particular, $K_X + B^+$ is $\mathbb{Q}$-Cartier, $n(K_X + B^+) \sim 0$, and

$$h^*(K_X + B^+) = \frac{1}{n}\operatorname{div}_W(\varphi) = K_W + C_W^+.$$

Since nK_X and $\operatorname{div}_X(\varphi)$ are divisors with integer coefficients, (10.2) implies that nB^+ has integer coefficients.

Step 2. In this step we show that (X, B^+) is sub-lc.

By Lemma 3.8(1)(2), applied to q, the sub-pair (W, C_W), the trivialization in **Step 1**, and the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor F, the sub-pair $(W, C_W + q^*F)$ is sub-lc if and only if the coefficients of $\Delta_{T''} + \rho^*F$ are at most 1 for any projective birational morphism $\rho : T'' \to T'$ from a normal variety T''. Since $\mathbf{M}$ descends to T', $\mathbf{M}_{T''} = \rho^*\mathbf{M}_{T'}$. Thus Set-up 10.2(6) implies

$$K_{T''} + \Delta_{T''} + \rho^*F = \rho^*(K_{T'} + \Delta_{T'} + \mathbf{M}_{T'}) - \mathbf{M}_{T''} + \rho^*F = \rho^*(K_{T'} + \Delta_{T'}^+).$$

Hence the coefficients of $\Delta_{T''} + \rho^*F$ are the coefficients of the crepant pullback of $K_{T'} + \Delta_{T'}^+$ to T'', which are at most 1 by (1) and [KM98, Lemma 2.30]. Thus (W, C_W^+) is sub-lc. Since $K_W + C_W^+ = h^*(K_X + B^+)$ by **Step 1**, (X, B^+) is sub-lc by [KM98, Lemma 2.30].

Step 3. In this step we show that

$$n \operatorname{mult}_P B^+ \geq n\lfloor c_P \rfloor + \lfloor (n+1)\{c_P\} \rfloor$$

for any prime divisor P on X, where $c_P := 1 - a(P, Y, A_Y)$.

Fix a prime divisor P on X, and let P_W be the strict transform of P on W. Since h is an isomorphism over the generic point of P, $\operatorname{mult}_P B^+ = \operatorname{mult}_{P_W} C_W^+$, and $c_P = \operatorname{mult}_{P_W} C_W$ by [KM98, Lemma 2.30]. Let $c_P^+ := \operatorname{mult}_P B^+$. By Set-up 10.2(3),

$$c_P - \operatorname{mult}_P A = \operatorname{mult}_{P_W}\left(h'^*(K_Y + A_Y) - h^*(K_X + A)\right) \geq 0.$$

Since (Y, A_Y) is lc, $c_P \leq 1$. Thus $0 \leq \operatorname{mult}_P A \leq c_P \leq 1$. By **Step 1**, nc_P^+ is an integer, and by **Step 2**, $c_P^+ \leq 1$.

We first show the following: if $c_P^+ = c_P$, then the inequality of this step holds at P. In fact, $nc_P = nc_P^+$ is an integer. If $c_P = 1$, then $n\lfloor c_P \rfloor + \lfloor (n+1)\{c_P\} \rfloor = n = nc_P^+$. If $c_P < 1$, then by Lemma 2.13(1),

$$n\lfloor c_P \rfloor + \lfloor (n+1)\{c_P\} \rfloor = \lfloor (n+1)c_P \rfloor = nc_P + \lfloor c_P \rfloor = nc_P = nc_P^+.$$

(**Case 1**) P dominates T. Since h is an isomorphism over the generic point of P and $g \circ \alpha = g \circ h' \circ h^{-1}$, the closed subset $\sigma(q(P_W)) = g(h'(P_W))$ is dense in T, hence equal to T. Thus the closed subset $q(P_W)$ of T' dominates T, so $q(P_W) = T'$. Thus $\operatorname{mult}_{P_W} D = 0$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on T'. In particular, $c_P^+ = c_P + \operatorname{mult}_{P_W} F = c_P$, and the inequality of this step holds at P by the previous paragraph.

From now on, assume that P does not dominate T. Let $Q := Q_P$ and $e := e_P$ be as in Definition 10.1, $\delta := \delta_P = \operatorname{mult}_Q \Delta_{T'}$, and $\beta := \operatorname{mult}_Q \Delta_{T'}^+$. Let t_Q be the lc threshold of Q, so that $\delta = 1 - t_Q$ by **Step 1**. Since (W, C_W) is sub-lc, $\delta \leq 1$ by Lemma 3.8(2), that is, $t_Q \geq 0$. Since h is an isomorphism over the generic point of P and $\sigma \circ q = g \circ h'$, the rational map $X \dashrightarrow T'$ is $q \circ h^{-1}$. Thus, by Definition 10.1, $q(P_W)$ is dense in Q and $\operatorname{mult}_{P_W} Q = e$. We compute t_Q on W itself (Definition 3.1(2)). For any real number $0 \leq t < t_Q$, $(W, C_W + tq^*Q)$ is sub-lc over a neighborhood of the generic point of Q, so $c_P + te \leq 1$. Since $c_P \leq 1$, we deduce

$$c_P + e(1 - \delta) = c_P + et_Q \leq 1. \tag{10.3}$$

Since $c_P \geq 0$ and $e \geq 1$, (10.3) implies $1-\delta \leq 1$, so $\delta \in [0,1]$. Near the generic point of Q, $F = (\beta-\delta)Q$. Thus

$$c_P^+ = c_P + e(\beta-\delta), \qquad c_P + e(1-\delta) = c_P^+ + e(1-\beta).$$

(**Case 2**) $\delta = 1$. By (3), $n\beta \geq n\lfloor 1\rfloor + \lfloor (n+1)\cdot 0\rfloor = n$, so $\beta \geq 1$. Since $(T', \Delta_{T'}^+)$ is sub-lc, $\beta \leq 1$. Thus $\beta = \delta = 1$ and $c_P^+ = c_P$. Hence the inequality of this step holds at P.

(**Case 3**) $c_P = 1$. By (10.3) and $e \geq 1$, $\delta = 1$, and we are in Case 2.

(**Case 4**) $c_P < 1$ and $\delta < 1$. Then $\lfloor \delta \rfloor = 0$, and (3) implies $n\beta \geq \lfloor (n+1)\delta \rfloor = (n+1)\delta - s$, where $s := \{(n+1)\delta\} \in [0,1)$. Since $e \geq 1$, we have

$$\begin{aligned}(n+1)c_P - nc_P^+ &= c_P^+ + e\left((n+1)\delta - n\beta - \beta\right) \leq c_P^+ + e(s-\beta)\\ &< c_P^+ + e(1-\beta) = c_P + e(1-\delta) \leq 1,\end{aligned}$$

where the last inequality is (10.3). Since nc_P^+ is an integer, $\lfloor (n+1)c_P \rfloor \leq nc_P^+$. Since $0 \leq c_P < 1$, Lemma 2.13(1) implies

$$n\lfloor c_P \rfloor + \lfloor (n+1)\{c_P\} \rfloor = \lfloor (n+1)c_P \rfloor \leq nc_P^+.$$

Cases 1–4 cover all prime divisors P on X, which proves the inequality of this step.

Step 4. In this step we show that $(X/Z \ni z, B^+)$ is an n-complement of $(X/Z \ni z, A)$.

Let P be a prime divisor on X and c_P as in **Step 3**. Since $0 \leq \operatorname{mult}_P A \leq c_P \leq 1$ by **Step 3**, Lemma 2.14(2) and **Step 3** imply

$$0 \leq n\lfloor \operatorname{mult}_P A \rfloor + \lfloor (n+1)\{\operatorname{mult}_P A\} \rfloor \leq n\lfloor c_P \rfloor + \lfloor (n+1)\{c_P\} \rfloor \leq n \operatorname{mult}_P B^+.$$

Thus $B^+ \geq 0$ and $nB^+ \geq n\lfloor A \rfloor + \lfloor (n+1)\{A\} \rfloor$. By **Step 2**, (X, B^+) is lc, and by (10.2), $n(K_X + B^+) \sim 0$. Thus $(X/Z \ni z, B^+)$ is an n-complement of $(X/Z \ni z, A)$. □

11. The hierarchy of truncations

In this section, we divide the proof of Theorem 12.2 into the three cases of Lemma 11.4, listed in the sketch of the proof in Section 1, and prove the results used in the last two of them.

11.1. **Index sets.** In this subsection, we fix the index sets for the proof of Theorem 12.2.

Construction 11.1. Let d and l be positive integers, $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ a finite set with $0, 1 \in \mathfrak{R}$, and $\Phi := \Phi(\mathfrak{R})$. Let $\mathcal{N}_1 := \mathcal{J}(d, 1, l)$ be the set of Proposition 4.3, and $\Phi_1 := \Gamma(\mathcal{N}_1, \Phi)$. For $i = 1, \dots, d-1$, inductively let $\mathfrak{R}_i \subset [0,1] \cap \mathbb{Q}$ be a finite set with $0, 1 \in \mathfrak{R}_i$ and $\Phi_i = \Phi(\mathfrak{R}_i)$ (Lemma 2.14(1)), $m_i \mid p_i$ the integers of Theorem 3.3 for d and Φ_i, $\mathcal{N}_{i+1} := \mathcal{J}(d-1, p_i, l)$, and

$$\Phi_{i+1} := \Gamma(\mathcal{N}_1 \cup \dots \cup \mathcal{N}_{i+1}, \Phi).$$

Let $\mathfrak{R}_d \subset [0,1] \cap \mathbb{Q}$ be a finite set with $0, 1 \in \mathfrak{R}_d$ and $\Phi_d = \Phi(\mathfrak{R}_d)$ (Lemma 2.14(1)). Let $n_0 := \operatorname{lcm}\{l, I(d, \Phi_d)\}$, and

$$\mathcal{N}(d, l, \mathfrak{R}) := \mathcal{N}_1 \cup \dots \cup \mathcal{N}_d \cup \{n_0\}.$$

The order in Construction 11.1 is not circular. At the i-th stage, $m_i \mid p_i$ are the integers of Theorem 3.3 for d and Φ_i, and by Proposition 9.3, the denominator p_i of the moduli $\boldsymbol{b}$-divisor on the common base of the i-th and the $(i+1)$-th truncations depends only on Φ_i. Only then are $\mathcal{N}_{i+1}$, whose members are divisible by p_i, and Φ_{i+1} defined. The set $\mathcal{N}_1$ is used in case (1) of Lemma 11.4, where the complement is taken on X itself, of dimension d, and the moduli part is zero. The integer n_0 is used in case (2), where Z is a point (Lemma 11.3). The set $\mathcal{N}_{i+1}$ is used in case (3), where the complement is taken on an exposing model of the common base, of dimension at most $d-1$, and the Cartier index of the moduli part divides p_i (Proposition 11.5). We define Φ_{i+1} with the union $\mathcal{N}_1 \cup \dots \cup \mathcal{N}_{i+1}$ rather than with $\mathcal{N}_{i+1}$ alone for two reasons. First, $\Gamma(\mathcal{N}, \Phi)$ increases with $\mathcal{N}$, so the sets Φ_i increase with i, which is the condition on the truncations in Lemma 11.4. Second, Proposition 11.5 uses Lemma 2.14(4) with $\mathcal{N}_1 \cup \dots \cup \mathcal{N}_{i+1}$ to lift an n-complement of $(X/Z \ni z, B_{i+1})$ to an n-complement of $(X/Z \ni z, B)$.

11.2. **The extreme cases.** In this subsection, we prove Lemma 11.4 and treat the case when the base is a point (Lemma 11.3). Lemmas 11.3 and 11.4 and Proposition 11.5 share the following setting.

Set-up 11.2. Let d, $\pi : X \to Z$, and (X, B) be as follows.

(1) d is a positive integer, and $\pi : X \to Z$ is a contraction such that Z is affine and X is $\mathbb{Q}$-factorial of dimension d.
(2) (X, B) is a klt pair such that B is a $\mathbb{Q}$-divisor and $K_X + B \sim_{\mathbb{Q},Z} 0$.

Lemma 11.3. *Notation and conditions as in Set-up 11.2. Let $\Gamma \subset [0,1] \cap \mathbb{Q}$ be a DCC set and $0 \leq B' \leq B$ a $\mathbb{Q}$-divisor such that $B' \in \Gamma$ and $\dim Z + \kappa(X/Z, B - B') = 0$. Then Z is a point and $I(d,\Gamma)(K_X + B) \sim 0$.*

Proof. Since $B - B' \geq 0$, we have $\kappa(X/Z, B - B') \geq 0$. Thus $\dim Z = 0$ and $\kappa(X/Z, B - B') = 0$. In particular, Z is a point and X is projective.

Since X is $\mathbb{Q}$-factorial, $B - B' \geq 0$ is $\mathbb{Q}$-Cartier, so there exists a rational number $\epsilon > 0$ such that $(X, B + \epsilon(B - B'))$ is klt [KM98, Corollary 2.35(2)]. Thus $(X, B + \epsilon(B - B'))$ is a $\mathbb{Q}$-factorial dlt pair, and since $K_X + B \sim_{\mathbb{Q},Z} 0$,

$$K_X + B + \epsilon(B - B') \sim_{\mathbb{Q},Z} \epsilon(B - B') \geq 0.$$

By Theorem 5.1, there exists a $(K_X + B + \epsilon(B - B'))$-MMP $\phi : X \dashrightarrow X'$ over Z which terminates with a $\mathbb{Q}$-factorial dlt pair $(X', B_{X'} + \epsilon(B - B')_{X'})$ such that $K_{X'} + B_{X'} + \epsilon(B - B')_{X'}$ is semi-ample over Z, where $B_{X'} := \phi_* B$, $B'_{X'} := \phi_* B'$, and $(B - B')_{X'} := \phi_*(B - B')$.

Since $K_X + B \sim_{\mathbb{Q},Z} 0$, ϕ is also a $(B - B')$-MMP over Z, and $K_{X'} + B_{X'} \sim_{\mathbb{Q}} 0$. Thus $(B - B')_{X'}$ is semi-ample, and by Lemma 2.20(2),

$$\kappa(X'/Z, (B - B')_{X'}) = \kappa(X/Z, B - B') = 0.$$

Hence there exists a positive integer r such that $r(B - B')_{X'}$ is a base point free Cartier divisor which defines a morphism from X' to a point, so $r(B - B')_{X'} \sim 0$. Since X' is a normal projective variety and $(B - B')_{X'} \geq 0$, we have $(B - B')_{X'} = 0$. Thus $B_{X'} = B'_{X'} \in \Gamma$, and $(X', B_{X'})$ is a projective lc pair of dimension d such that $K_{X'} + B_{X'} \sim_{\mathbb{Q}} 0$. By Theorem 2.21, there exists a rational function φ on X' such that

$$I(d,\Gamma)(K_{X'} + B_{X'}) = \operatorname{div}(\varphi).$$

Let $p : W \to X$ and $q : W \to X'$ be a common resolution of ϕ, and

$$E := p^*(K_X + B) - q^*(K_{X'} + B_{X'}).$$

Since ϕ does not extract any divisor, $q_* p^*(K_X + B) = K_{X'} + B_{X'}$, so $q_* E = 0$. Since $K_X + B \sim_{\mathbb{Q}} 0$ and $K_{X'} + B_{X'} \sim_{\mathbb{Q}} 0$, E is numerically trivial over X'. By the negativity lemma [KM98, Lemma 3.39(1)] applied to E and to $-E$, we have $E \geq 0$ and $-E \geq 0$, so $E = 0$. Thus

$$I(d,\Gamma) p^*(K_X + B) = I(d,\Gamma) q^*(K_{X'} + B_{X'}) = \operatorname{div}_W(\varphi).$$

Pushing forward by p, we obtain $I(d,\Gamma)(K_X + B) = \operatorname{div}(\varphi) \sim 0$. □

Lemma 11.4. *Notation and conditions as in Set-up 11.2. Let $\mathfrak{R}_1, \dots, \mathfrak{R}_d$ be finite subsets of $[0,1] \cap \mathbb{Q}$ containing 0 and 1 such that $\Phi(\mathfrak{R}_1) \subset \cdots \subset \Phi(\mathfrak{R}_d)$, let $B_i := B_{\Phi(\mathfrak{R}_i)}$, and let $k_i := \dim Z + \kappa(X/Z, B - B_i)$. Then $d \geq k_1 \geq \cdots \geq k_d \geq 0$, and one of the following holds:*

(1) $k_1 = d$, and X is of Fano type over Z;
(2) $k_d = 0$;
(3) $k_i = k_{i+1}$ and $0 < k_i < d$ for some $1 \leq i \leq d - 1$.

Proof. Since $\Phi(\mathfrak{R}_1) \subset \cdots \subset \Phi(\mathfrak{R}_d)$, we have $B_1 \leq \cdots \leq B_d \leq B$, so

$$B - B_1 \geq B - B_2 \geq \cdots \geq B - B_d \geq 0.$$

Since B is a $\mathbb{Q}$-divisor, $\Phi(\mathfrak{R}_j) \subset \mathbb{Q}$, and X is $\mathbb{Q}$-factorial, $B - B_j$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor for any j. By Lemma 2.20(1),

$$\kappa(X/Z, B - B_1) \geq \cdots \geq \kappa(X/Z, B - B_d) \geq \kappa(X/Z, 0) = 0.$$

Since $\kappa(X/Z, B - B_1) \leq \dim X - \dim Z = d - \dim Z$, we have $d \geq k_1 \geq \cdots \geq k_d \geq 0$.

Suppose that $k_1 = d$, that is, $\kappa(X/Z, B - B_1) = \dim X - \dim Z$. Let r be a positive integer such that $r(B - B_1)$ is Cartier. Let H_0 be a Cartier divisor on X which is ample over Z. Since $\kappa(X/Z, B - B_1) = \dim X - \dim Z$, Lemma 2.19(3) implies that the Cartier divisor $r(B - B_1)$ is π-big in the sense of [KMM87, Definition 0-3-2]. By Kodaira's lemma [KMM87, Corollary 0-3-4] applied to $r(B-B_1)$ and H_0, there exists a positive integer s such that $\pi_*\mathcal{O}_X(sr(B-B_1)-H_0) \neq 0$. Since Z is affine, there exists a Cartier divisor $F_0 \geq 0$ on X such that $F_0 \sim sr(B - B_1) - H_0$. Let $H := \frac{1}{sr}H_0$ and $F := \frac{1}{sr}F_0$. Then H is ample over Z, $F \geq 0$, and

$$B - B_1 \sim_{\mathbb{Q}} H + F.$$

In particular, $B - B_1$ is big over Z (Definition 2.9). Pick a rational number $0 < t \ll 1$ and let

$$\Theta := B - t(B - B_1) + tF = (1 - t)B + tB_1 + tF.$$

Then $\Theta \geq 0$. Since (X, B) is klt and $F \geq 0$ is $\mathbb{Q}$-Cartier, $(X, B + tF)$ is klt for $0 < t \ll 1$ [KM98, Corollary 2.35(2)]. Since $\Theta + t(B - B_1) = B + tF$ and $t(B - B_1) \geq 0$ is $\mathbb{Q}$-Cartier, (X, Θ) is klt [KM98, Corollary 2.35(1)]. Since $K_X + B \sim_{\mathbb{Q},Z} 0$,

$$K_X + \Theta \sim_{\mathbb{Q},Z} -t(B - B_1) + tF \sim_{\mathbb{Q}} -tH.$$

Thus $-(K_X + \Theta)$ is ample over Z, in particular nef and big over Z, so X is of Fano type over Z by Definition 2.9 (cf. [PS09, Lemma-Definition 2.6]). Hence (1) holds.

Suppose that $k_1 < d$ and $k_d > 0$. Then $k_1, \ldots, k_d \in \{1, \ldots, d - 1\}$. Since $\{1, \ldots, d - 1\}$ has $d - 1$ elements, there exist $1 \leq i < j \leq d$ such that $k_i = k_j$. Since $k_i \geq k_{i+1} \geq k_j$, we have $k_i = k_{i+1}$ and $0 < k_i < d$, so (3) holds. In the remaining case $k_d = 0$, and (2) holds. □

11.3. **Equal Iitaka dimensions.** In this subsection, we treat case (3) of Lemma 11.4, where X need not be of Fano type over Z. After the MMPs of Proposition 9.3, minus the generalized log canonical divisor which B_{i+1} induces on the common base is $\mathbb{Q}$-linearly equivalent over Z to the pullback of a divisor ample over Z (Lemma 9.2(2)), hence nef and big over Z. Thus, by Proposition 4.3, the generalized pair induced on an exposing model of the common base (Proposition 10.3) has an n-complement, which lifts to X by Proposition 10.4.

Proposition 11.5. *Notation and conditions as in Set-up 11.2 and Construction 11.1, and let $z \in Z$. Let $B_j := B_{\Phi_j}$ for $1 \leq j \leq d$, and assume that $\kappa(X/Z, B - B_i) = \kappa(X/Z, B - B_{i+1}) = k$ and $0 < \dim Z + k < d$ for some $1 \leq i \leq d - 1$. Then $(X/Z \ni z, B)$ has an n-complement for some $n \in \mathcal{N}_{i+1}$.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct the common base T_2 of the truncations B_i and B_{i+1}.

Since

$$\mathcal{N}_1 \cup \cdots \cup \mathcal{N}_i \subset \mathcal{N}_1 \cup \cdots \cup \mathcal{N}_{i+1},$$

we have $\Phi_i \subset \Phi_{i+1}$ (Definition 2.12), hence $0 \leq B_i \leq B_{i+1} \leq B$. Since $\Phi_i \cup \Phi_{i+1} \subset \mathbb{Q}$, the truncations B_i and B_{i+1} are $\mathbb{Q}$-divisors. Thus π, (X, B), B_i, B_{i+1}, and k satisfy the conditions of Set-up 9.1, with B_i and B_{i+1} in place of B_1 and B_2. Below, D_{X_j} is as in Set-up 9.1.

Recall that $0, 1 \in \mathfrak{R}_i$, $\Phi_i = \Phi(\mathfrak{R}_i)$, and $m_i \mid p_i$ are the integers of Theorem 3.3 for d and Φ_i (Construction 11.1). Moreover, $B_i \in \Phi_i$, $\dim X = d$, and (X, B) is klt. By Proposition 9.3 with $\mathfrak{R}_i$, B_i, and B_{i+1} in place of $\mathfrak{R}_1$, B_1, and B_2, there exist

- a $(B - B_{i+1})$-MMP $X \dashrightarrow X_1$ over Z such that $(B - B_{i+1})_{X_1}$ is semi-ample over Z and defines a contraction $f_1 : X_1 \to T_1$ over Z, and
- a $(B - B_i)_{X_1}$-MMP $X_1 \dashrightarrow X_2$ over T_1 such that $(B - B_i)_{X_2}$ is semi-ample over T_1 and defines a contraction $f_2 : X_2 \to T_2$ over T_1,

such that X_1 and X_2 are $\mathbb{Q}$-factorial, (X_1, B_{X_1}) and (X_2, B_{X_2}) are klt, and there exists a trivialization

$$K_{X_2} + B_{i,X_2} + \frac{1}{m_i}\operatorname{div}(\psi) = f_2^*(K_{T_2} + \Delta_i + \mathbf{M}_{T_2}) \tag{11.1}$$

of degree m_i over T_2, with discriminant Δ_i and moduli $\boldsymbol{b}$-divisor $\mathbf{M}$, such that $\mathbf{M}$ is nef over Z and $p_i\mathbf{M}_{V'}$ is Cartier for any smooth variety V' projective and birational over T_2. Let $\alpha : X \dashrightarrow X_2$ be the composition of $X \dashrightarrow X_1$ and $X_1 \dashrightarrow X_2$, $\tau : T_2 \to T_1$ the induced morphism, and $t_2 : T_2 \to Z$ the induced morphism.

By Lemma 9.2(1)–(3), with B_i and B_{i+1} in place of B_1 and B_2, τ is birational, $\dim T_2 = \dim Z + k$, and there exist a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $A_1 \geq 0$ on T_1 which is ample over Z and a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ on T_2 such that

$$(B - B_{i+1})_{X_2} = f_2^*\tau^* A_1 \quad \text{and} \quad B_{i+1,X_2} - B_{i,X_2} = f_2^* G.$$

By Lemma 9.2(5),

$$K_{X_2} + B_{i+1,X_2} + \frac{1}{m_i}\operatorname{div}(\psi) = f_2^*(K_{T_2} + \Delta_i + G + \mathbf{M}_{T_2}) \tag{11.2}$$

is a trivialization of degree m_i whose discriminant is $\Delta_i + G$ and whose moduli $\boldsymbol{b}$-divisor is $\mathbf{M}$. For any projective birational morphism $h : T'' \to T_2$ from a normal variety T'', let $\Delta_{T''}$ be the trace on T'' of the discriminant $\boldsymbol{b}$-divisor of (X_2, B_{i+1,X_2}) over T_2 defined by (11.2), so that

$$K_{T''} + \Delta_{T''} + \mathbf{M}_{T''} = h^*(K_{T_2} + \Delta_i + G + \mathbf{M}_{T_2})$$

(Definition 3.1).

Step 2. In this step, we show that $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, that $p_i\mathbf{M}$ is $\boldsymbol{b}$-Cartier, that

$$-(K_{T_2} + \Delta_i + G + \mathbf{M}_{T_2}) \sim_{\mathbb{Q},Z} \tau^* A_1, \tag{11.3}$$

and that $(T_2, \Delta_i + G + \tau^* A_1, \mathbf{M})/Z$ is generalized klt.

Since $B_{i,X_2} \leq B_{X_2}$ and (X_2, B_{X_2}) is klt, (X_2, B_{i,X_2}) is lc. Moreover, $B_{i,X_2} \in \Phi_i$, and $0 < \dim T_2 \leq \dim X_2 \leq d$ since $\dim T_2 = \dim Z + k > 0$. By (11.1),

$$K_{X_2} + B_{i,X_2} \sim_{\mathbb{Q}} f_2^*(K_{T_2} + \Delta_i + \mathbf{M}_{T_2}),$$

where $K_{T_2} + \Delta_i + \mathbf{M}_{T_2}$ is $\mathbb{Q}$-Cartier (Definition 3.1). By Theorem 3.3(2), applied to f_2, (X_2, B_{i,X_2}), and the trivialization (11.1) of degree m_i, with m_i and p_i in place of m and p, we have $\Delta_i \geq 0$ and $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier. Thus $\mathbf{M}$ descends to a smooth variety V' projective and birational over T_2. Since $p_i\mathbf{M}_{V'}$ is Cartier, $p_i\mathbf{M}$ is $\boldsymbol{b}$-Cartier.

Since α does not extract any divisor and $K_X + B \sim_{\mathbb{Q},Z} 0$, we have $K_{X_2} + B_{X_2} \sim_{\mathbb{Q},Z} 0$. By **Step 1**,

$$B_{X_2} = B_{i+1,X_2} + (B - B_{i+1})_{X_2} = B_{i+1,X_2} + f_2^*\tau^* A_1,$$

so by (11.2),

$$f_2^*(K_{T_2} + \Delta_i + G + \tau^* A_1 + \mathbf{M}_{T_2}) = K_{X_2} + B_{X_2} + \frac{1}{m_i}\operatorname{div}(\psi) \sim_{\mathbb{Q},Z} 0.$$

Thus there exist a positive integer r, a Cartier divisor L on Z, and a rational function φ on X_2 such that $R := r(K_{T_2} + \Delta_i + G + \tau^* A_1 + \mathbf{M}_{T_2}) - t_2^* L$ is Cartier and $f_2^* R = \operatorname{div}(\varphi)$. By Lemma 2.17(2), $R = \operatorname{div}(\varphi_0)$ for a rational function φ_0 on T_2. This implies (11.3). Since A_1 is ample over Z and τ is birational, $\tau^* A_1$ is nef and big over Z.

By Lemma 3.8(1) with $P = \tau^* A_1$,

$$K_{X_2} + B_{X_2} + \frac{1}{m_i}\operatorname{div}(\psi) = f_2^*(K_{T_2} + \Delta_i + G + \tau^* A_1 + \mathbf{M}_{T_2})$$

is a trivialization of degree m_i of $K_{X_2} + B_{X_2}$ over T_2, whose discriminant $\boldsymbol{b}$-divisor has trace $\Delta_{T''} + h^*\tau^* A_1$ on T'' for any $h : T'' \to T_2$ as in **Step 1**, and whose moduli $\boldsymbol{b}$-divisor is $\mathbf{M}$. Since

(X_2, B_{X_2}) is klt, by Lemma 3.8(3), the coefficients of $\Delta_{T''} + h^*\tau^*A_1$ are less than 1 for any such h. Since $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier and nef over Z, and

$$K_{T''} + \Delta_{T''} + h^*\tau^*A_1 + \mathbf{M}_{T''} = h^*(K_{T_2} + \Delta_i + G + \tau^*A_1 + \mathbf{M}_{T_2}),$$

the generalized log discrepancies of $(T_2, \Delta_i + G + \tau^*A_1, \mathbf{M})$ are positive (Definition 2.8). Since $\Delta_i \geq 0$, $G \geq 0$, and $A_1 \geq 0$, $(T_2, \Delta_i + G + \tau^*A_1, \mathbf{M})/Z$ is generalized klt.

Step 3. In this step, we construct an exposing model $\sigma : T' \to T_2$ for α, and an n-complement on T' for some $n \in \mathcal{N}_{i+1}$.

We show that X, X_2, Z, T_2, α, B_{i+1}, B_{i+1,X_2}, f_2, p_i, ψ^{p_i/m_i}, $\Delta_i + G$, and $\mathbf{M}$ satisfy the conditions of Set-up 10.2, with X_2, B_{i+1}, B_{i+1,X_2}, f_2, T_2, p_i, ψ^{p_i/m_i}, and $\Delta_i + G$ in place of Y, A, A_Y, g, T, p, ψ, and Δ_T. The map α is birational over Z, and $X_2 \to Z$ is a contraction, since each step of $X \dashrightarrow X_2$ is a divisorial contraction or a flip over Z (the pushforward of the structure sheaf to Z does not change under a divisorial contraction or a flip, by Zariski's main theorem [Har77, Corollary III.11.4]). Since X and X_2 are $\mathbb{Q}$-factorial, $K_X + B_{i+1}$ and $K_{X_2} + B_{i+1,X_2}$ are $\mathbb{Q}$-Cartier, and $\alpha_* B_{i+1} = B_{i+1,X_2}$. By Lemma 9.2(4),

$$\beta_2^*(K_{X_2} + B_{i+1,X_2}) \geq \beta^*(K_X + B_{i+1})$$

for a common resolution $\beta : W \to X$ and $\beta_2 : W \to X_2$ of α. Moreover, $\dim T_2 > 0$. Since $m_i \mid p_i$,

$$\frac{1}{p_i}\operatorname{div}(\psi^{p_i/m_i}) = \frac{1}{m_i}\operatorname{div}(\psi),$$

so (11.2) is also a trivialization

$$K_{X_2} + B_{i+1,X_2} + \frac{1}{p_i}\operatorname{div}(\psi^{p_i/m_i}) = f_2^*(K_{T_2} + \Delta_i + G + \mathbf{M}_{T_2})$$

of degree p_i, whose right-hand side is the right-hand side of (11.2). Hence the discriminant is $\Delta_i + G$, the moduli $\boldsymbol{b}$-divisor is $\mathbf{M}$, and the trace of the discriminant $\boldsymbol{b}$-divisor on T'' is $\Delta_{T''}$ for any $h : T'' \to T_2$ as in **Step 1** (Definition 3.1).

Since $B_{i+1,X_2} \leq B_{X_2}$, (X_2, B_{i+1,X_2}) is klt. By **Step 2**, $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, and by **Step 1**, $\mathbf{M}$ is nef over Z. By Proposition 10.3, there exists an exposing model $\sigma : T' \to T_2$ for α such that T' is $\mathbb{Q}$-factorial, $\Delta_{T'} \geq 0$, and $(T', \Delta_{T'}, \mathbf{M})/Z$ is generalized klt. In particular, $\Delta_{T'} \in [0, 1]$.

We check the hypotheses of Proposition 4.3 for $(T'/Z \ni z, \Delta_{T'}, \mathbf{M})$. Since $X_2 \to Z$, f_2, and σ are contractions ($\sigma_*\mathcal{O}_{T'} = \mathcal{O}_{T_2}$ since T_2 is normal), the induced morphism $T' \to Z$ is a contraction. By Lemma 9.2(1), $\dim T' = \dim T_2 = \dim Z + k \leq d - 1$. By **Step 2**, $p_i\mathbf{M}$ is $\boldsymbol{b}$-Cartier. Recall that $A_1 \geq 0$ is the divisor on T_1 of **Step 1** which is ample over Z. Since $\sigma^*\tau^*A_1 \geq 0$, we have $\Delta_{T'} + \sigma^*\tau^*A_1 \geq \Delta_{T'}$, and by (11.3),

$$K_{T'} + \Delta_{T'} + \sigma^*\tau^*A_1 + \mathbf{M}_{T'} = \sigma^*(K_{T_2} + \Delta_i + G + \tau^*A_1 + \mathbf{M}_{T_2}) \sim_{\mathbb{Q},Z} 0.$$

Since a Cartier divisor on Z is principal over a neighborhood of z,

$$K_{T'} + \Delta_{T'} + \sigma^*\tau^*A_1 + \mathbf{M}_{T'} \sim_{\mathbb{Q}} 0$$

over a neighborhood of z. Since $(T', \Delta_{T'} + \sigma^*\tau^*A_1, \mathbf{M})$ is the crepant pullback of $(T_2, \Delta_i + G + \tau^*A_1, \mathbf{M})$, it is generalized klt by **Step 2**. Thus $(T'/Z \ni z, \Delta_{T'} + \sigma^*\tau^*A_1, \mathbf{M})$ is a klt $\mathbb{R}$-complement of $(T'/Z \ni z, \Delta_{T'}, \mathbf{M})$ (Definition 2.11). Moreover, by (11.3),

$$-(K_{T'} + \Delta_{T'} + \mathbf{M}_{T'}) \sim_{\mathbb{Q},Z} \sigma^*\tau^*A_1,$$

which is nef and big over Z. By Proposition 4.3, there exist $n \in \mathcal{J}(d-1, p_i, l) = \mathcal{N}_{i+1}$ and an n-complement $(T'/Z \ni z, \Delta^+_{T'}, \mathbf{M})$ of $(T'/Z \ni z, \Delta_{T'}, \mathbf{M})$. By Definition 2.11(2), $n\mathbf{M}$ is $\boldsymbol{b}$-Cartier, and over a neighborhood of z,

- $(T', \Delta^+_{T'}, \mathbf{M})$ is generalized lc,
- $n(K_{T'} + \Delta^+_{T'} + \mathbf{M}_{T'}) \sim 0$, and

- the inequality
$$n\Delta^+_{T'} \geq n\lfloor \Delta_{T'} \rfloor + \lfloor (n+1)\{\Delta_{T'}\} \rfloor$$
holds.

Moreover, $p_i \mid n$.

Step 4. In this step, we lift $(T'/Z \ni z, \Delta^+_{T'}, \mathbf{M})$ to an n-complement of $(X/Z \ni z, B)$ and conclude the proof.

Possibly shrinking Z to an affine neighborhood of z, we may assume that $(T', \Delta^+_{T'}, \mathbf{M})$ is generalized lc and $n(K_{T'} + \Delta^+_{T'} + \mathbf{M}_{T'}) \sim 0$, and that the inequality of **Step 3** holds on T'.

Since $\mathbf{M}$ is $\boldsymbol{b}$-$\mathbb{Q}$-Cartier, there exists a projective birational morphism $\eta : V \to T'$ from a smooth variety V such that $\mathbf{M}$ descends to V. For any prime divisor P on X which does not dominate T_2, let Q_P be as in Definition 10.1 for σ, and let Q^V_P be the strict transform of Q_P on V. Since T' is normal, η is an isomorphism over the generic point of Q_P, so the induced rational map $X \dashrightarrow V$, which is the composite of $X \dashrightarrow T'$ and η^{-1}, maps P onto the prime divisor Q^V_P. Thus $\sigma \circ \eta : V \to T_2$ is an exposing model for α, with Q^V_P in place of Q_P (Definition 10.1). By **Step 1**,
$$K_V + \Delta_V + \mathbf{M}_V = \eta^*(K_{T'} + \Delta_{T'} + \mathbf{M}_{T'}),$$
and Δ_V is the $\mathbb{Q}$-divisor of Set-up 10.2(6). Let Δ^+_V be the $\mathbb{Q}$-divisor on V such that
$$K_V + \Delta^+_V + \mathbf{M}_V = \eta^*(K_{T'} + \Delta^+_{T'} + \mathbf{M}_{T'}).$$

We check the hypotheses of Proposition 10.4, with V and $\sigma \circ \eta$ in place of T' and σ. Since α is a composition of divisorial contractions and flips, α^{-1} does not contract any divisor. The pair (X_2, B_{i+1,X_2}) is klt by **Step 3**, V is smooth, $\mathbf{M}$ descends to V, and $p_i \mid n$.

(1) Let $\rho : V'' \to V$ be a projective birational morphism such that $\eta \circ \rho : V'' \to T'$ is a log resolution of $(T', \operatorname{Supp} \Delta^+_{T'})$, and write
$$K_{V''} + \Delta^+_{V''} + \mathbf{M}_{V''} := \rho^*(K_V + \Delta^+_V + \mathbf{M}_V).$$
Since $\mathbf{M}$ descends to V, $\mathbf{M}_{V''} = \rho^*\mathbf{M}_V$, so $K_{V''} + \Delta^+_{V''} = \rho^*(K_V + \Delta^+_V)$. Every prime divisor over T' is a prime divisor on such a V'', so
$$a(E, T', \Delta^+_{T'}, \mathbf{M}) = 1 - \operatorname{mult}_E \Delta^+_{V''} = a(E, V, \Delta^+_V)$$
for any prime divisor E over V (Definition 2.8). Since $(T', \Delta^+_{T'}, \mathbf{M})$ is generalized lc, (V, Δ^+_V) is sub-lc.

(2) We have
$$n(K_V + \Delta^+_V + \mathbf{M}_V) = \eta^*\big(n(K_{T'} + \Delta^+_{T'} + \mathbf{M}_{T'})\big) \sim 0.$$

(3) Let P be a prime divisor on X which does not dominate T_2. Since T' is normal, η is an isomorphism over the generic point of Q_P. Thus
$$\operatorname{mult}_{Q^V_P} \Delta_V = \operatorname{mult}_{Q_P} \Delta_{T'} =: \delta_P \quad \text{and} \quad \operatorname{mult}_{Q^V_P} \Delta^+_V = \operatorname{mult}_{Q_P} \Delta^+_{T'}.$$
By the inequality of **Step 3** at Q_P,
$$n \operatorname{mult}_{Q^V_P} \Delta^+_V \geq n\lfloor \delta_P \rfloor + \lfloor (n+1)\{\delta_P\} \rfloor.$$

By Proposition 10.4, $(X/Z \ni z, B_{i+1})$ has an n-complement $(X/Z \ni z, B^+)$. By Definition 2.10(2), $B^+ \geq 0$ is a $\mathbb{Q}$-divisor such that, over a neighborhood of z, (X, B^+) is lc, $n(K_X + B^+) \sim 0$, and
$$nB^+ \geq n\lfloor B_{i+1} \rfloor + \lfloor (n+1)\{B_{i+1}\} \rfloor.$$
Since (X, B) is klt, $B \in [0, 1]$. Moreover, $n \in \mathcal{N}_{i+1}$ and
$$B_{i+1} = B_{\Phi_{i+1}} = B_{\Gamma(\mathcal{N}_1 \cup \cdots \cup \mathcal{N}_{i+1}, \Phi)}.$$
By Lemma 2.14(4) with $B' = B_{i+1}$ and $\mathcal{N} = \mathcal{N}_1 \cup \cdots \cup \mathcal{N}_{i+1}$, $(X/Z \ni z, B^+)$ is an n-complement of $(X/Z \ni z, B)$. □

12. Proofs of the main theorems

In this section, we prove Theorems 1.3, 1.5, 12.2, and 1.6.

12.1. **Finite rational coefficients.** In this subsection, we prove Theorem 1.3 by combining Proposition 5.3 with Theorem 8.3, which also implies Theorem 1.5.

Proof of Theorem 1.3. Let $N := N(d, \Gamma)$ be the integer of Construction 8.2. Then N depends only on d and Γ, and $Nb \in \mathbb{Z}$ for any $b \in \Gamma$ since $\delta \mid N$.

Let $(X/Z \ni z, B)$ be an $\mathbb{R}$-complementary pair such that $\dim X \leq d$ and $B \in \Gamma$.

By Proposition 5.3, after replacing Z with an affine neighborhood of z, there exist a $\mathbb{Q}$-factorial dlt pair (Y, B_Y) with $B_Y \in \Gamma \cup \{1\}$, a birational map $X \dashrightarrow Y$ over Z, contractions $g : Y \to T$ and $T \to Z$, and a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H on T which is ample over Z, such that

$$-(K_Y + B_Y) \sim_{\mathbb{Q}} g^*H,$$

and such that, for any positive integer n, if $(Y/Z \ni z, B_Y)$ has a monotonic n-complement, then $(X/Z \ni z, B)$ has a monotonic n-complement.

The dlt pair (Y, B_Y) has finitely many lc centers, namely the irreducible components of the intersections of components of $\lfloor B_Y \rfloor$ [Kol13, Theorem 4.16]. Let $V_1, \dots, V_r$ be the lc centers of (Y, B_Y) whose images on T do not intersect the fiber T_z of $T \to Z$ over z, and let $\Sigma \subset Z$ be the union of the images of $V_1, \dots, V_r$ on Z. Since $Y \to Z$ is projective, Σ is closed, and $z \notin \Sigma$. Possibly shrinking Z to an affine neighborhood of z in $Z \backslash \Sigma$, we may assume that the image on T of any lc center of (Y, B_Y) intersects T_z.

The composite of the contractions $g : Y \to T$ and $T \to Z$ is a contraction $Y \to Z$, and $\dim Y = \dim X \leq d$. Thus $(Y/Z \ni z, B_Y)$, $g : Y \to T$, $T \to Z$, and H satisfy the conditions of Set-up 8.1 (with Y, B_Y, Z, z in place of X, B, U, u). By Theorem 8.3, applied to $(Y/Z \ni z, B_Y)$, $g : Y \to T$, $T \to Z$, and H, the pair $(Y/Z \ni z, B_Y)$ has a monotonic N-complement. Hence $(X/Z \ni z, B)$ has a monotonic N-complement by the last sentence of Proposition 5.3. □

Proof of Theorem 1.5. This follows from Theorem 1.3, [OAI26a, Theorem 11.1], and [HLS19, Theorem 8.25]. □

12.2. **Coefficients in the unit interval.** In this subsection, we prove Theorem 1.6. We first prove a stronger statement, Theorem 12.2, in which only the pair with the boundary replaced by a truncation is assumed to be klt $\mathbb{R}$-complementary, and the boundary itself need not be lc (cf. [CHX24, Remark 2.17(1)]). Both proofs pass to a small $\mathbb{Q}$-factorialization, and the following lemma pushes complements back down without changing the index.

Lemma 12.1. *Let $(X/Z \ni z, B)$ be a pair, $q : X' \to X$ a small projective birational morphism from a normal quasi-projective variety X', and B' the strict transform of B on X'. Then $(X'/Z \ni z, B')$ is a pair, and for any positive integer n and any n-complement $(X'/Z \ni z, B'^+)$ of $(X'/Z \ni z, B')$, $(X/Z \ni z, q_*B'^+)$ is an n-complement of $(X/Z \ni z, B)$.*

Proof. Since q has no exceptional divisor, $q^*(K_X + B) = K_{X'} + B'$ over a neighborhood of z, and $X' \to Z$ is a contraction ($q_*\mathcal{O}_{X'} = \mathcal{O}_X$ since X is normal). Thus $(X'/Z \ni z, B')$ is a pair. Let $(X'/Z \ni z, B'^+)$ be an n-complement of $(X'/Z \ni z, B')$. There exist an open neighborhood Z_1 of z in Z and a rational function φ on X' such that, over Z_1, (X', B'^+) is lc and $n(K_{X'} + B'^+) = \operatorname{div}_{X'}(\varphi)$, so pushing forward implies $n(K_X + q_*B'^+) = \operatorname{div}_X(\varphi)$, and pulling back again implies

$$q^*(K_X + q_*B'^+) = K_{X'} + B'^+.$$

Hence $(X, q_*B'^+)$ is lc over Z_1. The strict transform is a bijection between the prime divisors on X and those on X' which preserves the coefficients of B, B' and of $q_*B'^+$, B'^+, so the inequality $nq_*B'^+ \geq n\lfloor B \rfloor + \lfloor (n+1)\{B\} \rfloor$ follows from $nB'^+ \geq n\lfloor B' \rfloor + \lfloor (n+1)\{B'\} \rfloor$. □

Theorem 12.2. *Let d and l be positive integers, $\mathfrak{R} \subset [0,1] \cap \mathbb{Q}$ a finite set such that $0, 1 \in \mathfrak{R}$, and $\Phi := \Phi(\mathfrak{R})$. Then there exists a finite set $\mathcal{N}$ of positive integers divisible by l, depending only on d, l, and $\mathfrak{R}$, satisfying the following. Let $(X/Z \ni z, B)$ be a pair of dimension d such that $B \in [0,1]$ and $(X/Z \ni z, B_{\Gamma(\mathcal{N},\Phi)})$ is klt $\mathbb{R}$-complementary. Then $(X/Z \ni z, B)$ has an n-complement for some $n \in \mathcal{N}$.*

Proof of Theorem 12.2. The proof is long, so we divide it into several steps.

Step 1. In this step, we fix $\mathcal{N}$ and reduce to a $\mathbb{Q}$-factorial klt pair over an affine base.

Let $\mathcal{N} := \mathcal{N}(d, l, \mathfrak{R})$ be the set of Construction 11.1. Any member of $\mathcal{N}_1 = \mathcal{J}(d, 1, l)$ and of $\mathcal{N}_{i+1} = \mathcal{J}(d-1, p_i, l)$ is divisible by l (Proposition 4.3), and $l \mid n_0$. Thus any member of $\mathcal{N}$ is divisible by l, and $\mathcal{N}$ depends only on d, l, and $\mathfrak{R}$.

Let $(X/Z \ni z, B)$ be a pair as in the theorem, and $(X/Z \ni z, \Delta)$ a klt $\mathbb{R}$-complement of $(X/Z \ni z, B_{\Gamma(\mathcal{N},\Phi)})$.

Since $\Gamma(\mathcal{N}, \Phi) \subset \mathbb{Q}$, $B_{\Gamma(\mathcal{N},\Phi)}$ is a $\mathbb{Q}$-divisor. By Lemma 5.2 applied to $(X/Z \ni z, \Delta)$, there exists a $\mathbb{Q}$-complement $(X/Z \ni z, B_0^+)$ of $(X/Z \ni z, B_{\Gamma(\mathcal{N},\Phi)})$ such that, over some neighborhood of z, (X, B_0^+) is klt and $K_X + B_0^+ \sim_{\mathbb{Q}} 0$. Possibly shrinking Z to a neighborhood of z, we may assume that Z is affine, $K_X + B$ is $\mathbb{R}$-Cartier, (X, B_0^+) is klt, and $K_X + B_0^+ \sim_{\mathbb{Q},Z} 0$.

Since (X, B_0^+) is klt, there exists a small $\mathbb{Q}$-factorialization $q : X' \to X$ (q is projective, being constructed by an MMP over X in [BCHM10, Corollary 1.4.3]). Let $B' := q_*^{-1}B$ and $B_0'^+ := q_*^{-1}B_0^+$. Since q is small,

$$K_{X'} + B_0'^+ = q^*(K_X + B_0^+).$$

Thus $(X', B_0'^+)$ is klt, $B_0'^+$ is a $\mathbb{Q}$-divisor, and $K_{X'} + B_0'^+ \sim_{\mathbb{Q},Z} 0$. Hence d, the contraction $X' \to Z$, and $(X', B_0'^+)$ satisfy the conditions of Set-up 11.2 (with X' and $B_0'^+$ in place of X and B). In the rest of the proof, Lemma 11.4, Lemma 11.3, and Proposition 11.5 are applied with X' and $B_0'^+$ in place of X and B.

Step 2. In this step, we show that $(X'/Z \ni z, B_0'^+)$ has an n-complement $(X'/Z \ni z, B'^+)$ for some $n \in \mathcal{N}$.

Let $\mathfrak{R}_1, \dots, \mathfrak{R}_d$ be the finite sets of Construction 11.1. Then $0, 1 \in \mathfrak{R}_i$ and $\Phi_i = \Phi(\mathfrak{R}_i)$ for any i. For finite sets $\mathcal{N}' \subset \mathcal{N}''$ of positive integers, we have $\Gamma(\mathcal{N}', \Phi) \subset \Gamma(\mathcal{N}'', \Phi)$, by taking $u_n = 0$ for $n \in \mathcal{N}'' \backslash \mathcal{N}'$ in Definition 2.12. Since

$$\mathcal{N}_1 \subset \mathcal{N}_1 \cup \mathcal{N}_2 \subset \cdots \subset \mathcal{N}_1 \cup \cdots \cup \mathcal{N}_d,$$

we obtain

$$\Phi_1 \subset \Phi_2 \subset \cdots \subset \Phi_d.$$

For $1 \le i \le d$, let

$$k_i := \dim Z + \kappa\big(X'/Z, B_0'^+ - (B_0'^+)_{\Phi_i}\big).$$

By Lemma 11.4, one of the following three cases holds.

Case 1: $k_1 = d$, and X' is of Fano type over Z. Consider the generalized pair $(X'/Z \ni z, B_0'^+, \mathbf{M})$ with $\mathbf{M} = 0$. Then $X' \to Z$ is a contraction, $\dim X' = d$, $B_0'^+ \in [0,1]$, $1 \cdot \mathbf{M}$ is $\boldsymbol{b}$-Cartier, and $(X'/Z \ni z, B_0'^+, \mathbf{M})$ is klt $\mathbb{R}$-complementary, since $(X', B_0'^+)$ is klt and $K_{X'} + B_0'^+ \sim_{\mathbb{Q},Z} 0$. By Proposition 4.3 (with 1 in place of p), $(X'/Z \ni z, B_0'^+)$ has an n-complement for some $n \in \mathcal{J}(d, 1, l) = \mathcal{N}_1$ (as $\mathbf{M} = 0$, Definition 2.11(2) is Definition 2.10(2)).

Case 2: $k_d = 0$. By Lemma 2.14(1), $\Phi_d = \Phi(\mathfrak{R}_d) \subset [0,1] \cap \mathbb{Q}$ satisfies the DCC. The $\mathbb{Q}$-divisor $(B_0'^+)_{\Phi_d}$ satisfies $0 \le (B_0'^+)_{\Phi_d} \le B_0'^+$ and $(B_0'^+)_{\Phi_d} \in \Phi_d$. Since $k_d = 0$, Lemma 11.3 (applied to Φ_d and $(B_0'^+)_{\Phi_d}$) implies that Z is a point and

$$I(d, \Phi_d)(K_{X'} + B_0'^+) \sim 0.$$

Since $I(d, \Phi_d) \mid n_0$, $(X'/Z \ni z, B_0'^+)$ is an n_0-complement of $(X'/Z \ni z, B_0'^+)$, and $n_0 \in \mathcal{N}$.

Case 3: $k_i = k_{i+1}$ and $0 < k_i < d$ for some $1 \le i \le d-1$. Then

$$\kappa\big(X'/Z, B_0'^+ - (B_0'^+)_{\Phi_i}\big) = \kappa\big(X'/Z, B_0'^+ - (B_0'^+)_{\Phi_{i+1}}\big)$$

and $0 < k_i < d$. By Proposition 11.5, $(X'/Z \ni z, B_0'^+)$ has an n-complement for some $n \in \mathcal{N}_{i+1}$.

In any case, $(X'/Z \ni z, B_0'^+)$ has an n-complement $(X'/Z \ni z, B'^+)$ for some $n \in \mathcal{N}$.

Step 3. In this step, we show that $(X/Z \ni z, q_*B'^+)$ is an n-complement of $(X/Z \ni z, B)$.

Recall that $q : X' \to X$ is the small $\mathbb{Q}$-factorialization of **Step 1**, $B' = q_*^{-1}B$, and $B_0'^+ = q_*^{-1}B_0^+$. By Lemma 12.1, $(X'/Z \ni z, B')$ is a pair, and $B' \in [0,1]$. Since the truncation is taken coefficientwise and $B_0^+ \ge B_{\Gamma(\mathcal{N},\Phi)}$, we have

$$B_0'^+ \ge q_*^{-1}\big(B_{\Gamma(\mathcal{N},\Phi)}\big) = B'_{\Gamma(\mathcal{N},\Phi)},$$

and $B_0'^+ \in [0,1]$. Since $(X'/Z \ni z, B'^+)$ is an n-complement of $(X'/Z \ni z, B_0'^+)$, over some neighborhood of z, (X', B'^+) is lc, $n(K_{X'} + B'^+) \sim 0$, and

$$nB'^+ \ge n\lfloor B_0'^+ \rfloor + \lfloor (n+1)\{B_0'^+\} \rfloor.$$

By Lemma 2.14(4), applied to the pair $(X'/Z \ni z, B')$ and the $\mathbb{Q}$-divisor $B_0'^+$, $(X'/Z \ni z, B'^+)$ is an n-complement of $(X'/Z \ni z, B')$. By Lemma 12.1, $(X/Z \ni z, q_*B'^+)$ is an n-complement of $(X/Z \ni z, B)$. □

Proof of Theorem 1.6. Let $\mathfrak{R} := \{0,1\}$, $\Phi := \Phi(\mathfrak{R})$, and $\mathcal{N}$ the finite set of Theorem 12.2 for d, l, and $\mathfrak{R}$. Then any member of $\mathcal{N}$ is divisible by l, and $\mathcal{N}$ depends only on d and l.

Let $(X/Z \ni z, B)$ be a pair of dimension d with $B \in [0,1]$, and $(X/Z \ni z, B_0^+)$ a klt $\mathbb{R}$-complement of $(X/Z \ni z, B)$. Possibly shrinking Z to a neighborhood of z, we may assume that (X, B_0^+) is klt and $K_X + B_0^+ \sim_{\mathbb{R}} 0$ over Z. Since (X, B_0^+) is klt, there exists a small $\mathbb{Q}$-factorialization $q : X' \to X$ (q is projective, being constructed by an MMP over X in [BCHM10, Corollary 1.4.3]). Let $B' := q_*^{-1}B$ and $B_0'^+ := q_*^{-1}B_0^+$.

By Lemma 12.1, $(X'/Z \ni z, B')$ is a pair. Since X' is $\mathbb{Q}$-factorial and $B'_{\Gamma(\mathcal{N},\Phi)}$ is a $\mathbb{Q}$-divisor, $K_{X'} + B'_{\Gamma(\mathcal{N},\Phi)}$ is $\mathbb{Q}$-Cartier, so $(X'/Z \ni z, B'_{\Gamma(\mathcal{N},\Phi)})$ is a pair. Since q is small, we have

$$K_{X'} + B_0'^+ = q^*(K_X + B_0^+),$$

so $(X', B_0'^+)$ is klt and $K_{X'} + B_0'^+ \sim_{\mathbb{R}} 0$. Moreover,

$$B_0'^+ \ge B' \ge B'_{\Gamma(\mathcal{N},\Phi)}.$$

Thus $(X'/Z \ni z, B_0'^+)$ is a klt $\mathbb{R}$-complement of $(X'/Z \ni z, B'_{\Gamma(\mathcal{N},\Phi)})$. Since $\dim X' = d$ and $B' \in [0,1]$, Theorem 12.2 implies that $(X'/Z \ni z, B')$ has an n-complement $(X'/Z \ni z, B'^+)$ for some $n \in \mathcal{N}$. By Lemma 12.1, $(X/Z \ni z, q_*B'^+)$ is an n-complement of $(X/Z \ni z, B)$. □

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